



Cortical and subcortical networks involved in multisensory processing and sensory-motor transformations

Maxime Gaudet-Trafit

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**THESE de DOCTORAT DE
L'UNIVERSITE CLAUDE BERNARD LYON 1**

**Ecole Doctorale N° 476
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Maxime Gaudet-trafit

**Cortical and subcortical networks involved in
multisensory processing and sensory-motor
transformations**

-

**Réseaux corticaux et sous-corticaux impliqués
dans le traitement multisensoriel et les
transformations sensori-motrices**

Devant le jury composé de :

Pouget, Pierre	DR., CNRS, ICM, Paris	Président
Latinus, Marianne	CR, INSERM, CRCN Tours	Rapporteur
Durand, Jean-Baptiste	DR., CNRS CERCO, Toulouse	Rapporteur
Cristofori, Irene	Maître de conférences, ISCMJ/ UCBL, Lyon	Examinatrice
Pouget, Pierre	DR., CNRS, ICM, Paris	Examineur
Ben Hamed, Suliann	D.R., ISCMJ, Lyon	Directrice de thèse
Heed, Tobias	Professor., Paris Lodron University Salzburg	Co-directeur de thèse

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« Non quia difficilia sunt non audemus, sed quia non audemus difficilia sunt »

« Ce n'est pas parce que les chose sont difficiles que l'on n'essaie pas, c'est parce que l'on n'essaie pas qu'elles sont difficiles »

Sénèque, Lettres à Lucilius (Epistulae morales ad Lucilium), Lettre 104, 26

Résumé

Au quotidien, nous évoluons dans notre environnement grâce à la capacité de notre cerveau à caractériser les informations s’y trouvant. Pour cela, il intègre les signaux provenant de différentes modalités sensorielles et construit une représentation multisensorielle du monde. Par exemple, nous associons automatiquement la vue de la personne qui nous parle avec sa voix, au point que, dans l’effet ventriloque, si l’on présente le son d’une voix synchronisée aux mouvements de lèvres d’une personne, nous attribuons la voix à la personne en considérant que la position spatiale des deux informations est la même. Pour comprendre ces mécanismes d’intégration multisensoriels, il est important de déterminer les aires cérébrales impliquées dans ces traitements uni- et multisensoriels et leur coordination au sein de réseaux fonctionnels.

Alors que le rôle des aires corticales dans ces processus est de mieux en mieux connu, la contribution des régions sous-corticales l’est moins. Dans ce contexte, le thalamus et notamment son noyau principal, le pulvinar est une région de grand intérêt de par ses connections variées avec le reste du cerveau et son implication dans des traitements cognitifs de haut niveau tels que les processus attentionnels et multisensoriels. La première partie de ce travail de thèse s’articule autour du rôle du pulvinar dans le traitement unimodal visuel, auditif et tactile ainsi que multimodal visuo-tactile et audio-visuel chez l’humain en IRMf. Nous étudions l’implication du pulvinar dans les traitements des visages, des voix et des stimulations tactiles dans des tâches passives unisensorielles ou actives multisensorielles. Nous confirmons le rôle du pulvinar dans le traitement des visages et nous montrons son implication dans le traitement de l’information tactile. Nous montrons que le pulvinar adapte sa connectivité fonctionnelle avec le domaine cortical selon la tâche en cours. Par exemple, lorsque l’on regarde des visages, le pulvinar se déconnecte des régions visuelles primaires pour se connecter avec des régions visuelles impliquées dans le traitement des visages. De même, en fonction de la localisation de la stimulation tactile (visage, mains, pieds), le pulvinar montre des profils de connectivité différents avec le cortex. Ceci renforce l’hypothèse de son implication dans les traitements multisensoriels et dans la modulation d’aires corticales.

La perception de notre environnement est souvent associée à la production d’une réponse comportementale telle qu’une action motrice. Une stimulation tactile sur le bras peut induire un mouvement de l’autre bras pour toucher la partie stimulée, ou bien un mouvement des yeux, pour vérifier l’origine de la sensation. Pour ce faire, le codage initial en référence anatomique doit être transformé en codage allocentrique ou externe. Ce processus est grandement complexifié dans des configurations corporelles spécifiques, par exemple, bras croisés. La seconde partie de ce travail de thèse porte sur l’identification, par IRMf, des régions cérébrales impliquées dans ces transformations de coordonnées entre information tactile et visuelle en fonction de la posture corporelle. L’objectif est de comparer les activations cérébrales lors de saccades oculaires en direction de stimulations tactiles ou visuelles lorsque

l'information visuelle est alignée avec l'information tactile (bras non croisés) ou pas (bras croisés). Nous montrons un codage allocentrique (i.e. dans un repère externe) très précoce de l'information tactile grâce à l'implication du cortex pariétal supérieur et du cortex préfrontal.

En conclusion, nos résultats montrent que la perception de l'environnement et la préparation d'actions motrices sont largement supportées par des mécanismes multisensoriels complexes et dynamiques impliquant à la fois des bases neurales corticales et sous-corticales. Nous apportons ainsi des nouvelles connaissances pour la compréhension globale des boucles perception-action.

Mots-clés : Perception multisensorielle, Pulvinar, IRMf, transformations somato-motrices, posture corporelle, apprentissage profond, réseaux de neurones de convolution

Abstract

In our daily lives, we navigate our environment thanks to the brain's ability to process the information around us. It integrates signals from different sensory modalities and constructs a multisensory representation of the world. For example, we automatically associate the sight of the person speaking to us with their voice. In the ventriloquist effect, if the sound of a voice is synchronized with someone's lip movements, we attribute the voice to that person, assuming the spatial location of both is the same. To understand these multisensory integration mechanisms, it's essential to identify the brain areas involved in unisensory and multisensory processing and their coordination within functional networks.

While the role of cortical areas in these processes is becoming increasingly well understood, the contribution of subcortical regions is less known. In this context, the thalamus, and particularly its main nucleus, the pulvinar, is of great interest due to its varied connections with the rest of the brain and its involvement in high-level cognitive functions such as attention and multisensory processes. The first part of this thesis focuses on the role of the pulvinar in unimodal visual, auditory, and tactile processing, as well as visuo-tactile and audio-visual multimodal processing in humans, studied with fMRI. We investigate the pulvinar's involvement in processing faces, voices, and tactile stimuli in passive unisensory tasks or active multisensory tasks. We confirm the pulvinar's role in face processing and demonstrate its involvement in tactile information processing. We show that the pulvinar adapts its functional connectivity with cortical areas based on the task. For example, when viewing faces, the pulvinar disconnects from primary visual regions to connect with regions involved in face processing. Similarly, depending on the location of the tactile stimulation (face, hands, feet), the pulvinar exhibits different connectivity profiles with the cortex. This supports the hypothesis of its involvement in multisensory processing and modulation of cortical areas.

Perception of our environment is often associated with producing a behavioral response, such as a motor action. A tactile stimulus on the arm can trigger movement in the other arm to touch the stimulated area or an eye movement to check the source of the sensation. To do so, the initial anatomical coding must be transformed into an allocentric or external spatial coding. This process becomes much more complex in specific body configurations, such as crossed arms. The second part of this thesis aims to identify, via fMRI, the brain regions involved in these coordinate transformations between tactile and visual information depending on body posture. The objective is to compare brain activations during eye saccades directed toward tactile or visual stimuli when visual and tactile information is aligned (uncrossed arms) versus when it is not (crossed arms). We demonstrate early allocentric (i.e., external frame of reference) coding of tactile information, thanks to the involvement of the superior parietal cortex and prefrontal cortex.

In conclusion, our results show that the perception of the environment and the preparation of motor actions are largely supported by complex and dynamic multisensory mechanisms involving both cortical and subcortical neural bases. Thus, we provide new insights into the overall understanding of the perception-action loops.

Keywords: Multisensory perception, Pulvinar, fMRI, somatomotor transformations, body posture, deep learning, convolution neural network

Preface

The work done in this PhD has been realized with a high level of collaboration with team members and collaborators outside of the team, that is why the pronoun “we” will be used to present the experiment preparation, results and interpretations.

Propriété intellectuelle

La présente thèse de doctorat est organisée sur le mode d’une thèse sur article. L’introduction est la production propre du doctorant auteur de la présente thèse, sauf lorsque cette introduction est une revue de la littérature destinée à être publiée et associe plusieurs auteurs. L’ordre des auteurs en a été déterminé au préalable. Chaque chapitre empirique reflète un travail collectif et collaboratif, réalisé sur plusieurs années ayant vocation à être publié comme article scientifique. Plusieurs personnes ont contribué à chaque chapitre. L’ordre des auteurs avancé dans chaque chapitre est basé sur la contribution de chaque auteur de ce travail à la date de la soumission de la thèse. Cette contribution est détaillée à la fin de chaque chapitre. Les auteurs, leur ordre et leur contribution ont été discutés entre tous les auteurs et approuvés par eux. Selon les bonnes pratiques en intégrité scientifique, tous les collaborateurs (ont pu) pourront, (s’ils l’ont demandé) s’ils le demandent, avoir accès aux données et aux codes associées à chaque article scientifique (chapitre empirique). Auteurs, ordre et contribution peuvent occasionnellement être amené à changer après la soutenance de la thèse, en fonction de la vie de l’article. Ceci sera rediscuté entre toutes les parties impliquées. La discussion et la conclusion de la thèse est la propriété propre de l’auteur de la thèse. Les éventuels travaux scientifiques en cours qui pourrait être cités dans la discussion et la conclusion suivent les règles des publications scientifiques. Auteurs, ordre et contributions seront discutés en amont de la finalisation de chaque travail de recherche entre tous les contributeurs tout au long de sa vie. L’ordre des auteurs dans les éventuelles communications présentées à des conférences internationales, outils de formation et de motivation des étudiants ne reflètent pas nécessairement une contribution effective au travail finalisé, et peut donc être rediscuté.

Données et codes

Les données et les codes produits par le doctorant dans le cadre de ce travail sont une propriété partagée entre le doctorant et le CNRS. Les données et les codes non produits par le doctorant mais ayant été utilisés dans le cadre de ce doctorat et n’ayant pas fait l’objet d’une publication feront l’objet de discussions spécifiques. Les données permettant de reproduire les études ainsi que les codes associés seront publiés en même temps que les articles scientifiques.

Abbreviation list

ACC: Anterior Cingulate Cortex

AI: Primary Auditory Cortex

AIP: Anterior Intraparietal Area

AMD: Anterior Medial Dorsal

AMV: Anterior Medial Ventral

ASD: Autistic Spectrum Disorder

ATL: Anterior Temporal Lobe

BE: Background Elimination

BOLD: Blood Oxygen Level Dependent

CM: Centre Median

dACC: Dorsal Anterior Cingulate Cortex

DCN: Dorsal Column Nuclei

dIPFC: Dorsolateral Prefrontal Cortex

DMN: Default Mode Network

fMRI: Functional Magnetic Resonance Imaging

FP: Frontoparietal

fSTS: Face-Responsive Superior Temporal Sulcus

FFA: Fusiform Face Area

FEF: Frontal Eye Field

FOV: Field Of View

FPN: Frontoparietal Network

GLM: General Linear Model

INU: Intensity Non-Uniformity

IPS: Intraparietal Sulcus

ISI: Inter-Stimuli Interval

IIPS: Lateral Intraparietal Area

lpIPS: Lateral Posterior Intraparietal Area

LGN: Lateral Geniculate Nucleus

mIPS: Medial Intraparietal Sulcus

MIP: Medial Intraparietal Area

MGN: Medial Geniculate Nucleus

mpIPS: Medial Portion of the Intraparietal Sulcus

MPRAGE: Magnetization Prepared Rapid Gradient Echo

MVN: Medial Visual Network

NHP: Non-Human Primate

OFA: Occipital Face Area

OFC: Orbito-frontal Cortex

PET: Positron Emission Tomography

PI: Inferior Pulvinar

PL: Lateral Pulvinar

PM/PuM: Medial Pulvinar

PMD: Premotor Cortex, Dorsal

PPC: Posterior Parietal Cortex

PSC: Percent Signal Change

pSTS: Posterior Superior Temporal Lobe

PuA: Anterior Pulvinar

PuI: Inferior Pulvinar

PuL: Lateral Pulvinar

ROI: Region Of Interest

RT: Reaction Times

SC: Superior Colliculus

SDC: Susceptibility Distortion Correction

SI: Primary Somatosensory Cortex

SII: Secondary Somatosensory Cortex

SMA: Supplementary Motor Area

SN: Salience Network

SOA: Stimulus Onset Asynchrony

SPL: Superior Parietal Lobule

SPM: Statistical Parametric Mapping

SR: Sensimetrics Research

STC: Slice-Timing Correction

STS: Superior Temporal Sulcus

STG: Superior Temporal Gyrus

TA: Acquisition Time

TDT: The Decoding Toolbox

TE: Echo Time

TI: Time Interval

TMS: Transcranial Magnetic Stimulation

TOJ: Tactile Temporal Order

TPM: Tissue Probability Map

TR: Repetition Time

V1: Primary Visual Cortex

VIP: Ventral Intraparietal Area

VLPCF: Ventrolateral Prefrontal Cortex

VPL: Ventral Posterior Lateral

VT: Visuo-Tactile

WM: White Matter

List of publications

Under revision

1.. MacqD - A Deep Learning-based Model for Automatic Detection of Socially-housed Laboratory Macaques

Genevieve Moat, Maxime Gaudet-Trafit*, Julian Paul, Jaume Bacardit, Suliann ben Hamed, and Colline Poirier*

BioRxiv, <https://www.biorxiv.org/content/10.1101/2024.12.23.629644v1>

Scientific report

Contribution : Study design, neural network training, contribution to writing of original draft

In preparation

2. Human medial pulvinar is involved in face and tactile processing and dynamically adjusts its cortical connectivity based on ongoing stimulation

M. Gaudet-Trafit, M. Froesel*, S. Dali*, F. Lamberton and S. Ben Hamed (2024)*

BioRxiv, <http://biorxiv.org/lookup/doi/10.1101/2024.09.20.614101>

Contribution : Study and design, stimuli preparation, methodology, ethical document writing, data acquisitions, data analysis, writing - original draft.

3. Functional connectivity of pulvinar subnuclei to the brain vary as a function of sensory context

S. Dali, M. Froesel, M. Gaudet-Trafit, F. Lamberton and S. Ben Hamed*

Contribution : Study design, stimuli preparation, methodology, ethical document writing, data acquisitions, contribution to writing of original draft

4. Spatial transformations underlying saccadic eye movements to tactile and visual target stimuli

Celia Foster, Maxime Gaudet-Trafit, Valentin Marcon, Franck Lamberton, Wei-An Sheng, Suliann Ben Hamed, Tobias Heed*

Contribution : Study design, stimuli preparation, methodology, ethical document writing, data acquisitions, contribution to writing of original draft.

5. Role of human pulvinar in visuo-tactile context

Maxime Gaudet-Trafit*, M. Froesel, S. Dali, F. Lamberton and S. Ben Hamed

Contribution : Study design, stimuli preparation, methodology, ethical document writing, data acquisitions, data analysis, writing – original draft.

6. Effect of Fluoxetine and behavioral training on resting-state connectivity of the macaque visual cortex with the rest of the brain.

Maëva Gacoin, Mathilda Froesel, Simon Clavagnier, **Maxime Gaudet-Trafit**, Suliann Ben Hamed

Contribution : data acquisition

7. Effects of Fluoxetine and behavioral training on the retinotopic organization of the macaque visual cortex.

Maëva Gacoin, Mathilda Froesel, Simon Clavagnier, Justine Cléry, **Maxime Gaudet-Trafit**, Suliann Ben Hamed

Contribution : data acquisition

Presentations

1. Role of the human pulvinar in visuo-tactile and audio-visual integration during the perception of salient faces

M. Gaudet-Trafit & M. Froesel, S. Dali, F. Lamberton, S. Ben Hamed

NeuroFrance – Poster – 2023

2. Spatial transformations underlying saccadic eye movements to tactile and visual target stimuli

Celia Foster, **Maxime Gaudet-Trafit**, Valentin Marcon, Franck Lamberton, Wei-An Sheng, Suliann Ben Hamed, Tobias Heed

EuropeanConference on Visual Perception (ECPV) – Poster – 2023

3. Reward spatial contingencies reshape visual activations in the primate visual cortex.

Maëva Gacoin, Mathilda Froesel, Simon Clavagnier, Justine Cléry, Wei-An Sheng, **Maxime Gaudet-Trafit**, Suliann Ben Hamed

NeuroFrance – Poster – 2021

General introduction

We perceive and interact with our world thanks to the ability of our brain to build a construct based on our senses. It integrates and combines information from these different sensory sources. The sensory modalities are processed by sensory systems in the peripheral and central nervous systems and receive information from the world from specific receptors, the sensory neurons. Each type of sensory neuron has its specific sensitivity toward a physical feature which triggers its activation. Studying how we construct a unified perception of the world through our senses is a long journey and has kept generations of scientists busy. From the moment the neuron doctrine theory of Ramon y Cajal was consensually accepted versus the reticular theory of Golgi, the neurons became the smallest anatomical and functional unit of the nervous system (Yuste, 2015). From this view, the notion of populations of neurons acting as a functional unit emerged until reaching the concept of neural networks gathering multiple populations of neurons under one or multiple brain functions (Yuste, 2015). Sensory neurons usually have a sensitivity to stimuli in a restricted spatial location, called their receptive field. The receptive field notion is defined by Jose-Manuel Alonso and Yao Chen as follow: “The receptive field is a portion of sensory space that can elicit neuronal responses when stimulated” (Alonso and Chen, 2009). The first documented work on receptive fields is the work of Sherrington who published in 1906 a collection of ten lectures from the past two decades that aimed at explaining different body-part unification by the nervous system and brought a first definition as “an area of the body where stimulus could elicit a reflex”. Then, Hartline in 1938 brought the receptive field concept to the visual domain with his work on optic nerve as he said: “Responses can be obtained in a given optic nerve fiber only upon illumination of a certain restricted region of the retina, the receptive field of the fiber”. In the following years, the notion of the receptive field has been expanded to other neurons in the visual pathway and further to other sensory modalities. (Burke, 2006; Hartline, 1938; Levine, 2007). From the sensory neurons in the peripheral nervous system, information is then transmitted via nerves to the central nervous system. For example, in tactile processing, from the tip of the finger to the somatosensory cortex in the brain. Within the brain, from this basic sensory information, more complex computations are performed like combining information from multiple different sensory sources or preparing motor actions that take place in the spatial environment. To be able to achieve such processes, the cortical and subcortical structures interact closely and dynamically in large scale functional networks. Whereas the posterior half of the brain is dedicated to sensory information reception and perception, and is said to look in the past, the anterior subdivision is looking in the future and is involved in preparing goal-directed actions (Fuster, 2019).

In this present general introduction, we will describe the organization of the visual, auditory and tactile modalities in unisensory and multisensory processing as well as the implication of the pulvinar, a major thalamic subcortical structure, in such processes. Then, we will cover how the brain uses perceived

information to compute a goal-oriented response, such as an eye movement toward an external stimulus or a stimulus perceived on the body.

Sensory systems and pulvinar in unisensory processing

A. The Pulvinar nucleus of the Thalamus

The thalamus is classically modeled as the major relay from periphery sensory inputs to cortex and back from cortex to periphery motor outputs. Except for olfaction, all sensory inputs are relayed by the thalamus before reaching their primary sensory cortex. The thalamus is highly connected with the neocortex and also plays a role in higher order cognitive functions, putting it in a target position to identify thalamic markers and apply surgical treatments for human pathologies (Anne Morel et al., 1997; Wolff and Vann, 2019). The thalamus is composed by a multitude of subnuclei functionally organized along an antero-posterior axis (Kumar et al., 2017; Anne Morel et al., 1997). However, the thalamus is not just a relay and is known for being implicated in shaping perception and cognition through precise recurrent communication with the cortex. In particular, it has a predominant role modulating attention and perception in vision (Saalmann et al., 2011; Yuri B. Saalmann and Kastner, 2011).

Within the thalamus, the pulvinar is the largest and the most intriguing nucleus (Froesel et al., 2021a; Kalamatianos et al., 2023). Located posteriorly in the thalamus, medially and dorsally to the lateral geniculate nucleus (LGN), it belongs to the posterior nuclei group (Anne Morel et al., 1997). In humans, the pulvinar is classically subdivided in 4 subnuclei based on multiarchitectonic parcellation (cyto, myelo, chemoarchitectures) studies, the anterior (PuA), the inferior (PuI), the lateral (PuL) and the medial (PuM) (Anne Morel et al., 1997). The parcellation of the pulvinar is still being updated. The definition of pulvinar subnuclei is complex as it relies on anatomical and functional characteristics. Some studies divided it into 5 subnuclei using functional MRI connectivity. Barron et al defined the anterior, superior, lateral, inferior and medial pulvinar subdivisions (Barron et al., 2015a) whereas Guedj et al proposed a segmentation along the dorso-ventral axis with dorsomedial, lateral, anterior, inferior and ventromedial subdivisions (Guedj and Vuilleumier, 2020a). Thanks to NHP experiments and recent medical imaging in humans, the pulvinar dense anatomical and functional interconnections with cortex and subcortical areas have been assessed and documented in multiple studies (Asanuma et al., 1985a; Froesel et al., 2021a, 2024b; Kaas and Lyon, 2007a; Leh et al., 2008a; Shipp, 2003a; Vittek et al., 2023). Figure 1 from (Froesel et al., 2021a) illustrates the rich anatomical connectivity of the subdivisions of the pulvinar in NHP model. The pulvinar is largely connected with the entire brain according to a dorso-ventral gradient which makes it a perfect potential hub for area communications (Froesel et al., 2021a). Pulvinar activity has been reported in various cognitive and higher order functions like visual processing, spatial attention, memory, emotion regulation, sensorimotor processing, salience processing, social cognition, multisensory processing, etc. (Benarroch, 2015a; Fiebelkorn et al., 2019a; Froesel et al., 2021a; Guedj and Vuilleumier, 2020a; Yuri B. Saalmann and Kastner, 2011; Wen et al., 2023). In

NHP, the pulvinar has been subdivided into 3 parts, the inferior (PI), the medial (PM) and the lateral (PL) with subdivisions within each parts (Froesel et al., 2021a). Functionally, the PI has been associated with

visual processing, eye movements, auditory processing and hand movement. The PM has been associated with multisensory processing, simple and complex visual and auditory processing, somatosensation, olfactory processing and motor hand and eye movements. The PL has been associated with simple and complex visual stimuli, somatosensation, auditory, olfactory, processing, eye movement and saccade and motor reaching and grasping (Froesel et al., 2021a). If one segments the pulvinar between dorsal and ventral parts, ventral pulvinar has a retinotopic organization and is anatomically connected to superficial layers of the colliculus and visual cortex. Dorsal pulvinar has connections with cortical regions implicated in goal-directed actions such as the frontal eye field (FEF), dorsolateral prefrontal cortex (dlPFC), premotor area (PM), area 7, areas within posterior parietal cortex (PPC) and the superior temporal sulcus (STS) (Wilke and Kagan, 2023). For example, in an fMRI and white matter stimulation study on auditory and speech processing, the pulvino-temporal anatomical and functional connectivity has been described in patients. The authors observed an anomia after white matter stimulations in awake surgery of the auditory radiation like component, reinforcing the implication of pulvinar in lexical retrieval and picture naming (Maldonado, and Descoteaux, 2024). In addition to the dorsoventral functional connectivity gradient, a recent study highlighted the existence of an antero-posterior gradient (Froesel et al., 2024a). The authors also demonstrated the connectivity of pulvinar with the main sulci such as, intraparietal, cingulate, central, superior temporal, central sulci and prefrontal cortex follows the same gradient. For example, an antero-posterior connectivity gradient for intraparietal sulci (IPS) and pulvinar and a ventro-dorsal gradient for orbitofrontal sulci and pulvinar.

The role of the pulvinar in attention has been largely described in literature (Bourgeois et al., 2020a; Greenwood and Ward, 2024). Studies highlighted that clinical lesions of the pulvinar impact spatial orienting mostly contralesionally (I. Arend et al., 2008; Bourgeois et al., 2020a; Karnath et al., 2002; R D Rafal and Posner, 1987). For example Karnath et al identified a network for spatial neglect composed of right putamen, caudate nucleus, pulvinar and superior temporal gyrus (STG) in lesioned patients (Karnath et al., 2002). Lesions to the dorsal pulvinar resulted in deficits in reaching and grasping (Wen et al., 2023; Wilke et al., 2018a). Pulvinar has also been highlighted in NHP electrophysiological studies, as well as in patients with lesions, to play a role in sensory detection and selection and is specifically important to process visual saliency (Froesel et al., 2021a; Michael and Buron, 2005; Robinson and Petersen, 1992). The pulvinar has been identified in several pathologies and disorders (McTeague et al., 2020) like major depressive disorder (MDD) where its connectivity is altered for objective rather than subjective cognitive impairment in patients linking its role with evolution of dementia in depression (Yulug et al., 2024). Pulvinar functional connectivity has been shown to be changed in spinal cord injury patients (Karunakaran et al., 2020). The authors showed an increase of functional connectivity between the left pulvinar, and both left inferior frontal gyrus and left inferior parietal lobule in lesioned patients.

Due to the highlighted electrophysiological and imaging modifications of the pulvinar in epilepsy, its medial subdivision has been proposed as novel target for deep brain stimulation (DBS), supplementing the traditional anterior thalamic nuclei (Kalamatianos et al., 2023). Overall, the fact that the pulvinar is widely associated with numerous disorders has increased interest to consider it as a target for developing new treatments and to understand pathologies.

There is growing consensus that the pulvinar plays a general role in modulating the activity of cortical areas across the wide range of functions we just described above (Arcaro et al., 2018a; Benarroch, 2015a; Eradath et al., 2021; Fiebelkorn et al., 2019a; Froesel et al., 2021a; Greenwood and Ward, 2024; Guedj and Vuilleumier, 2023a; Yuri B. Saalman and Kastner, 2011). For example, Eradath assessed the causal role of pulvinar in NHP drug inactivation and electrophysiological recordings (Eradath et al., 2021). They found that pulvinar synchronized brain areas from large-scale networks passively, without requiring engagement in an active task. Other studies used a model-based approach to investigate the role of the pulvinar in synchronization of cortical areas during attention selection processes (Greenwood and Ward, 2024). These studies propose a model showing how pulvinar could synchronize gamma oscillations in cortical areas through generation and transmission of alpha oscillations. In NHP fMRI, the pulvinar has been recently highlighted in an audio-visual vocalization perception study (Froesel et al., 2024b). The authors showed activations within claustrum, dorsolateral amygdala and pulvinar for congruent semantic visual and auditory information. Particularly, they showed that the pulvinar and the two other structures expressed multisensory integration mechanisms. In another recent study in human positron emission tomography (PET), the authors highlighted the existence of dorsoventral and mediolateral gradients of the pulvinar interconnections with cortex (Basile et al., 2024). On the other hand, (Froesel et al., 2024a) emphasized the major role of dorso-ventral and antero-posterior gradients and a lesser role of the medio-lateral gradient in pulvinar organization.

Given the implication of pulvinar in multiple cognitive processes and the difficulty to establish a clear anatomo-functional organization, understanding the role of pulvinar is a challenge with potentially major impact on many fields of neuroscience. In the following sections, we will focus on the role of the pulvinar in unisensory visual, auditory, tactile and multisensory visuo-tactile processing.

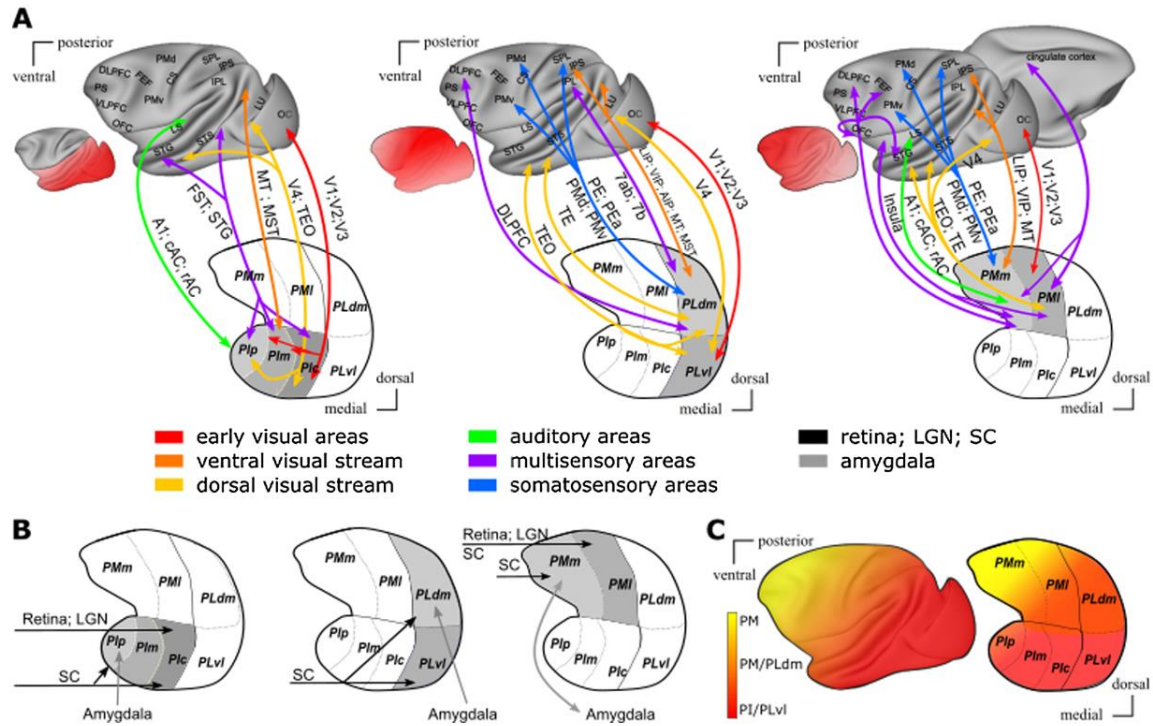


Figure 1: Anatomical connections between three subdivisions of NHP pulvinar and cortical, subcortical area and structures. From left to right, pulvinar inferior (PI), lateral (PL) and medial (PM) subdivisions are represented with their connections with cortical areas for panel A and subcortical structures for panel B. Panel C shows pulvinar connectivity gradient with the cortex. From Froesel et al 2021

B. Vision and face processing

Visual system

Vision is the most important sensory modality for humans and diurnal non-human primate species. Vision allows us to quickly process our environment by detecting a wide range of light wavelengths (from violet 400 to red 800 nanometers) and processing them in order to categorize objects and their location in space. This results in us creating a model of our surrounding world full of different elements with different properties such as shape and color. It also achieves perceptual stability in spite of the fact that the environment is dynamically changing as objects of the same category can have a different position, size, shape or lighting. Vision begins in the eye. At the back of the eye lies the retina, where a two-dimensional (2D) image of the observed object is formed. This image formation results from the cornea bending (refracting) incoming light rays. The retina is composed of 2 types of sensory neurons capturing light information, which are called photoreceptors. Rods and cones photoreceptors are reciprocally specialized in black and white vision (scotopic vision) and color (red, green, blue) vision. The distribution of photoreceptors is not uniform, for example a high concentration of them are gathered in the fovea, which is the central point of our vision with highest resolution. On the retina, observed objects are upside-down and left-right inverted. The output of the retina is in its outer cell layer (retinal ganglion cells) which gather their axons in the optic nerve to conduct information from peripheral to

central nervous system. (Dowiasch et al., 2022; Gottfried, 2011; Mahabadi and Al Khalili, 2023). Optic nerves from each eye meet at the optic chiasm, where fibers cross over and signal arrives in lateral geniculate nucleus (LGN) of the thalamus. LGN neurons relay signals to the primary visual cortex (V1) in the occipital lobe where V1 processes basic visual features like edges and orientations (Dowiasch et al., 2022). Organization in V1 is retinotopic, meaning that close points in visual field are represented by neighboring neurons. By doing so, the representation in V1 preserves the spatial relationship of elements in the visual scene. In addition to electrophysiological studies in NHP, fMRI methodology in humans was used to show the functional representation of a visual stimulus in V1, V2 and V3 and the general organization of primary visual cortex at large scale (Arcaro et al., 2009; Roger B. H. Tootell et al., 1998). FMRI has been used to highlight similarities between human and NHP models by studying multiple processes in healthy humans in a non-invasive way. The involvement of subcortical structures and the contralateral projections in the superior colliculus (Schneider and Kastner, 2005) and LGN (Schneider et al., 2004) has been highlighted thanks to this methodology. Visual information are further processed in higher visual areas like V4, middle temporal (MT) and others which extract more complex information in a hierarchical organization (Gottfried, 2011). The first proposition of a hierarchical organization of the visual system has been proposed by Hubel and Wiesel in the cat model (Felleman and Van Essen, 1991; Hubel and Wiesel, 1965). A substantial amount of knowledge about the visual system has been described thanks to non-human primates experiments (Gottfried, 2011). The left panel of figure 2 shows a schematic representation of the visual system from the eye in the periphery to V1. The visual information is then split in two distinct pathways, the dorsal stream and the ventral stream. The notion of two distinct visual processing pathways, each carried by different neural substrates, emerged from lesion studies in NHP (Ungerleider, 1982) and has been supported by years of research using multiple methodologies at different scales, such as neuropsychological, single-unit recording and functional imaging (Freud et al., 2016; Milner and Goodale, 2006; Ungerleider, 1982). A third pathway was more recently defined, the social pathway anchored laterally in temporal lobe and processing motion information about social stimuli (Pitcher, 2021). We will not discuss the third pathway further in this introduction. The dorsal pathway extends from occipital to parietal cortices and is referred to as the “where” or “how” pathway as it is involved in localizing the spatial position of the seen object and preparing visually guided actions (Freud et al., 2016; Gottfried, 2011; Pitcher, 2021). The ventral pathway extends from occipital to temporal cortices is referred as the “what” pathway as it is composed of areas processing object shapes and identities (Freud et al., 2016; Milner and Goodale, 2006; Pitcher, 2021; Ungerleider, 1982). The ventral pathway is notably composed of areas implicated in face perception (Freud et al., 2016; Milner and Goodale, 2006; Pitcher, 2021). Being able to perceive faces is important for social interactions and to be able to interact with others in our social world. A large functional network is implicated in face perception, it is composed of multiple brains areas in occipital, temporal, and frontal cortices. The Fusiform face area (FFA) located in the fusiform gyrus was one of the first area identified by fMRI to specifically process faces (Kanwisher et al., 1997). Before the fMRI

contribution, face processing has been identified by lesional, single-cell recordings neuropsychology and neurophysiology as a distinct process which has its own mechanism and neural substrate (Kanwisher et al., 1997). More recently, more regions have been identified to play a role in face processing, including the occipital face area (OFA), posterior superior temporal lobe (pSTS, also called face-responsive STS, fSTS), anterior temporal lobe (ATL) and orbito-frontal cortex (OFC) (Gratton et al., 2013; Haxby et al., 2000; Schobert et al., 2018; Troiani et al., 2016; Zhang et al., 2009). The right panel of figure 2 shows a representation of identified cortical areas for face processing.

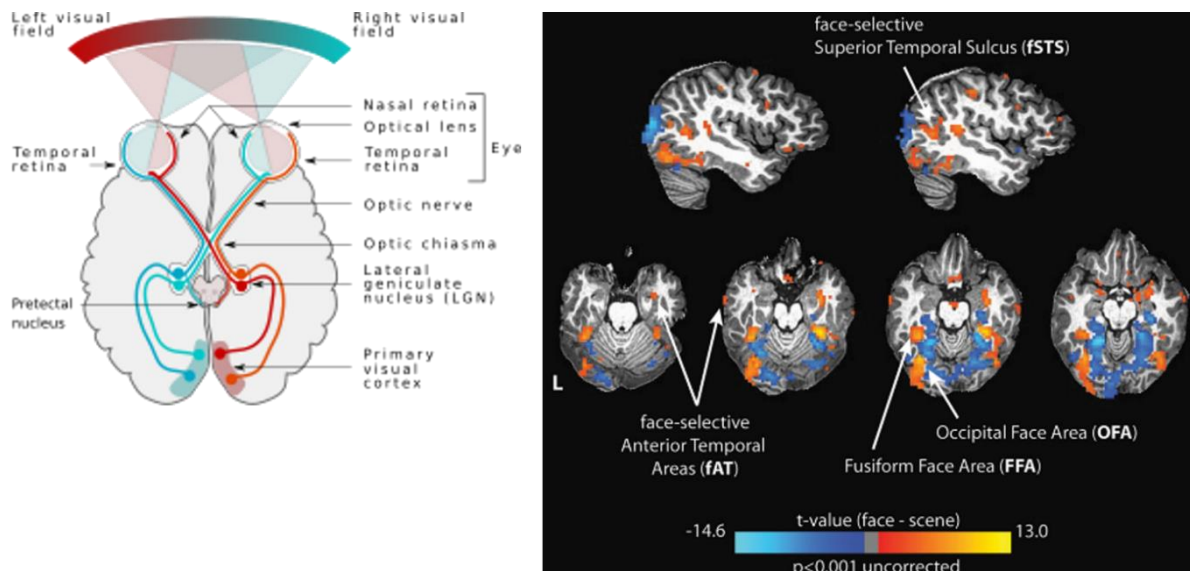


Figure 2: Description of the visual system and functional cortical areas for face processing from https://en.wikipedia.org/wiki/Visual_system (left panel) and (Gratton et al., 2013) (right panel)

Not only cortical but also subcortical structures have been identified to play a role in face processing. In the NHP model, superior colliculus (SC) has a population of neurons coding for face information, we will extend the role of the superior colliculus further on the manuscript (Nguyen et al., 2016).

Pulvinar in vision

The thalamus relays visual information via the LGN but also is composed of high-order nuclei like the pulvinar, its largest nucleus. The pulvinar has been recently highlighted to play a role in object categorization using fMRI (Arcaro et al., 2015a, 2018a; Benarroch, 2015a; Wen et al., 2023), electrophysiological recordings (Nguyen et al., 2016) and MEG recordings (McFadyen et al., 2017). Moreover, the pulvinar shares direct anatomical connections with both dorsal and ventral visual streams (Kaas and Lyon, 2007a). The central lateral nucleus of the inferior pulvinar and ventral lateral nucleus of the lateral pulvinar project to visual areas of dorsal and ventral stream (V1, V2) and specific areas of the ventral stream (V4) (Kaas and Lyon, 2007a). These projections are likely responsible for carrying signals for object categorization, within the ventral stream. Within visual object categorization, face perception is a central process for social cognition, enabling us to navigate our social

environment. The exact role of pulvinar in the face processing functional network remains unclear as the output of the studies show similar but different results. A popular hypothesis is to propose that the pulvinar has a role of modulator of cortical brain areas (Arcaro et al., 2018a, p. 201; Froesel et al., 2021a; Saalman et al., 2012a). In NHP, the role of pulvinar in face processing has been assessed using fMRI (Froesel et al., 2024b). Particularly, the authors showed anterior, medial and inferior pulvinar subnuclei activations in unimodal visual and multimodal audio-visual face perception contexts. The authors also suggested the existence of a multisensory network that modulate emotional visual and auditory processing of faces and vocalization. This network is composed of dorsolateral amygdala, claustrum and pulvinar.

The authors showed lateralized activation of the medio-dorsal and ventro-lateral parts of the left lateral pulvinar but not the right. In the study, the monkeys performed both audio-visual and pure visual tasks using the same face stimuli. The observed pulvinar activations for the visual stimuli between the two tasks were not located in the same place. For the audio-visual task, the visual only condition (same as pure visual task) activated only the ventro-lateral part of the lateral pulvinar like the pure visual task. These results suggest that the activation pattern within the pulvinar could be mediated by the task context and explain partially the different results highlighted in the literature. The pulvinar has also been demonstrated to be part of an extended face-processing network as it was connected to the face patch system of NHP along with other subcortical areas, the superior colliculus and the amygdala (Schwiedrzik et al., 2015). Arcaro et al (2018) showed that the ventral sub-division of pulvinar was associated with face processing and connected with occipito-temporal areas whereas the dorsal sub-division was associated with attention on orienting behaviors and connected to frontal and parietal area (Arcaro et al., 2018a). The authors studied the connectivity at rest showing the predictive connection of the ventral pulvinar with functional specific brain areas like FFA, OFA, ATL or pSTS. This showed that the pulvinar shares connection with face processing cortical areas at a basal level. On the other hand, it did not inform us about the dynamic connections when we engage in a face processing task. In another study, Wen et al (2023) showed a finer segmentation of object categorization within the pulvinar (Wen et al., 2023). They showed activations in both ventral and dorsal pulvinar for face perception but mainly showed distinct activation patterns for animal and tool object categories. They also studied the functional connectivity at rest and showed a strong connectivity for animal and tool categories between pulvinar clusters and cortical areas, namely fusiform and occipital cortex for animal and lateral occipitotemporal cortex, parietal, medial fusiform and occipital areas for tools. It would be interesting to assess dynamic pulvino-cortical connectivity during a face detection task and identify specific patterns related to object categorization, and how this connectivity changes along with the type of stimuli. For example, the connectivity between pulvinar and cortical areas may change in terms of connected areas if we process faces or large objects. We can expect that the pulvinar would strengthen its connections with a face-patch system during face perception. By doing so, the pulvinar could be a good candidate for the role of modulator hub in face processing as a stronger connectivity can be a marker of area response

enhancement and sensory processing facilitation. We propose to assess these questions in the first project of this PhD work. We will see the implication of pulvinar and its subdivisions in face processing and its resting-state and task-related functional connectivity with the rest of the brain.

C. Audition and voice processing

Auditory system

From bird chirping to discussion with a friend, the auditory sensory modality allows us to perceive a wide range of sounds in our surrounding environment. The outer ear collects vibrations from the environment that travel through the ear canal until the eardrum, which is composed of a vibrating membrane responding to the sound waves. Behind the eardrum lies the middle ear, a small cavity containing three tiny bones: the malleus, incus, and stapes. These bones amplify the vibrations from the eardrum and transmit them to the inner ear. In the inner ear, the cochlea is a structure with spiral shape filled with liquid and containing sensory neurons, hair cells, which transform vibrations into an electric neuronal signal. This signal is conducted from the cochlear nucleus to the inferior colliculus then to the medial geniculate nucleus (MGN) of the thalamus. The MGN then transmits information to the primary auditory cortex (AI), the core of which is located in the lower bank of the STS (Kaas and Hackett, 1998). This primary auditory cortex is surrounded by secondary areas involved in higher cognitive processing in a hierarchical organization, the belt and the parabelt (Gottfried, 2011; Kaas and Hackett, 1998; Saenz and Langers, 2014). The core, the belt and parabelt are strongly interconnected with a hierarchical organization such as the parabelt activity is related to complex auditory stimuli and even somatosensory and visual stimulations. The parabelt projects to cortical areas located in STG, STS and prefrontal cortex (Kaas and Hackett, 1998). These three regions all share interconnections with thalamic nuclei, like the MGN and pulvinar (Kaas and Hackett, 1998). An illustration of the auditory system is shown figure 3. From the brainstem to cortical domain, auditory information is spatially organized according to sound frequencies, this defines the tonotopic organization of the auditory system (Gottfried, 2011; Moerel et al., 2014; Saenz and Langers, 2014).

Studying the brain representation of conspecific vocalization is essential to understand social cognition. In humans, understanding voice perception and production allows to recognize person identity and to communicate via language. Non-human primate's vocalizations are seen as precursor of human language, which makes the research on this topic highly important to understand the evolution of language (Belin, 2006; Bodin and Belin, 2020; Moerel et al., 2014; Poremba et al., 2004). In NHP, one of the first studies highlighting the species specific vocalizations processing in auditory cortex has been conducted in electrophysiology in squirrel monkeys (Winter and Funkenstein, 1973). This study demonstrated the existence of cells specifically responding to conspecific vocalization in auditory cortex. Later, imaging studies have identified a voice patch system along the superior temporal gyrus (STG) and sulcus (STS) (Belin, 2006; Bodin and Belin, 2020; Pernet et al., 2015; Poremba et al., 2004). Three main voice patches were identified, one posterior, one median and one anterior region (Bodin and

Belin, 2020). The bottom panel of figure 3 illustrates the voice-patch system. The voice-patches network has also been described in intracranial implanted epileptic patients (Zhang et al., 2021). The study showed specificity of patches for voice versus non-voice sound processing and described a strong intrinsic connectivity within voice patches during resting state and active task. Posterior superior temporal sulcus, pSTS (fSTS) which has been identified as an important node for face processing is more generally implicated in social processing of both face and voice stimuli. A more medial region of STS has been identified to be unimodal auditory (Deen et al., 2020). This recent view has been reviewed recently has part a third visual pathway processing social information located laterally in the brain (Pitcher, 2021).

Pulvinar in audition

The anatomical interconnections between pulvinar and auditory cortex have been highlighted thanks to NHP studies using tracer injections in particular. PM is connected to the rostral and caudal auditory cortex (belt and parabelt), STG and STS (Cappe et al., 2009b; Froesel et al., 2021a; Kaas and Hackett, 1998). PI is connected to the primary auditory cortex AI (Froesel et al., 2021a). Functionally, neurons in the three subdivisions of the NHP pulvinar (PI, PL and PM) are sensitive to auditory stimulation, PI to simple stimuli, PL to socially meaningful sounds and PM to both simple and complex sounds (Froesel et al., 2021a; Ricardo Gattass et al., 1978; Magariños-Ascone et al., 1988a; Yirmiya and Hocherman, 1987a). Recently, three studies, including NHP electrophysiological recordings (Vittek et al., 2024, 2023), mice electrophysiological recordings (Chou et al., 2020) and NHP fMRI (Froesel et al., 2024b) showed that the pulvinar is responding to simple and complex auditory stimuli across species. Vittek and Froesel showed that the medial and lateral pulvinar were activated for simple and complex sounds and performed audio-visual multisensory integration. Chou et al demonstrated that the rodent homologue nucleus of the primate pulvinar modulated both frequency tuning and signal-to-noise ratio with a subtractive suppression of neurons in the auditory cortex through excitatory projection toward inhibitory neurons. In humans, there is less evidence of implication of the pulvinar in voice processing assessed by clinical studies. In patients, lesions in the posterior thalamus including MGN and pulvinar showed deficits in complex auditory processing as well as auditory attention deficit (Wester et al., 2001). One study in autistic spectrum disorder (ASD) patients in fMRI showed that the connectivity patterns for mild aversive auditory and tactile stimuli is reduced to pulvino-temporal lobe while neurotypical individual showed connectivity patterns with broader brains areas (Green et al., 2017). Finally, contralateral deficit in dichotic listening has been highlighted in a patient with a pulvinar lesion (Hugdahl and Wester, 1994). Anatomical and functional connectivity as well as active auditory task recordings in animal models and several clinical cases strongly suggest the involvement of the human pulvinar in simple and complex auditory stimuli processing. Moreover, as the pulvinar is regulating saliency in the visual domain, we can expect it may do the same in the auditory domain. No study directly assessed its role in healthy humans. We propose to investigate the fMRI activity of pulvinar in a voice localizer task, in chapter one of this manuscript.

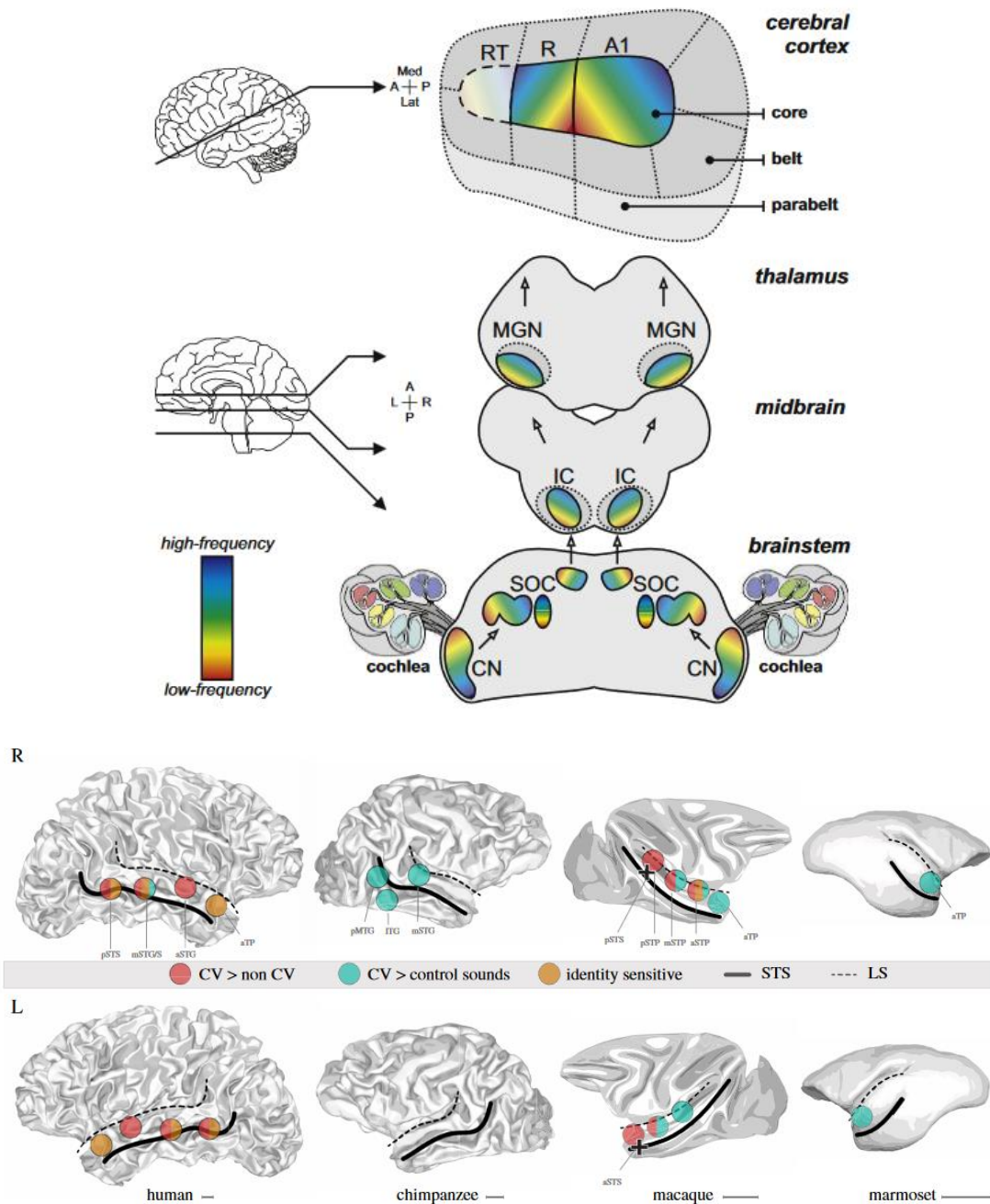


Figure 3: Organization of the auditory system and functional areas for voice processing. Taken from (Saenz and Langers, 2014) in top pane and from (Bodin and Belin, 2020) in bottom panel.

D. Somatosensation and touch processing

Somatosensory system

The sense of touch is carried by the somatosensory system, which processes information of an element that comes in contact with our own body. This information is not unimodal as the system can capture touch, itches, nociception (pain), thermoception, texture and shapes via the wide variety of sensory neurons located in the skin. This system has an important role in alerting us in case of imminent or

already existing danger that can undermine the integrity of the body itself by detecting foreign objects in contact with it, putting pressure on the skin or causing pain. The second important role is to interact with our environment by touch and manipulation in order to determine characteristics of objects composing our environment (Gottfried, 2011). In this PhD work, we will focus only on touch detection and perception. In the 1860 and 70s, starting with the work of Fritsch and Hitzig in electrical evoked motor responses in dogs (Fritsch and Hitzig, 1870) and Ferrier in monkeys (Ebbesen and Brecht, 2017; Ferrier, 1874), the understanding of motor function of the brain has continued to grow. In 1950, with their work in human model, Penfield and Rasmussen gave a first general view of segregation of somatosensorial cortex with specific allocation of areas to body part when the body part is stimulated (Penfield and Rasmussen, 1950). The sense of touch is based on the transformation of mechanical deformation into an electrical signal. Several types of mechanoreceptor neurons fulfill these functions (Delhay et al., 2018). Peripheral information is conveyed to spinal cord by sensory nerves that then arrive in the medulla where the dorsal column nuclei (DCN) contralaterally transfer the information in the ventroposterior complex of the thalamus. Then, the thalamus projects to the primary somatosensory cortex (SI) (Delhay et al., 2018; Sereno and Huang, 2014). Figure 4 illustrates this pathway. Like the two previous modalities, SI is following a hierarchic somatotopic organization. Touch stimulation of a part of the body contralaterally activates neurons at a specific location along the postcentral gyrus and sulcus, posteriorly to the central sulcus (Chen et al., 2008). A somatosensory body representation can be thus built, resulting in the famous somatosensory homunculus that maps the somatotopy of the tactile system. Whereas SI is involved in low level representations of tactile stimuli like type and intensity of the input, the secondary somatosensory cortex (SII) is implicated in more complex processing such as sensorimotor integration, attention, learning, memory, pain (Chen et al., 2008; Fujiwara et al., 2002). SII is located dorsally from the lateral sulcus in the parietal operculum (Chen et al., 2008).

Pulvinar in tactile processing

The pulvinar is anatomically connected to the cortical tactile areas by its anterior subdivision demonstrated in non-human primate (areas 1, 2, 7b and lateral parietal cortex) (Delhay et al., 2018) and by its medial and lateral subdivisions (Froesel et al., 2021a). Moreover, it has been shown to be responsive to tactile and pressure stimulations (Froesel et al., 2021a; Ricardo Gattass et al., 1978). In humans, the role of pulvinar has been identified in tactile and motor processing in high field MRI (Charyasz et al., 2023). The authors showed that the medial, lateral and inferior pulvinar subdivisions were activated for passive and active tactile tasks and only for several subjects over the whole cohort. The tactile stimulations were delivered on 4 fingers of the hands with an inflatable device causing skin deformation as the tactile stimulus. Assessing the activity pattern of the pulvinar for distinct body parts is needed in order to characterize a possible somatotopic organization within it. In addition, air-puff (pressured air) stimulation can be used as alternative mechanical stimulation, and this stimulation is processed differently from other mechanical skin deformation in a way that they represent a tactile stimulation without any object in contact with the body (Hashimoto, 1987; Kim et al., 2018). Air-puff

stimulation is lighter as it activates less units (Kuwabara et al., 2000). It has been postulated that a fast fear processing visual pathway exists involving pulvinar to amygdala direct connection (McFadyen et al., 2017). The pulvinar hypothetically being recruited to produce a fast response to a danger, we can hypothesize that it could also be recruited for an object touching our body to increase consciousness of that probable threat. Air-puff stimulation and direct contact activate different skin receptors and different neural pathways (Abraira and Ginty, 2013; Hashimoto, 1987; Kim et al., 2018). The first one activates hair follicle skin receptors that have projections to somatosensory cortex. In a context of tactile stimulation without direct contact with an object, like for air-puffs, it is unclear if the pulvinar is also activated. As the pulvinar is implicated in visual saliency regulation, we can expect that the pulvinar may produce early signals in anticipation of the incoming tactile stimuli to enhance the cortical response. We saw this kind of enhancement from the pulvinar is possible in an auditory context when the pulvinar increased the signal to noise ratio of auditory neurons (Chou et al., 2020). More, we can expect that the light and spread air-puff tactile stimulation mimics social touch. The pulvinar being a high order nucleus could play a role in this process. The topographical organization of tactile processing within the pulvinar has not been directly assessed yet given the reduced number of studies on this topic. In macaques, the functional connectivity is strong between anterior, medial regions of the pulvinar and somatosensory, sensorimotor cortices (Froesel et al., 2024a). Thus, the pulvinar could be implicated in multiple sensorimotor loops. In this thesis work, we will investigate the general role of the pulvinar in tactile processing and test a possible topographical organization within it for different body parts, namely face, hands and foot.

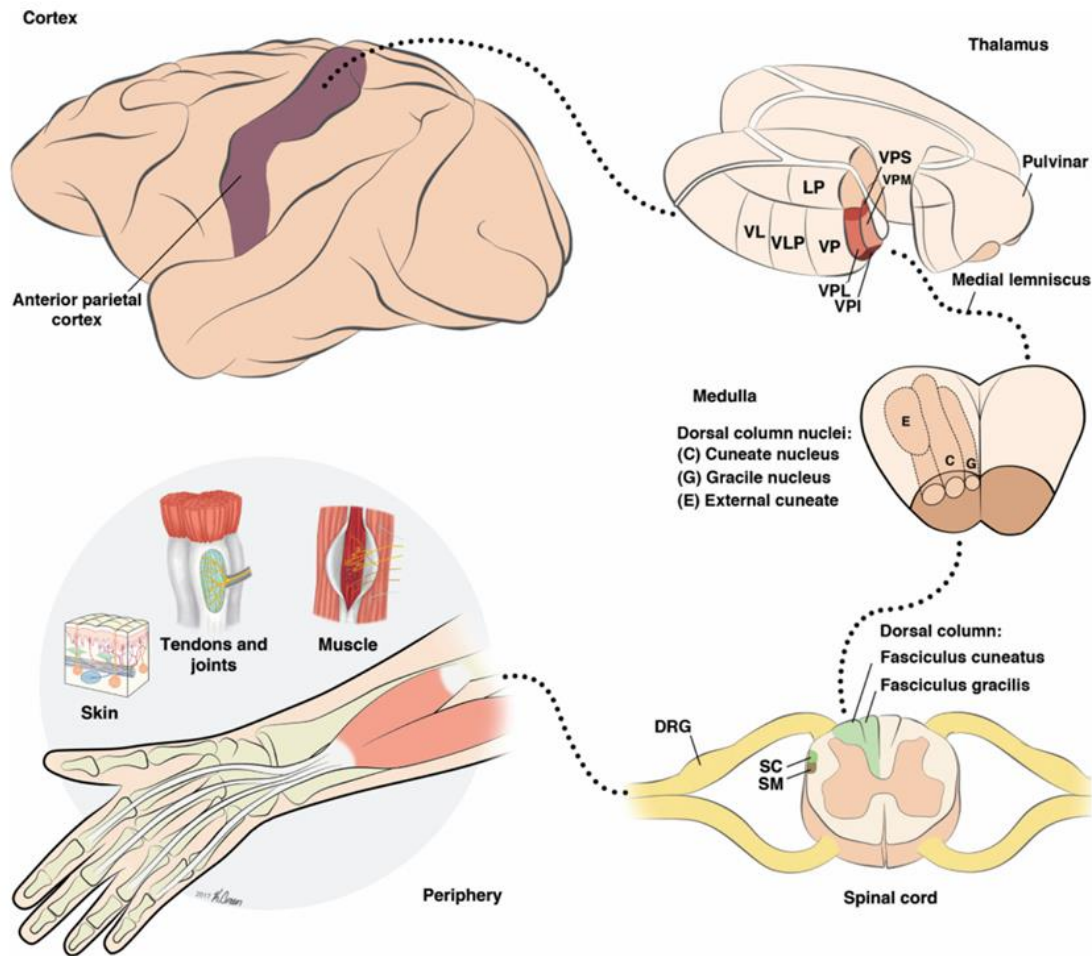


Figure 4: Primate's somatosensory system organization from periphery to central nervous system
Taken from (Delhaye et al., 2018)

Multisensory processing and Pulvinar

Studying the brain by considering sensory modalities in isolation is an oversimplification. The brain is intrinsically multisensory, not only in associative areas but also within primary sensory areas (Cappe et al., 2009b; Froesel et al., 2021a; Vittek et al., 2023). The role of multisensory processing is to enhance perception (accuracy, detection rates) and produce fast responses (faster reaction times) to the environment by building a unified construct of it (Cappe et al., 2009b; Froesel et al., 2021a; Murray and Wallace, 2012a). For example, we automatically associate the sight of the person speaking to us with their voice. In the ventriloquist effect, if the sound of a voice is synchronized with someone's lip movements, we attribute the voice to that person, assuming the spatial location of both is the same. Multisensory processing is mainly defined by convergence and integration principles. The convergence of inputs from distinct sensory modalities within a specific brain region, ultimately onto a single neuron, is an essential condition for multisensory processing (Murray and Wallace, 2012a). This convergence may allow further integration of sensory information from different sources into one percept. Historically, multisensory processing has been demonstrated at the neuronal level in superior colliculus

(SC) by Horn and Hill (Horn and Hill, 1966). Then, cortical associative areas have been identified, these areas are located between primary sensory cortices from which information converge. The monkey's superior temporal sulcus (STS) was one of the first cortical structures identified to contain such areas described as bimodal and trimodal and in which were identified neurons generating action potentials specifically to two (bimodal) or three (trimodal) modalities after stimulation (Bruce et al., 1981; Murray and Wallace, 2012a). Two other main associative areas are the parietal cortex and the frontal and prefrontal cortices. Both are related to spatial representation of external objects and spatially-guided behavior and motor actions (Murray and Wallace, 2012a). A schematic representation of cortical areas performing multisensory integration is shown in figure 5. Multisensory processing also occurs in unimodal sensory areas. A unisensory area can exhibit multisensory integration properties, notably through direct communication from other sensory areas. For example, visual, auditory and somatosensory cortices share anatomical interconnections which functionally make activity in the primary visual cortex to be modulated by tactile input through visuo-tactile multisensory processing (Cappe et al., 2009b; Guipponi et al., 2015) or auditory input to visual areas (Wang et al., 2008). At the neuronal level, multisensory processing can be studied with electrophysiological recordings where the activity of neurons allows us to classify them according to their response profile to one or multiple modalities. The integration process is defined by the fact that the recorded multisensory response is significantly superior or inferior to the sum of response in unimodal context leading to superadditive and subadditive effects, respectively (Avillac et al., 2007; Cléry et al., 2015; Froesel et al., 2021a; Murray and Wallace, 2012a; Stanford and Stein, 2007; Stein and Stanford, 2008; Vittek et al., 2023). These effects can be seen in fMRI by comparing the BOLD response (Calvert et al., 2000; Cléry et al., 2017; Murray and Wallace, 2012a). Cléry et al, showed in NHP multisensory integration superadditive properties of areas in visual, parietal and frontal cortices in a visuo-tactile task involving looming stimuli mimicking approach to the body. These areas expressed this superadditive effect when the impact on the body was predictable, when visual and tactile information share a congruent information about an approaching object to the body (Cléry et al., 2017). Although the cortical domain plays a significant role in multisensory processing, the subcortical domain is an important actor in these processes, the thalamus notably (Cappe et al., 2009b; Froesel et al., 2021a; Murray and Wallace, 2012a; Tyll et al., 2011a; Vittek et al., 2023). The involvement of thalamus nuclei in these multisensory processes is based on their high reciprocal connectivity with the cortex (Tyll et al., 2011a). Within the thalamus, several subnuclei showed multisensory neurons. The centre median (CM) and the parafascicular showed uni-, bi- or trimodal neurons for visual, auditory and tactile modalities (Matsumoto et al., 2001). The pulvinar is a place a convergence of sensory modalities with high connectivity with the rest of the brain and functionally involved in numerous high order processes (as seen previously). For example, a study looked at thalamocortical connection by injecting tracers in specific cortical areas, they found the most overlap across sensory modalities was in the medial pulvinar (Cappe et al., 2009a). These characteristics make it a relevant candidate for performing multisensory processing (Cappe et al., 2009b; Froesel et al.,

2021a; Murray and Wallace, 2012a; Vittek et al., 2023). Audio-visual subadditive effect has been recently shown in the medial pulvinar for audio-visual task in NHP, demonstrating existence of multisensory integration within the pulvinar (Vittek et al., 2024, 2023). In NHP fMRI, another study focused on auditory social stimuli in a visual social context. The authors showed multisensory integration properties in the medial pulvinar with greater activation when audio-visual stimuli are congruent with the visual context than unimodal visual and auditory (Froesel et al., 2024b). In a human fMRI study, the pulvinar has shown multisensory properties for audio-visual narrative speech perception (Ross et al., 2022a). But the activation cluster is wide and encompasses the MGN and LGN. Further investigation to confirm the specificity of pulvinar multisensory integration in speech processing will be needed. Given the rich convergence of multiple sensory modalities, the implication of pulvinar in higher order processes like face detection in unimodal context and the capability of pulvinar neurons to perform multisensory integration, we address the question of whether and how the pulvinar expresses visuo-tactile multisensory properties. In the next chapter, we will present an experiment assessing the cortical and subcortical activity during face detection in the presence or absence of tactile air-puff stimulation, in varying visual salience conditions.

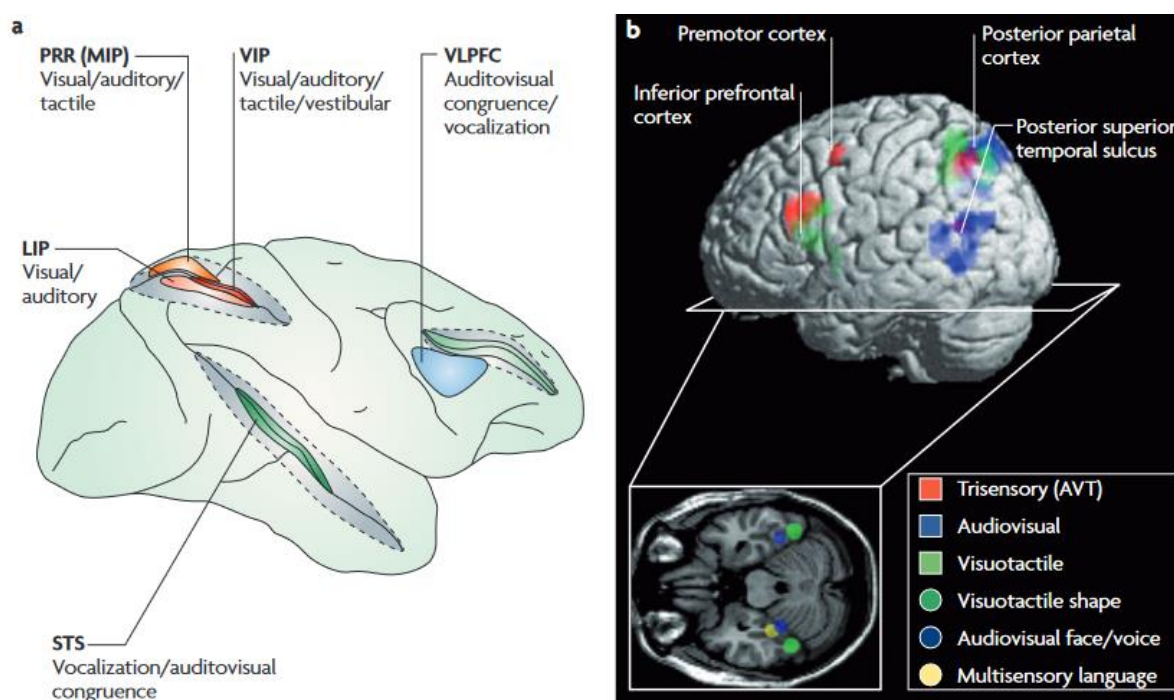


Figure 5: macaque and human cortical multisensory areas.

Taken from (Stein and Stanford, 2008)

Motor action and goal directed movement

A. Motor action

The brain perceives the multisensory characteristics of the environmental objects and determines what to do, most of the time by producing a motor action. The action can be simple, like a quick eye movement, or more complex, like navigating within a room and interacting with an object, which requires a series of coordinated movements. Such behavior requires a hierarchical organization of the brain and coordination of both frontal and parietal cortices. One important question is : how does the brain use and combine information from our senses to guide our actions, especially when the sense we use to perceive an object (e.g. tactile) is different from the sense we use to interact with it (e.g. visual)? In order to frame the position of a perceived or target object in space, we have to associate it with a spatial coordinate code, also called a spatial reference frame. It is defined by Cohen and Anderson as “a set of axes that describe the location of an object” (Cohen and Andersen, 2002). To localize an object in space, different reference frames are used depending on the sensory system that computes the information and the intention associated with the object (Andersen and Buneo, 2002; Cohen and Andersen, 2002). To localize the spatial position of elements around us, two representations exist, the allocentric and the egocentric ones. Allocentric representation is based on elements fixed to the environment and independent from the observer’s body. An object in this representation keeps the same spatial position within the cognitive map when the observer’s body moves. The hippocampus and the temporal lobe have been identified as major structures to use allocentric representation in spatial navigation (Burgess et al., 2002). Egocentric representation, on the other hand, is based on the body of the observer to set external object spatial position in a reference frame relatively to the position of the observer (Burgess et al., 2002). More precisely, the origin of the coordinates system can be set on different body parts like midline of the head or trunk, it can be centered on the retina or on motor effector like the hand or the foot (Burgess et al., 2002). For example, the position of a cocktail just brought to our table by a waiter is framed in eye-centered coordinates based on retinal information. Thus, we are able to investigate the color of the cocktail by doing a saccade directly toward its location, meaning we can give its spatial localization relative to ourselves (egocentric). We can also give its position relative to the configuration of the room’s layout, the cocktail is on a table close to the exit door (allocentric). Then, we may want to pick it up to taste it, its position is also framed in coordinates relative to our main hand that we can use to reach it. This is thanks to these coordinates that we can calculate how we need to move our arm to make the reaching movement. Finally, we may want to taste it, again we need to compute the relative position of the straw to our mouth. In some situations, to perform actions, it is important to recompute the location of the object because the reference frame used to perceive it is different from the one used to execute the action (Cohen and Andersen, 2002; Klautke et al., 2022). This process is carried out by the PPC which codes the spatial position of objects of interest within different coordinates systems. It makes the required coordinates transformation in order to be able to make the

action with the selected motor effector (Bencivenga et al., 2023; Cohen and Andersen, 2002; Klautke et al., 2022).

Any motor action starts with an intention. An intention is an abstract conception that can materialize by becoming an actual action or staying as an idea in our mind (Andersen and Buneo, 2002). Intentions are formed in the PPC (Andersen and Buneo, 2002). The intention defines a goal which must be reached through motor behavior. This behavior is composed of a sequence of muscle activations involving one or more motor effectors (Andersen and Buneo, 2002). Several areas have been identified in animal studies for specific motor behavior such as lateral intraparietal area (LIP) for saccade planning, medial intraparietal area (MIP) for reaching movement, anterior intraparietal area (AIP) for grasping and ventral intraparietal area (VIP) for multiple processes such as sensorimotor coding relative to the head and speed and direction representation (Andersen and Buneo, 2002; Foster et al., 2022; Rolls et al., 2023). From motor intentions, we build-up a motor action plan composed of all the elements required as sequences of muscles activation, limb torquing etc. (Andersen and Buneo, 2002). The action plan is encoded in the prefrontal cortex. It is hierarchically organized with dorsolateral prefrontal cortex (dlPFC) playing a specific role in action planning and coordinating with frontal cortex (Levy, 2024; Niedermeyer, 1998). Within the frontal cortex, multiple areas coordinates to produce the motor command, starting with dorsal anterior cingulate cortex (dACC), then pre-supplementary motor area (pre-SMA), supplementary motor area (SMA) and finally the motor cortex (M1) (Gordon et al., 2023). M1 is the last cortical structure transmitting motor sequences to the thalamus then the motor neurons in the spinal cord. This transmission from M1 to motor neurons is in majority indirect with projections to spinal interneurons or direct for a minority of projections (Ebbesen and Brecht, 2017). M1 is classically described as having global continuous somatotopic organization, such as the stimulation of a specific area activates the corresponding muscle (Ebbesen and Brecht, 2017; Niedermeyer, 1998). This view has now been challenged and updated. Motor cortex neurons express both motor map specific representation and capability of voluntary suppress movements, a volitional control (Ebbesen and Brecht, 2017). A recent study has revisited the continuous aspect of motor cortex (Gordon et al., 2023). The authors showed the existence of inter-effector regions within the classical M1 effectors specific areas (mouth, hand and foot). These inter-effector did not show effector specificity but were activated for action planning and broader body movement, such as abdomen contraction (Gordon et al., 2023). Their results highlighted the complexity of M1 subdivisions for processing both basic motor commands relative to an effector as well as higher planification processes. The pyramidal neurons also project to S1, striatum, cerebellum or even adrenal medulla for movement adaptation, learning and physiologic body response (Ebbesen and Brecht, 2017; Gordon et al., 2023). Thus, the motor cortex has a large motor representation of the body (Ebbesen and Brecht, 2017), not just in terms of body parts but also in terms of complex movement and motor plan coordination..

B. Sensory-motor coordinates transformations

A well-studied sensorimotor transformation is visuo-motor transformation in which visual information is used to guide motor action. For example, seeing a cup of coffee, reaching out for it with our hand and grabbing it. Recently, it has been demonstrated that frontoparietal functional networks are modulated by the choice of movement effector (Bencivenga et al., 2023). The authors identified two sub regions of IPS, lateral posterior IPS (lpIPS) and medial portion IPS (mpIPS), which respectively correlated with FEF during saccade planning and with premotor cortex (PMd) during pointing planning. Different effective connectivity patterns were found for ocular saccades and pointing (hand, foot) movement. More specifically, lpIPS was more specific to saccade and more connected to FEF whereas mpIPS was more connected to PMd and specific to pointing movement. The causal connectivity analysis revealed differences in connectivity patterns between mIPS and premotor cortex for hand and foot pointing movements, demonstrating the importance of the motor effector in shaping the dynamics of neural networks (Bencivenga et al., 2023).

Many studies have focused on movements directed toward external space location and less studies have explored self-directed movements. However, we perform this type of action quite often in our everyday lives for example swatting a mosquito on our arm or scratch an itch. Self-directed actions require the definition of the tactile target location relative to our body and specifically skin location. In order to make a motor movement toward this target, tactile location in 3D space has to be determined. This can be done thanks to the integration of multiple information involving body knowledge, proprioception and body posture (Medina and Coslett, 2010; Shore et al., 2002; Tamè et al., 2019; Yamamoto and Kitazawa, 2001).

One experimental manipulation has been designed to study the remapping from tactile to spatial reference frames, by crossing of the hands. In such a body posture configuration, a stimulus on one hand can be encoded in different reference frames depending on whether we consider its somatosensory or its spatial location. The somatosensory location depends on the anatomical reference (left hand stimulation) regardless of the position of the limb while the spatial location depends on the body centered reference (right spatial position) regardless of the stimulated hand (Heed and Röder, 2012; Wada et al., 2012).

Indicating which part of the body has been touched and more specifically which hand has been touched is prone to more errors when the hands are crossed in comparison to uncross posture configuration. When crossed, the left hand is located in the right side leading to a conflict in reference frame between anatomically-centered and externally-centered codes that can be used to choose stimuli location. Tactile stimulation is framed in anatomical codes when perceived but we need external code to point out to its location. This strong assumption derives from several studies describing consequences of a wrong remapping. As behavioral measure, judgement of temporal order of two consecutive tactile stimuli under cross or uncross body limb posture is heavily used. This paradigm is called tactile temporal order (TOJ) task (Heed and Azañón, 2014; Shore et al., 2002; Wada et al., 2004; Yamamoto et al., 2005; Yamamoto

and Kitazawa, 2001). In this paradigm, two fast and consecutive tactile stimuli are presented, one on each limb. The subject has to indicate which was stimulated first. They tend to make more errors when the limbs are crossed. It has been hypothesized that the cause is a conflict between anatomical and external-spatial reference frames. Azanon and Soto-Faraco proposed that external representation of tactile stimuli is encoded outside of somatosensory cortex in a network involving associative areas whereas anatomical representation is restricted to somatosensory cortex (Azañón and Soto-Faraco, 2008). Studies in fMRI confirmed that PPC encodes the external touch location of a tactile stimuli whereas somatosensory encodes its anatomical location (Klautke et al., 2022). Thus, different reference frames are computed and carried by distinct brain areas according to sensory and motor processing. Somatosensory cortex will code for stimuli from tactile perception whereas PPC will code for action preparation, intention and goal-oriented motor actions. However, the dynamic of these processes is not yet fully understood and several questions remain.

Does the brain dynamically disentangle between those two different ways of representing a location? Does the brain build external codes automatically or are these formed only when the intention of making a movement is decided? Are both anatomical and external codes kept in working memory or does one overwrite the other?

Maij et al., 2020 used a TOJ task to bring more evidence that external location is built on demand according to the chosen limb for the motor action. In this experiment, participants were asked to point out spatial location of the first tactile stimulus among two consecutive stimuli delivered on both hands under dynamic bimanual postural changes (crossing, uncrossing). In other words, participants started trials in a hand posture (cross or uncross) and moved their hands (or kept the hand at its location) to end up in a different (or same) posture. During the trial, each hand received a tactile stimulation with stimulus onset asynchrony (SOA) of 110ms. Once the trial was over, they had to lift up the perceived first stimulated finger and point towards the location of stimulation. Two studied metrics were first detected stimulus and position in space. The authors report that if the participant mistakenly selects the second stimulated hand, the pointed location corresponds to the position of the *other* hand at the time of the *first* stimulus. In other words, participants pointed to a location where no tactile stimulation was present during the second stimulation. Thus, they used the hypothetical first perceived hand and retroactively build back the position of this hand at the moment of the real first stimulation. This result questions the classical view of automatic reconstruction of tactile stimuli in space and supports a time reconstruction hypothesis in which stimulus time and history of body posture are integrated to reconstruct spatial position afterwards. In a series of five experiments published by Badde et al., (2019), the authors investigated in healthy adults the errors in attribution of touched limb when two tactile stimulations were delivered consecutively with a varying time interval (TOJ task) on different body parts. They showed no importance of external location of the stimuli on errors across different experimental conditions including body posture (cross, uncross limbs) and body part (hands, feet). Thus, external location of a

stimulus may not be used to identify which body part has been touched. Mainly, the errors were more attributed to the limbs' usual side of space rather than external-spatial location.

On the clinical side, patients with lesions within PPC demonstrate altered sensorimotor behaviors. For example, spatial neglect patients do not perceive half of their environment relative to them (in an egocentric representation) and contralaterally to the lesion whatever visual, tactile or auditory modality is tested. They do see this part of their environment as they can make saccades in this space but they cannot perform any detection or identification processes revealing a well-functioning retinotopic framing but non-functioning coding or transformation in body-centered reference frame (Bremmer et al., 2001; Kerkhoff, 1999). This also extends to the extinction phenomenon, described in patients when one stimulus in contralateral side of a lesion is not being perceived whereas another simultaneous stimulus positioned ipsilaterally is properly perceived, while each stimulus is perfectly detected or identified when presented in isolation (Làdavas et al., 1998).

The contribution of functional imaging to these questions allows to establish correlational links between activated areas and task events. In humans, homologue of NHP VIP has been proposed to play a central role in encoding stimuli location in an fMRI egomotion experiment, like the macaque VIP does. (Wall and Smith, 2008). Egomotion is applied to any movement of the environment relative to an observer. Dealing with egomotion requires the ability to defined and track location of the body in space and thus use external-spatial reference framing (Wall and Smith, 2008). In non-human primates fMRI and electrophysiology, VIP has been hypothesized to be an important actor for containing external spatial representations of environmental elements (Bremmer et al., 2001; Duhamel et al., 1997; Foster et al., 2022).

Some studies conducted experiments which incorporated more causality using transcranial magnetic stimulation (TMS). TMS allows stimulation of a selected part of the cortex by applying a local magnetic field of one pulse. This stimulation can either activate or deactivate targeted cortical area which makes it convenient to use to investigate cognitive functions. A group of studies focused on impairing ability to transform tactile location into a spatial location. They identified PPC as a major actors for such processes (Azañón et al., 2010; Bolognini and Maravita, 2007; Ruzzoli and Soto-Faraco, 2014). Azañón et al., (2010), performed a TMS stimulation on PPC encompassing putative human VIP homologous of monkey VIP in a task of tactile stimulation location judgment. Participants received two subsequent tactile stimulations, one on the face then on their arm positioned in a vertical position. They had to judge if arm stimulation was higher or lower than face stimulation. TMS stimulation occurred 60 or 250ms after arm stimulation. Subjects that received TMS stimulation on PPC overestimated the height of arm stimulus compared to control group. This experiment shows a strong implication of PPC in encoding spatial representation of tactile stimulus. It remains not clear whether this spatial representation is built in PPC or transferred to it. Moreover, it does not allows to assess information about precise localization within the PPC and also about the dynamic of remapping process.

Klautke et al., (2022) published an fMRI study in which participants had to point out the location of a tactile stimuli on one of their feet in either uncross or cross legs configuration. They showed that the location of a tactile stimulus is represented by both anatomical and external location coding and organized in antero-posterior axis within PPC. Anterior PPC was shown to code for anatomical reference frame in regions including primary sensory cortex S1 and superior parietal lobule (SPL). In contrast, posterior PPC and more specifically medial intraparietal sulcus (mIPS) encoded the external location of the stimulus. The authors also investigated the dynamic aspect of codes transformation and showed that PPC regions were implicated in both anatomical location coding and definition of motor goal spatial location. In other words, in the task paradigm, after movement instruction phase, the subjects could plan their final movement. Only the code for goal remained whereas sensory coding was not retained. More, the regions which previously coded anatomical or external location of the tactile stimulus coded for the task goal, namely bilateral SI, left SPL and right PMD. They showed the importance of one region in SPL coding for anatomical stimulus location, task rule and movement goal, thus this region is a good candidate position for being a major actor for the computation of coordinates conversion from tactile to motor spatial location (aka somato-motor transformation). Finally, the authors identified a broad network for coding movement goal towards targets on one own's body, which is similar to the network described in movement goal planification of visual targets suggesting the same process and regions are used for both.

These studies bring about the hypothesis that a tactile stimulus may not be automatically recoded in external space once perceived. Thus, the process of definition of target location occurs after conscious decisions to make a movement toward an anatomical target. It remains unclear how the brain establishes the 3D spatial coordinates of a touch target. Does it directly convert anatomical coordinates (the location of the touch on the body) into spatial coordinates (where the target is located in external space)? Or does it first categorize which limb the touch occurred on and then determine that limb's position in space? The first hypothesis implies the tactile stimulus coordinates to be recoded in 3D space. The second hypothesis implies that the tactile stimulus may never be recoded in 3D space because it would be associated with the body part used to determine movement target. This difference could be a major distinction between visual and tactile processing. The tactile reference frame has to be represented in a spatial frame to make a movement. This representation could be linked to different body parts, such as eyes, trunk arms, etc. It is not clear whether tactile and visual stimuli location share the same external reference frames in multisensory coding. In order to answer this question, in this thesis work, we used cross-classification of external-spatial coding from touch to vision and vision to touch. In Klautke et al experiment (Klautke et al., 2022), subjects had to point a location (feet) they could not see. It is possible that touch stimuli could be coded in early visual areas when they are visible.

The dynamic of somato-motor remapping is still unclear as many experiments show different results and focused on part of the transformation (sensory or motor). It is necessary to determine within the

PPC and prefrontal cortex which neural substrates hold and then transform reference frames when a tactile stimulus is taken as a motor target. To study these questions, fMRI is a powerful tool as it allows to study large scale networks. We will present a human fMRI experiment in which we seek to compare visual and tactile target coordinate transformation during eye movement planning and under different body postures (cross vs uncross hands).

PhD objectives

The introduction highlights the sensory and motor principles of the perception-action cycle; the knowledge gaps regarding the role of subcortical structures in sensory and multisensory processing for both low and high order processes; the knowledge gaps regarding the neural basis and dynamics of somatomotor transformation. In order to study these questions, we use functional magnetic resonance imaging (fMRI). FMRI allows identifying large-scale networks by recording whole brain hemodynamic activity. Thus, this methodology fits our objectives of identifying brain areas composing functional networks which are often spread over the entire brain.

In the first three chapters, we will describe the involvement of the pulvinar in the processing of face, voice and tactile stimulations and multisensory visuo-tactile processing. In the first study (Gaudet-Trafit*, Froesel*, Dali* et al 2024, in preparation), we will describe the activation of the pulvinar during the processing of visual categories such as faces, auditory processing and tactile processing. In the second study (Dali et al., in preparation), we will describe how the pulvinar adapts its functional connectivity with cortical areas depending on the ongoing task. Overall, our results reinforce the hypothesis of pulvinar as a cortical modulator nucleus. In a third study (Gaudet-Trafit et al, in preparation), we will describe the activation of the pulvinar in a face detection task in a visuo-tactile context. We will show that pulvinar is activated in multisensory and unisensory tactile contexts. In particular, we will show that the pulvinar co-activated with the cortical attentional network during the processing of the least salient sensory conditions. Our results support the hypothesis that the pulvinar is specifically recruited to enhance the processing of challenging sensory stimuli requiring high demanding implication such as detection and discrimination.

In the last chapter, we will describe the cerebral dynamics behind somatomotor actions and how body posture influences the perception and execution of a saccade directed toward a tactile stimulus. The objective is to better identify the neural basis and dynamic underlying somatomotor transformations of tactile coordinates in cross or uncross arms posture. Two fMRI studies have been conducted, one in human (Foster et al, in preparation) and one in macaques which will not be presented here. The goal is to map the reference frames used to encode visual and tactile stimuli in macaque PPC for different arm configurations. We will analyze the regions involved in anatomical and external-spatial coding of tactile

and visual stimuli, as well as the transformation of these sensory signals into planning of saccade eye movements.

Independently from these studies, we also participated in the development of an algorithm to automatically label macaque behaviors on videos. This work is presented as Annex 1 (Moat*, Gaudet-Trafit* et al, 2024 in preparation). Our deep-learning based algorithm achieves higher detecting and tracking of laboratory-housed macaque monkeys relative to the current state of the art. This algorithm will be used both to enhance animal welfare monitoring and to study the neural bases of social cognition in primates.

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Chapter I : Implication of Pulvinar in social and Multisensory processes

Study 1: Human medial pulvinar activity and connectivity patterns for, voice and tactile processing

In this study, we investigated the role of human pulvinar in visual, auditory and tactile processing using fMRI. The role of the pulvinar is well documented for visual category processing such as face and objects , (Arcaro et al., 2015a; Froesel et al., 2024b; Wen et al., 2023). However, the results are inconsistent regarding which pulvinar subnuclei are involved. Human fMRI studies identified activation sites in left ventro-medial (Arcaro et al., 2015a), dorso-medial and ventro-medial (Wen et al., 2023) parts of the pulvinar. One monkey fMRI study identified activation in the inferior, medial and anterior subnuclei of the pulvinar (Froesel et al., 2024b). In addition, the pulvino-cortical functional connectivity during ongoing visual object perception has been assessed only by resting state analysis (Arcaro et al., 2015a). Both macaque fMRI (Froesel et al., 2024b) and electrophysiological (Vittek et al., 2024, 2023) studies highlighted pulvinar activity in audio-visual context for conspecific vocalizations stimuli. However, no evidence has been highlighted in human voice processing. Few studies investigated the implication of pulvinar in auditory processing. Finally, the role of the pulvinar in tactile processing has been assessed in monkey studies with identification of pulvinar neurons for wide somatic stimulation (Ricardo Gattass et al., 1978). In humans fMRI, activations cluster have been identified in multiple pulvinar subnuclei for tactile stimulation on the hand (Charyasz et al., 2023). However, the study shows a high interindividual variability. Moreover, no studies investigated the existence of a fine topographical organization within the pulvinar. For the three visual, auditory and tactile modalities, a clear view of pulvino-cortical functional connectivity during visual, auditory and tactile processing is missing. In this study, we characterized the pulvinar subdivisions activated for face, voice and assessed activation maps of tactile processing by stimulating 3 body parts (face, hand and feet). We also provide a description of the pulvino-cortical connectivity during ongoing task for the three modalities.

Human medial pulvinar is involved in face and tactile processing and dynamically adjusts its cortical connectivity based on ongoing stimulation

M. Gaudet-Trafit⁺⁺¹, M. Froesel⁺⁺¹, S. Dali⁺⁺¹, F. Lamberton² and S. Ben Hamed^{*1}

¹ Institut des Sciences Cognitives Marc Jeannerod, CNRS, Lyon 69500 bron, France

² CERMEP-imagerie du vivant, 59 Boulevard Pinel, 69677 Bron Cedex, France

Corresponding author: benhamed@isc.cnrs.fr

⁺these authors contributed equally to this work

ABSTRACT

Optimal processing of our surrounding environment relies on our ability to detect and integrate external information through multiple sensory modalities to build a coherent representation of the world. Interactions between cortical and subcortical structures contribute to this process. In this context, there is growing evidence that the pulvinar plays a crucial role in visual and face processing. In contrast, its role in auditory, voice and tactile processing has been poorly investigated. Here, we used fMRI localizer tasks to describe pulvinar functional organization and functional connectivity with the cortex during face, voice and tactile processing. We reproduced the activation of the ventral part of the medial pulvinar in face perception and we described an increased pulvinar connectivity with face processing areas and a decreased connectivity with low level visual areas during face processing. In addition, we describe activations of the medial pulvinar during air-puff face, hands and feet tactile stimulations and changes in pulvino-cortical connectivity as a function of which body part is being stimulated. No activations are observed during either voice or non-voice stimulations. Overall, this supports a role of medial pulvinar in multisensory processing and the modulation of cortical areas as a function of the sensory context.

Keywords:

fMRI, Human cognition, Pulvinar, face processing, tactile processing, auditory processing

Introduction

Optimal processing of the surrounding environment and the identification of the most relevant incoming information is crucial for navigating and adapting in a complex dynamic environment. Sensory processing is carried out by both cortical and subcortical areas of the central nervous system (Mesulam, 1998; Murray and Wallace, 2012b). The processing of social sensory signals involved in primate

communication is mainly carried out by visual and auditory brain systems. In the visual system, the processing of faces is crucial to social cognition. A large network of regions spanning from the occipital to frontal cortices including the fusiform (FFA) and occipital face area (OFA), posterior superior temporal lobe (pSTS, also called face-responsive STS, fSTS), anterior temporal lobe (ATL) and orbito-frontal cortex (OFC) has been associated with visual face processing (Gratton et al., 2013; Haxby et al., 2000; Schobert et al., 2018; Troiani et al., 2016; Zhang et al., 2009). In the auditory system, in both human and non-human primates, a voice patch system similar to the face patch system was recently described (Bodin and Belin, 2020; Pernet et al., 2015). Three main voice patches have been identified along the superior temporal gyrus and sulcus, a posterior voice patch, a median voice patch and an anterior voice patch with a large interindividual variability (Bodin and Belin, 2020; Pernet et al., 2015).

In this context, it was initially proposed that subcortical regions such as the thalamus are mere sensory relays, while cortical areas are responsible for the implementation of higher order cognitive functions such as social cognitive functions. This view has been challenged over the last years. Notably, there is growing evidence that the pulvinar, the largest thalamic nucleus, plays a crucial role in higher cognitive functions and prioritizes incoming sensory information including social-related sensory information (Arcaro et al., 2018a; Froesel et al., 2021a; Wen et al., 2023). In addition to visual and audio-visual processing (Froesel et al., 2024a; Vittek et al., 2023) the pulvinar has been shown to be involved in sensorimotor processing (Charyasz et al., 2023). While somatotopic information from periphery to cortex, has been reliably described to be relayed by the ventro-posterior complex of the thalamus (Delhaye et al., 2018), in non-human primates, the anterior subdivision of the pulvinar has anatomic connections to parietal areas 1, 2 and 7b, lateral parietal cortex (Delhaye et al., 2018) and has been shown to be responsive to puff stimulations on the face as well as to visuo-tactile stimulation (Froesel et al., 2021).

The pulvinar is a posterior thalamic nucleus covering 30 % of the thalamus. It exhibits strong interconnections with the primary and associative cortices (Froesel et al., 2021a; Shipp, 2003a). It has a complex and rich structural and functional connectivity organization with the cortex (Asanuma et al., 1985; Froesel et al., 2021; Froesel et al., 2024b; Leh et al., 2008; Shipp, 2003; Vittek et al., 2023). This extensive connectivity has led this nucleus to be considered as a cortical hub for various cognitive processes such as attention, salience or social cognition (Benarroch, 2015; Saalmann and Kastner, 2011; Froesel et al 2021). These studies highlight the role of pulvinar as a coordinator and modulator of perception primarily focusing on vision (Benarroch, 2015a; Fiebelkorn et al., 2019a; Yuri B. Saalmann and Kastner, 2011). In particular, dorsal pulvinar has been associated with attention while ventral pulvinar has been associated with visual temporal structure processing and spatial position and categorization in the context of visual face and scene processing. Accordingly, these subdivisions are functionally connected with specific cortical areas (Arcaro et al., 2018a). In primates, the pulvinar is classically divided in several sub-regions based on their specific cytoarchitectonic and

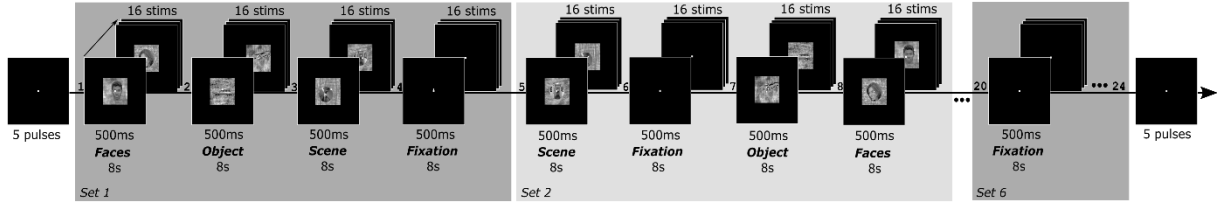
chemoarchitectonic properties, namely the anterior, medial, lateral and inferior subdivisions, each containing further subdivisions (Gutierrez et al., 2000, 1995; Morel et al., 1997; Olszewski, 1952; Stepniewska et al., 1999; Walker, 1938). Aside from this anatomical description of the pulvinar, a meta-analysis of task-related neuroimaging studies identified five pulvinar clusters: inferior, lateral, medial, anterior and superior pulvinar cluster (Barron et al., 2015a). In contrast, pulvinar clustering based on functional connectivity studies splits the pulvinar into a dorso-medial, a ventro-medial, a lateral, an anterior and an inferior cluster (Guedj and Vuilleumier, 2020a). Thus, the parcellation of this nucleus is challenging, three different methods resulting in three different parcellations.

Although the role of the pulvinar in vision processing has been well studied, its involvement in auditory and tactile processing remains less understood. The exploration of these sensory modalities is essential to comprehensively elucidate the pulvinar's multifaceted functions. Additionally, investigating the functional connectivity patterns of the pulvinar across different sensory modalities and sensory contexts can provide deeper insights into how this thalamic structure integrates and coordinates the processing of sensory modalities. In this study, we aim to investigate the functional role of pulvinar in face, voice and tactile processing and elucidate the functional connectivity of the responsive pulvinar regions of interest (seeds) with the rest of the brain. Participants performed three fMRI localizer tasks. First, a face localizer task in which participants were exposed to human faces, objects and scenes. Second, a voice localizer task in which participants were exposed to human voices and different non-voice sounds. Third, a tactile localizer task in which the face (cheeks), the hands and the feet of participants were stimulated unilaterally by air puffs. Our results revealed a localized activation of medial pulvinar in both face and tactile tasks but no response during the auditory task. Specifically, the central part of medial pulvinar was involved in face processing and medial regions were involved in tactile processing. We further characterize the dynamic connectivity patterns of pulvinar with the cortical domain as a function of the ongoing sensory stimulation. In particular, we identify specific pulvino-cortical connectivity patterns for visual face processing as well as for tactile processing.

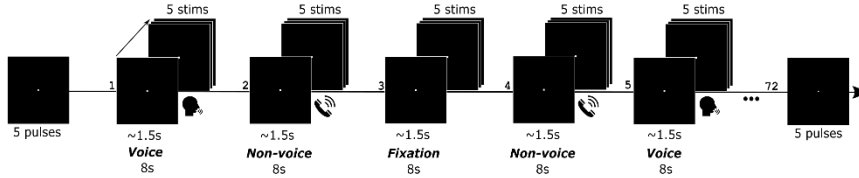
Results

The main purpose of this study was to explore the role of pulvinar in face, voice and tactile processing. In a first step, we validated our localizer tasks (figure 1) by describing the cortical and subcortical network that they activated. In a second step, we performed group-level statistical analyses of whole-brain BOLD activity to investigate pulvinar activation during face, voice, and tactile localization tasks. In a last step, we used the pulvinar activations as seeds for a seed-based analysis, in order to characterize their functional connectivity patterns with the rest of the brain both during resting state and during the different localizer tasks.

A Face localizer run



B Voice localizer run



C Tactile localizer run

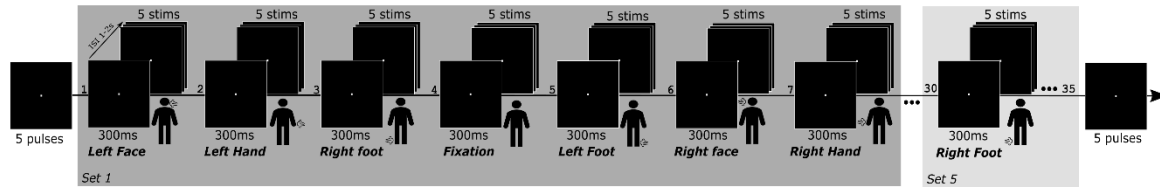


Figure 1: Face, tactile and voice localizer tasks. A) Face localizer. Each run was composed of 6 sets of 4 blocks (total of 24 blocks). Each set was composed of one face block, one object block, one scene block, and one fixation block presented in a pseudo-randomized order. Each block lasted 8s and involved the presentation of 16 randomized stimuli of 500ms. B) Voice Localizer. Each run was composed by 72 randomized blocks of voice stimuli, non-voice stimuli and fixation. Each block involved the repetition of 5 auditory stimuli of ~1.5s separated by 15ms for a total duration of 8s. C) Tactile localizer. Each run was composed of 5 sets of 7 blocks each (total of 35 blocks). Each set was composed of a pseudorandomized presentation of left and right face, hand and feet blocks as well as one fixation blocks. Each block lasted 1.5s and involved the presentation of 5 repetitions of a 300ms stimulation.

Face localizer cortical and subcortical activations

Two main contrasts were defined to analyze the activations during the Face localizer task: 1) a contrast comparing the face, object and scene conditions to fixation (face+object+scene>fixation a.k.a all>fixation); 2) a specific contrast comparing the face condition to the non-face conditions (face>object+scene). For the all>fixation contrast, cortical activations were located within a large cortical network encompassing bilateral occipital cortex, fusiform cortex and parietal cortex (figure 2, top panel and table S1.1). Subcortical activations were observed in the thalamus, superior colliculus, lateral and medial geniculate nuclei (figures 2 and 3B, and table S1.1) and hippocampus (figure 2 and table S1.1), bilaterally. For the face>object+scene contrast, we identified a large cortical activation cluster encompassing bilateral occipital cortex and fusiform cortex in addition to the ventral part of the right medial pulvinar and bilateral hippocampus. All activations for both contrasts were reported with FWE correction for multiple comparisons at voxel level $p < 0.05$. Thus, the Face localizer robustly activated a cortical-subcortical network involved in visual processing. During face processing, a more restricted network is identified, coinciding with the classically reported face areas (Gratton et al., 2013; Haxby et al., 2000; Schwiedrzik et al., 2015; Tsao et al., 2008; Zhang et al., 2009).

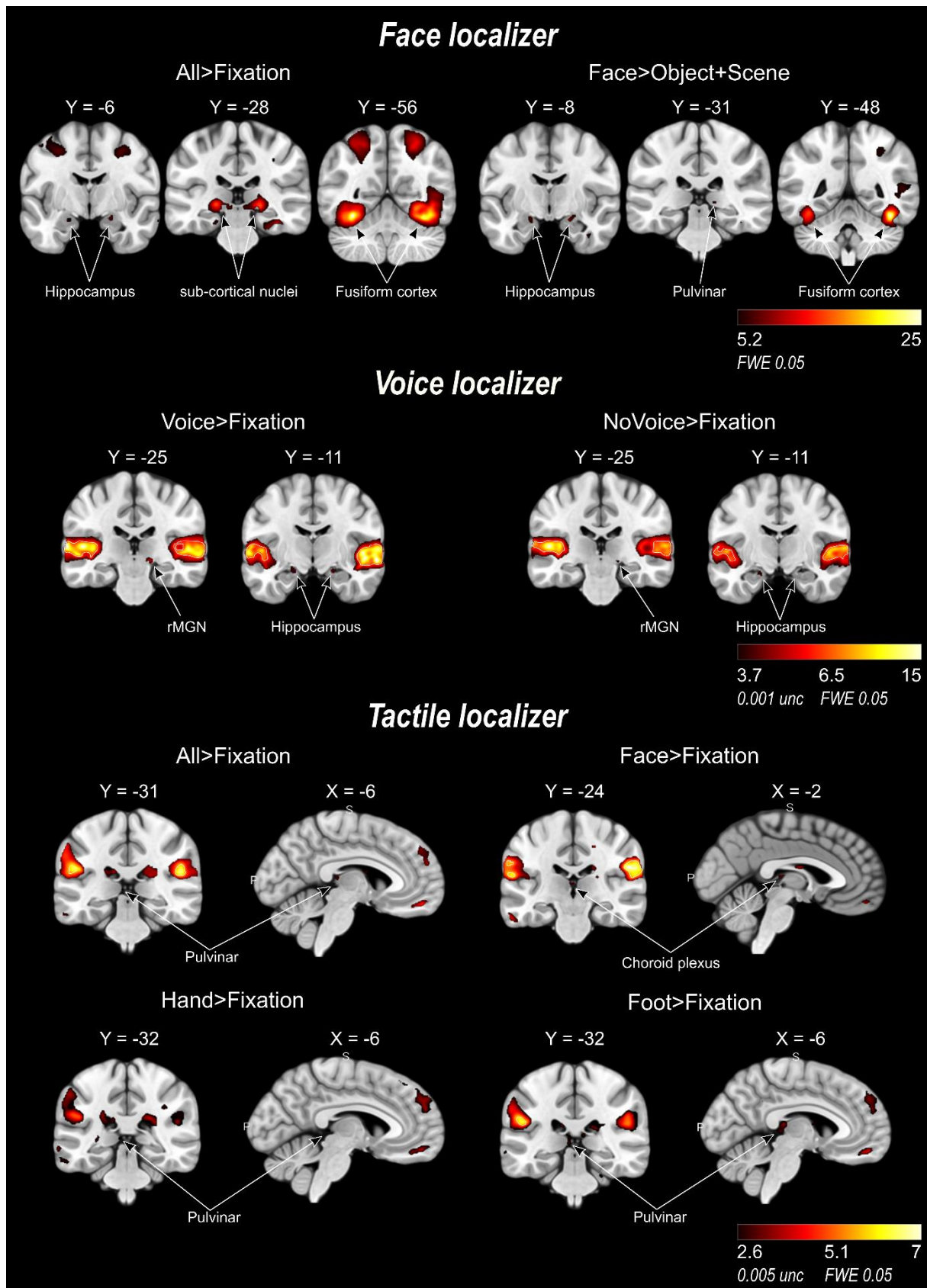


Figure 2: Whole brain contrasts activation for face, vocal and tactile localizer tasks. (Top) Face localizer. Coronal view sections of group level activation maps thresholded at $p < 0.05$ FWE at voxel level and for all>fixation and face>object+scene contrasts. (Middle) Voice localizer. Coronal view sections of group level activation maps thresholded at $p < 0.05$ FWE at voxel

level (white contour on activation maps) and at $p < 0.001$ uncorrected for voice>fixation and non-voice>fixation contrasts. (Bottom) Tactile localizer. Coronal view sections of group level activation maps thresholded at $p < 0.05$ FWE at voxel level (white contour on activation maps) and at $p < 0.001$ uncorrected for all>fixation, face>fixation, hand>fixation and foot>fixation contrasts. MNI coordinates are displayed above each section.

Voice localizer cortical and subcortical activations

Three main contrasts were defined to analyze the activations during the Voice localizer task: 1) a voice>non-voice contrast; 2) a voice>fixation contrast; 3) a non-voice>fixation contrast. Fixation blocks were silent blocks in which no sound was displayed. In the voice>non-voice contrast, significant activations were observed within three cortical clusters in the left temporal mid gyrus, the right superior temporal gyrus and the right temporal mid gyrus ($p < 0.05$, FWE correction for multiple comparison at voxel level, figure 2, Table S1.2). In the voice>fixation contrast, activations were observed mainly in the left and right temporal cortex ($p < 0.05$, FWE correction for multiple comparison at voxel level, figure 2, Table S1.2). At uncorrected $p < 0.001$ threshold at voxel level, we additionally identified activations in the right medial geniculate nucleus (MGN) and bilateral hippocampus (figure 2, Table S1.2). In the non-voice>fixation, two large clusters in the left and right temporal lobe were significantly activated at $p < 0.05$, FWE correction for multiple comparison at voxel level (figures 2, Table S1.2). At uncorrected threshold $p < 0.001$ at voxel level, activations were observed in the medial geniculate nucleus (MGN) and bilateral hippocampus (figures 2, Table S1.2). In summary, voice and no voice stimuli activated cortical temporal areas. Voices specifically activates the voice temporal patches. Right MGN and bilateral hippocampus were activated at lower threshold with no specificity for voice processing. No activations within the pulvinar nucleus were observed at these statistical thresholds. Thus, the Voice localizer robustly activated a cortical-subcortical network involved in auditory processing. During voice processing, a more restricted network is identified, coinciding with the classically reported voice areas (Belin et al., 2000; Bodin and Belin, 2020; Moerel et al., 2014; Pernet et al., 2015; Zhang et al., 2021).

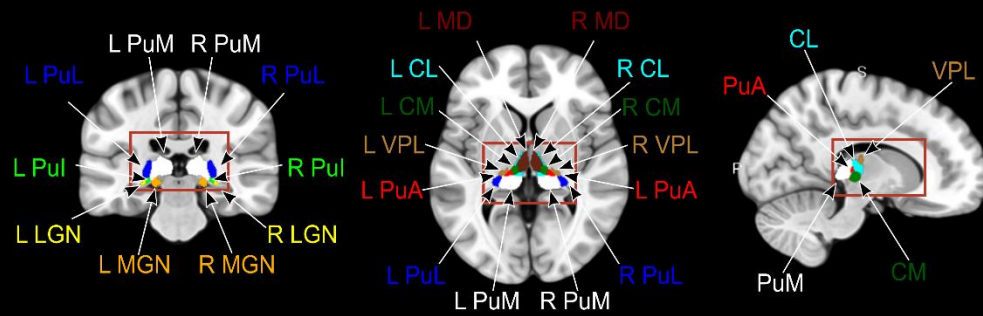
Tactile localizer cortical and subcortical activations

Four main contrasts were defined to analyze the activations during the Tactile localizer task: 1) a face>fixation contrast; 2) a hand>fixation contrast; 3) a foot>fixation contrast; 4) all stimulation conditions vs contrast, a.k.a. all>fixation contrast. For the all>fixation contrast, at a threshold of $p < 0.05$ FWE corrected, we observed activations within the superior temporal gyrus, the supramarginal area and the Rolandic operculum bilaterally (figure 2, Table S1.3). We also observed activations in the left angular gyrus and in the ventral part of the left postcentral gyrus. At an uncorrected threshold of $p < 0.005$, we identified a cluster of 99 voxels in the medial part of medial pulvinar exhibiting 2 peaks (figures 2 and 3C, Table S1.3). The anterior peak activity was identified as possibly originating from the posterior choroid artery of the third ventricle whereas the posterior peak was identified as a medial pulvinar activity.

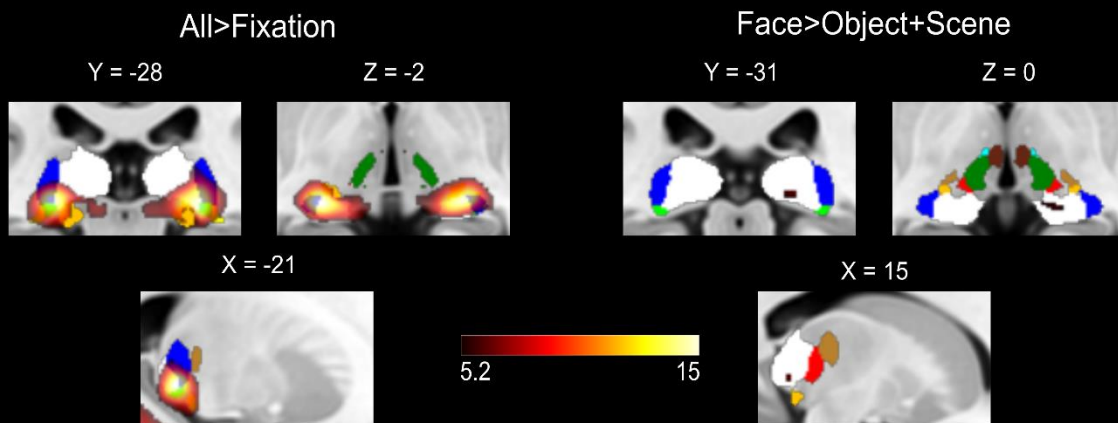
In the face>fixation contrast, at $p < 0.05$ FWE corrected threshold, activations were located in the superior temporal gyrus, supramarginal area, Rolandic operculum bilaterally and in the left postcentral gyrus as well as left insula (figure 2, Table S1.3). At uncorrected $p < 0.005$ threshold, a cluster of 56 voxels was observed in the third ventricle and was considered as activity from posterior choroid artery of the third ventricle (figure 2, bottom). A smaller cluster of 21 voxels was also identified in medial pulvinar. In the hand>fixation contrast, at a $p < 0.05$ FWE corrected threshold, we identified activity in the left angular gyrus and right Rolandic operculum. At uncorrected $p < 0.005$ threshold, we identified one cluster in the third ventricle (originating from posterior choroid artery) and one cluster in the medial pulvinar (figures 2 and 3C, Table S1.3). In the foot>fixation contrast, at $p < 0.05$ FWE corrected threshold, activations were observed in the left superior temporal gyrus, the Rolandic operculum and the left angular gyrus (figures 2 and 3C, Table S1.3). At uncorrected $p < 0.005$ threshold, we identified a cluster of 181 voxels. This cluster encompassed both voxels from the part of the medial pulvinar as well as voxels within the third ventricle. It exhibited 3 peaks. Two peaks were identified as bilateral activity from posterior choroid artery of the third ventricle (Li et al., 2018; Neau and Bogousslavsky, 1996) and one peak was identified as medial pulvinar. Additionally, at this same threshold (uncorrected $p < 0.005$), we identified activations in the caudate nucleus for all contrasts.

Thus, overall, the Tactile localizer robustly activated a cortical-subcortical network involved in tactile processing (Chen et al., 2008; Grahm et al., 2008; Huang et al., 2012; Neau and Bogousslavsky, 1996).

A Thalamic subnuclei



B Thalamic activation in face localizer



C Thalamic activation in tactile localizer

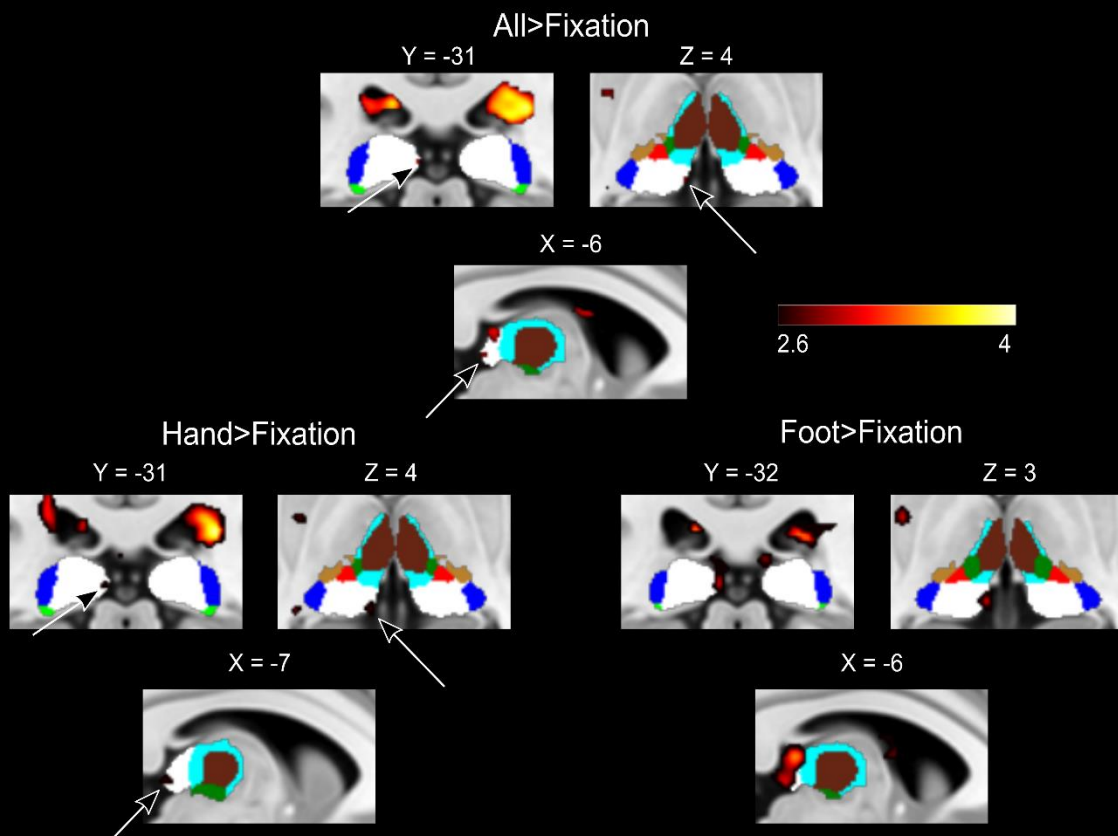


Figure 3: Thalamic activation in face and tactile localizer tasks. A) Thalamic subnuclei location based on the Morel human thalamic atlas (Morel et al., 1997). B) Face localizer. Coronal, axial and sagittal sections of group level activation thresholded maps ($p < 0.05$ FWE at voxel level), for all>fixation and face>object+scene contrasts. C) Tactile localizer. Coronal, axial and sagittal sections of group level activation thresholded maps ($p < 0.005$ uncorrected at voxel level), for all>fixation, hand>fixation and foot>fixation contrasts. Thalamic nuclei and subcortical structures are shown in different colors: PuI: inferior pulvinar (green), PuA: anterior pulvinar (red), PuL: lateral pulvinar (blue), PuM: medial pulvinar (white), MD: mediodorsal nucleus (brown), CM: centre median nucleus (dark green), CL: central lateral nucleus (cyan), VPL: ventral posterior lateral nucleus (gold), MGN: medial geniculate nucleus (orange), LGN: lateral geniculate nucleus (yellow). MNI x, y, z coordinates are displayed above each section.

The pulvinar is activated by visual face and non-face stimuli and tactile hand and foot stimuli but not by auditory voice and non-voice stimuli

The goal of our study was to investigate the sensory properties of the pulvinar during passive sensory stimulation. Figure 3 summarizes the observed activations. The precise segmentation of the pulvinar was achieved thanks to the human Morel atlas (Morel et al., 1997). Visual activations in the All>Fixation contrast were widespread and encompassed the Inferior pulvinar, the ventral part of the lateral pulvinar as well as the ventral part of the medial pulvinar ($p < 0.05$, FWE correction for multiple comparison at voxel level, figure 3B, left panel). The activation also included the MGN. These activations were bilateral. The specific face>object+scene activated a cluster of 34 voxels located in the right medial pulvinar, in its ventral anterior portion ($p < 0.05$, FWE correction for multiple comparison at voxel level, figure 3B, right panel).

Tactile activations in the All>Fixation contrast were focal and activated a small cluster of 19 voxels in the posterior part of the left medial pulvinar (uncorrected $p < 0.005$, figure 3C, top panel). The specific face>fixation contrast activated a cluster of 21 voxels located in this posterior part of the left medial pulvinar (Table S1.3). The specific hand>fixation contrast activated a cluster of 97 voxels located in this posterior part of the left medial pulvinar (Figure 3C, bottom left panel, Table S1.3). The specific foot>fixation contrast activated a cluster of 160 voxels located in the posterior part of the left medial pulvinar (Figure 3C, bottom right panel, Table S1.3) and a much smaller cluster of 4 voxels in the posterior part of the right medial pulvinar (Table S1.3). Two peaks could be identified in the hand and foot left activation clusters. These peaks were at identical location for both hand and foot stimulations. The peak of the face activation was located in between the two hand/foot peaks.

In order to further investigate the medial pulvinar responses to tactile stimulation, we quantified the percent signal change (PSC) in a spherical seed of 4mm of radius created from the activated cluster extracted from the All>fixation contrast analysis (MNI coordinates, $x = -7$, $y = -30$, $z = 5$, figure 4, right panel). The goal was to compare PSCs across stimulation conditions. We tested six lateralized tactile stimulation conditions: left face, right face, left hand, right hand, left foot and right foot. Figure 4 shows a significant PSC relative to baseline for left hand stimulations (0.0971%, $p < 0.01$, Wilcoxon signed rank test), left foot stimulations (0.0787%, $p < 0.01$), and right face stimulations (0.0975%, $p < 0.008$)

conditions (Figure 4). Thus, the seed located in the left medial pulvinar showed significant activation by ipsilateral left hand and foot and contralateral right face stimulations. However, no significant difference was found between the six conditions (Friedman test, $p=0.16$, $X=27.8857$).

Overall, it is thus unclear whether the pulvinar holds a tactile body map and how ipsilateral and contralateral stimulations are processed. This will have to be explored with higher resolution methods such as fast ultrasound imaging, intracortical recordings or high-field 7T imaging.

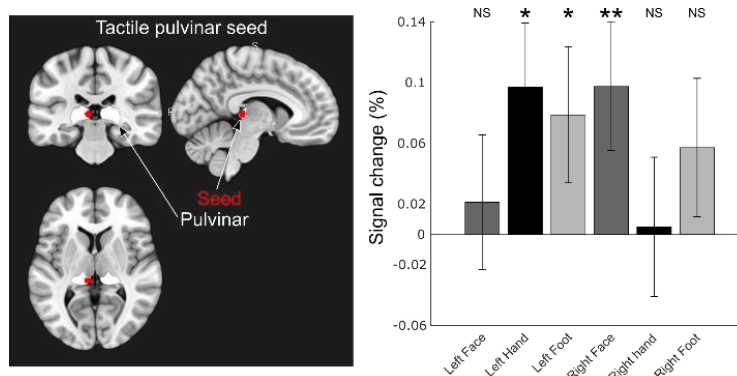


Figure 4: PSC of left medial pulvinar for tactile localizer conditions. Right panel. Location of the 4mm radius spherical seed in left medial pulvinar (Seed, red, MNI coordinates, $x=-7$, $y=-30$, $z=5$; pulvinar, white). Left panel. Percent signal change for left, right face, hand and foot tactile stimulation conditions (Wilcoxon signed rank test, * : p -value <0.05 , ** : p -value <0.01 , corrected for multiple comparisons -Dunnnett's test).

Right ventral medial pulvinar specifically connects to the contralateral temporo-frontal ventral visual pathway during face processing

In order to assess the intrinsic connectivity affiliation of pulvinar with the rest of the brain during the face localizer task, we employed seed-based connectivity analyses using previously identified activation clusters extracted from the face localizer task as seeds. For the all>fixation contrast, two medial pulvinar clusters (one left and one right) were used as seeds and for the face>object+scene contrast, the right medial pulvinar cluster was used as seed.

Overall, in the all>fixation contrast (figure 5, left panel, tops rows, figure S1), the left seed was mostly connected to the left lateral and medial prefrontal cortex and disconnected from bilateral medial posterior parietal cortex. The right seed was likewise connected to the right lateral and medial prefrontal cortex. It was additionally connected to right the temporal cortex and the medial part of the central sulcus. Prefrontal and central sulcus activations were grossly bilateral. More specifically, the left seed was positively correlated with left frontal gyrus, left frontal pole and left precentral gyrus (figure 5, left panel, left columns, top rows, Table S2.1). It was also negatively correlated with precuneus cortex and cingulate gyrus (figure 5, left panel, left columns, bottom rows, Table S2.1). The right seed was positively correlated with right precentral gyrus, frontal gyrus, frontal pole, supplementary motor cortex, temporal gyrus, parahippocampal gyrus, temporal pole, hippocampus, orbitofrontal cortex, thalamus,

and temporal gyrus as well as to left lingual gyrus (figure 5, left panel, right columns, top rows, Table S2.2).

The face>object+scene contrast mostly highlighted an increase in the right seed connectivity with the left prefrontal, occipitotemporal and temporal areas and a slight decrease in its connectivity with the medial occipital cortex bilaterally. More specifically, this right medial pulvinar seed was mainly positively correlated with left frontal pole, left frontal gyrus, left orbitofrontal cortex, left occipital cortex, left angular gyrus, left paracingulate gyrus, left temporal gyrus and left temporal pole (figure 5, right panel, top rows, Table S2.3). It was negatively correlated with different regions located mainly in the left hemisphere including the precuneus cortex, intracalcarine cortex, lingual gyrus, occipital cortex, in addition to right intracalcarine cortex (figure 5, right panel, bottom rows, Table S2.3). Thus, the seed was contralaterally connected to prefrontal, temporal and higher occipital visual areas, highlighting the ventral visual stream and it was disconnected from bilateral or contralateral lower-level occipital visual areas.

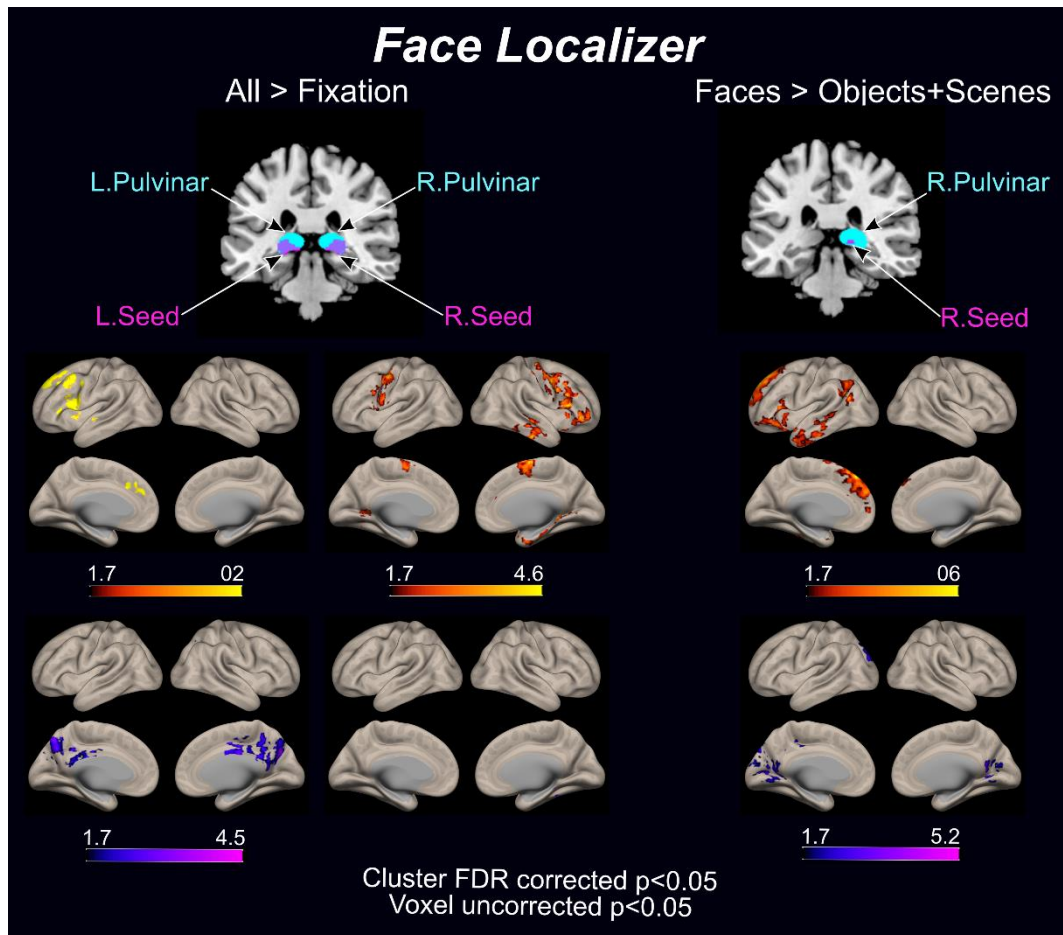


Figure 5: Seed-based connectivity of the pulvinar with the cortex in face localizer tasks. Top panel. Pulvinar nucleus (cyan), pulvinar seed locations in purple. Bottom panel. Seed-to-voxel connectivity analysis displayed on inflated brain representations for all>fixation contrast (left, left seed and right seed) and face>object+scene contrast (right, right seed). Maps thresholded at cluster level FDR correction $p<0.05$ and voxel level uncorrected $p<0.05$.

Connectivity of the left medial medial pulvinar shifts dynamically as a function of the location of the tactile stimulation on the body

We proceeded in the same way to assess the intrinsic connectivity affiliation of pulvinar with the rest of the brain during the tactile localizer task. We thus employed seed-based connectivity analysis using the previously identified left medial pulvinar cluster activation extracted from the tactile localizer all>fixation contrast as seed. In the all>fixation contrast, there was no notable connectivity between the seed and the rest of the brain (figure 6). This could be due to an actual absence of functional connectivity between the seed and the cortex. This is unlikely. Alternatively, this could be due to the fact that each tactile stimulation activated a different cortical network. The analysis of the seed to whole brain connectivity for each of the face>fixation, hand>fixation and feet>fixation independently confirmed this latter hypothesis (figure 6, cluster level FDR correction $p < 0.05$ and voxel level uncorrected $p < 0.05$). Indeed, in the face contrast, the seed showed enhanced contralateral connectivity with occipital regions encoding the periphery of the visual field, and inferior parietal cortex in addition to the precuneus. In the hand contrast, the seed showed enhanced bilateral connectivity with the posterior cingulate cortex and the precuneus. In the foot contrast, the seed showed enhanced ipsilateral connectivity with motor regions in the precentral and postcentral gyrus, inferior frontal gyrus and parietal cortex. The specific foot>face contrast described enhanced connectivity with ipsilateral prefrontal cortex (inferior frontal gyrus and middle frontal gyrus) and anterior cingulate cortex in addition to the precentral gyrus and supplementary motor cortex. The specific foot>hand contrast described enhanced connectivity particularly with inferior frontal gyrus and decreased bilateral connectivity with the paracingulate cortex gyrus, anterior cingulate cortex and left inferior temporal gyrus.

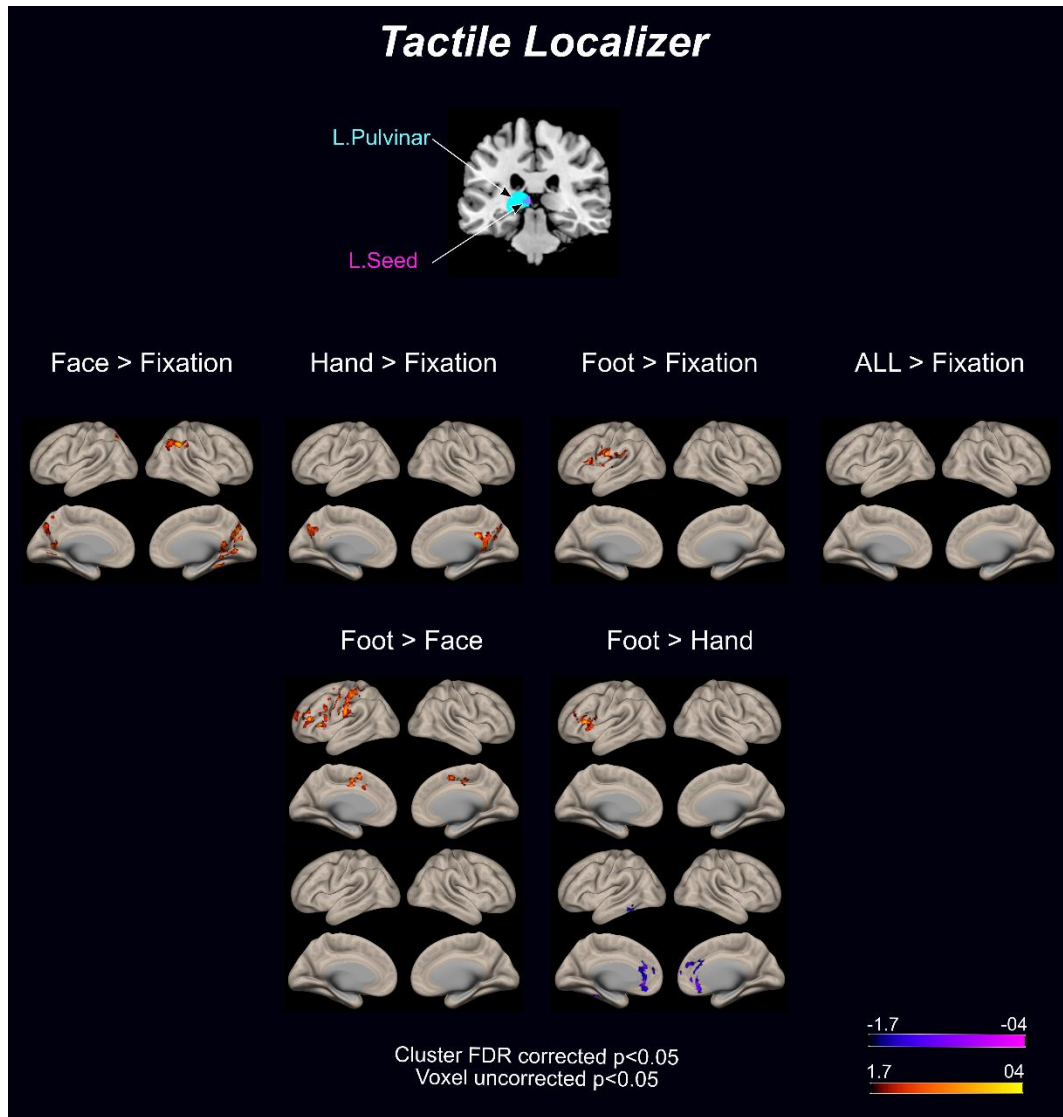


Figure 5: Seed-based connectivity of the pulvinar with the cortex in face localizer tasks. Top panel. Pulvinar nucleus (cyan), pulvinar seed locations in purple. Bottom panel. Seed-to-voxel connectivity analysis displayed on inflated brain representations for Face>Fixation, Hand>Fixation, Foot>Fixation, Foot>Face and Foot>Hand. Maps thresholded at cluster level FDR correction $p < 0.05$ and voxel level uncorrected $p < 0.05$.

Discussion

The present study investigated the involvement of the pulvinar in sensory processing. Stimulating independently three different sensory modalities (i.e. visual, auditory and tactile) thanks to three distinct localizer tasks and using a voxel resolution (1.6x1.6x2.5mm) two or three time higher than standard acquisitions (e.g. 2.5x2.5x2.5mm or 3x3x3mm) with a field of view covering the entire brain, we confirm the pulvinar's significant role in visual processing. We additionally demonstrated its involvement in tactile processing associated with very localized activations. Unexpectedly, we showed no significant role of the pulvinar in auditory processing. Interrogating the functional connectivity between the identified visual and tactile pulvinar regions of interest and the rest of the brain we showed that, within a given localizer task (visual or tactile), the pulvinar changed its connectivity pattern with

the cortex as a function of the ongoing stimulations, thus dynamically adjusting to the sensory demand. For example, during face processing, the pulvinar exhibited decreased connectivity with low level visual processing areas and increased connectivity with cortical regions specialized in face recognition, as well as frontal areas as compared to when processing non-facial stimuli.

Medial pulvinar face activations

Our results corroborate the role of pulvinar in face processing as previously demonstrated in various studies (Arcaro et al., 2018; Benarroch, 2015; Froesel et al., 2024a; Kaas and Lyon, 2007; Wen et al., 2023). In the cortical domain, face processing induced a large activation in occipital and temporal cortices comprising both bilateral fusiform face area (FFA) and orbital face area (OFA). No activations were observed in pSTS area which has been described as a face patch by Deen and colleagues (Deen et al., 2020). This could be due to the specificity of pSTS for face movement and motion, and to the fact that our stimuli did not have those characteristics. The ventral posterior activations we report within the pulvinar are associated with object categorization, and more specifically face processing, aligns with previous findings (Arcaro et al., 2018a; Schwiedrzik et al., 2015; Wen et al., 2023).

Dynamic coupling of medial pulvinar with face-related cortical areas

Our functional connectivity results bring new insights on the dynamic connectivity of pulvinar with the cortex during sensory processing. Indeed, the pulvinar showed different connectivity patterns with the brain depending on the ongoing sensory modality and stimuli category. During face processing, the pulvinar increased its connectivity with frontal and ventral visual stream areas and decreased its connectivity with low level visual processing areas. The prefrontal cortex is crucial in complex cognitive tasks, including attention, working memory, and decision-making. All these functions are engaged. This increased connectivity likely reflects the coordination between sensory processing and cognitive evaluation required for recognizing and processing faces, given their social and emotional significance (Arend and Zimmer, 2011). The ventral visual stream, particularly regions like the fusiform face area (FFA), is specialized for object and face recognition. The enhanced connectivity between the pulvinar and the ventral stream indicates that this nucleus might facilitate the detailed processing of facial features, integrating sensory inputs to construct a coherent percept of the face. This supports the role of the pulvinar in selectively modulating visual information flow to optimize face perception and recognition (Bridge et al., 2016a). The enhanced connectivity with the prefrontal cortex supports the possible engagement of attention, working memory, and decision-making during face recognition, encoding and identification (Fuster, 2019; Levy, 2024). The reduced connectivity with low-level visual processing areas, such as the primary visual cortex (V1), suggests a shift in the pulvinar's role from basic visual processing to more specialized, higher-order tasks during face recognition. This decrease may indicate a gating mechanism where the pulvinar suppresses irrelevant or redundant information from early visual areas to prevent sensory overload, thus allowing for more efficient processing of complex visual stimuli like faces. This selective connectivity is essential for focusing cognitive resources on the

most relevant aspects of the visual input. This connectivity dynamics is expected to enhance the role of pulvinar in modulating visual information flow and coordinating between sensory and cognitive regions during face recognition (Yuri B. Saalman and Kastner, 2011). The pulvinar's dynamic connectivity changes reflect its function as a hub that prioritizes and integrates relevant sensory information while filtering out less pertinent data, facilitating efficient and accurate face recognition (Shipp, 2003a).

Auditory activations in the MGN and hippocampus but not in the pulvinar

During the voice localizer task, a large cortical activation along bilateral superior temporal gyrus and sulcus was observed, coinciding with the three main voice patches (Bodin and Belin, 2020). No pulvinar activation was observed while subcortical rMGN and hippocampus were activated for both voice and non-voice conditions. This lack of activation in the pulvinar is surprising knowing that the pulvinar has been shown to be a part of the auditory system (Kaas and Hackett, 2000) and is activated by vocalizations in marmoset (Jafari et al., 2023) and macaques (Froesel et al., 2024a). The absence of pulvinar activation might be attributed to the passive nature of the task, not requiring any specific cognitive engagement from the subjects except for fixation. Alternatively, this could be due to the fact that the sounds were not delivered in a multisensory context. This point will have to be explored in future experiments involving tasks requiring active auditory discrimination. Indeed, it has been proposed that the pulvinar is more likely to be recruited during more challenging tasks in order to enhance stimuli perception (Froesel et al., 2021a).

The rMGN is part of the auditory thalamus and plays a crucial role in the relay and processing of auditory information from the ear to the auditory cortex. Its activation in both voice and non-voice conditions is consistent with its well-established function in basic auditory signal transmission and initial processing (Lee and Sherman, 2011; Winer, 2006). Activation of bilateral hippocampus for both face and voice localizer tasks is consistent with previous literature. Sliwa et al, 2016 (Sliwa et al., 2016), identified neurons responding for social visual and auditory stimuli in macaque hippocampus. The identified cluster at lower statistical threshold for voice localizer than face localizer is also consistent with their findings of predominance for visual stimuli. The activation peaks we reported were close to hippocampus/amygdala border, fitting with an fMRI visual localizer study that reports interindividual variability in the localization of visual activations in the amygdala or the hippocampus (Schwarz et al., 2019). The hippocampus is known for its role in forming and retrieving memories (Suzuki and Eichenbaum, 2000). The activation of the hippocampus in response to both voice and non-voice stimuli possibly indicates the engagement of memory processes for the recognition and contextualization of auditory stimuli, regardless of their nature (Poremba et al., 2004; Sliwa et al., 2016).

Medial pulvinar activations are similar for face, hand and foot tactile stimulations

The tactile localizer robustly activated cortical regions involved in somatosensory processing. We report activations in parietal and temporal areas, as well as bilateral activations of secondary somatosensory

cortex (SII) in agreement with previous studies (Chen et al., 2008; Fujiwara et al., 2002; Hoechstetter et al., 2000). We did not observe any activation in SI. This is usually the case in studies requiring some level of attention to the tactile stimulation (Chen et al., 2008; Fujiwara et al., 2002; Hoechstetter et al., 2000), and matches our instruction to the subjects to focus on the tactile stimulations as they were presented.

The caudate nucleus of the striatum was activated across all conditions in tactile localizer at low statistical threshold. This nucleus is one of the main input to basal ganglia and receives connections from a large part of the cortex including somatosensory and motor cortex (Grahn et al., 2008). Functionally, it is implicated in high level cognitive process like goal planning, sensorimotor processes and oculomotor circuit (Grahn et al., 2008). During the tactile localizer task, participants had to focus on the stimulated body part and press a response button when fixation point was changing color. Thus, activation of the caudate nucleus in this task might be due to either its anatomical connections with somatosensory cortex or the oculomotor demands or a combination of both.

Our first whole brain contrast analysis showed a weak activation cluster in the medial part of medial pulvinar during tactile processing of air puffs. The PSC analysis confirmed the increase in activity in medial pulvinar during face, hand, and foot stimulations and showed no difference between the three conditions. This result is, to our knowledge, the first demonstration of pulvinar activation following puff stimulation in humans and reveals no body-part specific organization within the pulvinar. These results are in line with a recent fMRI study (Charyasz et al., 2023) using both passive and active tactile tasks to investigate the sensorimotor activations in the thalamus. The authors found activation in PuA, PuM and PuL in both of these tasks involving either movement or stimulation of the right finger. We add to this study PuM activations following the stimulation of other body parts. They observed pulvinar single subject activations in a restricted number of subjects for the 4 subdivisions of pulvinar but with dominance of the left over the right pulvinar. A convergent observation between the two projects is that both observed activations in left medial pulvinar during tactile stimulation. However, Charyasz et al. observed only few voxels activated in medial, lateral and inferior pulvinar and a high number of voxels in anterior pulvinar whereas we only observed activations in medial pulvinar. The location of the observed voxels within medial pulvinar across the two studies is also different, our activations being located close to the medial border of the pulvinar whereas they observed voxels more laterally and anteriorly. This might be due to the difference in the type of tactile stimulation applied, one study using air puffs and the other study using air pulses through an inflatable finger clip.

The identified pulvinar tactile activation is located within the pulvinar and the peak of the cluster falls within the Morel pulvinar ROI (Anne Morel et al., 1997). This is intriguing as no direct anatomical connections exist between the pulvinar and the somatosensory cortex. Three hypotheses may account for this activity. First, the activity might be mislocated in the pulvinar while originating from a nearby

structure. Two main candidates would be ventral posterior lateral (VPL) and central lateral (CL) nuclei of thalamus (Delhay et al., 2018; Kumar et al., 2017; Mai and Majtanik, 2019). The VPL which relays somatosensory information to the cortex, can be dismissed due to its distance from the tactile cluster location (Delhay et al., 2018). The CL part of the internal medullary lamina complex (Kumar et al., 2017) is closer to our tactile cluster than VPL. However, the spatial resolution of our fMRI scans allows us to unambiguously describe no overlap between CL and the reported tactile cluster. Additionally the CL is involved in pain perception (Nowacki et al., 2023) and air-puff stimulation used in the tactile localizer did not elicit any pain perception in none of the subjects. Thus, we can rule out the hypothesis of adjacent structures accounting for the activity observed in the posterior ventral cluster. Another possibility is that projections between our pulvinar cluster and somatosensory system may exist but have not yet been described. The cluster is located close to the third ventricle and injections and recordings are difficult to target at that location. Yet, the most plausible hypothesis is that of an indirect activation of medial pulvinar through secondary projections possibly related to awareness of tactile perception. Pulvinar is indeed implicated in visual attention and saliency. Participants were instructed to focus on the tactile stimulations. Thus, the pulvinar may have been activated due to the deployment of attentional processes or to the intrinsic salience of the unpredictable tactile stimuli (Froesel et al., 2021a; Guedj and Vuilleumier, 2020a). This possibly suggests that the pulvinar responds to general sensory salience rather than to the specific nature of the stimulus. This aligns with the pulvinar's role in modulating attention towards stimuli that require immediate processing, regardless of their specific sensory characteristics highlighting its function in integrating diverse sensory information and directing attention towards salient stimuli, irrespective of their specific nature. Interestingly, we found activation in the medial part of the medial pulvinar and not in anterior pulvinar (PuA) in spite of the fact that anatomical connections between PuA and lateral and anterior parts of the parietal cortex and area 1 have previously been described (Delhay et al., 2018). This might be due to the fact that our localizer did not activate these cortical regions.

Dynamic coupling of the tactile medial pulvinar ROI with cortical areas

Connectivity analysis showed that the tactile medial pulvinar seed cluster exhibited distinct connectivity patterns depending on the location of the tactile stimulation. Indeed, while in the all>fixation contrast, no significant connectivity was observed, the separate contrasts (face>Fixation, Hands>Fixation and feet>Fixation) resulted in distinct non-overlapping cortical functional connectivity patterns with this tactile medial pulvinar ROI. This indicates that tactile stimulation resulted in different pulvino-cortical connectivity patterns as a function of which body part was stimulated.

Within the face>Fixation contrast, the left medial pulvinar exhibited enhanced contralateral connectivity with occipital regions involved in visual field periphery processing (Roger B H Tootell et al., 1998), the inferior parietal cortex, and the precuneus. These findings align with the pulvinar's role in integrating multisensory information, particularly in coordinating visual and tactile inputs (Benarroch, 2015a;

Fiebelkorn et al., 2019a; Yuri B. Saalmann and Kastner, 2011), as well as with privileged convergence between tactile information and visual peripheral information (Guipponi et al., 2015). The connection with the inferior parietal cortex, a region associated with attention and sensorimotor integration (Behrmann et al., 2004; Sereno and Huang, 2014), suggests that the pulvinar may facilitate the alignment of visual and somatosensory maps, supporting efficient cross-modal processing.

The same seed showed for Hands>Fixation contrast, enhanced bilateral connectivity with the posterior cingulate cortex and the precuneus. The posterior cingulate cortex is a key node in the default mode network (DMN), which is implicated in self-referential thought and internally focused attention (Raichle, 2015). The precuneus, another DMN region, is involved in visuospatial imagery and attention (Raichle, 2015). The pulvinar's connectivity with these regions during hand stimulation may reflect the integration of tactile information with internally directed cognitive processes, potentially facilitating the mental representation of hand movements or positions.

In the Feet>Fixation contrast, this left medial cluster demonstrated an enhanced ipsilateral connectivity with motor regions (precentral and postcentral gyri), the inferior frontal gyrus, and the parietal cortex. The involvement of motor regions suggests that the pulvinar plays a role in linking tactile inputs from the feet with motor planning and execution areas, which may be crucial for tasks requiring coordination between sensory input and motor output, such as walking or balance. The connectivity with the inferior frontal gyrus, a region associated with motor planning (Dick et al., 2019; Niedermeyer, 1998), further supports the pulvinar's role in preparing and executing motor responses based on tactile information.

The specific contrasts between foot and face, as well as foot and hand, revealed additional insights into the functional specialization of the pulvinar's connectivity. The foot>face contrast showed enhanced connectivity with ipsilateral prefrontal regions (inferior and middle frontal gyri), the anterior cingulate cortex, and motor areas (precentral gyrus and supplementary motor area). These connections highlight the pulvinar's involvement in integrating tactile information from the feet with higher-order executive functions, such as decision-making and motor planning. The anterior cingulate cortex's involvement suggests a role in monitoring and adjusting behavior in response to tactile stimuli from the feet, which is critical for maintaining balance and posture. In the foot>hand contrast, there was enhanced connectivity particularly with the inferior frontal gyrus and decreased bilateral connectivity with the paracingulate gyrus, anterior cingulate cortex, and left inferior temporal gyrus. This pattern indicates a shift in the pulvinar's role from integrating tactile information with motor planning (in the case of foot stimulation) to modulating higher cognitive functions, such as attention and decision-making, when the tactile input comes from the hands. The decreased connectivity with regions involved in self-referential processing and visual memory (paracingulate gyrus and inferior temporal gyrus) suggests that the pulvinar's connectivity is finely tuned to the specific demands of the tactile task at hand.

Relation with previous work

Guedj and Vuilleumier proposed five clusters functional parcellation of the pulvinar based on fMRI resting-state data from 100 healthy subjects (Guedj and Vuilleumier, 2020a). According to their parcellation, our face localizer activation cluster is located in their dorsomedial cluster. Our tactile localizer activation cluster is located in their ventromedial cluster. This supports the hypothesis of a process dependent organization within pulvinar. Guedj and Vuilleumier further specify resting-state connectivity of each cluster with the rest of the brain as well as task-related connectivity. The dorsomedial cluster was identified to be connected with cortical regions involved in default mode and saliency networks. The ventromedial cluster showed a widespread connectivity pattern including the ventral visual extrastriate areas and medial temporal lobe which belong to the visual ventral stream important for object recognition including face recognition. Our activations are thus not consistent with the connectivity of the corresponding cluster described by Guedj and Vuilleumier.

Taken together, these results advance our understanding of the pulvinar's multisensory role in the brain, highlighting its role in face processing and tactile processing possibly supporting its role in integrating and modulating sensory information to support higher-order cognitive functions and attentional processes. Future studies should address the functional parcellation of pulvinar and their contribution to different cognitive functions so as to clarify the role of this nucleus in human cognition.

Method

Participants and ethical approval

Twenty-five subjects aged between 20 to 33 years old (12 males and 13 females) with mean age of 28.08 years old (SD = 3.8) participated in the study after signing an informed written consent. All subjects were healthy and had no history of psychiatric or neurological disorders and had normal or corrected-to-normal vision. Edinburgh Handedness Inventory laterality test (doi: <https://doi.org/10.1016%2F0028-3932%2871%2990067-4>) is used to determine a lateralization score for each participant (17 right-handed, 1 left-handed and 7 ambidextrous). The present study was approved by the French ethical comity CPP sud-Med IV (Comité de Protection des Personnes Sud-Méditerranée IV - 19.00758.190401-MS02 – 2018-A03438-47).

Experimental setup

During MRI sessions, participants were lying on their back in a supine position. A mirror was positioned above their head and fixated to the head coil. A plastic translucent screen of 30 cm of radius was set at 1 meter from the eyes/mirrors. Visual stimuli were retro-projected onto this screen. Tactile stimulation was delivered by air pressure at 2.3 bars (air puffs) through plastic tubes rigidly maintained close to the target body locations. This level of pressure induced in all participants a clear and unambiguous perception of tactile stimulation on the target area. Air pressure was created by a compressor (*SIP Industrial*, SIP QT 50/10 Low Noise Direct Drive Compressor) and adjusted by pressure regulators

placed outside of the scanner room. Auditory stimuli were delivered using an MRI compatible device (Sensimetric Store, Model S14 - Insert Earphones for fMRI Research). Volume adjustment was checked during installation of the participant. Eye position (x and y for right or left eye) was recorded using an eye tracking system (Eyelink 1000, SR Research™) and sampled at 1000Hz. Stimulus control for all modalities as well as online eye position control was achieved using Presentation software (Neurobehavioral systems™) coupled with and I/O control Arduino-based setup.

Behavioral tasks

Subjects performed three different localizer tasks. In all three localizer tasks, in order to have participants keep focusing on the stimuli during the entire runs, they were asked to fixate the central point displayed on top of the stimuli and to answer by pressing a button when the point change color.

Face localizer task (Fig. 1, top panel). The face localizer task was composed of four conditions: fixation, human faces, objects (cars) and scenes (corridors). Stimuli dataset is based on the open fLoc (<http://vpnl.stanford.edu/fLoc/>) functional localizer package (Stigliani et al., 2015). During the task, subjects were asked to maintain their gaze on a fixation point presented at the center of the screen, overlapping the presented images. Subjects had to perform four runs of the face localizer. Each run was composed of a total of 24 blocks organized in 6 sets of the 4 conditions or blocks of stimuli (Fixation, Faces, Objects, Scenes). Conditions were pseudo-randomized within each set. Each block was composed of 16 different consecutive images each displayed for 500ms and selected randomly from a set of 144 images. Images were presented in the center of the screen and were 24° x 24°. In each run, the task started after 5 pulses to insure stability of magnetic field.

Tactile localizer task (Fig 1. Bottom panel). The tactile localizer task is composed of seven conditions: fixation, tactile stimulation to the left cheek, to the right cheek, to the left hand, to the right hand, to the left and right feet. During this task, participants were asked to perform four runs each composed by a total of 35 blocks organized in 5 sets or repetitions of the 7 conditions pseudo-randomized within each set. One block was composed by 5 consecutives tactile stimulations (air puffs) of 300ms duration with a random inter-stimulation interval between 1000 and 2000ms. Tactile stimulation was not associated with any simultaneous auditory stimuli as compressor and regulators were placed outside of the scanner room. Targeted locations were maintained as similar as possible across subjects: below the cheekbone for cheeks stimulation, half way between the ear and the nose; between index and thumb for hand stimulation and on the top of the foot for feet stimulation, aligned with the shin, and half way between the ankle and the fingers.

Voice localizer task (Fig. 1, Middle row). The voice localizer task is composed of three conditions: fixation, voices and non-voice sounds. Stimuli dataset (one dataset of voice, one dataset of non-voice, 20 audio sequences per set) was taken from (Pernet et al., 2015) and (Belin et al., 2000). For the description of the stimuli in the blocks see Table 1 of Pernet et al., 2015. Participant were asked to

perform one run of 72 blocks lasting 8 seconds. The three conditions were played equally (24 voice blocks, 24 non-voice blocks, 24 fixation blocks) and were presented randomly during the run.

Image acquisition

MRI data were acquired with a Siemens 3T PRISMA scanner and using a circularly polarized head coil of 64-channels (Siemens Healthineers, Erlangen, Germany). High-resolution anatomical images were acquired: T1-weighted images MPRAGE sequence for the T1-weighted (sagittal incidence, GRAPPA factor=2, FOV=180x224x256 mm, 0.8 mm isotropic voxel size, TR=3.0 s, TE=3.7ms, TI=1.1s, flip angle =8°). High-resolution 3D-T2-weighted anatomical images were acquired with SPACE sequence (sagittal incidence, GRAPPA factor=3, FOV=180x224x256 mm, 0.8 mm isotropic voxel size, TR=3.0s, TE_{eff}=299 ms). Functional images were acquired with EPI sequence covering whole cortex (50 axial slices, multiband factor=2, FOV=204x204x125, voxel size=1.6x1.6x2.5 mm with phase encoding along AP direction, TR = 2.3s; TE = 36 ms; FA = 80°). To correct for geometrical distortion, a magnetic field map was acquired before the functional runs with Siemens `gre_field_mapping` sequence (44 axial slices, FOV=225x225x132, voxel size=2.5x2.5x3.0 mm TR =500ms, TEs = 4.59 and 7.05ms, FA = 60°). The face localizer task involves 4 runs of 100 volumes each, the tactile localizer task involves 4 runs 120 volumes and the voice localizer task involved 1 run of 264 volumes.

Image preprocessing

Image pre-processing was performed using Statistical Parametric Mapping (SPM12) (Friston et al., 1994) software (Wellcome Centre for Human Neuroimaging, University College London, UK; <http://www.fil.ion.ucl.ac.uk/spm>). First, the first six volumes of the task were discarded allowing for T1 equilibrium. To adjust within-volume time differences, temporal misalignment between different slices of the functional data, introduced by the sequential nature of the fMRI acquisition protocol, is corrected using SPM12 slice-timing correction (STC) procedure (Henson et al. 1999), where the functional data is time-shifted and resampled using sinc-interpolation to match the time in the middle of each TA (acquisition time). Functional data were then realigned using SPM12 realign & unwarp procedure (Andersson et al. 2001), where all scans are resampled to a reference image (first scan of the first session) using b-spline interpolation. This procedure also addresses potential susceptibility distortion-by-motion interactions by estimating the derivatives of the deformation field with respect to head movement and resampling the functional data to match the deformation field of the reference image. Finally, functional scans were co-registered with the 3D structural image and spatially normalized against the standard, stereotactic space of the MNI template. Spatial smoothing was performed with a 6-mm full width at half maximum Gaussian kernel.

Functional MRI GLM analyses

Functional images were analyzed using statistical parametric mapping software (SPM12) (Friston et al., 1994, Wellcome Centre for Human Neuroimaging, University College London, UK;

<http://www.fil.ion.ucl.ac.uk/spm>). The first five functional volumes were discarded to allow for T1 equilibrium. At single subject level, a fixed-effect model was created for each individual participant in order to perform random-effect analyses with FWE $p < 0.05$ corrected for multiple comparisons. The motion parameters were included in the General Linear Model (GLM) as regressors to correct the head movement. In order to assess brain activations specific to the experimental conditions, different contrasts were computed for each task. For the face localizer task, activations related to the 3 visual conditions were assessed by all>fixation contrasts. Activations specifically related to the face condition was assessed by the following contrasts: face>objects+scenes. For the voice localizer task, non-specific, voice-specific and non-voice-specific auditory activations were assessed by the following contrasts: voice>fixation, non-voice>fixation and voice>non-voice. For the tactile localizer task, activations related to tactile stimulation of face (cheek), hands, and feet were assessed by the following contrasts: face>fixation, hand>fixation, foot<fixation and all>fixation. For the face and the tactile localizer tasks, in the GLM, contrasts were replicated and rescaled over runs accounting for inter-run variability by applying the same contrast weights to the corresponding regressors across the multiple runs and scaling them appropriately. This option allows to ensure that the contrasts are appropriately normalized and average the effects across the runs, enhancing the statistical power to detect consistent effects. A second-level random-effects analysis was then conducted and statistical results were thresholded using a voxel-wise family-wise error (FWE) corrected height threshold of $p < 0.05$. For tactile localizer percent signal change was calculated. A 4mm radius spherical region of interest was created using marsbar matlab toolbox (Brett et al., 2002) and located at MNI coordinates of $x=-7$, $y=-30$, $z=5$, to encompass the pulvinar All>fixation tactile activation cluster identified in whole brain analysis..

Seed Based analyses

In order to establish the intrinsic connectivity network affiliation of the pulvinar within each localizer (Face, Voice and Tactile), a seed to voxel analysis using the CONN19b toolbox (Whitfield-Gabrieli and Nieto-Castanon, 2012a) was used. First, we extracted pulvinar activation clusters by applying the Morel pulvinar mask at FWE $p < 0.05$ corrected for multiple comparisons (Anne Morel et al., 1997; Niemann et al., 2000). For face localizer task, seeds from the following contrasts were extracted: all>fixation (left seed, $x=-21$, $y=-28$, $z=-2$, size=232 voxels, right seed, $x=21$, $y=-28$, $z=-3$, size=278 voxels), face>object+scenes ($x=15$, $y=-32$, $z=0$, size=24 voxels). Then, in the CONN toolbox, additional denoising steps were carried out. Potential outlier scans were identified from the observed global BOLD signal as well as the amount of subject-motion in the scanner. Principal components were brought out from white matter and cerebrospinal fluid time-series to reduce noise in reference to the anatomical CompCor approach (Component-based Noise Correction) for each participant (Behzadi et al., 2007). These components, in addition to head movements parameters were used as confounds in the denoising step. The signal BOLD time-series were extracted and band pass-filtered for each participant (0.008–0.09 Hz). Finally, all pulvinar clusters extracted from SPM contrasts within each localizer task were

implemented in the CONN toolbox as ROIs. These were then used unilaterally as seeds in the seed to voxel analysis during which correlation maps on voxel-level were computed by correlating the BOLD-signal time course, averaged over all voxels within identified regions with the BOLD-signal time-course of every single voxel in the brain over the duration of the task for each participant. For the group-level analysis the Conn toolbox used ReML estimation of covariance components (which are evaluated through F-statistical parameter maps) to implement contrasts for analyses at the voxel level as repeated-measures analyses (Mueller et al., 2015). We reported the correlation maps within the contrasts that have served to extract the seed clusters. A threshold at the clusters $p < 0.05$ FDR corrected and peak voxel threshold $p < 0.05$ uncorrected were used.

Abbreviations

PuI: inferior pulvinar (green), PuA: anterior pulvinar (red), PuL: lateral pulvinar (blue), PuM: medial pulvinar (white), MD: mediodorsal nucleus (brown), CM: centre median nucleus (dark green), CL: central lateral nucleus (cyan), VPL: ventral posterior lateral nucleus (gold), MGN: medial geniculate nucleus (orange), LGN: lateral geniculate nucleus (yellow).

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Author contributions statement

Conceptualization: S.B.H., M.F.; Data Curation: MGT.M., M.F. and S.D.; Formal Analysis: MGT.M., M.F. S.D. and S.B.H; Funding Acquisition: S.B.H.; Investigation: MGT.M., M.F., S.D. and S.B.H.; Methodology: MGT.M., M.F., S.D. and S.B.H.; Supervision: S.B.H.; Writing—Original draft: MGT.M., M.F and S.D and S.B.H.; and Writing—review & editing: MGT.M., M.F., S.D. and S.B.H.

Additional information

Data and code are available upon request.

Competing interests

Authors declare no conflicts of interest.

Supplementary material

		Cortical activations				Pulvinar						Hippocampus					
						Left			Right			Left			Right		
		cortical area	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size
Face Localizer	All>fix FWE p<0.05	left fusiform	-33/-55/-16	25.33	50543	-21/-28/-2	15.52	672	21/-28/-3	15.04	737	-20/-8/-18	6.04	31	21/-4/-15	7.26	43
	Face>Objects+Scenes FWE p<0.05	right fusiform	40/-48/-19	25.42	8118	-	-	-	15/-31/0	6.40	34	-18/-8/-16	7.25	88	18/-8/-16	9.40	174

Supplementary Table S1.1. fMRI activations in face localizer task. Cortical, pulvinar and hippocampal regions displaying cluster activations in the face localizer task, for the All>Fixation contrast and the Face>Objects+Scenes contrast. Cluster peak location, peak t-score and cluster size are indicated. Star (*) indicates presence of two local peaks within the same cluster.

Cortical activations					Pulvinar						Hippocampus					
					Left			Right								
	cortical area	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size
Voice Localizer	V>Non-V unc p<0.001	left middle temporal	-68/-24/6	6.80	900	-	-	-	-	-	-	-	-	-	-	-
		right superior temporal	60/-10/0	6.44	592	-	-	-	-	-	-	-	-	-	-	-
		right middle temporal	58/-27/0	6.80	900	-	-	-	-	-	-	-	-	-	-	-
		right superior temporal	63/-12/-6	14.50	4778	-	-	-	-	-	-18/-9/-16	4.61	132	20/-9/-18	5.29	96
		left superior temporal	-64/-16/6	14.26	4498	-	-	-	-	-	-	-	-	-	-	-
	Voice>Fix unc p<0.001	left superior temporal	-52/-24/9	12.55	3177	-	-	-	-	-	-16/-10/-20	4.23	56	20/-12/-20	3.76	5
		right superior temporal	52/-6/-2	10.18	3385	-	-	-	-	-	-	-	-	-	-	-

Supplementary Table S1.2. fMRI activations in voice localizer task. Cortical, pulvinar and hippocampal regions displaying cluster activations in the voice localizer task, for the Voice>non-voice contrast (V>Non-V), Voice>Fixation contrast (Voice>Fix) and Non-Voice>Fixation contrast (Non-V>Fix). Cluster peak location, peak t-score and cluster size are indicated. Star (*) indicates presence of two local peaks within the same cluster.

Cortical activations						Pulvinar						Hippocampus					
						Left			Right								
		cortical area	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size
Tactile Localizer	All>Fix unc p<0.005	left superior temporal	-48/-34/24	7.03	330	-3/-27/9	5.21	19	-	-	-	-	-	-	-	-	-
		right rolandic operculum	51/-28/22	6.99	492	-6/-33/3	6.02	19	-	-	-	-	-	-	-	-	-
		left angular	-50/-64/33	5.25	64	-	-	-	-	-	-	-	-	-	-	-	-
		left postcentral	-58/-16/18	5.12	37	-	-	-	-	-	-	-	-	-	-	-	-
	Face>Fix unc p<0.005	left superior temporal	-50/-34/22	7.96	1532	-3/-25.5/9	5.33	21	-	-	-	-	-	-	-	-	-
		right superior temporal	51/-32/20	7.00	1038	-	-	-	-	-	-	-	-	-	-	-	-
		left insula	-38/-3/-2	5.99	32	-	-	-	-	-	-	-	-	-	-	-	-
	Hand>Fix unc p<0.005	left angular	-50/-64/32	5.53	36	-3/-27/9	6.56	97*	-	-	-	-	-	-	-	-	-
		right rolandic operculum	52/-26/22	5.31	19	-6/-31.5/3	5.9	97*	-	-	-	-	-	-	-	-	-
	Foot>Fix unc p<0.005	left superior temporal	-51/-33/24	6.27	145	-3/-27/9	7.36	160*	6/-30/4.5	4.80	4	-	-	-	-	-	-
		right rolandic operculum	51/-28/24	5.95	139	-6/-31.5/3	6.72	160*	-	-	-	-	-	-	-	-	-
		left angular	-50/-64/33	5.41	29	-	-	-	-	-	-	-	-	-	-	-	-

Supplementary Table S1.3. fMRI activations in tactile localizer task. Cortical, pulvinar and hippocampal regions displaying cluster activations in the tactile localizer task, for the All>Fixation contrast, Face>Fixation contrast, Hand>Fixation contrast and Foot>Fixation contrast. Cluster peak location, peak t-score and cluster size are indicated. Star (*) indicates presence of two local peaks within the same cluster.

Positive connectivity				Negative connectivity		
All >fixation Left Seed	Brain area	# voxels	%	Brain area	# voxels	%
	left-middle frontal gyrus	540	18	precuneus cortex	1424	25
	left inferior frontal gyrus pars opercularis	401	52	cingulate gyrus posterior division	481	20
	left frontal pole	272	4	cingulate gyrus anterior division	102	4
	left precentral gyrus	254	6	-	-	-
	left superior frontal gyrus	217	8	-	-	-
	inferior frontal gyrus left pars triangularis	99	15	-	-	-

Supplementary Table S2.1. Connectivity results for All>Fixation face localizer contrast, left seed. Positive and negative brain connectivity for selected seed, number of voxels and % of areas activated.

Positive connectivity				Negative connectivity		
All >fixation Right Seed	Brain area	# voxels	%	Brain area	# voxels	%
	bilateral precentral gyrus	786/921	18/22	-	-	-
	right middle frontal gyrus	653	24	-	-	-
	right frontal pole	425	5	-	-	-
	supplementary motor cortex bilateral juxtapositional lobule	143/344	22/48	-	-	-
	right middle temporal gyrus posterior division	311	23	-	-	-
	right inferior frontal gyrus pars opercularis	295	43	-	-	-
	right parahippocampal gyrus anterior division	207	32	-	-	-
	right temporal pole	202	8	-	-	-
	right hippocampus	153	22	-	-	-
	right orbitofrontal cortex	151	10	-	-	-
	right superior frontal gyrus	134	5	-	-	-

	right thalamus	116	9	-	-	-
	right inferior frontal gyrus pars triangularis	106	19	-	-	-
	left lingual gyrus	103	7	-	-	-
	right inferior temporal gyrus posterior division	101	11	-	-	-

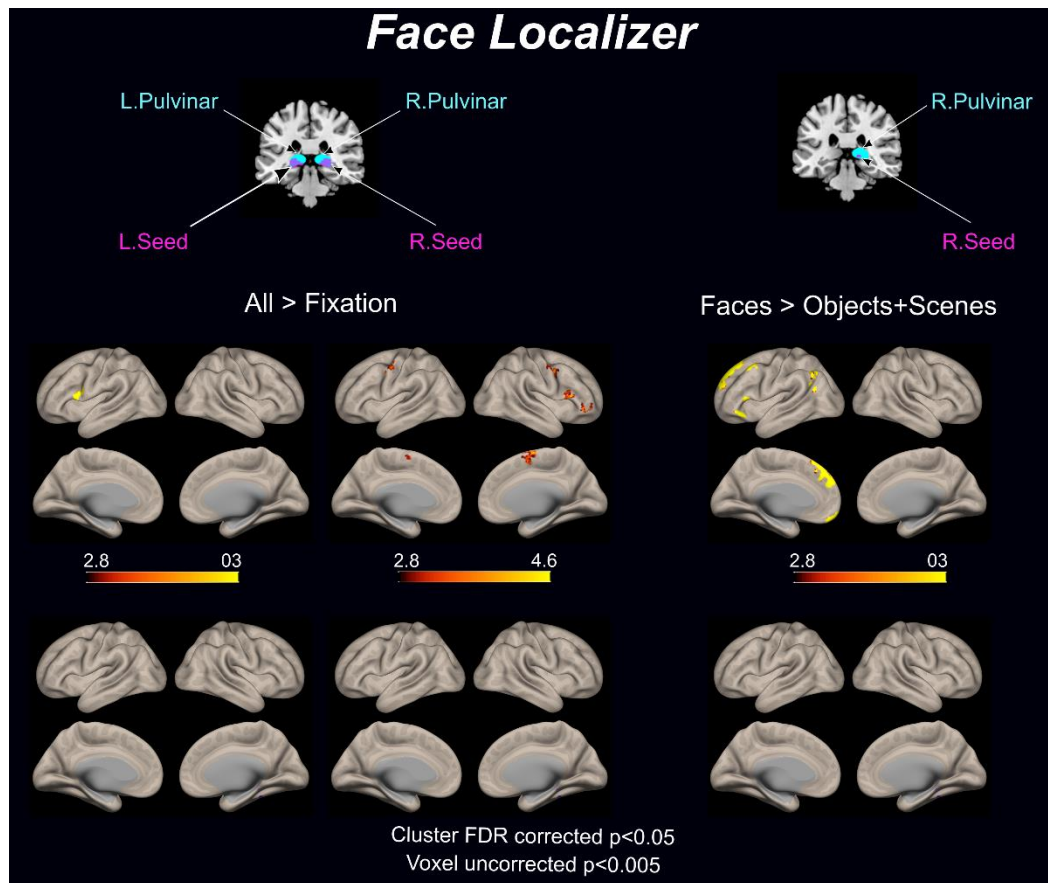
Supplementary Table S2.2. Connectivity results for All>Fixation face localizer contrast, right seed. Positive and negative brain connectivity for selected seed, number of voxels and % of areas activated.

Positive connectivity				Negative connectivity		
	Brain area	# voxels	%	Brain area	# voxels	%
Fac es> obj ect s+s cen es	left frontal pole	1466	21	precuneus Cortex	314	6
	left superior frontal gyrus	1006	36	left intracalcarine Cortex	167	26
	left orbitofrontal cortex	520	31	left lingual gyrus	150	10
	left lateral occipital cortex superior division	479	10	left superior division lateral occipital cortex	146	3
	left angular gyrus	417	44	right intracalcarine cortex	117	16
	left paracingulate gyrus	324	25	-	-	-
	left inferior temporal Gyrus posterior division	302	30	-	-	-
	left temporal pole	294	12	-	-	-
	left middle and inferior temporal gyrus posterior division	289	21	-	-	-
	left middle and inferior temporal gyrus anterior division	177	53	-	-	-
	left middle temporal gyrus anterior division	147	33	-	-	-
	left inferior frontal gyrus (pars triangularis & opercularis)	280	22/18	-	-	-

Supplementary Table S2.3. Connectivity results for Face>object+scene face localizer contrast, right seed. Positive and negative brain connectivity for selected seed, number of voxels and % of areas activated.

Positive connectivity				Negative connectivity		
	Brain area	# voxels	%	Brain area	# voxels	%
Faces > Fix	precuneous	910	16	-	-	-
	right angular gyrus	359	24	-	-	-
	right lateral occipital cortex superior division	294	6	-	-	-
	right supramarginal gyrus, posterior division	245	20	-	-	-
	Right lingual Gyrus	210	12	-	-	-
Hands > Fix	precuneous	625	11	-	-	-
	cingulate gyrus, posterior division	225	9	-	-	-
Feet > Fix	left Inferior frontal gyrus	209	27	-	-	-
	left postcentral gyrus	207	7	-	-	-
	left precentral gyrus	156	4	-	-	-
	left parietal operculum cortex	141	25	-	-	-
	left central opercular cortex	140	14	-	-	-
Feet > Hands	left Inferior frontal gyrus	485	68	cingulate gyrus, anterior division	324	12
				left inferior temporal gyrus	170	24
				right paracingulate gyrus	159	12
				left paracingulate gyrus	110	8
Feet > Faces	left Inferior frontal gyrus	891	24	-	-	-
	left frontal pole	521	8	-	-	-
	left precentral gyrus	244	6	-	-	-
	left Inferior frontal gyrus	180	23	-	-	-
	left middle frontal gyrus	151	5	-	-	-
	left supplementary motor cortex	118	18	-	-	-
	cingulate gyrus, anterior division	107	4	-	-	-

Supplementary Table S3. Connectivity results for Faces>Fixation, Hands>Fixation, Feet>Fixation, Feet>Hands and Feet>Faces for Tactile localizer contrast, left seed. Positive and negative brain connectivity for selected seed, number of voxels and % of areas activated.



Supplementary figure S1. Seed-based connectivity of the pulvinar with the cortical domain (cluster FDR $p < 0.05$ and voxel uncorrected $p < 0.005$). Same data as in Figure 5, except threshold is more restrictive in figure S1.

Study 2: Functional connectivity of pulvinar subnuclei to the brain vary as a function of sensory context

There is growing evidence to support the hypothesis of the pulvinar playing the role of a cortical modulator in several brain functions. Multiple anatomical and functional parcellations of the pulvinar has been described in the literature in order to assess its complex anatomo-functional organization (Barron et al., 2015a; Guedj and Vuilleumier, 2020a; Anne Morel et al., 1997). The variability in pulvinar segmentation across studies indicates a need for a unified and clearer representation of pulvinar segmentation that integrates both its anatomical and functional properties. Moreover, no studies investigated the dynamic of pulvinar connectivity in both resting state and active tasks. In this study, we investigated the functional connectivity of pulvinar subnuclei at rest and during visual, auditory and tactile processing. By doing so we can assess the dynamics of pulvino-cortical connectivity when we engage for a specific task from basic resting state status. This study refers to the pulvinar section of the general introduction.

Functional connectivity of pulvinar subnuclei to the brain vary as a function of sensory context

S. Dali¹, M. Froesel¹, M. Gaudet-Trafit¹, F. Lamberton² and S. Ben Hamed¹

¹Institut des Sciences Cognitives Marc Jeannerod, CNRS, Lyon 69500 bron, France

²CERMEP-imagerie du vivant, 59 Boulevard Pinel, 69677 Bron Cedex, France

Corresponding authors: Souhir Dali, souhir.dali@isc.cnrs.fr, Suliann Ben Hamed, benhamed@isc.cnrs.fr

Abstract

The pulvinar, a large thalamic structure, plays a critical role in regulating and coordinating communication between cortical and subcortical areas of the brain. Its subnuclei are thought to mediate sensory processing and attentional control, but the dynamic nature of its connectivity across different functional states and sensory modalities remains underexplored. In this study, we conducted fMRI scans of twenty-five participants during resting-state sessions and three sensory-specific localizer tasks each involving the stimulation of a distinct sensory modality (visual, auditory, and tactile). We used functional connectivity analyses to investigate the dynamic communication between human pulvinar subnuclei and cortical and subcortical regions across resting-state and tasks. Our findings reveal a complex and diverse set of connectivity patterns across the pulvinar subregions. Some connectivity profiles were consistent across both resting state and sensory tasks, reflecting intrinsic, stable networks, while others were distinctly task-dependent. In addition, our results provide novel insights into the functional specialization within the medial pulvinar, revealing distinct connectivity patterns across its anteroposterior, mediolateral and dorsoventral extent. Overall, these results support the classical view of the pulvinar as a dynamic hub, adapting its connectivity to meet specific sensory demands. and suggest a precisely coordinated specialization in processing different sensory inputs, refining our understanding of the pulvinar's role in sensory integration and attentional modulation.

Introduction

The primate cortex constitutes a complex network of functionally specialized regions. A fundamental goal in neuroscience is to elucidate the mechanisms underlying their activity coordination in a highly demanding multisensory environment. Based on the literature, the thalamus has been positioned as a pivotal hub facilitating interregional communication with the cortex (Jones, 2001). Given its extensive connections with cortical areas (Sherman and Guillery, 2006; Shipp, 2003b), the thalamus, by virtue of its anatomical positioning, exerts considerable influence over cortical functions. Within the posterior thalamus, the pulvinar nucleus has emerged as core to various cognitive processes, including salience detection (Snow et al., 2009), spatial orienting (R. D. Rafal and Posner, 1987) emotional and motor functions (Bertini et al., 2018; Lucas et al., 2019; Wilke et al., 2018b). Anatomically, the pulvinar is characterized by its extensive reciprocal connections with the cortex, as demonstrated in nonhuman primates (Adams et al., 2000; Kaas and Lyon, 2007b; Shipp, 2003b), which position it as a central hub within large-scale neural networks, contributing to the integration of activities across diverse cortical areas.

Traditionally, the pulvinar has been subdivided in four subregions, namely, the anterior, medial, lateral and inferior subdivisions based on their distinct cytoarchitectonic properties (C. Gutierrez et al., 2000; Gutierrez et al., 1995b; A. Morel et al., 1997; Olszewski, 1952; Stepniewska and Kaas, 1997; Walker, 1938). However, based on functional connectivity, it has been subdivided into five distinct functional clusters—namely, the inferior, lateral, anterior and medial ventral and dorsal regions (Guedj and Vuilleumier, 2020b). Each of these functional clusters exhibits a unique connectivity pattern and is associated with specific cognitive and sensory domains (Arcaro et al., 2018b; Barron et al., 2015b; Bridge et al., 2016b; Grieve et al., 2000). Most cortical areas, including early visual, somatosensory, and auditory processing regions, as well as higher-order associative areas in the parietal, temporal, premotor, and prefrontal cortices (Asanuma et al., 1985b; C. Cappe et al., 2009a; Leh et al., 2008b) ; demonstrate reciprocal connections with the pulvinar (Froesel et al., 2021b). In particular, the anterior pulvinar is essentially engaged with somatosensory regions while both the lateral and the inferior pulvinar segments project extensively to the occipital and temporal regions thus connecting with both of the dorsal and ventral visual processing streams (Kaas and Lyon, 2007b). The medial part of the pulvinar mainly connects with multimodal sensory association regions, along with the prefrontal and cingulate cortices. This medial pulvinar is one of the largest subnuclei of the pulvinar complex, occupying a significant portion of the posterior thalamus. Its size relative to the other pulvinar subnuclei, such as the anterior and inferior pulvinar, reflects its broad involvement in the integration of sensory, cognitive, and attentional processes. The medial pulvinar's extensive connections with cortical areas, including the prefrontal cortex, temporoparietal regions, and occipital areas, further

highlight its central role in higher-order processing. Based on functional connectivity analyses, the medial pulvinar has been subdivided into a dorsal and a ventral pars, each with potentially distinct functions (Guedj and Vuilleumier, 2020b). The dorsal part of the medial pulvinar is thought to be more involved in cognitive processes, such as attention and the integration of sensory information from different modalities. In contrast, the ventral portion is more closely associated with visual processing and has strong connections to sensory cortices (Yuri B. Saalmann and Kastner, 2011).

This general, specific anatomical and functional architecture of the pulvinar underscores its role as a central node in the communication across brain areas, particularly in sensory information processing (Froesel et al., 2024c, 2021b; Gattass et al., 2018; Kanai et al., 2015; Sherman, 2016). Extensive research has elucidated its multifaceted role in shaping various aspects of visual cognition, ranging from attentional modulation to perception (Benarroch, 2015b; Bourne and Morrone, 2017; Bridge et al., 2016b). Through its connections with cortical and subcortical regions, particularly with the primary visual cortex and higher-order associative areas, through the dorsal and ventral visual streams, the pulvinar has been shown to facilitate the selective processing of visual inputs and serve as a crucial node in the network underlying the integration of visual inputs from different modalities (Saalmann, 2014). This integrative function enables the pulvinar to contribute to the formation of coherent percepts and the extraction of information from complex visual scenes, thereby helping in the allocation of attentional resources to behaviorally relevant stimuli (Saalmann et al., 2012b).

Beyond its traditional association with visual processing, emerging evidence from neuroimaging, electrophysiological, and lesion studies has highlighted the pulvinar's involvement in auditory and tactile modalities. Neuroimaging research in both nonhuman primates and humans has identified pulvinar activation during auditory tasks, indicating its participation in auditory attention and salience detection (C. Cappe et al., 2009a; Froesel et al., 2024c). Electrophysiological studies have further demonstrated that pulvinar neurons are responsive to auditory stimuli, their activity being modulated by spatial and temporal auditory cues (Tyll et al., 2011b). Anatomically, pulvinar connections with cortical and subcortical auditory regions support its role in the integration of auditory information across various processing stages (Bridge et al., 2016b). In the somatosensory domain, the pulvinar also plays a crucial role in tactile processing and multisensory integration. Research involving nonhuman primates has shown that the pulvinar is activated by tactile stimuli, particularly during tasks requiring attentional focus or spatial discrimination (Benarroch, 2015b). Furthermore, lesion studies have revealed deficits in tactile attention and discrimination following pulvinar damage, emphasizing its significance in tactile processing (Bourne and Morrone, 2017). Finally, the pulvinar connectivity with somatosensory cortical regions and other multisensory areas suggests a role in the integration of tactile information with inputs from other sensory systems (Benarroch, 2015b).

While extensive knowledge exists regarding the anatomical pulvino-cortical connectivity and its implication in sensory processing from different modalities, whether and how its functional coupling with cortical areas dynamically adjusts to sensory contexts remains unclear. The pulvinar functional connectivity has been studied in both human (Guedj and Vuilleumier, 2020b) and macaques (Froesel et al., 2024c). Despite its extensive connectivity with the cortex, this nucleus does not systematically activate in response to sensory stimulations and might better be specifically recruited during perceptually challenging contexts (Guedj and Vuilleumier, 2023b). We have recently shown a task-dependent activation of the pulvinar, suggesting that this nucleus is differentially recruited depending on the sensory context, both in macaques (Froesel et al., 2024c) and in humans (Gaudet-Trafit et al., 2024). In particular, in Gaudet-Trafit et al. (2024), we describe localized pulvinar activations in response to visual (face) and tactile stimulations as well as context-dependent functional connectivity between these pulvinar functional regions and cortical areas. These results support the assumption that the pulvinar may orchestrate network recruitment depending on the context.

In this study, we aim to describe how the functional connectivity between major anatomical pulvinar landmarks and the rest of the brain vary as a function of sensory context. To do so, we employed functional magnetic resonance imaging (fMRI) in humans and we scanned 25 subjects during resting state as well as a face localizer task, a voice localizer task and a tactile face/hand/foot localizer task. Then, using functional connectivity analyses, we investigated the dynamic interplay between the different distinct subregions of the pulvinar and the cortex during each sensory context. Overall, we show a dynamic modulation of pulvinar functional connectivity with the rest of the brain varying as a function of both the sensory context and the specific pulvinar subregion. Overall, this work supports the view of the pulvinar as a hub facilitating interregional communication across cortical regions, and provides a functional map of how this is achieved.

Method

Subjects

Twenty-five healthy subjects with no history of psychiatric or neurological disorders participated in the study after signing their written informed consent (12 males and 13 females; mean age of 28.08, SD = 3.8). All subjects underwent the Edinburgh Handedness Inventory laterality test (doi: <https://doi.org/10.1016%2F0028-3932%2871%2990067-4>) to determine a lateralization score for each participant (17 right-handed, 1 left-handed and 7 ambidextrous). The experimental protocol was approved by French ethical comity CPP sud-Med IV (Comité de Protection des Personnes Sud-Méditerranée IV -19.00758.190401-MS02 – 2018-A03438-47).

Experimental setup

The entire experiment consisted on five fMRI sessions. Different tasks were performed during these sessions. A first session started with a resting state run of 15 minutes, and was followed by face localizer runs and T1 and T2 anatomical scans. A second session involved tactile localizer runs as well as an active visuo-tactile task not described here. The third and fourth sessions have the same organization and included a resting state run of 15 minutes, voice localizer runs as well as an active audio-visual task not described here. A fifth session included two resting state runs of 15 minutes as well as an active audio-visual task not described here. During the MRI sessions, participants were laying down in the scanner in a supine position, with a mirror fixed above their head and attached to the head coil. Positioned at a distance of 1 meter from their eyes, a plastic translucent screen served as the projection surface for the visual stimuli of the face localizer task, each presented as squares measuring 30cm. Auditory stimuli were displayed using an MRI-compatible device (Sensimetric Store, Model S14 - Insert Earphones for fMRI Research). Tactile stimulations consisted in air-puffs and were generated by a compressor (SIP Industrial, SIP QT 50/10 Low Noise Direct Drive Compressor) and adjusted by pressure regulators situated outside the scanner room, and were delivered at 2.3 bars air pressure from a compressor located outside the scanner room, conveyed through plastic tubes attached to an MRI-compatible frame. Oculomotor movements were recorded using an eye tracking system at 1000Hz with Eyelink1000™. Generated Stimuli and eye movements were controlled using the Presentation software (Neurobehavioral systems™) coupled with a custom software for online adjustment of task parameters.

Tasks

To maintain concentration in all three localizer tasks, participants were instructed to focus on a central point always visible on the screen. They were asked to press a button when this point changed color.

Tactile Localizer Task

The tactile localizer task included seven conditions: fixation (no air-puff), air-puff stimulation on left or right cheek (in independent blocks), left or right hands (in independent blocks), and left or right feet (in independent blocks). Participants were required to complete four such runs, each comprising 35 blocks with each condition repeated five times. Each block consisted of five consecutive tactile stimulations using air puffs lasting 300 milliseconds each, with an inter-stimuli interval of 1000 to 2000 milliseconds (Fig1.A). The stimulation targeted consistent locations across subjects: below the cheekbone for cheek stimulation, between the index finger and thumb for hand stimulation, and the top of the feet for foot stimulation.

Face Localizer Task

The face localizer task consisted in four different conditions: fixation, human faces, objects (specifically cars), and scenes (corridors). The stimuli used were sourced from the open fLoc functional localizer package, as described by Stigliani and colleagues (Stigliani et al., 2015). Throughout the task, participants completed four runs, with each run comprising six repetitions of four distinct blocks. Each block, representing one condition, featuring 16 consecutive images shown for 500 milliseconds each, with a brief interstimulus interval of 30 milliseconds. A total of 144 images were randomly selected for each condition and were centrally displayed on the screen, maintaining a visual angle of 12 degrees at the edges. The sequence of the blocks was randomized in each set to prevent consecutive repetition of the same block condition between sets (i.e., the last block of one set and the first block of the subsequent set could not both be face blocks) (Fig1.B). Each run began after five pulses to ensure the stability of the magnetic field.

Voice Localizer Task

The voice localizer task contained three conditions: fixation, voices, and non-voice sounds. The stimulus set includes two categories (voice and non-voice), each containing 20 auditory sequences lasting 8 secs each and composed of a mixture of either vocal sounds or non-vocal sounds sourced from studies by (Pernet et al., 2015) (<https://datashare.ed.ac.uk/handle/10283/818>). Participants undertook a single run composed of 72 blocks. The conditions were evenly distributed and randomized during the run, with 24 blocks for voices, 24 for non-voices, and 24 for fixation, i.e. silent blocks (Fig1.C).

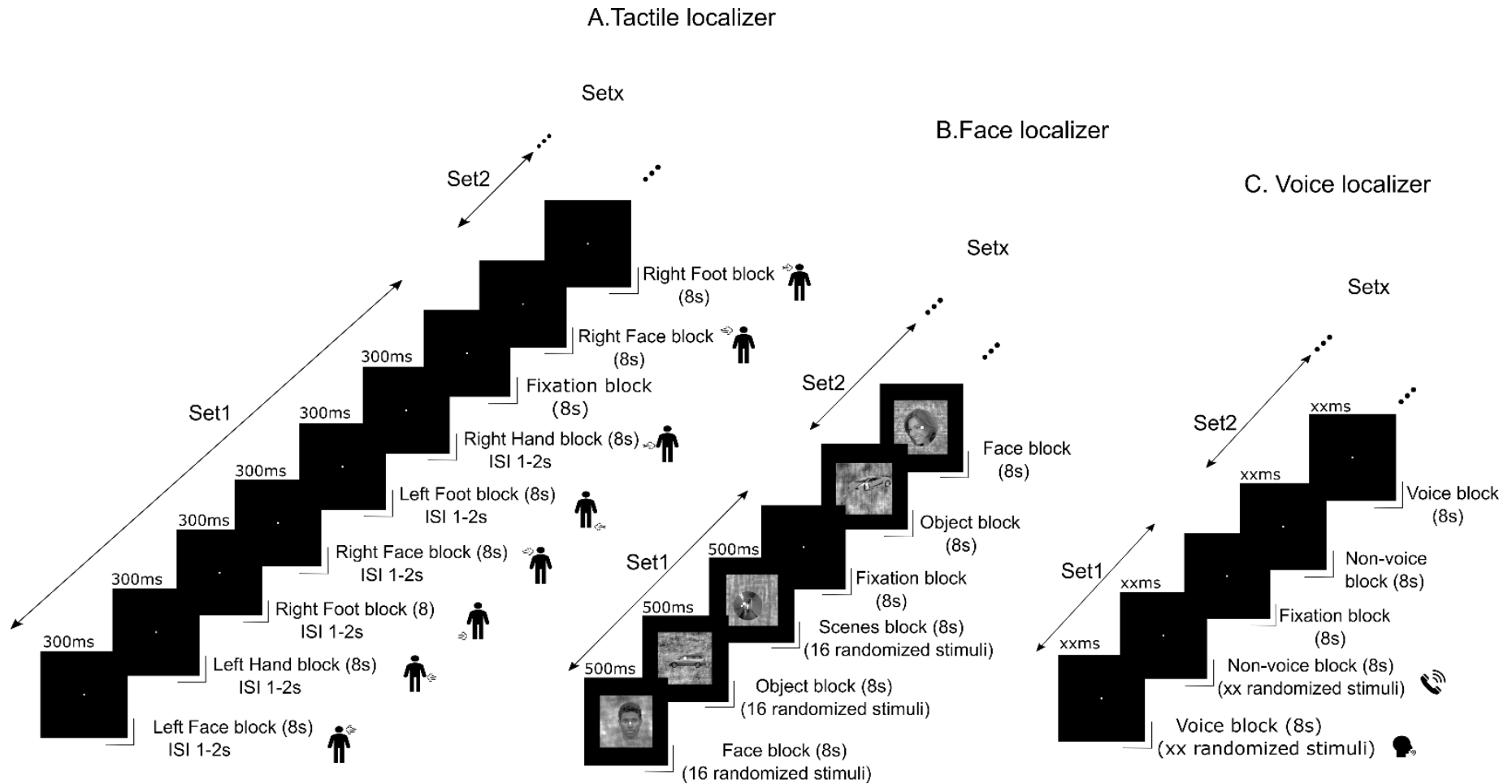


Figure 1: Tactile, face, and voice localizer tasks. A) Tactile localizer. Each run was composed of 5 sets of 7 blocks each (total of 35 blocks). Each set was composed of a pseudorandomized presentation of left and right face, hand and feet blocks as well as one fixation blocks. Each block lasted 1.5s and involved the presentation of 5 repetitions of a 300ms stimulation. B) Face localizer. Each run was composed of 6 sets of 4 blocks (total of 24 blocks). Each set was composed of one face block, one object block, one scene block, and one fixation block presented in a pseudo-randomized order. Each block lasted 8s and involved the presentation of 16 randomized stimuli of 500ms. C) Voice Localizer. Each run was composed by 72 randomized blocks of voice stimuli, non-voice stimuli and fixation. Each block involved the repetition of 5 auditory stimuli of ~1.5s separated by 15ms for a total duration of 8s.

MRI acquisitions

MRI data were acquired with a Siemens 3T PRISMA scanner and using a circularly polarized head coil of 64-channels (Siemens Healthineers, Erlangen, Germany). High-resolution anatomical images were acquired: T1-weighted images MPRAGE sequence for the T1-weighted (sagittal incidence, GRAPPA factor=2, FOV=180x224x256 mm, 0.8 mm isotropic voxel size, TR=3.000 ms, TE=3.7 ms, TI=1.1 s, flip angle =8°). High-resolution 3D-T2-weighted anatomical images were acquired with SPACE sequence (sagittal incidence, GRAPPA factor=3, FOV=180x224x256 mm, 0.8 mm isotropic voxel size, TR=3000 s, TE_{eff}=299 ms). Resting State images were acquired with an EPI sequence covering whole cortex (53 axial slices, multiband factor=2, FOV=204x204x125, voxel size=1.6x1.6x2.5 mm with phase encoding along AP direction, TR = 1600 ms; TE = 30 ms; FA = 75°). Functional images were acquired with EPI sequence covering whole cortex (50 axial slices, multiband factor=2, FOV=204x204x125, voxel size=1.6x1.6x2.5 mm with phase encoding along AP direction, TR = 2300 ms; TE = 36 ms; FA = 80°). To correct for geometrical distortion, a magnetic field map was acquired before the functional runs with Siemens gre_field_mapping sequence (44 axial slices, FOV=225x225x132, voxel size=2.5x2.5x3.0 mm TR =500 ms, TE_s = 4.59 and 7.05 ms, FA = 60°). Face localizer task is composed by 4 runs of 100 volumes each, tactile localizer task is composed by 4 runs 120 volumes and voice localizer task is composed by 1 run of 264 volumes.

Data Analysis

Functional Data Preprocessing

Functional images pre-processing was performed using Statistical Parametric Mapping (SPM12) (Friston, 1994) software (Wellcome Centre for Human Neuroimaging, University College London, UK; 204 <http://www.fil.ion.ucl.ac.uk/spm>). Initially, the first five volumes of the task were discarded to achieve T1 equilibrium. To mitigate temporal differences within volumes and slice misalignment inherent in the fMRI acquisition protocol, the Slice-Timing Correction (STC) procedure of SPM12 (Henson et al., 1999) was applied. This procedure involves time-shifting and resampling the functional data using sinc-interpolation to align with the midpoint of each acquisition time (TA). Subsequently, realignment and unwarping were executed using the SPM12 realign & unwarp procedure (Andersson et al., 2001), wherein all scans were resampled to a reference image (the first scan of the first session) via b-spline. This step also accounted for potential susceptibility distortion-by-motion interactions by estimating derivatives of the deformation field in relation with head movements and aligning the functional data with the deformation field of the reference. Finally, functional scans were then co-registered with the 3D structural image and spatially normalized in the standard stereotactic space of the MNI template. Spatial smoothing was then performed with a 6-mm full-width at half maximum Gaussian kernel.

Resting State Data Preprocessing

Image preprocessing was conducted using the default MNI pipeline of the CONN toolbox in its Conn19.c version. The preprocessing sequence included the following steps: functional realignment and unwarping, slice timing correction, structural segmentation and normalization, and functional normalization. Additionally, outlier detection was performed. The inhomogeneity of the magnetic field within the scanner (fieldmap) was assessed and applied in the Susceptibility Distortion Correction (SDC) during the unwarping phase. This correction resampled the functional data along the phase-encoded direction to address the distortions in the reference image due to field inhomogeneities. Data points showing a framewise displacement greater than 0.9 mm or global BOLD signal fluctuations exceeding five standard deviations were identified as potential outliers. Framewise displacement was calculated for each time point by using a 140x180x115 mm bounding box around the brain to identify the largest movement among six control points at the center of each face of the bounding box. Global BOLD signal changes were measured at each time point by evaluating the fluctuation in the average BOLD signal intensity within SPM's global-mean mask, scaled to standard deviation units to coincide with the midpoint of each acquisition time. Both functional and anatomical data were normalized and segmented into grey matter, white matter, and CSF tissue classes using the SPM12 unified segmentation and normalization method, as described by Ashburner and Friston in 2005 (Ashburner and Friston, 2005). This process involves iterative tissue classification based on the intensity values of the reference images and registration that optimizes the non-linear spatial transformation to align the posterior and prior tissue probability maps (TPMs), continuing until convergence is reached. Direct normalization applied this unified segmentation and normalization method separately to the functional and structural data, using the mean BOLD signal and the raw T1-weighted volume as reference images, respectively. The data were resampled into a standard 180x216x180 mm bounding box, with functional data at 2 mm isotropic voxels and anatomical data at 1 mm isotropic voxels, using 4th order spline interpolation. Finally, to enhance the BOLD signal-to-noise ratio and minimize the impact of residual variability in functional and gyral anatomy across subjects, the functional data underwent spatial smoothing with a Gaussian kernel of 6 mm FWHM.

ROIs of interest

To determine the intrinsic connectivity network association of the pulvinar subdivisions during the task-free baseline data (Resting state) and during each sensory contexts (Face, Voice, and Tactile localizer tasks), we applied the seed-to-voxel analysis method available in the CONN19b toolbox (Whitfield-Gabrieli and Nieto-Castanon, 2012b). Pulvinar subdivision seeds were selected independently for the Right and Left pulvinar based on the Morel anatomical atlas (A. Morel et al., 1997). Four subdivisions were used: an anterior pulvinar seed (PuA), a lateral pulvinar seed (PuL), an

inferior pulvinar seed (Pul) and a medial pulvinar seed (PuM). These seeds were sourced from (Ilinsky et al., 2018). The dorsal and ventral medial parts of PuM have been described to have different functional connectivity patterns (Guedj and Vuilleumier, 2020b). Independently, the macaque pulvinar has been shown to express an antero-posterior gradient of connectivity with the cortex (Froesel et al., 2024c). In order to have a clearer understanding of the organization of medial pulvinar to cortical connectivity during sensory processing, we segmented the PuM into eight subregions combining the anteroposterior, mediolateral and dorsoventral anatomical orientations. We thus defined an anterior lateral dorsal PuM (ALD), an anterior lateral ventral PuM (ALV), an anterior medial dorsal PuM (AMD), an anterior medial ventral PuM (AMV), a posterior lateral dorsal PuM (PLD), a posterior lateral ventral PuM (PLV), a posterior medial dorsal PuM (PMD) and a posterior medial ventral PuM (PMV). This procedure was performed separately for the Right and Left PuM, resulting in eight unilateral PuM ROIs.

Global Seed-based Connectivity Analyses

In order to explore the global intrinsic connectivity patterns during each localizer task, all conditions were averaged per localizer (Faces+Objects+Scenes+Fixation for Face localizer, Voices+Non Voices+Fixation for Voice localizer and Faces+Hands+Feet+Fixation for Tactile localizer). By averaging the conditions, we aimed to capture overarching connectivity patterns devoid of specific sensory contextual effects related to individual localizer conditions, allowing us to examine the broad connectivity affiliations of the pulvinar subdivisions during these tasks. For resting-state data, the five resting-state sessions were pooled together. To ascertain the intrinsic connectivity network associated with the pulvinar subdivisions described above, right and left clusters were implemented separately as ROIs in the CONN toolbox. Then, a seed correlation analysis (SCA) was conducted to compute correlation maps at the voxel level. This involved correlating the BOLD-signal time courses, averaged across all voxels within the identified regions, with the BOLD-signal time courses of each voxel in the rest of the brain independently for the resting-state runs and the three localizer runs. For group-level analysis, the Conn toolbox utilized ReML estimation of covariance components, evaluated via F-statistical parameter maps, to execute contrasts for voxel-level analyses as repeated-measures analyses (Muller et al., 2016).

Exploring connectivity maps

In order to investigate the connectivity maps derived from the seed-based analysis across the localizer tasks and the task-free sessions (resting-state), we first started by retrieving these maps in order to quantify both spatial extend and functional connectivity strength between the selected pulvinar ROIs and major subcortical and cortical regions, respectively computing the percentage of significantly connected voxels and the corresponding functional connectivity scores (z-scores).

Overlap Analysis of Connectivity Maps

In order to better characterize the overlap and divergence between connectivity maps derived from the resting state and the different localizer tasks across all pulvinar cluster seeds, we conducted an overlap analysis using FSL (FMRIB Software Library) (FSL,RID:birnlex_2067; Jenkinson et al., 2012 <http://fsl.fmrib.ox.ac.uk/fsl/fslwiki/>) which facilitates mathematical manipulation of image data and allows to provide insights into the consistency and variability of brain connectivity patterns under these different conditions. First, individual ROI-to-whole brain maps were initially thresholded and binarized (significant correlation between the seed and the cortex cluster $p < 0.05$ FDR corrected and voxel $p < 0.001$ uncorrected). Then, for each ROI, the binary masks from each task condition were compared using FSL's `fslmaths`, and the intersection and union of these masks were computed. The intersection map indicates the cortical and subcortical voxel with consistent significant connectivity across all compared maps. The union map provides a comprehensive view of all regions identified as significantly connected to the pulvinar region in at least one of the maps. `fslmaths` thus provides a single, interpretable figure that quantifies the extent of agreement between connectivity maps.

Results

Connectivity profiles of pulvinar subdivisions during Resting-State and localizer Tasks

To explore the intrinsic functional connectivity profiles of pulvinar subdivisions during resting-state, we conducted a seed-based analysis using the unilateral Morel atlases. This analysis focused on four pulvinar subregions: anterior (PuA), inferior (PuI), lateral (PuL), and medial (PuM). Figure 2 illustrates the connectivity patterns of these subregions with the rest of the brain presented on inflated brain maps (Figure 2A), along with the percentage of voxels covering the target atlases presented as histograms (Figure 2B). The analysis was performed using a cluster threshold of $p < 0.05$ (FDR corrected) and a voxel-level threshold of $p < 0.001$ (uncorrected). We describe unique functional connectivity patterns between each pulvinar subdivision and the rest of the brain.

The anterior pulvinar (PuA) exhibited predominant ipsilateral positive connectivity with the somatosensory (SMN) (postcentral gyrus + precentral gyrus) and the salience networks (SN) (including the insular cortex and anterior cingulate cortex) in addition to the Default Mode Network (DMN) (composed of the precuneus, posterior cingulate gyrus and medial frontal cortex). Less spread connectivity was observed with the frontoparietal network (FPN) (Middle frontal gyrus + superior frontal gyrus + superior and inferior parietal lobule) and the visual network (medial (MVN) and lateral (LVN)). At the subcortical level, extensive positive connectivity was observed with the rest of the thalamus, and to a lesser extent with the hippocampus and the amygdala. Connectivity profiles were

very comparable for left and right PuA. The *lateral pulvinar (PuL)* primarily connected ipsilaterally with the MVN (including the calcarine cortex and lingual gyrus) and the LVN (superior and inferior divisions), along with the DMN, and the thalamus as well as, to a lesser extent with the SN and the hippocampus. No connectivity was observed with the SMN, FPN, nor amygdala. Connectivity profiles were very comparable for left and right PuL. The *right inferior pulvinar (Pul)* exhibited strong ipsilateral positive connectivity with the MVN, as well as, to a lesser extent with the rest of networks (LVN, SMN, SN, and DMN) and subcortical areas (thalamus, and hippocampus). Similarly, to PuL, Pul presented no connectivity with the SMN, FPN, nor amygdala. Connectivity profiles were very comparable for left and right Pul. The *right PuM* displayed widespread ipsilateral connectivity with the MVN, LVN, SN, DMN, and all three investigated subcortical structures: the thalamus, hippocampus, and amygdala. Weaker connectivity was observed with the SMN and FPN. Connectivity profiles were very comparable for left and right PuM. Overall, the anterior pulvinar appears to be specialized in sensorimotor processing, the inferior and lateral subdivisions involved mainly in visual processing, whereas the medial pulvinar functions appears as a hub integrating diverse information across a larger amount of cortical and subcortical regions.

The same seed-based analysis were performed on the localizer tasks data, averaging all conditions per localizer. This allows to specifically capture the changes in pulvinar functional connectivity with the rest of the brain as a function of the sensory context. At same statistical thresholds (cluster $p < 0.05$ FDR corrected and voxel $p < 0.001$ uncorrected, Figures S1, S2 and S3), PuA is primarily connected to the SMN, SN, and thalamus across all three localizers. The PuL mainly connects with MVN (especially on the right), the thalamus and hippocampus in addition to slight connections with the DMN restricted to voice and tactile localizers. The Pul demonstrates relatively balanced connectivity with the MVN, SN, DMN, hippocampus and thalamus across the three localizers, with additional connectivity to the amygdala specifically in the face and voice localizers. Lastly, the PuM was mainly connected to the DMN, MVN, thalamus, and hippocampus in all three tasks, with additional connectivity to the SN and amygdala in the voice and face localizers. Only few minor differences are observed between the connectivity of left and right subnuclei, related to the percentage of voxels in the target regions rather than to the global connectivity per se. Overall, there is thus a marked refinement of pulvinar functional connectivity with the brain during the localizer tasks. This is characterized by a decreased and more specific overall functional connectivity with the brain. These results emphasize the functional heterogeneity of pulvinar subdivisions, each contributing in a unique manner to sensory, cognitive, and affective processes and underscores the pulvinar's critical role in modulating and coordinating brain-wide network dynamics.

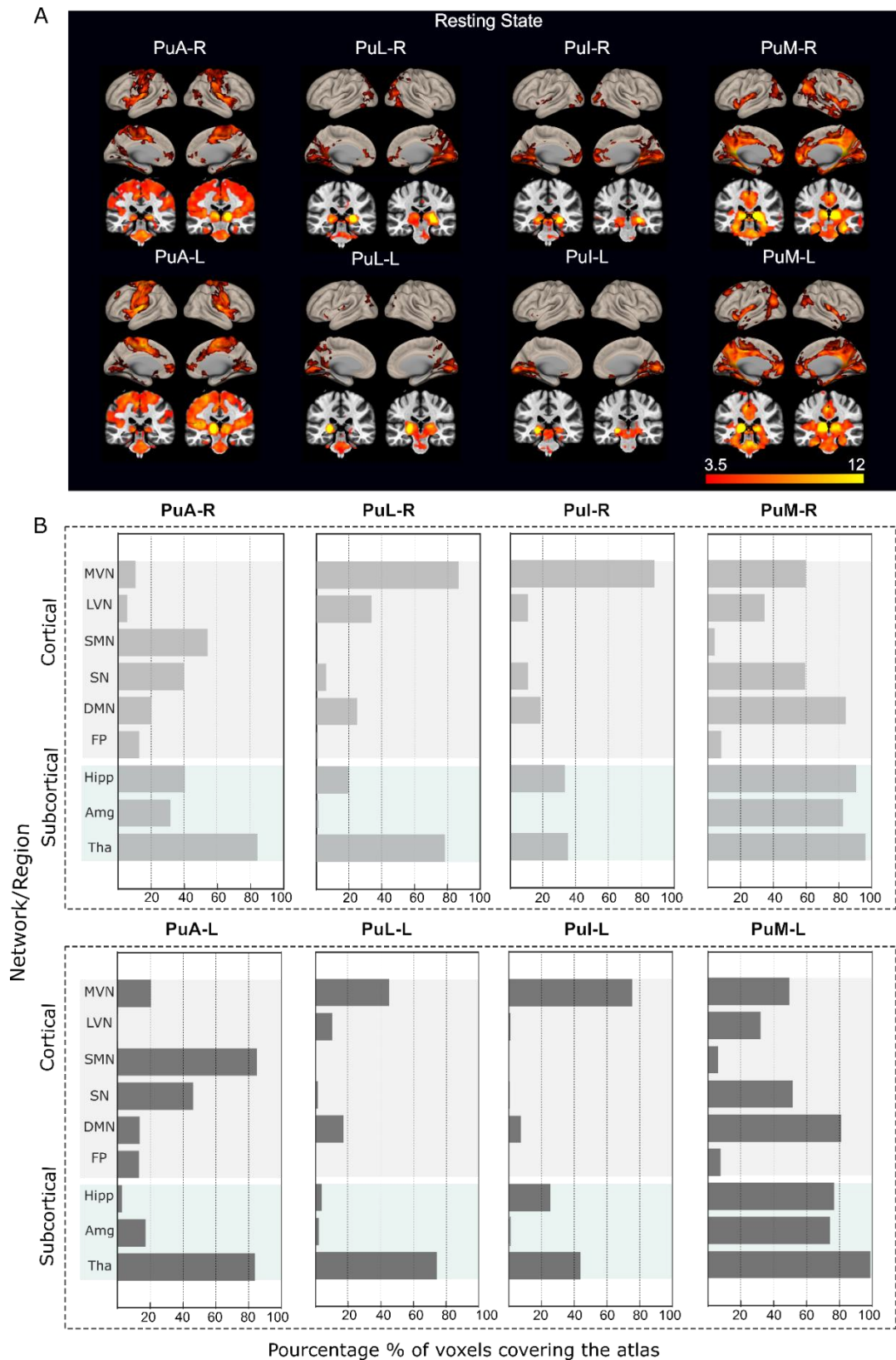


Figure 2. Pulvinar functional connectivity with major brain regions during resting state. (A) Whole brain functional connectivity maps of right (top) and left (bottom) PuA, PuL, Pul and PuM, projected on lateral and medial views of left and right brain hemispheres. (B) Percentage of voxels connected to right (top) and left (bottom) PuA, MuL, Pul and PuM for major cortical and subcortical networks or regions.

Preserved pulvinar-to-whole brain connectivity across resting and localizer tasks

Functional connectivity of pulvinar subdivisions with whole brain during task-free sessions and the three localizer tasks show remarkable core similarities in spatial patterns. To quantitatively assess these similarities, we overlaid the four correlation maps (cluster $p < 0.05$ FDR corrected and voxel $p < 0.005$ uncorrected). Figure 3 shows the spatial overlap of connectivity maps on color coded inflated brain maps. Yellow to red shades indicate strong to weak overlap. The spider plots presented in figure 4 capture the percentage of voxels that are functionally connected to the different pulvinar subnuclei as a function of context of interest and table S1 provides a quantification of these observations. Results reveal that each pulvinar subnucleus is functionally connected with a distinct specific core brain network. PuA core connectivity network was restricted to the somatosensory network (SMN) and thalamus. When considering overlap between two or three maps, this network additionally included DMN and FPN. PuL core connectivity network was restricted to the default-mode network (DMN). When considering overlap between two or three maps, this network additionally included MVN, thalamus and hippocampus. PuL core connectivity network was similar to that of PuL, except that when considering overlap between two or three maps, this network additionally included SN. Last, PuM core connectivity network encompassed all of DMN, MVN, LVN, SN, thalamus and hippocampus. When considering overlap between two or three maps, this network additionally included SMN and amygdala. This was true for both left and right pulvinar seeds.

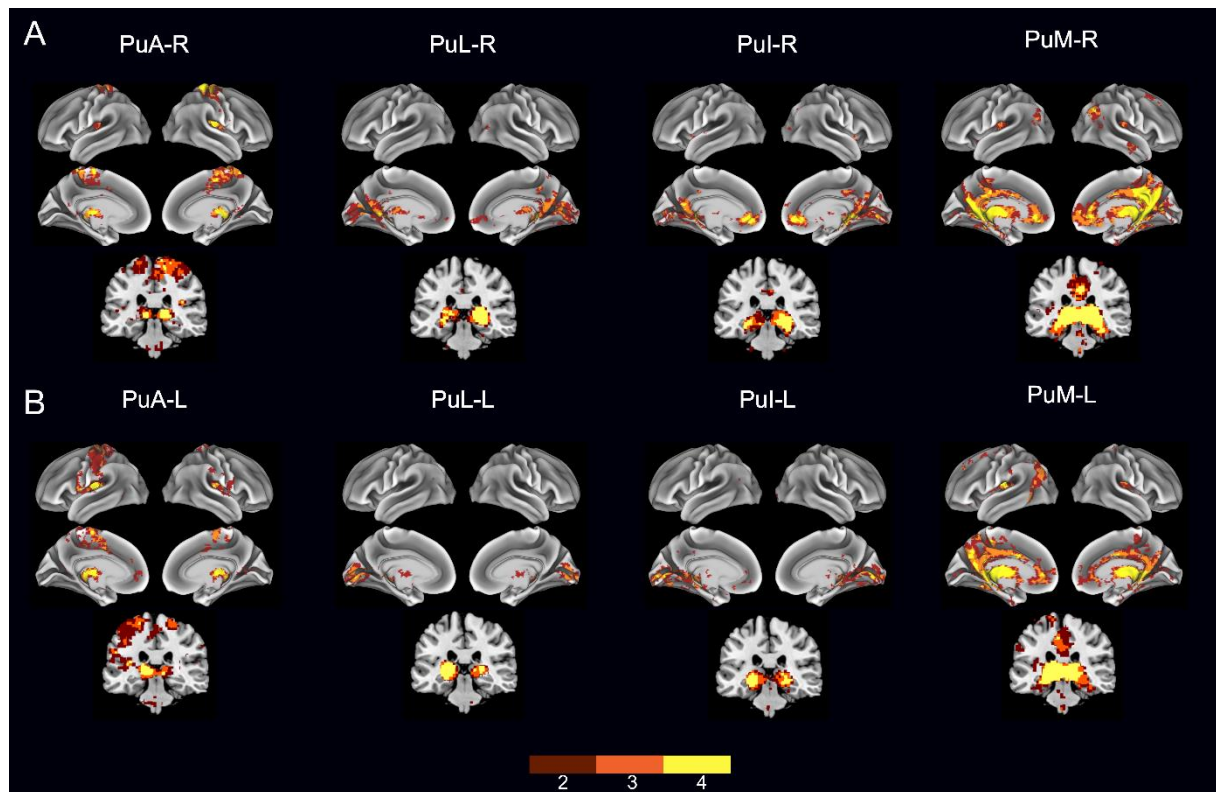


Figure 3. Overlap of pulvinar functional connectivity with major brain regions across resting state and localizer tasks for right (A) and left (B) major pulvinar subnuclei (PuA, PuL, PuL and PuM).

Overall, this indicates that pulvinar subnuclei functional connectivity with the brain is organized around specific core connectivity patterns that are further refined as a function of sensory context. How specific sensory content within a given sensory modality further refines these functional connectivity patterns will be investigated later on.

Changes in strength of functional connectivity between pulvinar and major brain regions across resting-state and localizer tasks

The previous section highlights similarities and differences in the spatial pattern of pulvinar subnuclei connectivity with major brain regions across resting-state and localizer tasks. The changes are accompanied by changes in connectivity strength between pulvinar and brain across tasks (Figure 4). In order to statistically assess the statistical significance of these changes in functional connectivity strength, we performed a factorial analysis taking as factor pulvinar subnuclei (PuA, PuL, PuI, PuM), task (Resting, Face, Voice, Tactile) and target brain region (MVN, LVN, SMN, SN, DMN, FPN, Hippocampus, Amygdala and Thalamus), correlation scores as dependent variable, with subjects included as a random factor. Results revealed significant differences in how specific pulvinar subdivisions connected with target brain regions across tasks. Functional connectivity strength and associated statistics were very similar for right and left pulvinar subnuclei ROIs (Figure 4, Table S2 and Table S3). As a result, we will focus on the description of the observation for the right pulvinar ROIs.

For PuA, we report a main effect of Task ($F(3)=36.91$, $p<0.001$) and a main effect of Target region ($F(8)=201.77$, $p<0.001$). A significant interaction was also observed between Task and Target ($F(24) = 6.45$, $p<0.001$), indicating that the connectivity with each target varied depending on the Task being performed. Post-hoc comparisons using Bonferroni correction revealed that connectivity between the PuA and the somatosensory network (SMN) was significantly stronger during the face and voice localizer tasks compared to the tactile localizer. Additionally, PuA correlation with the Salient Network was notably stronger during resting, voice and face localizers compared to the tactile localizer. In the voice localizer, PuA showed significantly stronger connectivity to the fronto-parietal network (FPN) compared to the other conditions. Last, connections to the thalamus were stronger during the three localizers compared to resting, and also stronger in face localizer compared to the tactile. For PuL, we report a significant main effect of Task ($F(3) = 16.72$, $p < 0.001$) and of Target ($F(8) = 178.46$, $p < 0.001$), and a significant interaction between Task and Target ($F(24) = 15.25$, $p < 0.001$). Post-hoc comparisons revealed that connectivity to the medial visual network (MVN) was stronger during, rest, face and voice localizers compared to the tactile condition. Connectivity to the lateral visual network (LVN) was stronger during the resting compared to the three localizers. PuL showed higher correlation with the DMN in the resting, voice and tactile localizers compared to the face localizer. Additionally, connectivity to the hippocampus and thalamus was stronger across all three localizers compared to

the resting state. Finally, PuL connectivity to the amygdala was stronger during the tactile localizer compared to the face localizer, and stronger during the face localizer compared to the voice localizer and resting state. For right PuL, the multiway anova showed as PuA and PuL, a significant main effect of Task $F(3) = 74.06$, $p < 0.001$, Target $F(8) = 213.31$, $p < 0.001$, and an interaction between Task and Target $F(24) = 9.98$, $p < 0.001$. Post-hoc comparisons revealed that right PuL exhibited stronger connectivity to the MVN and SN in the voice condition comparing to the resting state condition. Also, stronger correlation with the DMN was observed during the resting, voice and face localizer compared to tactile condition, and during voice condition compared to resting. Increased connectivity with hippocampus and thalamus was observed across all three localizers compared to the resting state. Lastly, connectivity to amygdala was stronger in both face and voice in comparison to both resting and tactile conditions. For PuM, only a significant effect of Target ($F(8) = 393.85$, $p < 0.001$) was observed in addition to a significant interaction between Task and Target ($F(24) = 5.94$, $p < 0.001$). Specifically, PuM showed stronger connectivity to the MVN and SN across resting, face and voice conditions compared to tactile, stronger connectivity to the frontoparietal network and lateral visual network during resting compared to the three localizers and stronger connectivity to the hippocampus across all localizers in comparison to resting state condition. Connectivity to amygdala was higher during the voice localizer compared to tactile localizer.

Overall, these results underscore the fact that functional connectivity of the pulvinar is subnuclei- and task-specific and support a dynamic refinement of pulvino-cortical functional connectivity as a function of context. This enforces the idea that the pulvinar is not merely a passive relay station, but an active participant in dynamically regulating large-scale brain networks, modulating its connectivity to optimize information processing in a context-specific manner.

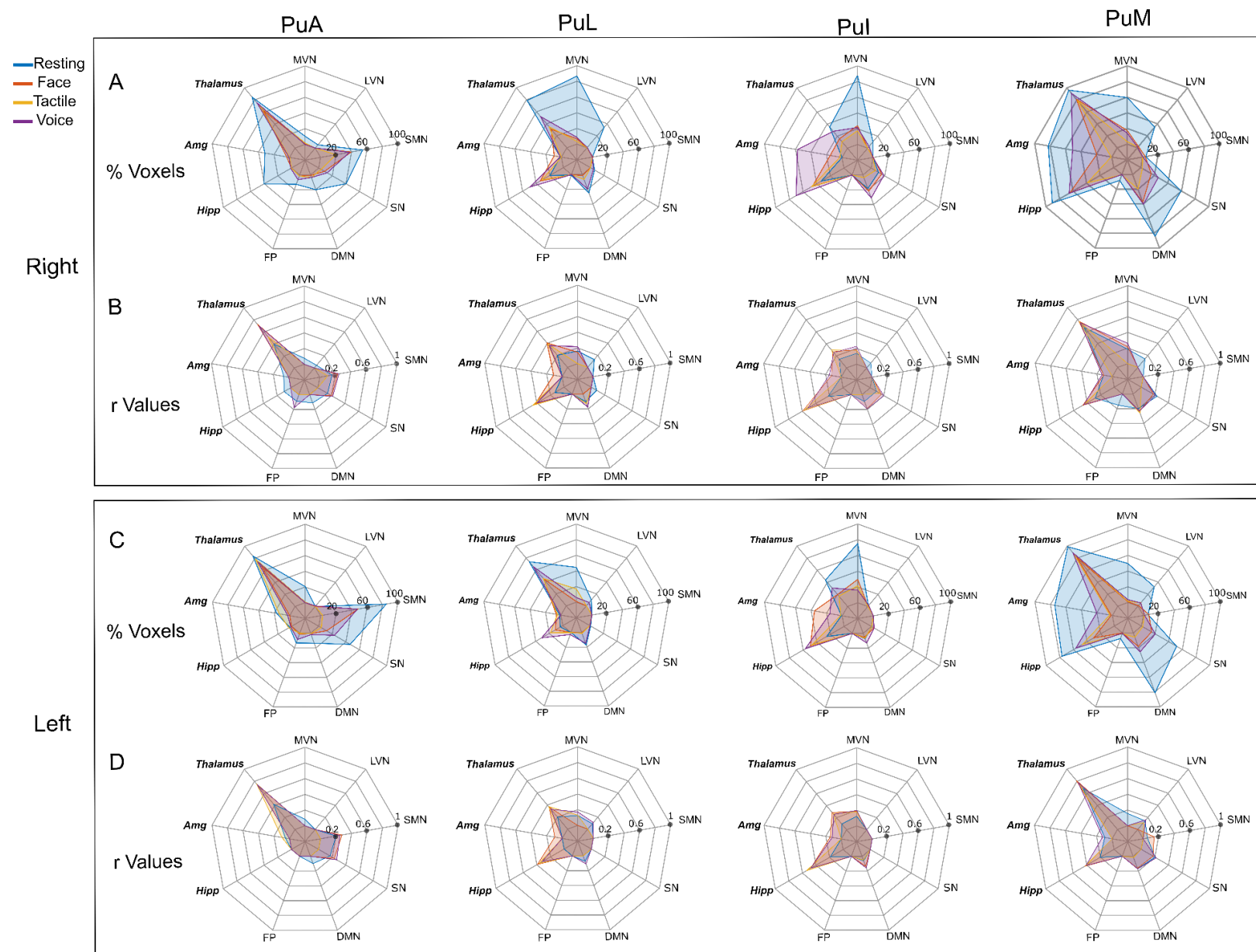


Figure 4. Pulvinar functional connectivity with major brain regions across resting state and localizer tasks. Changes in % of significantly correlated voxels for right (A) and left (C) pulvinar ROIs; Changes in r-scores with major brain regions for right (B) and left (D) pulvinar ROIs.

Connectivity profiles of medial pulvinar subdivisions during Resting-State and localizer Tasks

PuM is the largest pulvinar subdivision. Based on this and the observation that different part of PuM have different functional properties, we decided to divided it into 8 subregions of equivalent size, based on the three canonical anatomical orientations (anteroposterior, mediolateral and dorsoventral). These were named after their anatomical location: anterior-lateral-dorsal (ALD), anterior-lateral-ventral (ALV), anterior-medial-dorsal (AMD), anterior-medial-ventral (AMV), posterior-lateral-dorsal (PLD), posterior-lateral-ventral (PLV), posterior-medial-dorsal (PMD), and posterior-medial-ventral (PMV). In a first step, we performed a seed-to-whole brain analysis on these ROIs and we computed the percentage of voxels with significant functional connectivity in the resting-state condition (Figure 5). Right and left PuM subdivisions displayed very similar functional connectivity patterns. The ALD primarily exhibited widespread connectivity with both medial and lateral visual networks (MVN + LVN), the DMN, and the thalamus, and to a lesser extent with the SMN, SN, FP, and hippocampus. It only displayed minimal functional connectivity with the SMN and amygdala. ALV showed a similar pattern to ALD but with less spread connectivity to the DMN and FP. AMD displayed spread connections to the DMN, SN, hippocampus, and thalamus, and to a lesser extent with the MVN, LVN, and FP. Functional connectivity to the SMN and amygdala was extremely weak. AMV presented a similar connectivity profile to AMD but with more restricted connectivity with the DMN, along with more widespread connectivity with the amygdala. PLD demonstrated rich connectivity with the DMN, hippocampus, and thalamus, and to a lesser extent with the MVN, LVN, SN, FP, and amygdala. No connectivity to the SMN was observed. PLV and PMV shared similar functional connectivity patterns, showing extensive connections with the three subcortical structures (hippocampus, amygdala, and thalamus), more limited connectivity with the DMN, SN, and MVN, and nearly absent connectivity with the SMN and FP. Notably, LVN was uniquely functionally connected with the PLV and not with PMV. Lastly, PMD exhibited a distinct functional connectivity profile, showing functional connectivity only with the hippocampus and thalamus.

These distinct connectivity profiles of the medial pulvinar subdivisions confirm functional specialization within the PuM, with different subregions contributing to different cognitive processes such as visual processing, salience, memory, emotional regulation, and higher-order cognitive functions. The lack of involvement of PuM in sensorimotor processing further highlights its specialized role in cognitive integration.

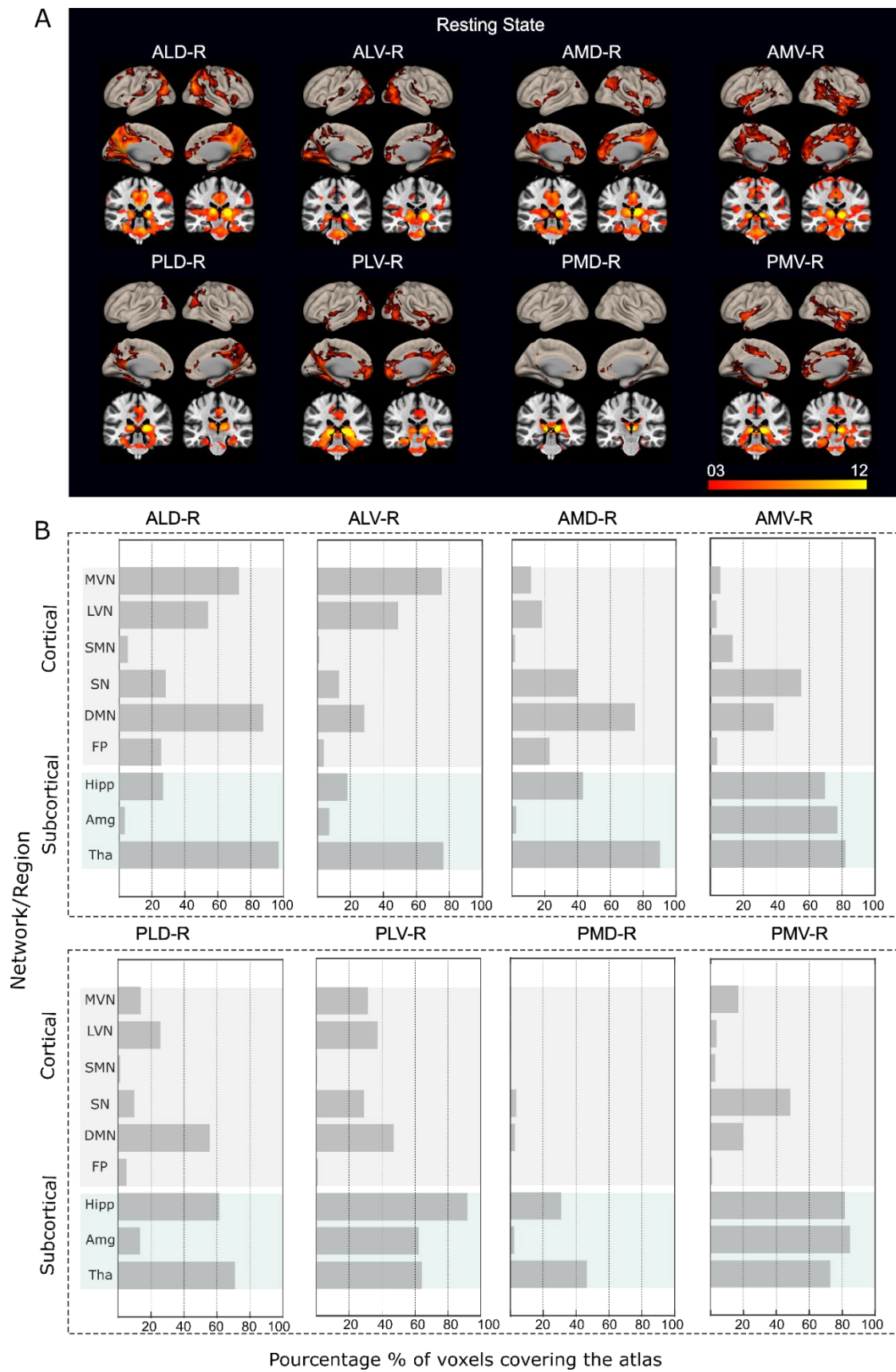


Figure 5. Right medial pulvinar functional connectivity with major brain regions during resting state. (A) Whole brain functional connectivity maps of the 8 anatomical subdivisions of right PuM, projected on lateral and medial views of left and right brain hemispheres. (B) Percentage of voxels connected to these 8 anatomical subdivisions of right PuM for major cortical and subcortical networks or regions.

Preserved PuM-to-whole brain connectivity across resting and localizer tasks

Figure 6 replicates figure 3 with the PuM subdivisions, illustrating the spatial overlap of connectivity maps on color coded inflated brain maps for each PuM subdivision. Yellow to red shades indicate strong to weak overlap. The spider plots presented in figure 7 further capture the percentage of voxels that are functionally connected to the different PuM subdivision as a function of context of interest and tables S4 and S5 provide a quantification of these observations. Results reveal that each PuM subdivision is functionally connected with a distinct specific core brain network. Again, left and right PuM subdivisions had similar connectivity patterns to the rest of the brain (see Figure S4 for the left PuM connectivity maps). ALD exhibited common functional connectivity patterns across all conditions with the DMN and thalamus, while ALV exhibited functional connectivity with the hippocampus and thalamus. ALD and ALV showed common functional connectivity maps in three conditions with the MVN, DMN, and thalamus, and ALV with the hippocampus. When considering common functional connectivity patterns across only two conditions, ALD and ALV exhibited a more extensive connectivity pattern with the visual network (primarily MVN, and to a lesser extent, LVN), DMN, hippocampus, and thalamus, and ALV specifically with the amygdala. A modest connectivity was also observed with the SMN and SN. AMD core connectivity network was restricted to the hippocampus, thalamus, and DMN. When considering overlap between two or three maps, this network additionally included SN and to a lesser extent to MVN, LVN and FPN. AMV core connectivity network was restricted to the hippocampus and thalamus. When considering overlap between two or three maps, this network additionally included the SMN, SN, DMN and amygdala. PLD core connectivity network was restricted to the DMN, hippocampus and thalamus. When considering overlap between two or three maps, this network additionally included a weak functional connectivity with the visual network and SN. Like AMV, PMD core connectivity network was restricted to the hippocampus and thalamus. When considering overlap between two or three maps, this network additionally included the DMN, SN, amygdala. Last, PLV and PMV core connectivity network was restricted to the hippocampus and thalamus, with the addition of the DMN for PLV and the amygdala for PMV. When considering overlap between two or three maps, this network additionally included functional connectivity with the MVN, DMN, SN, amygdala, hippocampus, and thalamus, and LVN specifically for PLV.

Overall, PuM subdivisions appear to share a core functional connectivity with the hippocampus, thalamus and DMN. In addition, they each have distinguishing feature in the different sensory contexts, although some PuM subregions share similar overlap patterns such as ALD and ALV, and PLV and PMV. This supports the hypothesis of a functional specialization of these different PuM subdivisions, at the same time that this suggests the possible existence of multiple functional gradients within this subnucleus.

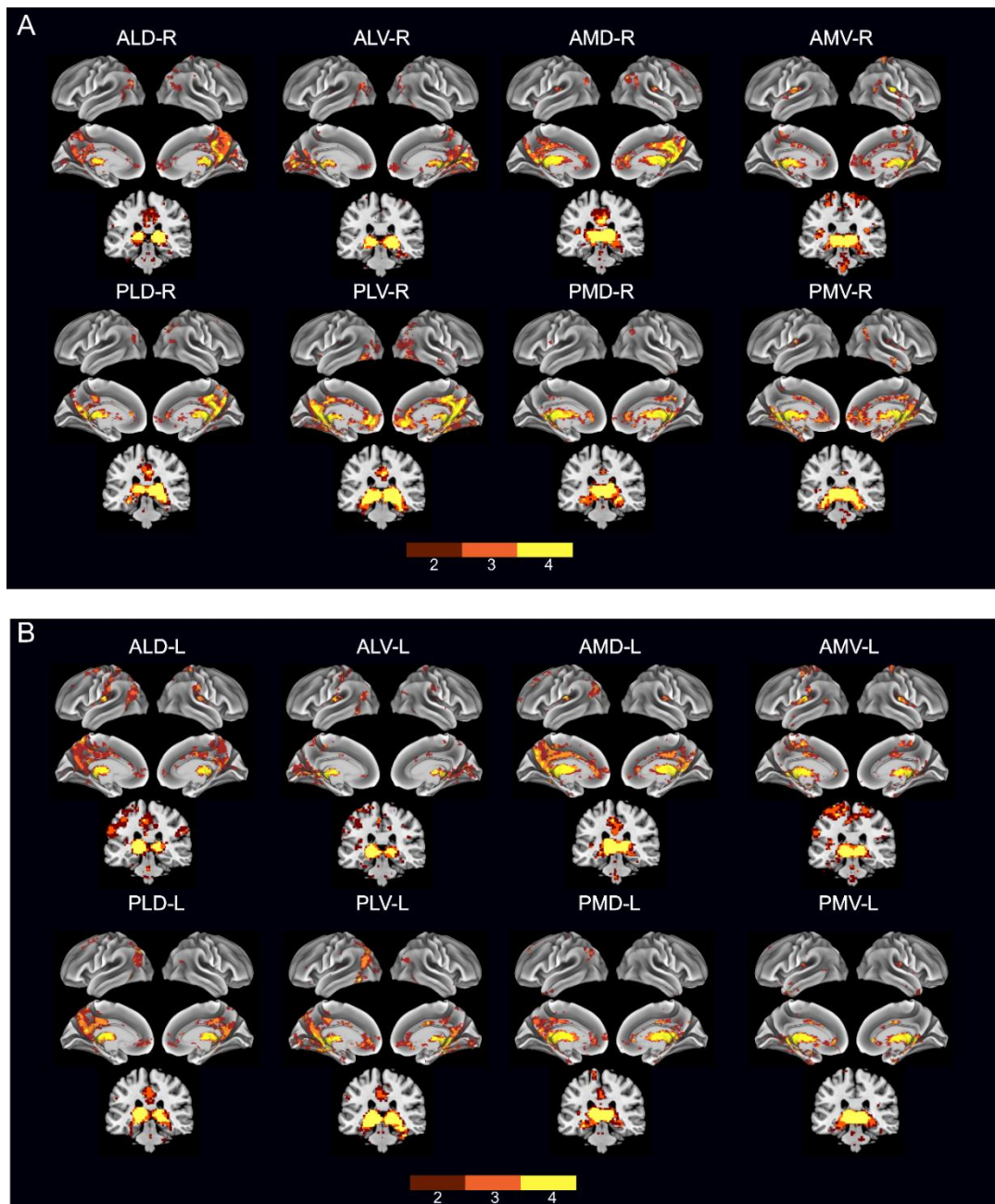


Figure 6. Overlap of medial pulvinar functional connectivity with major brain regions across resting state and localizer tasks for right (A) and left (B) medial pulvinar subdivisions.

Changes in strength of functional connectivity between pulvinar and major brain regions across resting-state and localizer tasks

The differences in spatial connectivity patterns described above are accompanied by changes in connectivity strength between pulvinar and brain across tasks (Figure 7). In order to statistically assess the statistical significance of these changes in functional connectivity strength, we performed a factorial analysis taking as factor the 8 PuM subdivisions (ALD, ALV, AMD, AMV, PLD, PLV, PMD, PMV), task (Resting, Face, Voice, Tactile) and target brain region (MVN, LVN, SMN, SN, DMN, FPN, Hippocampus, Amygdala and Thalamus), correlation scores as dependent variable, with subjects

included as a random factor. Results revealed significant differences in how specific PuM subdivisions connected with target brain regions across tasks. Functional connectivity strength and associated statistics were very similar for right and left pulvinar subnuclei ROIs. As a result, we will focus on the description of the observation for the right pulvinar ROIs. The results for left Pulvinar are presented in Figure S5. Brain connectivity of right ALD shows a main effect of task ($F(3) = 58.19$, $p < 0.001$) and target ($F(8) = 680$, $p < 0.001$) as well as an interaction between task and target factors ($F(24) = 21.34$, $p < 0.001$). Post-hoc comparisons showed a stronger functional connectivity with the medial visual network in the face ($p = 0.001$) and tactile ($p < 0.001$) localizers as compared to the voice localizer condition. Connectivity to the sensorimotor network was strong in the face condition but nearly absent in the voice and tactile conditions ($p < 0.001$). In addition, connectivity to the salience, lateral visual and frontoparietal networks as well as hippocampus was higher in resting-state as compared to the three localizers (all $p < 0.001$). In contrast, connectivity to thalamus was higher in all localizers compared to resting-state (all $p < 0.001$). Brain connectivity of right ALV shows a main effect of task ($F(3) = 6.71$, $p = 0.0002$) and target ($F(8) = 405.14$, $p < 0.001$) as well as an interaction between task and target effects ($F(24) = 19.56$, $p < 0.001$). Post-hoc comparisons revealed high functional connectivity to the default mode network in voice ($p < 0.001$) and tactile ($p < 0.001$) localizers compared to the face localizer. Similarly to ALD, connectivity to salience and lateral visual networks was higher in resting-state as compared to the three localizers (all $p < 0.001$). Brain connectivity of right AMD shows a main effect of task ($F(3) = 29.85$, $p < 0.001$) and target ($F(8) = 687.88$, $p < 0.001$) as well as an interaction between task and target ($F(24) = 19.72$, $p < 0.001$). Post-hoc comparisons indicated a high connectivity with the salience network and hippocampus in the voice and face conditions compared to the tactile condition ($p < 0.001$). Connectivity to the visual networks (lateral and medial) as well as to the frontoparietal network was higher in resting-state as compared to the three localizers (all p values were < 0.001). Brain connectivity of right AMV shows a main effect of task ($F(3) = 6.26$, $p = 0.0004$) and target ($F(8) = 671.18$, $p < 0.001$) as well as an interaction between task and target ($F(24) = 24.42$, $p < 0.001$). Post-hoc comparisons revealed, strong connectivity with the sensorimotor network in the face condition which was almost absent in the voice and tactile conditions (all $p < 0.001$). High connectivity to the salience network and amygdala was observed in the voice condition and was almost absent in face and tactile condition (all $p < 0.001$). Last, higher connectivity was observed in the resting-state as compared to the three localizer tasks in the default mode network ($p < 0.001$) and the medial visual network ($p = 0.01$). Brain connectivity of right PLD shows a main effect of task ($F(3) = 3.15$, $p = 0.02$) and target ($F(8) = 479.34$, $p < 0.001$) as well as an interaction between task and target ($F(24) = 22.89$, $p < 0.001$). Post-hoc comparisons show high connectivity with the salience network in the face condition compared to voice and tactile conditions for which connectivity was nearly absent (all $p < 0.001$). Higher connectivity was also observed with the frontoparietal network in the resting-state compared to three localizers tasks.

Brain connectivity of right PLV shows a main effect of task ($F(3) = 47.83$, $p < 0.001$) and target ($F(8) = 400.69$, $p < 0.001$) as well as an interaction between task and target ($F(24) = 20.74$, $p < 0.001$). Post-hoc comparisons show strong connectivity with the medial visual network, the salience network and the amygdala in the face and voice conditions compared to the tactile condition. Connectivity to the lateral visual network was observed in the voice condition and was almost absent in the face and tactile conditions. Brain connectivity of right PMD shows a main effect of task ($F(3) = 84.18$, $p < 0.001$) and target ($F(8) = 602.6$, $p < 0.001$) as well as an interaction between task and target ($F(24) = 27.75$, $p < 0.001$). Post-hoc comparisons showed high connectivity to the salience and default mode networks in the voice and face localizers as compared to the tactile and resting-state conditions. Connectivity to default mode network was stronger in the voice localizer compared to face localizer. Also, higher connectivity to the amygdala was observed in the voice localizer compared to the face and tactile localizers (all $p < 0.001$). Brain connectivity of right PMV shows a main effect of task ($F(3) = 28.51$, $p < 0.001$) and target ($F(8) = 315.42$, $p < 0.001$) as well as an interaction between task and target ($F(24) = 13.24$, $p < 0.001$). Post-hoc comparisons show stronger connectivity to the medial visual cortex in the tactile condition compared to face condition and in all localizers compared to the resting-state conditions. Last, high connectivity was observed with the salience network and amygdala in the voice and face conditions whereas connectivity was extremely weak in the tactile condition (all $p < 0.001$). Last, for all PuM subdivisions, except ALD, connectivity to thalamus and hippocampus was higher in all localizers as compared to resting-state (all p values were $p < 0.001$). Overall, this supports a functional reorganization of the fine-grained connectivity of PuM as a function of the sensory context.

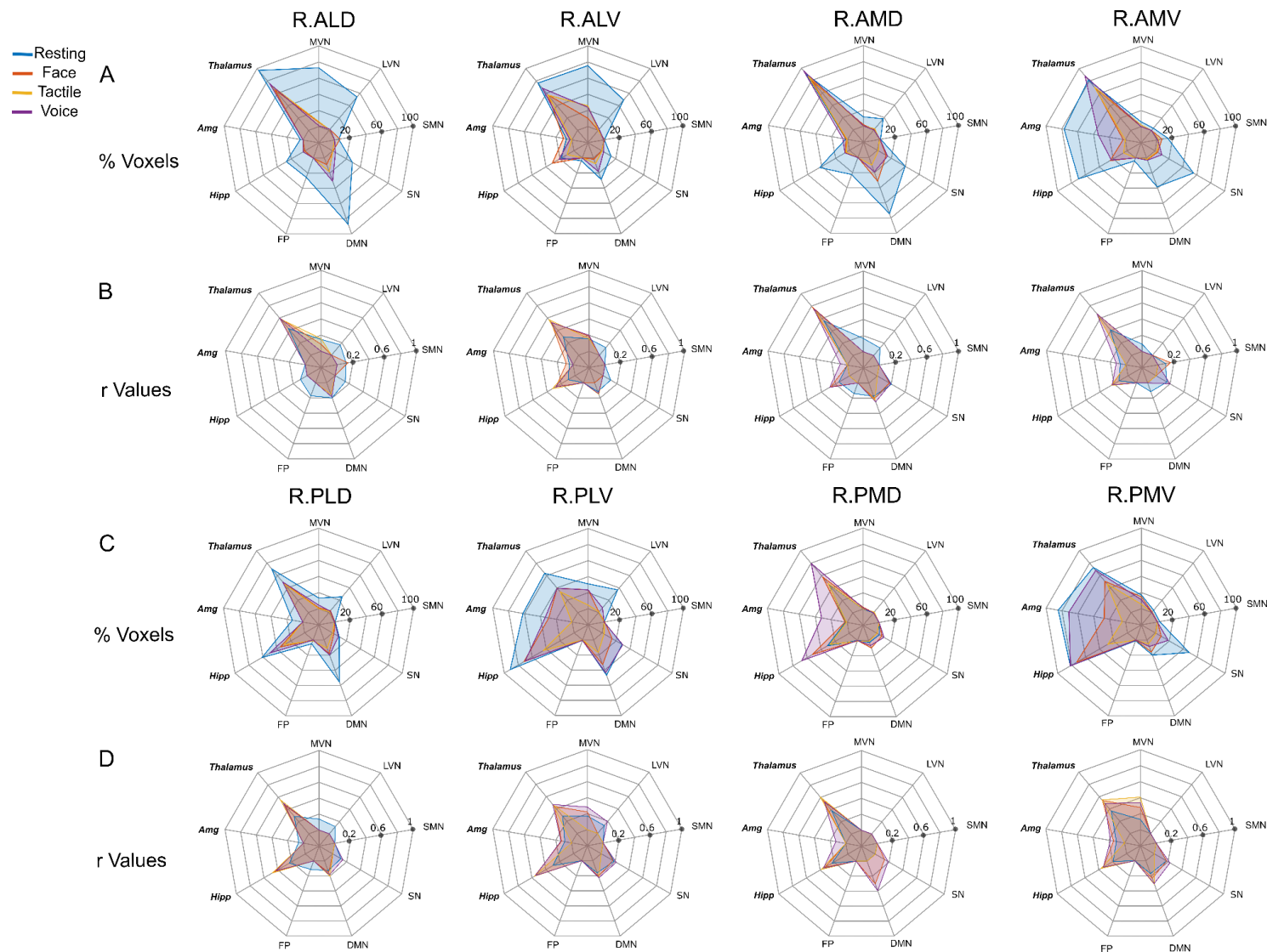


Figure 7. Right medial pulvinar functional connectivity with major brain regions across resting state and localizer tasks. Changes in % of significantly correlated voxels (A and C) pulvinar ROIs; Changes in r-scores with major brain regions for right (B and D) right PuM ROIs.

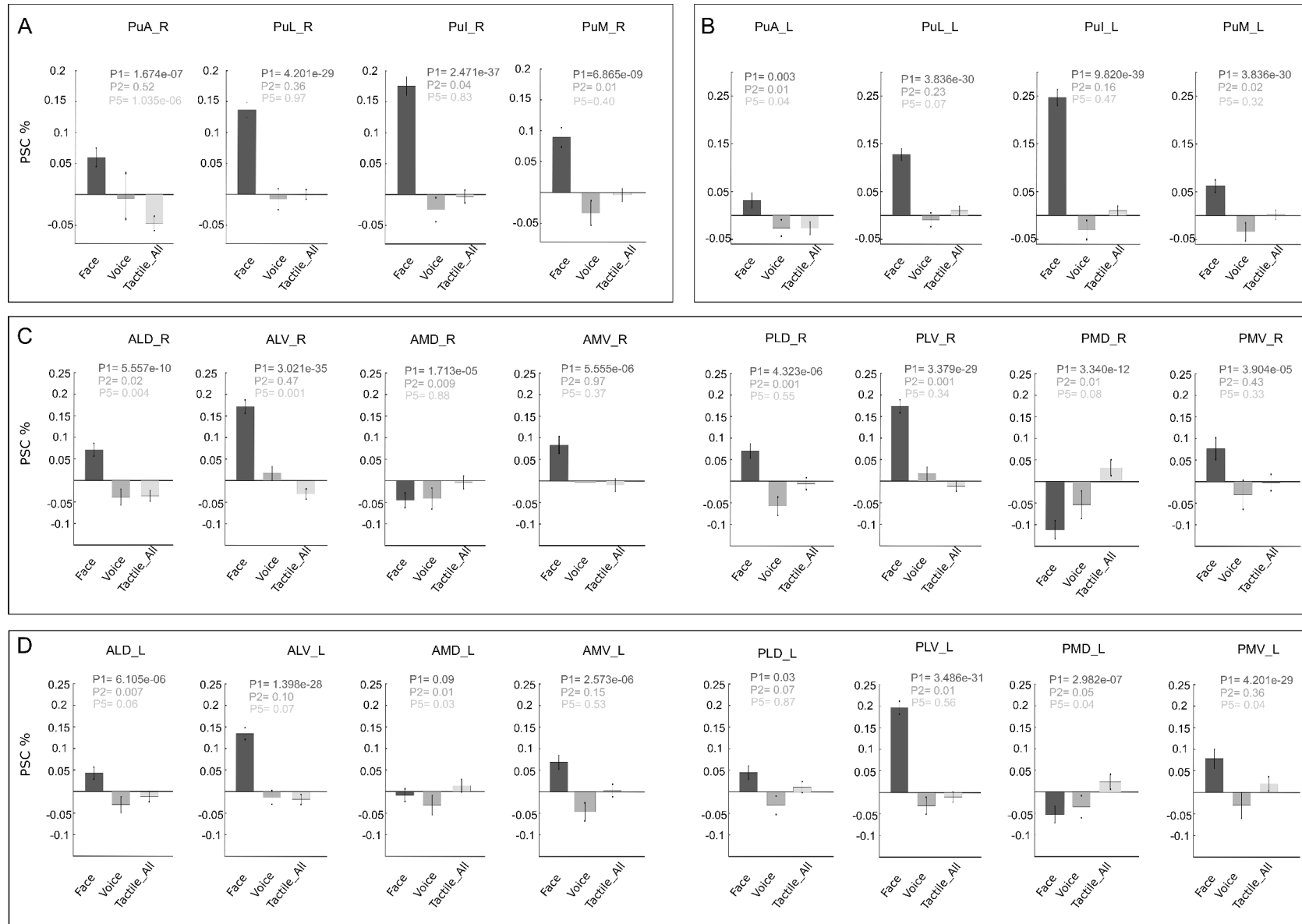


Figure 8. Percentage signal change across all pulvinar subdivision during the three localizer tasks. (A) Right and (B) left pulvinar subnuclei. (C) Right and (D) Left medial pulvinar subdivisions.

Percentage of signal change of all pulvinar subnuclei across three localizers.

Changes in spatial extent and strength of functional connectivity do not necessarily coincide with changes in functional responses to sensory input in the different experimental conditions. In order to quantify functional activations within each pulvinar subdivision and gain insight into their specific neural engagement across the different sensory contexts, we calculated the percentage of signal change (%SC) in all subnuclei and subdivisions during the three localizer tasks (Face, Voice, and Tactile, Figure 8). By comparing %SC across these tasks, we aimed to reveal task-dependent variations in pulvinar activity, a possible marker of functional specialization. Distinct patterns of %SC were observed in the pulvinar subnuclei across sensory conditions. In the face condition, all four right pulvinar subnuclei showed a significant positive %SC compared to the baseline (all $p < 0.001$). In the voice condition, only the PuM exhibited a significant negative %SC compared to baseline ($p = 0.03$). For the tactile condition, a significant negative %SC was found exclusively in PuA ($p < 0.001$). In the left pulvinar subnuclei, a similar pattern was observed. Specifically, for the face condition, all subregions showed a significantly positive %SC compared to baseline (all $p < 0.001$). In the voice condition, a significant negative %SC was observed in PuA ($p = 0.03$), with a trend toward significance in PuM ($p = 0.06$). No significant %SC was observed across any of the pulvinar subnuclei in the tactile condition. These results indicate a clear functional specialization within pulvinar subregions depending on sensory modality. The pulvinar's engagement with face stimuli was robust and bilateral, whereas its involvement in auditory (voice) and tactile processing was more selective and asymmetrical. The differences in activation between the right and left pulvinar subregions, especially in the voice condition, indicated a possible functional lateralization in sensory processing.

For the medial pulvinar subdivisions, the analysis of %SC in the face condition revealed significant positive %SC in the right ALD, ALV, AMV, PLD, PLV, and PMV subdivisions (all $p < 0.001$), while significant negative %SC compared to baseline was observed in AMD and PMD. In the voice condition, significant negative %SC was found in AMD ($p = 0.02$), PLD ($p = 0.003$), and PMD ($p = 0.03$), but a significant positive %SC was detected in PLV ($p = 0.003$). For the tactile condition, significant negative %SC compared to baseline was observed only in ALD ($p = 0.01$) and ALV ($p = 0.003$). In the left pulvinar subregions, the face condition showed similar results to those of the right side. For the voice condition, significant negative %SC was observed in AMD ($p = 0.03$) mirroring the right subdivisions, as well as in PLV ($p = 0.03$). Lastly, no significant %SC was found across any subregions in the tactile condition. These findings reveal a complex and condition-specific pattern of activations suggesting that the medial pulvinar subdivisions also exhibit modality-specific neural responses. The latero-dorsal, latero-ventral and medioventral subregions are more engaged in processing face-related stimuli, while the mediodorsal subdivisions appear to be inhibited in face processing. Also, voice stimuli lead to a more

mixed response with both negative and positive %SC in different regions with a specific inhibitory pattern in the dorsal subregions and more positive implication in the ventral subregion. In comparison to visual and auditory stimuli, the tactile condition showed minimal activations which were restricted to the lateral subdivisions.

Discussion

Core pulvino-cortical networks and associated context-dependent dynamics

Seed-based analyses in resting-state show distinct functional connectivity patterns reflecting functional specialization across pulvinar subnuclei, thereby replicating prior observations from human and non-human primate studies. Specifically, we found that the anterior pulvinar (PuA) is mainly functionally connected to somatosensory cortices (Froesel et al., 2021b; Guedj and Vuilleumier, 2020b), in agreement with its involvement in motor and postural control (Schneider et al., 2019) as well as motor planning (Cléry et al., 2018; Grieve et al., 2000). In particular, a study by (Haegens et al., 2011) demonstrated an impact on behavioral performance during visuo-tactile discrimination tasks related to a modulation of the oscillatory power in the alpha band in the pulvinar on spike timing and firing rates in somatosensory, premotor, and motor regions. In addition to the sensory-motor network, anterior pulvinar shows connectivity to the salience network (SN), the default mode network (DMN) and the frontoparietal network (FPN). Subcortically, it has extensive functional connectivity with the thalamus as well as weaker connectivity with the hippocampus and amygdala. Lateral and inferior Pulvinar (PuL and PuI) were predominantly functionally connected with the visual network (Arcaro et al., 2015b; Saalmann et al., 2012b), in agreement with their role in interfacing with the dorsal and ventral visual streams (Cortés et al., 2024), and the description of visual neurons in these pulvinar subnuclei (Froesel et al., 2021b; Mathers and Rapisardi, 1973). In our study, we observed additional connectivity of PuL and PuI to DMN and slightly to SN in addition to thalamus and hippocampus. The medial pulvinar exhibited widespread connectivity predominantly with the DMN, SN (Fox et al., 2005; Guedj and Vuilleumier, 2020b; C. Gutierrez et al., 2000; Homman-Ludiye et al., 2020) and visual cortices (Cappe et al., 2007; C. Cappe et al., 2009b; Homman-Ludiye et al., 2020) in addition to somatosensory cortices (Froesel et al., 2024c) as well as subcortical structures (thalamus, hippocampus, and amygdala). These observations are closely in line with previous studies on functional connectivity of pulvinar subclusters during resting state (Guedj and Vuilleumier, 2020b).

Overall, our observations emphasize the distinct functional roles of pulvinar subdivisions, PuA being mostly involved in somatosensory processing, PuI and PuL in the visual processing and PuM encompassing different modalities (vision, somato-sensation..) through its broad connectivity, emphasizing its role as a multimodal integration center participating in higher sensory and cognitive-

emotional-information processing (Froesel et al., 2021b; Homman-Ludiye et al., 2020). The recurrent connectivity to salience network observed across all pulvinar subnuclei demonstrates its role in salience detection (Snow et al., 2009) and attention (Bourgeois et al., 2020b). Additional connectivity observed with subcortical structure such amygdala and hippocampus further suggest the involvement of pulvinar in memory and emotional processing (Bertini et al., 2018).

In order to link brain networks connected to pulvinar subregions to specific sensory modalities, we further investigated the pattern of connectivity of each pulvinar subnucleus in each of the three localizers (Face, Voice and Tactile Localizers). Resulting functional connectivity patterns show strong similarities with the functional networks identified in the resting state, as well as additional functional connectivity patterns associated with a specific sensory context. These results demonstrate (i) a context independent core functional connectivity network between pulvinar subnuclei and the rest of the brain and (ii) sensory context-dependent refinements around this core functional connectivity. This is in agreement with previous studies reporting homogeneous, cluster-specific, functional connectivity patterns between pulvinar subcluster and whole brain in resting-state and during task performance, in addition to restricted refinements in these connectivity patterns (Guedj and Vuilleumier, 2020b). Overall, this provides a better understanding on how the context-dependent regulatory role of the pulvinar is achieved through a dynamic adjustment of the functional connectivity of its subnuclei with the cortex. These results emphasize the role of the pulvinar in integrating sensory, cognitive, and attentional processes across different states of brain and in the coordination and/or transition between different sensory states (Arcaro et al., 2018b; Bourgeois et al., 2020b)

Sensory-context-dependent changes in pulvino-cortical functional connectivity and activations

In spite of a stable core functional connectivity between the pulvinar subnuclei and the rest of the brain, specific changes in this connectivity can be identified in the different sensory contexts. These changes support a clear functional modulation depending on sensory context.

For anterior pulvinar, connectivity to somatosensory cortex and salience network was stronger in the face and voice condition compared to tactile condition. This result is surprising as the somatosensory nature of the tactile condition and the privileged functional connectivity of PuA with the SMN would have predicted a stronger connection between these two regions. This would be due to the weak saliency of our tactile stimuli (air puffs) as compared to the visual and auditory stimuli. The increased connectivity to the salience network emphasizes this hypothesis and may reflect attentional capture in these contexts. Additionally, the voice localizer condition demonstrated strong connectivity to the frontoparietal network that was nearly absent in the face and tactile localizers. The frontoparietal

network is often associated with higher-order processing tasks, suggesting that the auditory processing of voices might require more complex cognitive engagement.

For the lateral pulvinar, connectivity to the visual network was stronger during the face and voice localizers compared to tactile condition. Neurons within lateral pulvinar respond to different categories of visual stimuli including simple oriented and dynamic visual stimuli and complex visual stimuli like geometrical figures (Froesel et al., 2021b). The ventral part of PuL is involved in auditory processing (R. Gattass et al., 1978; Magariños-Ascone et al., 1988b; Yirmiya and Hocherman, 1987b). PuL is also responsive to tactile stimulations (Acuña et al., 1990; R. Gattass et al., 1978; Mathers and Rapisardi, 1973; Yirmiya and Hocherman, 1987b). Thus, despite of its responsiveness in the three sensory modalities, in our study, the PuL appears to be more involved in voice and face processing rather than in tactile processing. This may indicate a sensory specialization of PuL or may be explained by the specific type of tactile stimulations used. Our results show also that connectivity of PuL to the default mode network was high during voice and tactile localizer and nearly nonexistent in the face condition. Given the fact that PuL is strongly connected to the visual network in the face condition, this finding suggests that when visual input is less emphasized, PuL's connectivity with the DMN increases. The DMN is typically associated with mind-wandering or self-referential thinking and this stronger connectivity in non-visual conditions (resting, voice, tactile) could indicate that PuL shifts its engagement towards more generalized or internally directed functions when visual demands are lower. This pattern highlights a potential modulation of PuL's role depending on the sensory context. Finally, **connectivity of PuL and amygdala** was strong during the **face localizer** compared to the **tactile and voice localizer**. The amygdala is heavily involved in emotional processing, particularly in detecting and responding to emotionally salient stimuli, including fear and threat. This enhanced connectivity in the face condition may indicate that **visual information** is more closely integrated with emotional and sensory processing in certain contexts, possibly reflecting the role of the faces in signaling important environmental or emotional cues.

For inferior pulvinar, stronger correlation with the DMN was observed during voice and face localizer compared to tactile condition. The face and voice conditions, which involve complex social stimuli, likely require the DMN's participation for social cognition and emotion processing. Thus, Pul may be integrating these visual and auditory cues into broader self-referential or socially meaningful contexts, explaining the stronger connectivity during these conditions. The tactile condition, being primarily somatosensory, likely requires less engagement of the DMN, as tactile processing is possibly less associated with internal or social-cognitive functions, which could explain the weaker correlation in this condition. In contrast to PuL, connectivity between Pul and amygdala was stronger in both face

and voice localizers in comparison to tactile condition. It seems that, depending on the sensory context, there is a shift in amygdala connectivity from PuL to Pul.

For the medial pulvinar, interestingly, no main effect of task was observed for PuM indicating that PuM connectivity is not fully task-dependent. This may be explained by the fact that PuM is the most highly multisensory subregion of the pulvinar (Froesel et al., 2021b) and is described as a multimodal associative nucleus (Homman-Ludiye et al., 2020). It could thus be involved equally in the three localizers and this constitutes one of the main arguments of subdividing the PuM to multiple subregions to investigate in depth its functional connectivity profiles.

Interestingly the thalamus and hippocampus were strongly connected to all pulvinar subregions, except the anterior pulvinar, compared to resting. No difference between localizers was observed. This result suggests that the thalamus and hippocampus maintain a stable and robust functional relationship with most of the pulvinar subregions, regardless of sensory modality. This pattern could reflect the thalamus's role as a major relay center for sensory and motor inputs, and the hippocampus's involvement in memory processing across all subdivisions. However, the anterior pulvinar's weaker connectivity to hippocampus might indicate a functional specialization that differentiates it from the other pulvinar subregions. This anterior region may focus more on higher-order cognitive or social functions, making it less reliant on hippocampal inputs.

When looking at the percentage of signal change within the four subdivisions of the pulvinar across the three sensory conditions (Face, Voice, and Tactile), we observe a general specialization of pulvinar responses with all subdivisions responding to the visual context, the medial part being completely inhibited during the voice context and the anterior subregion also significantly inhibited during tactile context. Overall, these results indicate that the pulvinar is highly dynamic and has a context-dependent role in integrating sensory information, with different subregions showing specialized connectivity patterns based on the type of stimulus being processed. It thus appears as a flexible integrator of sensory, emotional, and cognitive information, adapting its connectivity based on the salience and complexity of the stimuli. This highlights the pulvinar's critical function in not only sensory processing but also in attention, emotional regulation, and social cognition.

Functional Specialization Within the Medial Pulvinar assessed through resting-state resting state

The seed-based analysis during resting-state of the eight PuM subdivisions showed distinct connectivity profiles among the eight subregions.

The anterior-latero-dorsal (ALD) and anterior-latero-ventral (ALV) subnuclei showed extensive connectivity to the visual network and thalamus and moderate connections to the rest of the brain,

with additional connectivity to DMN in ALD. The anterior-medio-dorsal (AMD) and anterior-medio-ventral (AMV) subnuclei showed widespread connections to the default mode network (DMN) and salience network (SN) in addition to hippocampus and thalamus, with additional connectivity to amygdala for AMV. The posterior subregions mirror somewhat the patterns observed in the anterior part except within the medio-dorsal subregion which exhibit a different connectivity profile compared to its anterior homolog with connectivity restricted only to hippocampus and thalamus. In addition, PLD and PLV are less connected to the visual network compared to their anterior homolog, and additional connection to amygdala was observed in PLV. These observations suggest a functional specialization within the medial pulvinar with the anterior-lateral part being more involved in visual processing and the integration of visual information with other sensory modalities than the posterior regions. The ALD's more spread connectivity to the DMN points toward a role in linking visual perception with cognitive processes. The additional amygdala connectivity observed in PLV points to its involvement in emotion-related-information processing. The anterior-medial portion appears to be linked more to integrating sensory information with higher cognitive functions, including attention, and memory (DMN, SN, hippocampus). The additional amygdala connectivity observed in AMV also indicate a possible involvement in emotionally relevant sensory processing, especially in contexts requiring emotional or threat evaluation. Posterior medio-dorsal (PMD) subregion has a more specialized role, potentially related to memory consolidation and sensory integration (Hippocampus and thalamus). In summary, the medial pulvinar subregions demonstrate differentiated roles in integrating sensory, cognitive, and emotional processes. This functional diversity highlights the medial pulvinar's complex involvement in linking sensory input to higher-order brain functions.

Sensory-context-dependent changes in PuM-to-cortical functional connectivity and activations

Statistical comparison of connectivity strength between medial pulvinar subregions and the rest of the brain across the different sensory modalities reveals different patterns of functional modulation.

In the right anterior latero-dorsal (ALD) subregion, robust connectivity was detected with the medial visual network during both the face and tactile conditions, but it was almost absent during the voice condition. The ALD's strong engagement with the visual network during the face condition aligns with its known role in visual processing, as the pulvinar is generally recognized for its involvement in guiding visual attention and processing visual stimuli, particularly those related to social cues, such as faces (Arcaro et al., 2015b). Its involvement in the tactile condition suggests that ALD might also participate in processing visual-tactile integration, which may be critical for tasks requiring spatial awareness and coordination between sensory modalities (Fiebelkorn et al., 2019b). The absence of connectivity during the voice condition indicates that ALD is less involved in auditory processing. Additionally, this

subregion displayed strong connectivity with the sensorimotor network in the face condition, which was absent in both the voice and tactile conditions. This observation emphasizes its role in visual-motor coordination. Previous research suggests that the pulvinar plays a key role in coordinating eye and body movements in response to visual stimuli, particularly in the peripersonal space (Cléry et al., 2018; Grieve et al., 2000). This would explain the enhanced ALD-sensorimotor network connectivity during face processing, as recognizing and responding to faces often requires coordinated eye and facial movements.

For the right anterior latero-ventral (ALV) subregion, high connectivity with the default mode network was present during the voice and tactile localizers, while no such connectivity was observed in the face localizer. This result could indicate that, in these former sensory modalities ALV has enhanced control over internal cognitive and social processes. For example, voices often carry emotional or social information, which may require deeper internal reflection or social cognitive processing, aligning with the DMN's role in these functions. In contrast, the absence of connectivity between ALV and the DMN during the face localizer may reflect that face processing is more externally focused, requiring visual attention and immediate social interaction rather than internal, reflective processes. The pulvinar's role in guiding visual attention toward socially relevant stimuli like faces (Arcaro et al., 2015b) likely diminishes the need for ALV's engagement with the DMN during face processing, as it becomes more engaged with networks associated with visual or attentional processes rather than introspective or socially reflective functions.

In the right anterior medio-dorsal (AMD) subregion, strong connections to the salience network and hippocampus were noted during the voice and face conditions, but no such connections were found in the tactile condition. This finding suggests that this region is particularly involved in processing socially salient and emotionally heavy stimuli. The salience network is crucial for detecting and filtering relevant stimuli from the environment (Menon and Uddin, 2010a), and its activation during voice and face conditions implies that these sensory inputs are perceived as more socially and emotionally relevant than tactile stimuli. The hippocampal connectivity likely reflects the involvement in memory processing, especially in recalling or encoding social and emotional information (Squire, 2004), which is prominent in both voice and face processing, possibly due to the recognition of individuals or emotional content associated with these modalities. In contrast, the absence of these connections in the tactile condition suggests that tactile stimuli may not be perceived as socially or emotionally salient enough to activate this network in AMD. This emphasizes the region's role in processing high-salience, socially relevant inputs, such as facial expressions and vocal tones, which are essential for communication and emotional interaction.

Similarly, to the ALD, the right anterior medio-ventral (AMV) subregion exhibited strong connectivity with the sensorimotor network during the face condition, which was absent during both voice and tactile conditions. This may indicate that ALD and AMV share some common functional roles regarding the visual-motor coordination. Furthermore, the AMV showed connectivity with the salience network and amygdala during the voice condition, which was not present in the face or tactile conditions. This pattern suggests that AMV is sensitive to the emotional and salience aspects of vocal stimuli, while its sensorimotor connectivity is primarily reserved for visual (face) stimuli.

In the right posterior latero-dorsal (PLD) subregion, there was strong connectivity with the salience network during the face condition, with no connections detected in the voice or tactile conditions. This observation indicates that this subregion may play a key role in detecting socially and emotionally salient visual stimuli, such as facial expressions, and suggest that PLD might be integral to filtering and prioritizing socially relevant visual stimuli for further processing. This selective engagement in the face condition underscores PLD's likely role in visually-guided social cognition, where recognizing facial expressions or gaze direction is critical for social interactions.

The right posterior latero-ventral (PLV) subregion demonstrated strong connectivity with the medial visual network, salience network, and amygdala during the face and voice conditions, compared to tactile condition. Additionally, the PLV showed connectivity with the lateral visual network during the voice condition, which was not present in the face or tactile conditions. These findings, indicate that PLV is key in integrating visual and emotional aspects of social stimuli, whether they are facial expressions or vocal cues. The medial visual network is known for its role in processing complex visual stimuli, such as facial recognition and visual scenes (Fox et al., 2005). The concurrent connectivity with the salience network and amygdala during face and voice conditions suggests as mentioned previously, that PLV is involved in processing not only the visual aspects of faces but also the emotional significance of both faces and voices. The amygdala's role in emotional processing, particularly in recognizing and responding to emotionally charged stimuli like facial expressions and vocal intonations (Schafe et al., 2000) aligns with this connectivity. Therefore, PLV's involvement with both the visual and emotional networks indicates its importance in social and emotional perception. Interestingly, the PLV also demonstrated connectivity with the lateral visual network specifically during the voice condition, which is unique and absent in the face and tactile conditions. The lateral visual network is associated with processing motion and object recognition (Goodale and Milner, 1992), suggesting that during voice processing, PLV may be involved in visual attention or visual imagery related to auditory input. This could reflect a cross-modal integration process, where listening to voices triggers visual representations or enhances attention to surrounding visual cues. The absence of these connectivity patterns in the tactile condition further reinforces that the PLV subregion is more specialized for

processing complex, socially relevant visual and auditory stimuli, while it is less engaged in purely somatosensory (tactile) inputs.

The right posterior medio-dorsal (PMD) subregion displayed high connectivity with the salience and default mode networks during the voice and face conditions, compared to tactile condition. In the voice localizer, the connectivity to the salience network was stronger than in the face localizer, and connectivity to the amygdala was also observed in the voice condition but absent in the face and tactile conditions. The PMD's strong connectivity to both the salience network (SN) and default mode network (DMN) during the voice and face conditions indicates its involvement in socially and emotionally relevant stimuli processing. The PMD appears to be a key in integrating social and emotional information, potentially for evaluating the relevance of these stimuli and linking them to self-referential or social-cognitive processes. The fact that the PMD's connectivity to the salience network was even stronger in the voice condition compared to the face condition indicates that vocal stimuli, perhaps due to their dynamic and emotionally rich nature, are particularly strong in capturing attention and engaging the salience network. This enhanced salience network engagement during voice processing may reflect the importance of vocal cues in human communication and social interaction, where intonation, pitch, and other auditory features are critical for interpreting emotions and intentions. Moreover, the observed connectivity to the amygdala during the voice condition but not during the face or tactile conditions further supports the idea that PMD plays a key role in processing emotional stimuli, especially auditory stimuli.

Lastly, the right posterior medio-ventral subregion exhibited stronger connectivity with the medial visual cortex during the tactile condition compared to the face condition. This is an interesting finding, as it suggests that during tactile stimulation, the PMV may play a role in cross-modal sensory integration, potentially linking tactile and visual inputs. The medial visual cortex is typically involved in processing visual stimuli related to space and orientation, so this connectivity could indicate that tactile input is being processed in conjunction with spatial or visual representations, possibly enhancing the perception of touch in a spatial context. Additionally, it demonstrated strong connectivity to the salience network and amygdala during the voice and face conditions, with no connectivity observed during the tactile condition. This again highlights the role of the amygdala in processing emotionally salient stimuli, especially in social contexts like vocal or facial expression recognition. The connectivity with the salience network reinforces the idea that both voice and face stimuli are processed as socially or emotionally significant, requiring attentional and emotional responses facilitated by the salience network and amygdala. However, the lack of connectivity in the tactile condition suggests that PMV's engagement in emotional and social processing is limited to visual and auditory stimuli, rather than somatosensory inputs.

The overall findings from the functional connectivity analysis of the medial pulvinar subregions reveal a distinct pattern of specialization and sensory integration, supporting the idea that each subregion within the medial pulvinar plays a different role in processing sensory, emotional, and social information. This functional segregation emphasizes the medial pulvinar's crucial role in integrating multiple modalities and contributing to higher cognitive functions like attention, salience detection, and emotional regulation.

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Author contributions statement

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Additional information

Data and code are available upon request to Suliann Ben Hamed or Souhir Dali, the corresponding authors.

Competing interests

Authors declare no conflicts of interest.

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Supplementary Material

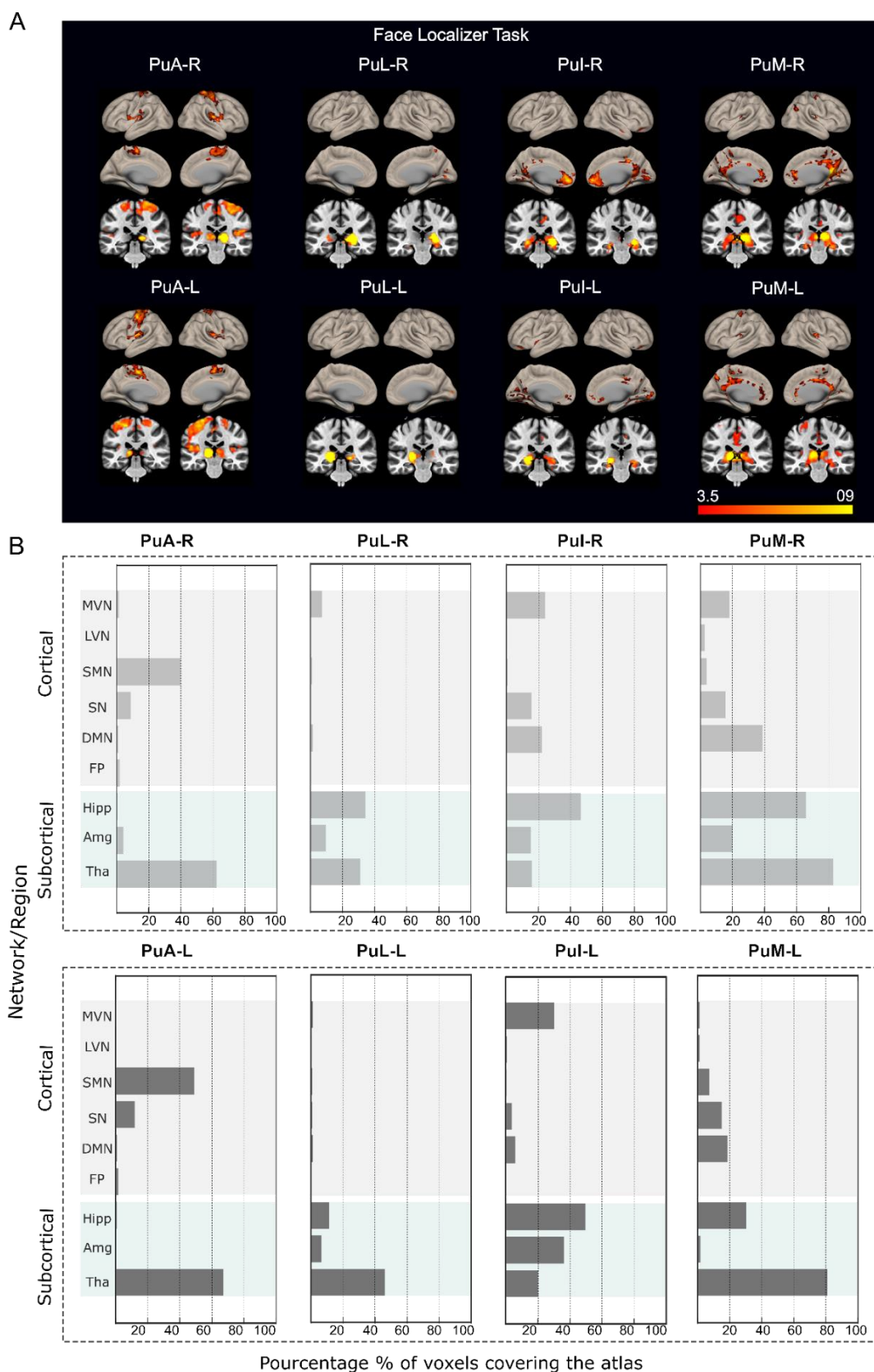


Figure S1. Pulvinar functional connectivity with major brain regions during Face localizer task. (A) Whole brain functional connectivity maps of right (top) and left (bottom) PuA, PuL, PuL and PuM, projected on lateral and medial views of left and right brain hemispheres. (B) Percentage of voxels connected to right (top) and left (bottom) PuA, MuL, PuL and PuM for major cortical and subcortical networks or regions.

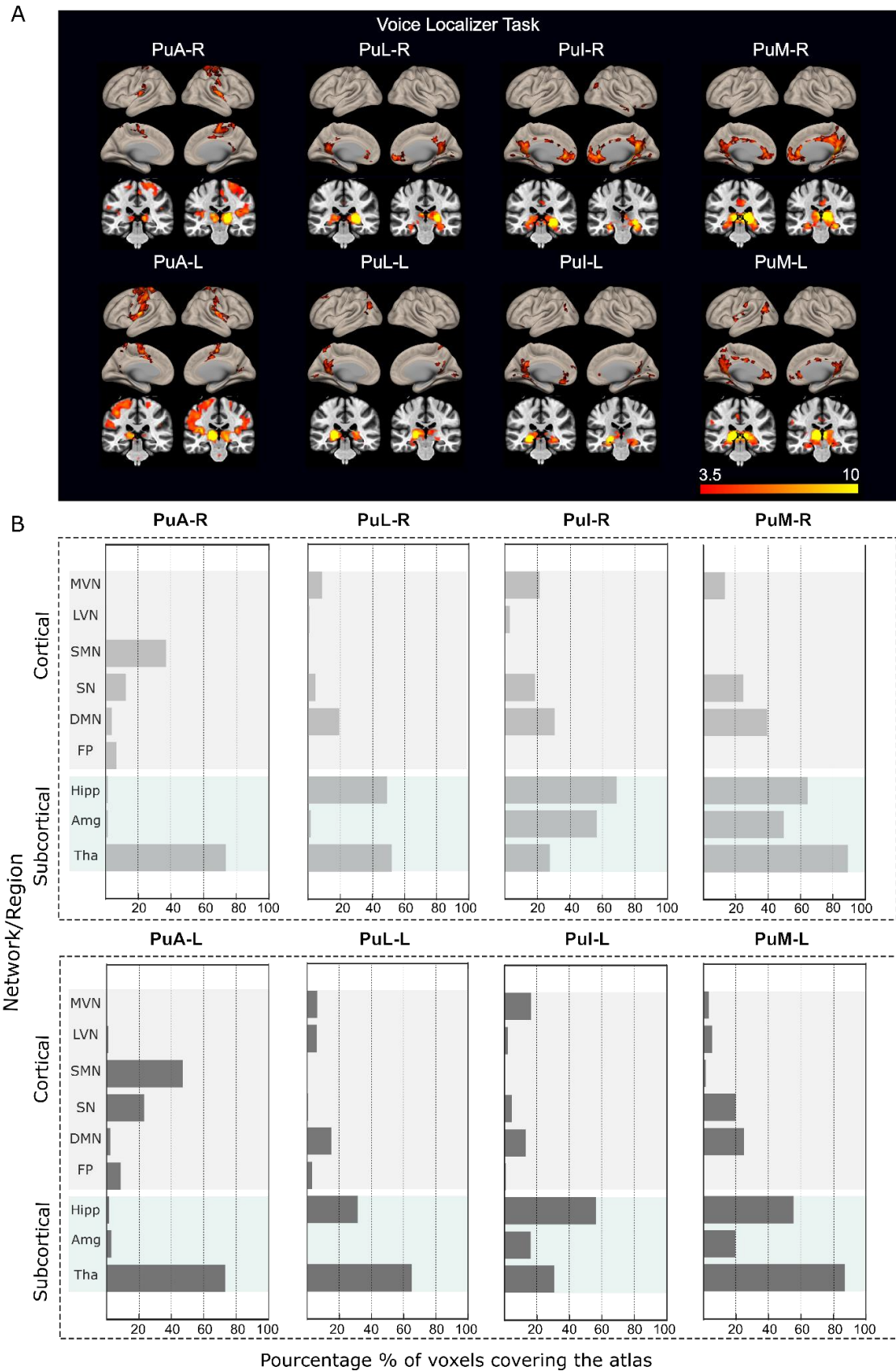


Figure S2. Pulvinar functional connectivity with major brain regions during Voice localizer task. (A) Whole brain functional connectivity maps of right (top) and left (bottom) PuA, PuL, PuL and PuM, projected on lateral and medial views of left and right brain hemispheres. (B) Percentage of voxels connected to right (top) and left (bottom) PuA, MuL, PuL and PuM for major cortical and subcortical networks or regions.

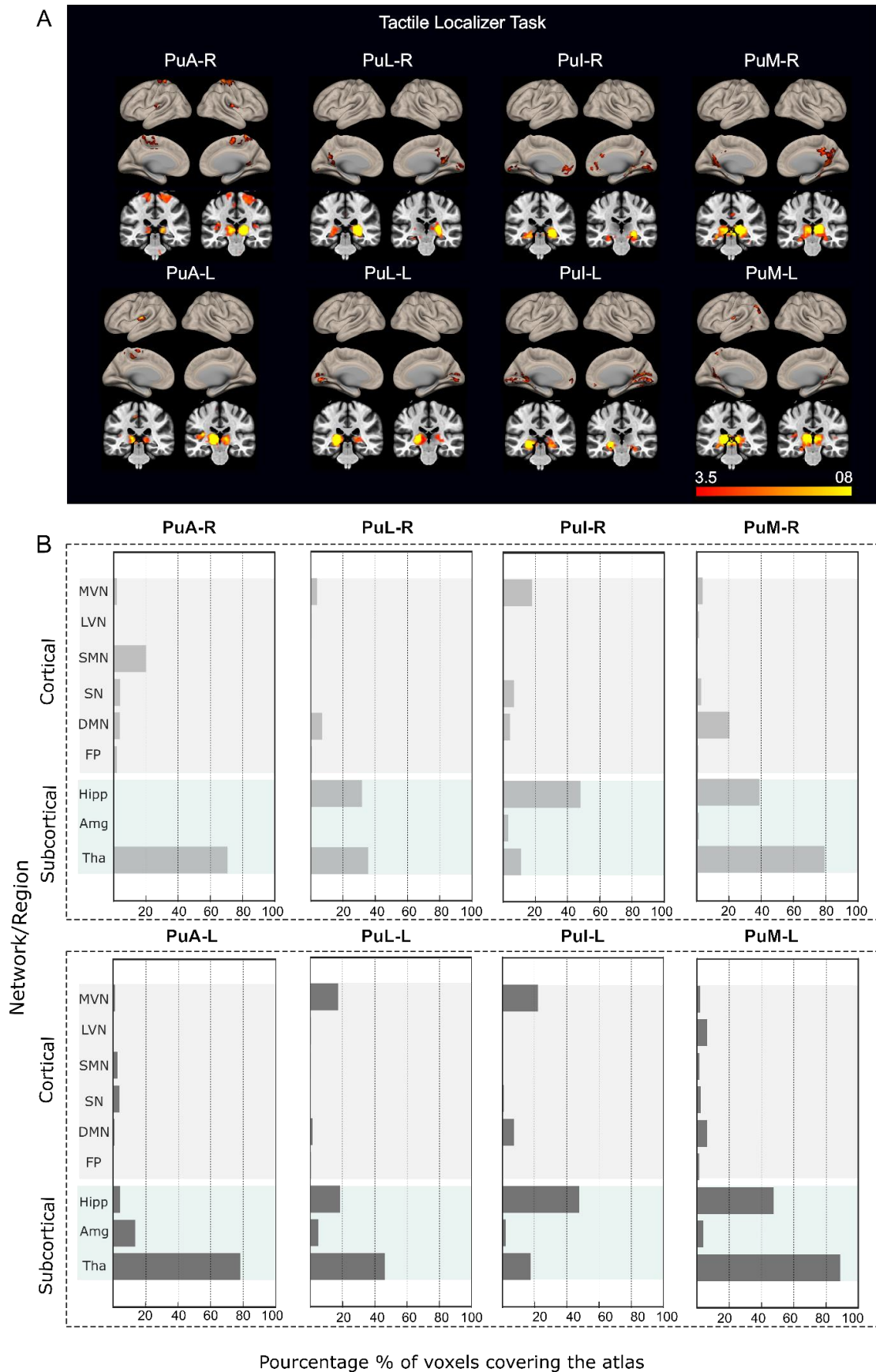


Figure S3. Pulvinar functional connectivity with major brain regions during Tactile localizer task. (A) Whole brain functional connectivity maps of right (top) and left (bottom) PuA, PuL, PuI and PuM, projected on lateral and medial views of left and right brain hemispheres. (B) Percentage of voxels connected to right (top) and left (bottom) PuA, MuL, PuI and PuM for major cortical and subcortical networks or regions.

	<i>PuA</i>			<i>PuL</i>			<i>PuI</i>			<i>PuM</i>		
	ROI	% Voxels		ROI	% Voxels		ROI	% Voxels		ROI	% Voxels	
		Right	Left		Right	Left		Right	Left		Right	Left
<i>O v e r l a p p i n g 2 c o n d i t i o</i>	MVN	1.6		MVN	46.28	27.9	MVN	42.8	47.8	MVN	39.6	11.5
	LVN	-		LVN	1.5	-	LVN	-	-	LVN	12	20
	SMN	41.8	48.1	SMN	-	-	SMN	-	-	SMN	3.4	6.2
	SN	11.5	19.60	SN	-	-	SN	11	1	SN	37.8	40.5
	DMN	5.9	4	DMN	20.13	2.9	DMN	23.2	6.7	DMN	67.4	54.7
	FPN	3.1	3.5	FPN	-	-	FPN	-	-	FPN	2.1	3.4
	Amygdala	14.9	20.8	Amygdala	6.4		Amygdala	20.1	5.2	Amygdala	64	48.6
	Hippocampus	6.1	1.5	Hippocampus	43.8	11.3	Hippocampus	60.3	42.3	Hippocampus	85.8	66
	Thalamus	76.1	83.5	Thalamus	55.6	58.8	Thalamus	24.5	29	Thalamus	96.5	96.6

n s O v e r l a p p i n g 3 c o n d i t i o												
	MVN	-	-	MVN	13.9	9	MVN	10	12.2	MVN	16.2	3.9
	LVN	-	-	LVN	-	-	LVN	-	-	LVN	2.3	4.3
	SMN	20.5	8.5	SMN	-	-	SMN	-	-	SMN	-	-
	SN	3.5	7.1	SN	-	-	SN	5.6	-	SN	20	16.3
	DMN	-	-	DMN	8.5	-	DMN	9.3	1.2	DMN	45.9	26.4
	FPN	1.4	-	FPN	-	-	FPN	-	-	FPN	-	-
	Amygdala	-	1.5	Amygdala	-	-	Amygdala	5.8	-	Amygdala	25.4	19.8
	Hippocampus	-	-	Hippocampus	30	5.2	Hippocampus	48.1	26	Hippocampus	71.2	53.35
	Thalamus	63.3	66.8	Thalamus	28.9	28.5	Thalamus	15.6	15.4	Thalamus	91.4	91.8

n s O v e r l a p p i n g c o n d i t i o												
	MVN	-	-	MVN	-	-	MVN	2	1.3	MVN	6.6	-
	LVN	-	-	LVN	-	-	LVN	-	-	LVN	21	-
	SMN	6.8	-	SMN	-	-	SMN	-	-	SMN	-	-
	SN	1.3	2.3	SN	-	-	SN	2.4	-	SN	4.1	2.6
	DMN	-	-	DMN	-	-	DMN	1.5	-	DMN	26.4	5.3
	FPN		-	FPN	-	-	FPN	-	-	FPN	-	-
	Amygdala	-	-	Amygdala	-	-	Amygdala	-	-	Amygdala	3.2	-
	Hippocampus	-	-	Hippocampus	17.3	2.3	Hippocampus	35.3	16.6	Hippocampus	46.7	37.8
	Thalamus	52.1	56.1	Thalamus	16.7	14.5	Thalamus	8.9	7.4	Thalamus	80	84.2

Table S1. Percentage of voxels functionally connected to PuA, PuL, Pul and PuM, in two conditions, three conditions or four conditions, per region of interest (ROI).

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Table S2. Percentage of voxels functionally connected to PuA, PuL, Pul and PuM, in two conditions, three conditions or four conditions, per region of interest (ROI).

<i>PuA-R</i>			<i>PuL-R</i>			<i>PuI-R</i>			<i>PuM-R</i>		
<i>ROI</i>	F(24) = 6.45, p<0.001		<i>ROI</i>	F(24) = 15.25, p < 0.001		<i>ROI</i>	F(24) = 9.98, p < 0.001		<i>ROI</i>	F(24) = 5.94, p < 0.001	
	Tasks	P		Tasks	P		Tasks	P		Tasks	P
<i>SMN</i>	Tactile vs. Voice	6,20E-08	<i>MVN</i>	Voice vs. Tactile	1,02E-20	<i>MVN</i>	Voice vs. Resting	2,39E-01	<i>MVN</i>	Tactile vs. Voice	8,07E-18
	Tactile vs. face	2,18E-05		Tactile vs. Face	7,09E-08	<i>SN</i>	Voice vs. Resting	1,21E-02		Tactile vs. Face	3,43E-12
<i>SN</i>	Voice vs. Tactile	1,53E-13		Tactile vs. Resting	2,11E-11	<i>DMN</i>	Voice vs. Tactile	5,30E-14		Tactile vs. Resting	5,34E-09
	Tactile vs. Face	1,52E-12	<i>LVN</i>	Voice vs. Resting	6,77E-07		Voice vs. Resting	2,49E-02	<i>LVN</i>	Voice vs. Resting	8,47E-05
	Tactile vs. Resting	1,40E-05		Tactile vs. Resting	6,77E-07		Tactile vs. Face	6,14E-14		Tactile vs. Resting	8,47E-05
<i>DMN</i>	Voice vs. Resting	8,02E-03		Face vs. Resting	6,77E-07		Tactile vs. Resting	5,64E-01		Face vs. Resting	8,47E-05
	Face vs. Resting	8,02E-03	<i>DMN</i>	Voice vs. Face	2,81E-13		Face vs. Resting	2,72E-02	<i>SN</i>	Tactile vs. Voice	1,06E-10
<i>FPN</i>	Voice vs. Tactile	7,01E-11		Tactile vs. Face	1,02E-07	<i>Hippo</i>	Voice vs. Resting	1,58E-32		Tactile vs. Face	1,16E-07
	Voice vs. Face	7,01E-11		Face vs. Resting	6,52E-04		Tactile vs. Resting	8,83E-44		Tactile vs. Resting	1,57E-14
	Voice vs. Resting	2,29E-02	<i>Hippo</i>	Voice vs. Resting	3,54E-29		Face vs. Resting	6,11E-48	<i>FPN</i>	Voice vs. Resting	1,10E-05
<i>Thalamus</i>	Voice vs. Tactile	1,70E-12		Tactile vs. Resting	2,15E-44	<i>Amygdala</i>	Voice vs. Tactile	2,77E-08		Face vs. Resting	1,10E-05
	Voice vs. Resting	5,68E-28		Face vs. Resting	1,55E-33		Voice vs. Resting	2,77E-08		Tactile vs. Resting	1,10E-05
	Tactile vs. Face	9,90E-15	<i>Amygdala</i>	Voice vs. Face	1,96E-04		Tactile vs. Face	1,89E-03	<i>Hippo</i>	Voice vs. Resting	3,50E-07
	Tactile vs. Resting	2,37E-02		Tactile vs. Face	1,96E-04		Face vs. Resting	1,89E-03		Tactile vs. Resting	6,95E-08

	Face vs. Resting	7.46E-31		Face vs. Resting	1,96E-04		Voice vs. Resting	3,26E-03		Face vs. Resting	1,42E-08
-	-	-	Thalamus	Voice vs. Resting	3,62E-12	Thalamus	Tactile vs. Resting	6,66E-09	Amygdala	Voice vs. Tactile	3,01E-03
-	-	-		Tactile vs. Resting	1,47E-17		Face vs. Resting	4,60E-05		-	-
-	-	-		Face vs. Resting	2,34E-17		-	-	-	-	-

<i>PuA-L</i>			<i>PuL-L</i>			<i>PuI-L</i>			<i>PuM-L</i>		
<i>ROI</i>	F(24) = 6.45, p<0.001		<i>ROI</i>	F(24) = 15.25, p < 0.001		<i>ROI</i>	F(24) = 9.98, p < 0.001		<i>ROI</i>	F(24) = 5.94, p < 0.001	
	Tasks	p		Tasks	p		Tasks	p		Tasks	p
<i>SMN</i>	Voice vs. Tactile	3,82E-25	<i>MVN</i>	Voice vs. Tactile	6,10E-13	<i>DMN</i>	Voice vs. Tactile	2,60E-03	<i>MVN</i>	Voice vs. Resting	3,60E-06
	Face vs. Tactile	1,19E-32		Tactile vs. Face	5,81E-08		Voice vs. Resting	2,76E-02		Tactile vs. Resting	3,60E-06
	Resting vs. Tactile	1,17E-16		Face vs. Resting	2,44E-07		Tactile vs. Face	4,15E-01		Face vs. Resting	3,60E-06
	Face vs. Resting	4,30E-02	<i>LVN</i>	Voice vs. Tactile	1,79E-04	<i>Hippo</i>	Voice vs. Resting	8,64E-34	<i>LVN</i>	Voice vs. Face	2,68E-07
<i>SN</i>	Voice vs. Tactile	1,95E-28		Tactile vs. Face	1,79E-04		Tactile vs. Resting	6,50E-49		Tactile vs. Face	1,47E-02
	Voice vs. Resting	3,69E-03		Tactile vs. Resting	6,89E-03		Face vs. Resting	1,82E-37		Face vs. Resting	4,41E-05
	Tactile vs. Face	6,06E-23		Face vs. Resting	6,89E-03	<i>Amygdala</i>	Voice vs. Tactile	1,30E-05	<i>SMN</i>	Voice vs. Face	1,07E-05
<i>DMN</i>	Tactile vs. Resting	1,60E-11	<i>DMN</i>	Voice vs. Tactile	4,89E-05		Voice vs. Resting	1,30E-05	<i>SN</i>	Tactile vs. Voice	5,23E-14
	Voice vs. Resting	3,76E-03		Voice vs. Face	4,89E-05		Tactile vs. Face	1,45E-08		Tactile vs. Face	7,65E-08
	Tactile vs. Resting	3,76E-03	<i>Hippo</i>	Voice vs. Face	3,20E-02		Face vs. Resting	1,45E-08		Tactile vs. Resting	3,62E-11
<i>Amygdala</i>	Voice vs. Tactile	2,51E-03		Voice vs. Resting	9,45E-43	<i>Thalamus</i>	Voice vs. Resting	6,87E-11	<i>DMN</i>	Tactile vs. Voice	1,91E-06
	Tactile vs. Face	2,51E-03		Tactile vs. Resting	1,51E-55		Tactile vs. Resting	6,71E-13		Tactile vs. Face	4,15E-05
<i>Thalamus</i>	Voice vs. Resting	5,41E-41	<i>Amygdala</i>	Face vs. Resting	1,81E-62		Face vs. Resting	7,55E-18		Tactile vs. Resting	8,59E-06
	Tactile vs. Resting	2,31E-44		Voice vs. Face	2,19E-03	-	-	-	<i>Hippo</i>	Voice vs. Resting	5,47E-12
				Tactile vs. Face	2,19E-03	-	-	-		Tactile vs. Resting	1,14E-12

Table S3. Percentage of voxels functionally connected to PuA, PuL, PuI and PuM, in two conditions, three conditions or four conditions, per region of interest (ROI).

			<i>Thalamus</i>						<i>Amygdala</i>		
-	-	-		Tactile vs. Resting	2,05E-14	-	-	-		Voice vs. Face	3,89E-03
-	-	-		Face vs. Resting	8,03E-14	-	-	-	<i>Thalamus</i>	Face vs. Resting	2,09E-04

	ALD			ALV			AMD			AMV		
	ROI	% Voxels		ROI	% Voxels		ROI	% Voxels		ROI	% Voxels	
		Right	Left		Right	Left		Right	Left		Right	Left
Overview	MVN	22.9	6.9	MVN	66	46.8	MVN	5.1	2.9	MVN	5	2.1
	LVN	11.5	17.3	LVN	6	9.7	LVN	8.9	11	LVN	-	-
	SMN	3.6	15.7	SMN	1.8	8.6	SMN	1	3.8	SMN	13.2	24.8
	SN	5.7	27.6	SN	3	8.4	SN	30.8	35.1	SN	23.4	27.6
	DMN	63.4	56.5	DMN	30.7	17.2	DMN	58	51	DMN	14	8
	FPN	2.5	11.6	FPN	-	4.1	FPN	4.8	5.9	FPN	-	-
	Amygdala	-	-	Amygdala	12.5	6.1	Amygdala	7	-	Amygdala	55.2	70
	Hippocampus	10.1	7.5	Hippocampus	32.2	16	Hippocampus	28.8	34.7	Hippocampus	54.4	48.6
	Thalamus	91.6	95.1	Thalamus	76	77.3	Thalamus	97.7	96.4	Thalamus	90.9	87.3

n s O v e r l a p p i n g 3 c o n d i t i o												
	MVN	5.4	1.1	MVN	38.1	9.6	MVN	-	-	MVN	-	-
	LVN	1	2.8	LVN	-	-	LVN	-	1.7	LVN	-	-
	SMN	-	3.2	SMN	-	1.1	SMN	-	1	SMN	2.9	8.8
	SN	-	5.1	SN	-	1.6	SN	11.6	10.8	SN	6.2	7.7
	DMN	35	16.7	DMN	12.9	2.9	DMN	33	24.2	DMN	4.7	1.8
	FPN	-	1.8	FPN	-	-	FPN	-		FPN	-	-
	Amygdala	-	-	Amygdala	2.3	-	Amygdala	3.2	-	Amygdala	6.7	21.7
	Hippocampus	3.5	-	Hippocampus	17	11.5	Hippocampus	14	13.5	Hippocampus	28	21.2
	Thalamus	81.7	89	Thalamus	65.7	68	Thalamus	93.4	92.2	Thalamus	81	81.3

n s O v e r l a p p i n g c o n d i t i o												
	MVN	-	-	MVN	12.3	-	MVN	-	-	MVN	-	-
	LVN	-	-	LVN	-	-	LVN	-	-	LVN	-	-
	SMN	-	-	SMN	-	-	SMN	-	-	SMN	-	1.3
	SN	-	1	SN	-	-	SN	2.2	-	SN	1.3	1.3
	DMN	6.7	1.7	DMN	2.6	-	DMN	14.5	3.4	DMN	-	-
	FPN	-	-	FPN	-	-	FPN	-	-	FPN	-	-
	Amygdala	-	-	Amygdala	-	-	Amygdala	-	-	Amygdala	-	
	Hippocampus	-	-	Hippocampus	12.3	7.8	Hippocampus	3.7	4.2	Hippocampus	5.7	5
	Thalamus	70	76	Thalamus	55.8	53.4	Thalamus	84.4	84.1	Thalamus	69.4	66.2

Table S4. Percentage of voxels functionally connected to ALD, ALV, AMD and AMV, in two conditions, three conditions or four conditions, per region of interest (ROI).

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Table S5. Percentage of voxels functionally connected to PLD, PLV, PMD and PMV, in two conditions, three conditions or four conditions, per region of interest (ROI).

	<i>PLD</i>			<i>PLV</i>			<i>PMD</i>			<i>PMV</i>		
	ROI	% Voxels		ROI	% Voxels		ROI	% Voxels		ROI	% Voxels	
		Right	Left		Right	Left		Right	Left		Right	Left
<i>O v e r l a p p i n g 2 c o n d i t i o</i>	MVN	8.4	1.6	MVN	44.6	39	MVN	2.6	1.1	MVN	25	7.9
	LVN	15	19	LVN	22.9	29.5	LVN	-	4.6	LVN	2.4	-
	SMN	-	1.7	SMN	-	1.7	SMN	-	1.5	SMN	-	-
	SN	8.5	4.8	SN	34	16.4	SN	12.2	11.2	SN	32	16.7
	DMN	51	55.5	DMN	54.3	37	DMN	12.5	19.5	DMN	23.1	5.5
	FPN	2.2	6.5	FPN	-	-	FPN	-		FPN	-	-
	Amygdala	-	-	Amygdala	67.2	23.8	Amygdala	14.3	6.7	Amygdala	87.1	82.8
	Hippocampus	60.1	38	Hippocampus	91.8	70.1	Hippocampus	80	80.6	Hippocampus	92.1	80
	Thalamus	67.1	85	Thalamus	55.4	70	Thalamus	74.6	77	Thalamus	77.1	81.2

n s O v e r l a p p i n g 3 c o n d i t i o												
	MVN	2		MVN	21.5	8.3	MVN	-	-	MVN	13.2	2.1
	LVN	3.3	4.3	LVN	2.4	12.8	LVN	-	-	LVN	-	-
	SMN	-	-	SMN	-	-	SMN	-	-	SMN	-	-
	SN	2.6	-	SN	15	4.1	SN	4.4	2.5	SN	10	4
	DMN	26.5	20.2	DMN	38.8	15.1	DMN		5.9	DMN	12.5	1.5
	FPN	-	1.2	FPN	-	-	FPN	-	-	FPN	-	-
	Amygdala	-	-	Amygdala	31.8	3.6	Amygdala	4.6	-	Amygdala	47	48.6
	Hippocampus	47.4	22.3	Hippocampus	76	58.8	Hippocampus	58.4	48	Hippocampus	81	55.8
	Thalamus	55.9	74.1	Thalamus	45.4	55.6	Thalamus	57.9	67.8	Thalamus	65.2	71.2

n s O v e r l a p p i n g c o n d i t i o												
	MVN	-	-	MVN	6.2	1.3	MVN	-	-	MVN	5.1	-
	LVN	-	-	LVN	-	1.3	LVN	-	-	LVN	-	-
	SMN	-	-	SMN	-	-	SMN	-	-	SMN	-	-
	SN	-	-	SN	5.1	-	SN	-	-	SN	-	-
	DMN	10.8	-	DMN	24	2.3	DMN	1.1	-	DMN	5.8	-
	FPN	-	-	FPN	-	-	FPN	-	-	FPN	-	-
	Amygdala	-	-	Amygdala	4.6	-	Amygdala	-	-	Amygdala	16	13.1
	Hippocampus	33.5	6.4	Hippocampus	48.5	48.5	Hippocampus	31.8	25.7	Hippocampus	38	27
	Thalamus	46.7	57.5	Thalamus	34.8	44.4	Thalamus	45	53.1	Thalamus	49.6	63

<i>n</i>												
<i>s</i>												

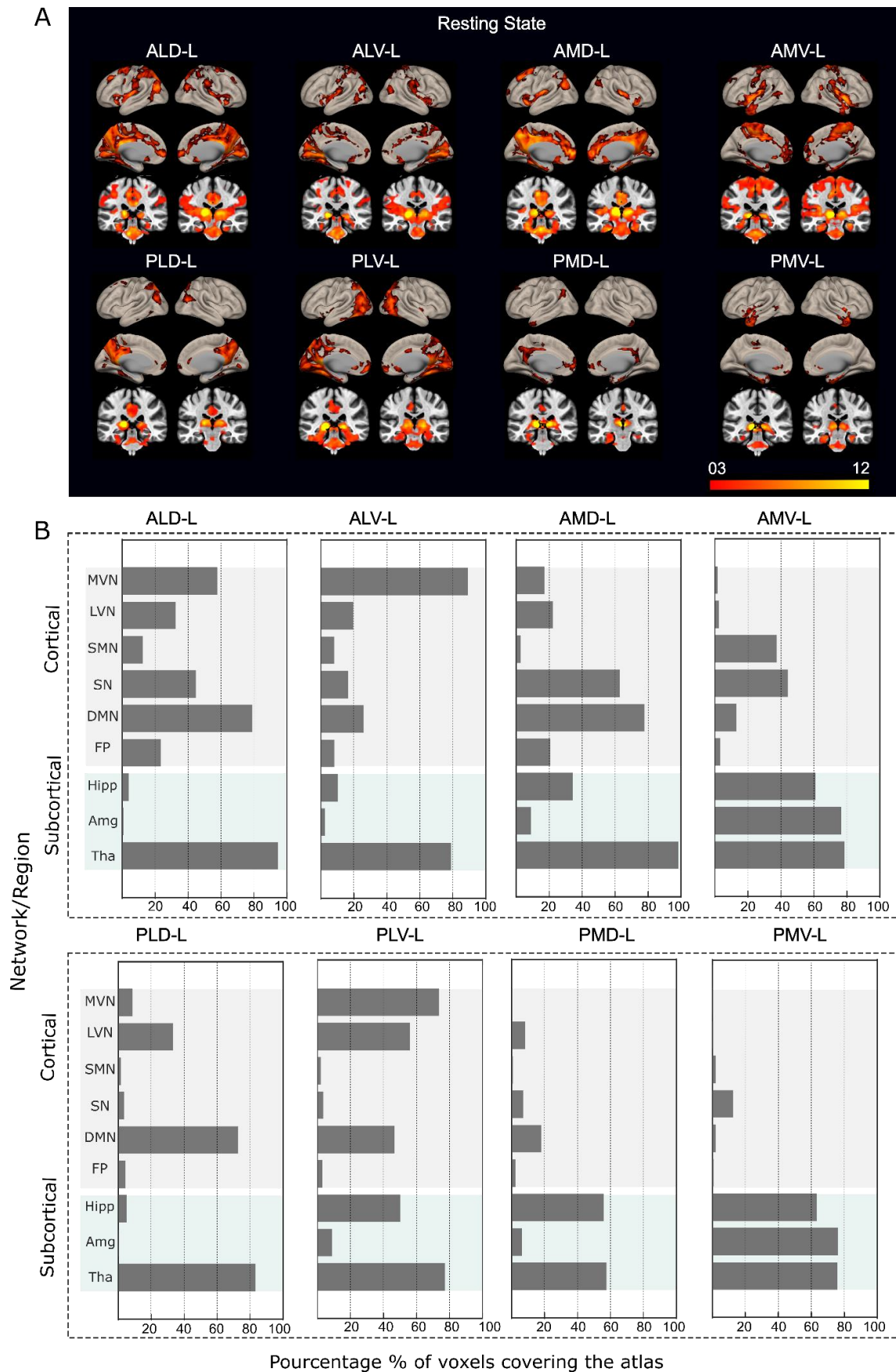


Table S4. Left medial pulvinar functional connectivity with major brain regions during resting state. (A) Whole brain functional connectivity maps of the 8 anatomical subdivisions of left PuM, projected on lateral and medial views of left and right brain hemispheres. **(B)** Percentage of voxels connected to these 8 anatomical subdivisions of left PuM for major cortical and subcortical networks or regions.

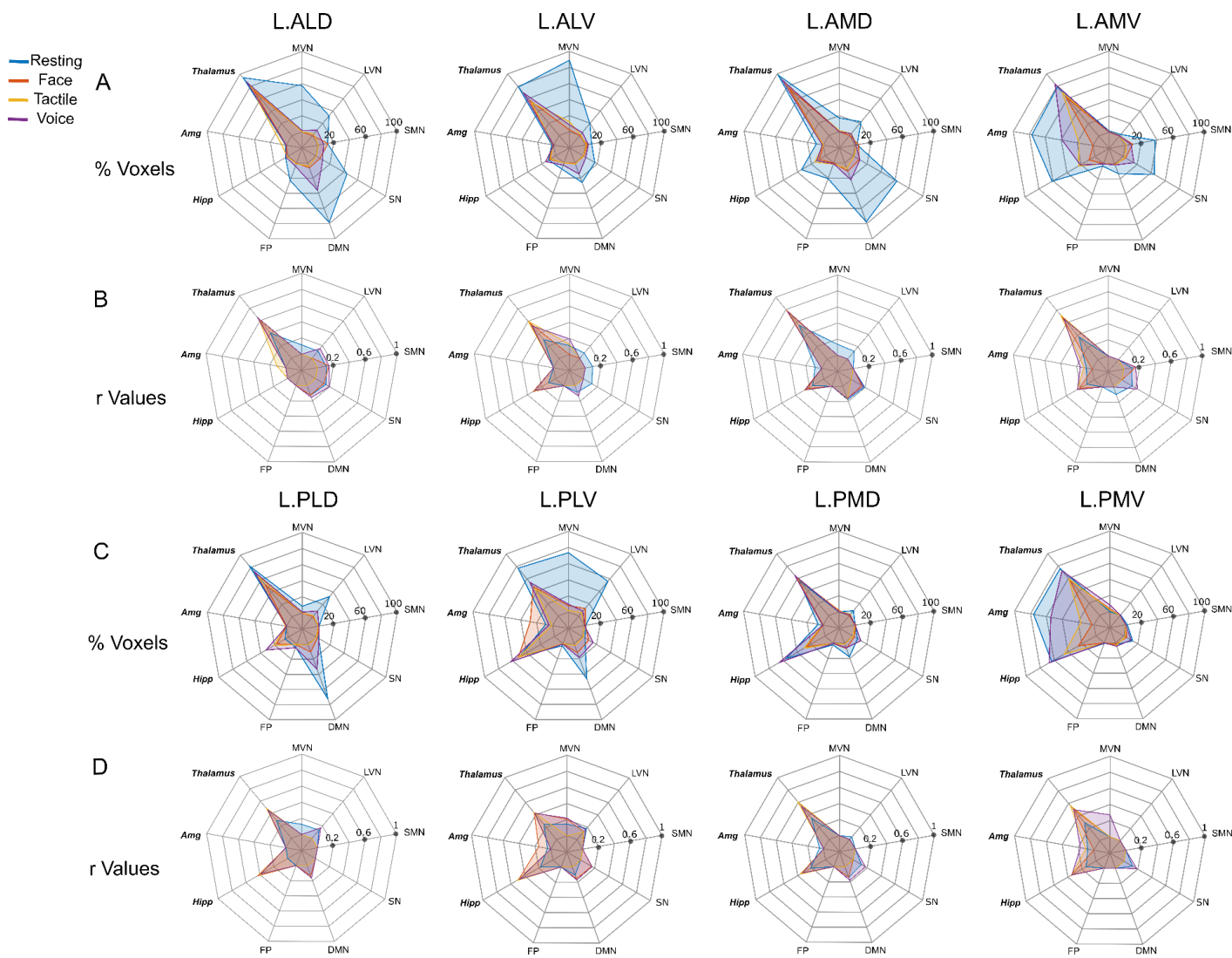


Figure S5. Left medial pulvinar functional connectivity with major brain regions across resting state and localizer tasks. Changes in % of significantly correlated voxels (A and C) pulvinar ROIs; Changes in r-scores with major brain regions for right (B and D) left PuM ROIs.

Study 3 : Role of human pulvinar in visuo-tactile context

The pulvinar has been shown to play a role in multiple cognitive functions including spatial attention, visual saliency and shows multisensory integration properties. For example, it is implicated in selecting and gating salient stimuli (Bourgeois et al., 2020b; Robinson and Petersen, 1992). It is also implicated in detection, discrimination and selection processes of attention (Fiebelkorn et al., 2019a; Pestilli et al., 2011). Both salience and attention functions are carried out by functional networks identified in the literature (Mengotti et al., 2020; Schimmelpfennig et al., 2023). Additionally, the pulvinar has shown multisensory integration properties in both electrophysiological and fMRI recordings in monkeys in an audio-visual social context (Froesel et al., 2024c; Vittek et al., 2024, 2023). Visual saliency, spatial attention and multisensory integration enhances stimuli detection, discrimination and selection. These brain functions are activated under specific contextual conditions and can exhibit interactions (Bourgeois et al., 2020a; Fiebelkorn et al., 2019a; Murray and Wallace, 2012b; Pestilli et al., 2011; Robinson and Petersen, 1992). Previous studies in this thesis highlighted the implication of the pulvinar in unisensory visual, auditory and tactile processing for low demanding tasks. We reported fMRI activation for visual and tactile processing and described functional connectivity of pulvinar subdivisions during visual, auditory and tactile processing. Our results for pulvinar functional connectivity supported the hypothesis of a pulvinar as a cortical modulator. We hypothesize that low demanding tasks lead to low or no recruitment of pulvinar whereas higher demanding tasks will recruit pulvinar along with cortical functional networks. Pulvinar implications in both salience and attention has been studied independently. We hypothesize that interactions exists between pulvinar, salience and attention networks such that specific activations patterns exist for specific contexts depending on saliency and multimodality properties of the stimulus. For example, a difficult trial is expected to recruit several cortical networks as well as pulvinar to perform the task. .In this third study, we investigated the role of pulvinar in multisensory processing for visuo-tactile context of face detection and discrimination in a human fMRI task.

Role of human pulvinar in visuo-tactile context

M. Gaudet-Trafit*¹, M. Froesel*¹, S. Dali*¹, F. Lamberton² and S. Ben Hamed*¹

(work in progress, all aspects of the work are susceptible to changes including author order and contribution, with agreement of all contributors)

1 Institut des Sciences Cognitives Marc Jeannerod, CNRS, Lyon 69500 bron, France

2 CERMEP-imagerie du vivant, 59 Boulevard Pinel, 69677 Bron Cedex, France

Keywords : pulvinar, multisensory processing, fMRI, human

Abstract

Multi-sensory processing is crucial to build representations of our surrounding environment by combining inputs from different sensory modalities. This process improves detection and discrimination of external stimuli and allows to associate or segregate information from common or different sensory sources. The contribution of cortical regions to multisensory integration has been well studied, however, the specific contribution of subcortical structures to multisensory integration is still under-investigated. In particular, the pulvinar, the largest nucleus of the thalamus, has been proposed to act as a functional hub of cortical processes due to its large reciprocal connectivity with the cortex, especially during salient stimuli processing. Despite The extensive knowledge about its reciprocal connection with the cortex, understanding its precise involvement in multisensory processing is still an unreached goal. In this study, we use functional magnetic resonance imaging (fMRI) to characterize the role of the pulvinar in processing and integrating bimodal visual-tactile information. For this purpose, 25 participants participated in this study and were asked to detect an outlier face among similar faces in the presence or absence of spatially congruent or incongruent somatosensory stimulations (a tactile air puff on the left or right cheek). The target face could either be in color (high salience) or black and white (low salience). Display of faces was brief and participants had to keep fixating a central point without making saccades. Overall, our results show behaviorally, significant effects of face saliency and visuo-tactile congruency on reaction times. Reaction times were shorter in salient and congruent conditions compared to non-salient and incongruent conditions. At the neural level, we show that pulvinar activations were modulated by the spatial congruency of the tactile stimuli with respect to the location of the salient face. We also described, a large cortio-subcortical network including several areas from salience and attentional networks that co-activated with the pulvinar. These findings support the role of the pulvinar in multisensory integration during the gating of salient and congruent information.

Introduction

Our brain perceives the external world through multiple senses. This multisensory processing improves speed and accuracy of detecting and distinguishing objects, enhancing overall perception (Stein and Stanford, 2008). For example, when crossing a road and noticing an approaching car, if we also hear the car arriving, information from both visual and auditory modalities aligns allowing to quickly assess the danger and move out of harm's way. Multisensory processing relies on two main concepts, the convergence of sensory input into the same single neuron or a population of neurons within a functional area. Subsequent integration of information results in a greater or smaller output response in multisensory context than the sum of the unisensory inputs independently (Murray and Wallace, 2012a; Stein and Stanford, 2008; Stevenson et al., 2014). This non-linearity of integration response can be assessed with single neurons recording or fMRI (Murray and Wallace, 2012a; Stanford and Stein, 2007; Stevenson et al., 2014). Multisensory processing has been extensively studied in the cortical domain and some subcortical structures like superior colliculus (Stein and Stanford, 2008). For instance, in a study in non-human primate using fMRI to investigate visuo-tactile looming effect (Cléry et al., 2017), authors showed a super-additive effect on bold signal for specific areas in visual, parietal and frontal cortices. More evidence suggests that the pulvinar, the largest nucleus of the thalamus is implicated in multisensory processing (Cappe et al., 2009b; Froesel et al., 2021a; Vittek et al., 2023) as it is reciprocally connected to a large part of the cortex and other subcortical structures (Cappe et al., 2009b; Froesel et al., 2021a). Furthermore, a recent study showed the ability of pulvinar to perform multisensory integration in an audio-visual task in non-human primates (NHP) (Vittek et al., 2023). The authors highlighted neurons with suppressive and subadditive activity within the medial part of the pulvinar. Functionally, the pulvinar has been demonstrated to be involved in a multitude of cognitive processes such as visual object categorization, visual attention, visual salience, memory, emotion regulation, social processing, multisensory processing (Benarroch, 2015a; Fiebelkorn et al., 2019a; Guedj and Vuilleumier, 2023a; Saalman et al., 2011; Wen et al., 2023). Among visual object categorization, pulvinar has been identified to play a role in face detection (Arcaro et al., 2018a; Benarroch, 2015a; Wen et al., 2023). The wide anatomical architecture of the pulvinar as well as its functional contributions to several cognitive and sensory domains positioning it as a major cortical modulator (Arcaro et al., 2018a; Benarroch, 2015a; Fiebelkorn et al., 2019a; Froesel et al., 2021a; Greenwood and Ward, 2024; Guedj and Vuilleumier, 2023a; Yuri B. Saalman and Kastner, 2011). The multisensory properties of pulvinar have been investigated in NHP electrophysiological recording (Vittek et al., 2024, 2023) and fMRI (Froesel et al., 2024b), mainly on audio-visual contexts, and nearly not investigating in human and in visuo-tactile context (Froesel et al., 2021a).

In this study, we aim to explore the role of pulvinar in face detection, both during salient and non-salient conditions, in multisensory context. Specifically, we examine how tactile stimuli, when spatially aligned (congruent) or misaligned (incongruent) with the target face, influence this detection process, and how the pulvinar contributes to integrating visual and tactile information to enhance saliency

perception. Our results showed implication of pulvinar in saliency and visuo-tactile multisensory processing during a face detection task.

Material and Method

Participants and ethical approval

Twenty-five subjects aged from 20 to 33 years (12 males and 13 females) with mean age of 28 years old (SD = 3.8) participated to the study. All subjects were healthy and had no history of psychiatric or neurological disorders and had normal or corrected-to-normal vision. Edinburgh Handedness Inventory laterality test (doi: <https://doi.org/10.1016%2F0028-3932%2871%2990067-4>) is used to determine a lateralization score for each participant (17 right-handed, 1 left-handed and 7 ambidextrous). The present study was approved by the French ethical comity CPP sud-Med IV (Comité de Protection des Personnes Sud-Méditerranée IV - 19.00758.190401-MS02 – 2018-A03438-47).

Experimental setup

During MRI sessions, participants were lying on their back in a supine position. A mirror was positioned above their head and fixated to the head coil. A plastic translucent screen of 30 cm of radius was set at 1 meter from the eyes/mirrors. Visual stimuli were retro-projected onto this screen. Tactile stimulation was delivered by air pressure at 2.3 bars (air puffs) through plastic tubes rigidly maintained close to the target body locations. This level of pressure induced in all participants a clear and unambiguous perception of tactile stimulation on the target area. Air pressure was created by a compressor (*SIP Industrial*, SIP QT 50/10 Low Noise Direct Drive Compressor) and adjusted by pressure regulators placed outside of the scanner room. Eye position (x and y for right or left eye) was recorded using an eye tracking system (Eyelink 1000, SR Research™) and sampled at 1000Hz. Stimulus control for all modalities as well as online eye position control was achieved using Presentation software (Neurobehavioral systems™) coupled with an I/O control Arduino-based setup.

Visual Stimuli

The experiment's face images stimuli were taken from the Karolinska Directed Emotional Faces (KDEF) database's neutral faces ("Lundqvist; Flykt; Öhman," 1998), available on <https://kdef.se/home/aboutKDEF>. This database contains a total of 4900 pictures of human facial expressions from 70 individuals and has been validated by Goeleven et al. (2008). From that database, we selected neutral expression 32 of individuals (16 men, 16 women) both in black and white and colored version. From these 32 individuals 2 pairs (4 individuals) were randomly formed at the beginning of each run and then displayed by pairs during the run such that each pair is used for all the experimental conditions.

Task paradigm

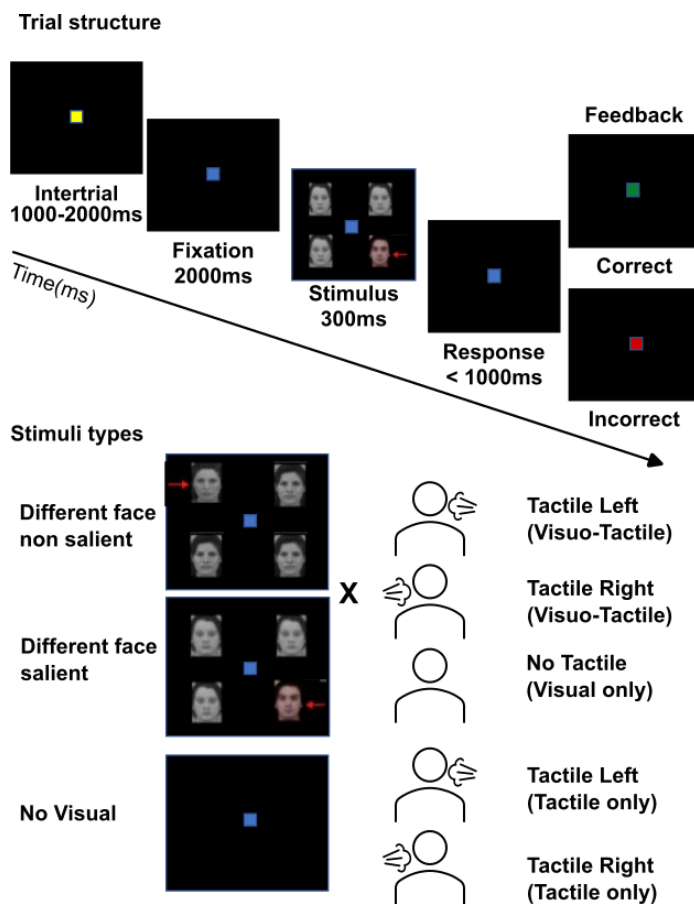


Figure 1. Task Paradigm. Top Panel. Time course of one trial. Bottom panel. Example of visuo-tactile, visual and tactile conditions.

The task was composed of 3 factors, face position, salience and tactile position split in 4 kinds of trials, visuo-tactile (VT), visual only (VO), tactile only (TO) and no stim (CT). The face position factor was associated with 4 conditions (up left, up right, bottom left, bottom right). The salience factor was associated 2 conditions (salient, non-salient). The tactile position factor was associated with 2 conditions (left and right). Each run was composed of 27 conditions repeated twice for a total of 54 trials. The 27 conditions were split as follow: 16 VT (4 positions x 2 salience x 2 tactile), 8 VO (4 positions x 2 salience), 2 TO (2 tactile) and 1 CT. Each trial was composed of a fixation period of 2s, followed by the display of 4 faces on the screen during 300ms (6 degrees from the center), a response window between 300ms and 1000ms from stimulus onset and a feedback period which consisted in the fixation point turning green for correct answer and red for non-correct. Two consecutive trials were separated by an inter-stimuli interval randomly ranging between 1000ms and 2000ms.

During all task duration, participants had to fixate a central fixation point without making saccades towards the images. The fixation point changed color depending on the task period: blue (fixation or response period); green or red (feedback), yellow (inter-stimuli interval). The aim of the participants during the task was to detect and localize the distinct face among four faces. Indeed, among the four

faces displayed on the screen, three of them belonged to the same individual while the fourth face was from a different individual. To answer, subjects had to press the correct key among 4 possible answer keys split in 2 controllers, one on each hand. Thumb and index fingers were used to answer top and bottom target, respectively, and controller was held in a such a way as to have thumb on top and index in bottom relative to participant head. The 3 identical faces were in grayscale color while the “target face”, i.e. the one with a different identity could be either in RGB color (salient condition) or in grayscale color (non-salient condition). In addition of these purely visual trials, multisensory and pure tactile trials could be presented. The multisensory or bimodal trials were built such as visual and tactile stimulations are delivered concomitantly. Tactile stimulation was delivered by one air-puff (2.3 bars) either on the right or the left cheek of the participant. A congruent condition was defined as visual position of the target being the same position as tactile stimulus (example: target face on the right side – air-puff on the right cheek) and incongruent condition was defined as the opposite (example: target face on the right side – air puff on the left cheek). In the TO trials only an air puff stimulation was delivered to the participant which had to retrieve the position (left or right) of this puff using either the upper or the lower keys on the proper controller. Four types of answers were categorized. Correct trials when the correct button was pressed in a range between 300ms and 1000ms after target onset. Wrong trials were produced when incorrect button in pressed within same 300-1000ms range. False alarm trials were produced when a response button was pressed either in the fixation period or within the first 300ms after stimulus onset. Miss trials were produced when no button was pressed or when a button was pressed later than 1000ms after target onset. To analyze behavioral and fMRI results, we defined analysis conditions from the correct trials. Unisensory conditions covered trials with one modality, VO and TO. Multisensory conditions covered trials with visual and tactile stimuli (VT). The salience condition covered VO an VT trials with salient (RGB color) faces. The non-salient condition covered VO and VT trials with non-salient (grayscale color) faces. The congruent condition covered VT trials with face target in the same left-right position as tactile air-puff. The incongruent condition covered VT trials with face target in the opposite left-right position as tactile air-puff.

Image acquisition

MRI data were acquired with a Siemens 3T PRISMA scanner and using a circularly polarized head coil of 64-channels (Siemens Healthineers, Erlangen, Germany). High-resolution anatomical images were acquired: T1-weighted images MPRAGE sequence for the T1-weighted (sagittal incidence, GRAPPA factor=2, FOV=180x224X256 mm, 0.8 mm isotropic voxel size, TR=3.0 s, TE=3.7ms, TI=1.1s, flip angle =8 °). High-resolution 3D-T2-weighted anatomical images were acquired with SPACE sequence (sagittal incidence, GRAPPA factor=3, FOV=180x224X256 mm, 0.8 mm isotropic voxel size, TR=3.0s, TE_{eff}=299 ms). Functional images were acquired with EPI sequence covering whole cortex (50 axial slices, multiband factor=2, FOV=204x204x125, voxel size=1.6x1.6x2.5 mm with phase encoding along AP direction, TR = 2.3s; TE = 36 ms; FA = 80°). To correct for geometrical distortion, a magnetic field map was acquired before the functional runs with Siemens gre_field_mapping sequence (44 axial slices,

FOV=225x225x132, voxel size=2.5x2.5x3.0 mm TR =500ms, TEs = 4.59 and 7.05ms, FA = 60 °). The task is composed by 12 runs of 134 volumes each.

fMRI preprocessing

Results included in this manuscript come from preprocessing performed using fMRIPrep 22.1.1 (Esteban, Markiewicz, et al. (2018); Esteban, Blair, et al. (2018); RRID:SCR_016216), which is based on Nipype 1.8.5 (K. Gorgolewski et al. (2011); K. J. Gorgolewski et al. (2018); RRID:SCR_002502).

Preprocessing of B0 inhomogeneity mappings

A B0 nonuniformity map (or fieldmap) was estimated from the phase-drift map(s) measure with two consecutive GRE (gradient-recalled echo) acquisitions for each of the 5 sessions performed by the subjects. The corresponding phase-map(s) were phase-unwrapped with prelude (FSL 6.0.5.1:57b01774).

Anatomical data preprocessing

For each subject, the T1-weighted (T1w) image was corrected for intensity non-uniformity (INU) with N4BiasFieldCorrection (Tustison et al. 2010), distributed with ANTs 2.3.3 (Avants et al. 2008, RRID:SCR_004757), and used as T1w-reference throughout the workflow. The T1w-reference was then skull-stripped with a Nipype implementation of the antsBrainExtraction.sh workflow (from ANTs), using OASIS30ANTs as target template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using fast (FSL 6.0.5.1:57b01774, RRID:SCR_002823, Zhang, Brady, and Smith 2001). Volume-based spatial normalization to one standard space (MNI152NLin2009cAsym) was performed through nonlinear registration with antsRegistration (ANTs 2.3.3), using brain-extracted versions of both T1w reference and the T1w template. The following template was selected for spatial normalization: ICBM 152 Nonlinear Asymmetrical template version 2009c [Fonov et al. (2009), RRID:SCR_008796; TemplateFlow ID: MNI152NLin2009cAsym].

Functional data preprocessing

For each of the 12 runs performed by the subjects, the following preprocessing was performed. First, a reference volume and its skull-stripped version were generated using a custom methodology of fMRIPrep. Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation parameters) are estimated before any spatiotemporal filtering using mcflirt (FSL 6.0.5.1:57b01774, Jenkinson et al. 2002). BOLD runs were slice-time corrected to 1.09s (0.5 of slice acquisition range 0s-2.19s) using 3dTshift from AFNI (Cox and Hyde 1997, RRID:SCR_005927). The BOLD time-series (including slice-timing correction when applied) were resampled onto their original, native space by applying the transforms to correct for head-motion. These resampled BOLD time-series will be referred to as preprocessed BOLD in original space, or just

preprocessed BOLD. The BOLD reference was then co-registered to the T1w reference using `mri_coreg` (FreeSurfer) followed by `flirt` (FSL 6.0.5.1:57b01774, Jenkinson and Smith 2001) with the boundary-based registration (Greve and Fischl 2009) cost-function. Co-registration was configured with six degrees of freedom. Several confounding time-series were calculated based on the preprocessed BOLD: framewise displacement (FD), DVARS and three region-wise global signals. FD was computed using two formulations following Power (absolute sum of relative motions, Power et al. (2014)) and Jenkinson (relative root mean square displacement between affines, Jenkinson et al. (2002)). FD and DVARS are calculated for each functional run, both using their implementations in Nipype (following the definitions by Power et al. 2014). The three global signals are extracted within the CSF, the WM, and the whole-brain masks. Additionally, a set of physiological regressors were extracted to allow for component-based noise correction (CompCor, Behzadi et al. 2007). Principal components are estimated after high-pass filtering the preprocessed BOLD time-series (using a discrete cosine filter with 128s cut-off) for the two CompCor variants: temporal (tCompCor) and anatomical (aCompCor). tCompCor components are then calculated from the top 2% variable voxels within the brain mask. For aCompCor, three probabilistic masks (CSF, WM and combined CSF+WM) are generated in anatomical space. The implementation differs from that of Behzadi et al. in that instead of eroding the masks by 2 pixels on BOLD space, a mask of pixels that likely contain a volume fraction of GM is subtracted from the aCompCor masks. This mask is obtained by thresholding the corresponding partial volume map at 0.05, and it ensures components are not extracted from voxels containing a minimal fraction of GM. Finally, these masks are resampled into BOLD space and binarized by thresholding at 0.99 (as in the original implementation). Components are also calculated separately within the WM and CSF masks. For each CompCor decomposition, the k components with the largest singular values are retained, such that the retained components' time series are sufficient to explain 50 percent of variance across the nuisance mask (CSF, WM, combined, or temporal). The remaining components are dropped from consideration. The head-motion estimates calculated in the correction step were also placed within the corresponding confounds file. The confound time series derived from head motion estimates and global signals were expanded with the inclusion of temporal derivatives and quadratic terms for each (Satterthwaite et al. 2013). Frames that exceeded a threshold of 0.5 mm FD or 1.5 standardized DVARS were annotated as motion outliers. Additional nuisance time series are calculated by means of principal components analysis of the signal found within a thin band (crown) of voxels around the edge of the brain, as proposed by (Patriat, Reynolds, and Birn 2017). The BOLD time-series were resampled into standard space, generating a preprocessed BOLD run in MNI152NLin2009cAsym space. First, a reference volume and its skull-stripped version were generated using a custom methodology of fMRIPrep. All resamplings can be performed with a single interpolation step by composing all the pertinent transformations (i.e. head-motion transform matrices, susceptibility distortion correction when available, and co-registrations to anatomical and output spaces). Gridded (volumetric) resamplings were performed using `antsApplyTransforms` (ANTs), configured with Lanczos interpolation to minimize the

smoothing effects of other kernels (Lanczos 1964). Non-gridded (surface) resamplings were performed using `mri_vol2surf` (FreeSurfer).

Many internal operations of fMRIPrep use Nilearn 0.9.1 (Abraham et al. 2014, RRID:SCR_001362), mostly within the functional processing workflow. For more details of the pipeline, see the section corresponding to workflows in fMRIPrep's documentation.

Functional MRI GLM analyses

Functional images were analyzed using statistical parametric mapping software (SPM12) (Friston et al., 1994, Wellcome Centre for Human Neuroimaging, University College London, UK; <http://www.fil.ion.ucl.ac.uk/spm>). The first five functional volumes were discarded to allow for T1 equilibrium. At single subject level, a fixed-effect model was created for each individual participant in order to perform random-effect analyses with FWE $p < 0.05$ corrected for multiple comparisons. The motion parameters were included in the General Linear Model (GLM) as regressors to correct for head movement. In order to assess brain activations specific to the experimental conditions, different contrasts were computed. Activations related to unimodal visual and tactile conditions, i.e. visual only and tactile only, were assessed by visual only versus tactile only (VO>TO) and tactile only versus visual only (TO>VO) contrasts, respectively. Activations related to saliency in unimodal visual conditions were assessed by non-salient visual only versus salient visual only (VONS>VOS) and salient visual only versus non-salient visual only (VOS>VONS) contrasts. Activations related to saliency in multimodal visuo-tactile conditions were assessed by visuo-tactile non salient versus visuo-tactile salient (VTNS>VTS) and visuo-tactile salient versus visuo-tactile non-salient (VTS>VTNS) contrasts. Activations related to interaction between saliency and congruency were assessed with visuo-tactile congruent salient vs incongruent salient (VTCS>VTIS) and inverse (VTIS>VTCS) contrasts and with visuo-tactile congruent salient versus visuo-tactile congruent non-salient (VTCS>VTCNS) and inverse (VTCNS>VTCS) contrasts. With visuo-tactile incongruent non-salient versus visuo-tactile incongruent salient (VTINS>VTIS) and inverse (VTIS>VTINS). In the GLM, contrasts were replicated and rescaled overruns accounting for inter-run variability by applying the same contrast weights to the corresponding regressors across the multiple runs and scaling them appropriately. This option allows us to ensure that the contrasts are appropriately normalized and average the effects across the runs, enhancing the statistical power to detect consistent effects. A second-level random-effects analysis was then conducted and statistical results were thresholded using a voxel-wise family-wise error (FWE) corrected height threshold of $p < 0.05$.

Behavioral Analysis

Behavioral data was analyzed with custom Matlab Software (9.6.0.1072779 (R2019a)) over 25 subjects. Reaction times (RTs) were collected for each trial and performance (accuracy) was computed as the proportion of correct trials within all trials. The correct trials are grouped as follow. The visuo-tactile

congruent salient (VTCS)VTIS (visuo-tactile incongruent salient), VTCNS (visuo-tactile congruent non salient), VTINS (visuo-tactile incongruent non-salient), VS (visual only salient), VNS (visual only non-salient), T (tactile only). A random ANOVA 2 factors (congruency x saliency) was computed between the 6 visual and visuo-tactiles conditions, VTCS, VTIS, VTCNS, VTINS, VS, VNS (David Stern (2024). randanova2 (<https://www.mathworks.com/matlabcentral/fileexchange/44308-randanova2>), MATLAB Central File Exchange) (Anderson, 2001). As post-hoc test following random ANOVA and to examine the independent effects of congruency and salience on cognitive processing, two differences were calculated: the congruency effect (incongruent RT - congruent RT) and the salience effect (non-salient RT - salient RT). We performed a wilcoxon signed ranked test (difference to 0) on 3 independent conditions for salience difference: VT congruent, VT incongruent, Visual only. We performed a wilcoxon signed ranked test (difference to 0) on 2 independents conditions for congruency difference: salient, non-salient. Each time, p-values were corrected with bonferroni-holm correction.

Results

Behavioral results

The objective of behavioral analysis was to test the saliency and the multisensory effects on facilitation of response. The facilitation was assessed by comparing RTs between congruent and incongruent trials and salient and non-salient trials. For this visuo-tactile task, RTs were significantly different between salient and non-salient conditions (Random ANOVA analysis:($p\text{-value}<0.001$, figure 2, left panel). The RTs were significantly different for congruent compared to incongruent conditions (Random ANOVA analysis: ($p\text{-value}<0.05$, figure 2, left panel). No interaction between congruency and saliency factors was observed ($p\text{-value}>0.05$). We performed post-hoc analyses by comparing the RTs difference to 0 for the salient and congruency factors, independently, using a Wilcoxon signed rank test. For salient condition, we computed non-salient RTs minus salient RTs for visuo-tactile congruent (VTC), visuo-tactile incongruent (VTI) and visual only (VO) correct trials. The difference to 0 was significant for VTC ($p\text{-value}<0.001$), VTI ($p\text{-value}<0.001$) and VO ($p\text{-value}<0.001$, figure 2, middle panel). Thus, RTs were significantly faster for salient as compared to non-salient condition. For congruency conditions, we computed incongruent RTs minus congruent RTs for salient (S) and non-salient (NS) correct trials. The difference to 0 was significant for both S ($p\text{-value}<0.001$) and NS conditions ($p\text{-value}<0.05$, figure 2, right panel). Thus, the RTs were faster for congruent than incongruent factors. Unimodal tactile trials were not directly statistically compared to visual and visuo-tactile as the task rule for this type of trial required to provide a two-choice response (left and right tactile stimulations) instead of 4 choice response (face positions). Unimodal tactile responses were faster than non-salient trials and slower than salient trials (figure 2, left panel).

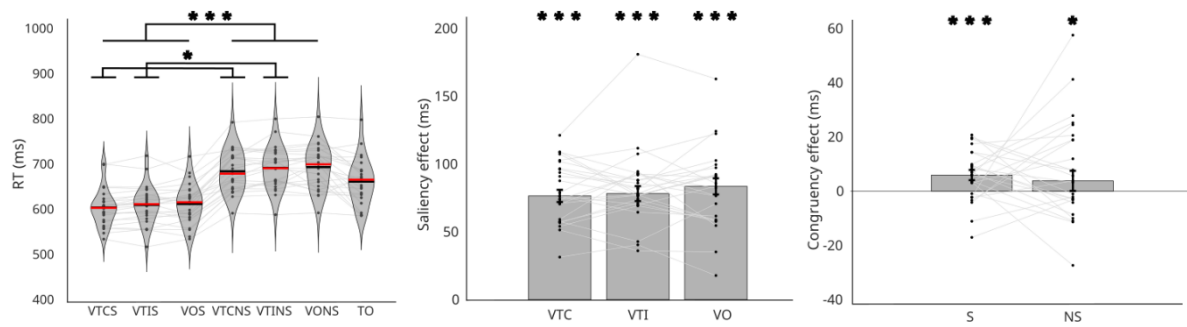


Figure 2: Behavioral results. Left Panel. Reaction times (RT) for visuo-tactile (VT), visual only (VO) and tactile only (TO) trials. VTCS (VT congruent salient), VTIS (VT incongruent salient), VTCNS (VT congruent non-salient), VTINS (VT incongruent non-salient), VS (VO salient), VNS (VO non-salient), T (TO) (random ANOVA test, $\ast = p\text{-value} < 0.05$, $\ast\ast\ast = p\text{-value} < 0.001$). Medians of each condition is represented by the red horizontal bar of each violin plot, mean is the black one. **Middle Panel.** Reaction times (RT) in ms for the salience factor, non-salient RT minus salient RT for correct trials. (Wilcoxon signed-rank test with Bonferroni-holm correction for multiple comparison, $\ast = p\text{-value} < 0.05$, $\ast\ast\ast = p\text{-value} < 0.001$). **Right Panel.** Reaction times (RT) in ms for the congruency factor, incongruent RT minus congruent RT for correct trials (Wilcoxon signed-rank test with Bonferroni-holm correction for multiple comparison, $\ast = p\text{-value} < 0.05$, $\ast\ast\ast = p\text{-value} < 0.001$).

Brain activations Group analysis

The main objective was to investigate the role of pulvinar in a multisensory face detection task. We explored its role in visual saliency in unimodal and multimodal visuo-tactile context. We explored its role in multisensory processing of spatial congruency between a target face and a tactile air-puff delivered concomitantly. First, we described fMRI activations for unimodal visual and tactile modalities. Then we described fMRI activations for unimodal and multimodal visual salient and non-salient conditions. Finally, we described fMRI activations for multimodal spatial congruent and incongruent conditions.

For the VO>TO contrast, cortical activations were located in occipital cortex, fusiform cortex, parietal superior and inferior cortices and frontal cortex. A subcortical cluster was found encompassing pulvinar and medial geniculate nucleus (MGN) (figure 3, top left panel and supplementary table). As activation in the MGN for visual, tactile or motor processing had never been shown in literature, we attributed this activation cluster to the pulvinar. For the inverse contrast (TO>VO), activations were located in temporal, parietal (postcentral gyrus), superior temporal and frontal (insula) cortices ($p < 0.05$, FWE correction for multiple comparison at voxel level, figure 3, top right panel and supplementary table). In unimodal visual and tactile trials contrasted against each other, the activations are compatible with unisensory processing within the brain. Indeed, visual stimulations activated occipital and fusiform areas as well as superior parietal lobule (SPL) and frontal areas. These activations in the SPL and frontal

implicated in visuo-motor actions and considering participants have to press a response key are congruent with previously described action planning networks (Rolls et al., 2023). Tactile stimulations activated postcentral and superior temporal areas as well as insula are also in agreement with previous literature (Chen et al., 2008; Menon and Uddin, 2010b). We did not find activations in insula for visual trials probably because of the salient and non-salient trials were mixed in the contrast.

In order to investigate cortical and pulvinar activity for visual salient trials, we analyzed the following contrasts: visual only salient vs non-salient (VOS>VONS) and visual only non-salient vs salient (VONS>VOS). For VOS>VONS contrast, we identified activations in insula, superior and middle and superior temporal gyri and supramarginal gyrus ($p<0.05$, FWE correction for multiple comparison at voxel level, figure 3 bottom right and supplementary table). For VONS>VOS contrast, activations were located in occipital superior cortex, parietal superior lobule, precentral gyrus, anterior cingulum cortex and frontal superior and orbitofrontal gyri, two bilateral subcortical clusters were observed encompassing pulvinar and MGN (uncorrected $p<0.001$ at voxel level, figure 3, bottom left panel and supplementary table). Like the previous contrasts, we attributed this activation to pulvinar rather than MGN for which no implication in vision has been reported.

Then we investigated multimodal activations for saliency and congruency effects. We investigated visuo-tactile salient versus non salient (VTS>VTNS) and inverse (VTNS>VTS) contrasts. For VTNS>VTS contrast, activations were observed in middle cingulate, insula, supplementary motor area (SMA), middle frontal gyrus (MFG), inferior and superior parietal lobules (IPL, SPL), and precentral gyrus ($p<0.05$, FWE correction for multiple comparison at voxel level, figure 4 top left and supplementary table). For the inverse contrast, VTS>VTNS, only middle cingulate, precuneus and angular gyrus were activated ($p<0.05$, FWE correction for multiple comparison at voxel level, figure 4 top right and supplementary table).

No activation clusters were identified for direct comparison between congruent and incongruent conditions in VTCS>VTIS and VTIS>VTCS contrasts. In a next step, we compares congruency conditions depending on saliency conditions. For VTCS>VTCNS contrast, activations were observed in anterior cingulate, MFG, precentral gyrus, postcentral gyrus, inferior operculum, IPL, SPL, occipital and fusiform cortices and a cluster in pulvinar has been observed ($p<0.001$, uncorrected threshold at voxel level, figure 4 middle right). For the invers contrast (VTCNS>VTCS), activations were located in temporal superior and supramarginal cortices. For visuo-tactile incongruent non-salient vs salient (VTINS>VTIS) contrast, we observed activations in insula, SMA, IPL, SPL, angular gyrus, precentral gyrus, frontal inferior operculum ($p<0.05$, FWE correction for multiple comparison at voxel level, and supplementary table). For the inverse contrast (VTIS>VTINS), we observed activations in the precuneus ($p<0.05$, FWE correction for multiple comparison at voxel level, and supplementary table).

Interestingly in the multimodal visuo-tactile context, we observed a strong insula activation for non-salient conditions but no insula activation for salient condition. This is mainly the case for incongruent

context whereas in congruent context, insula was never activated by either salient or non-salient condition. This was a reverse pattern relative to what we observed in the unimodal context where both visual salient and tactile condition activated insula.

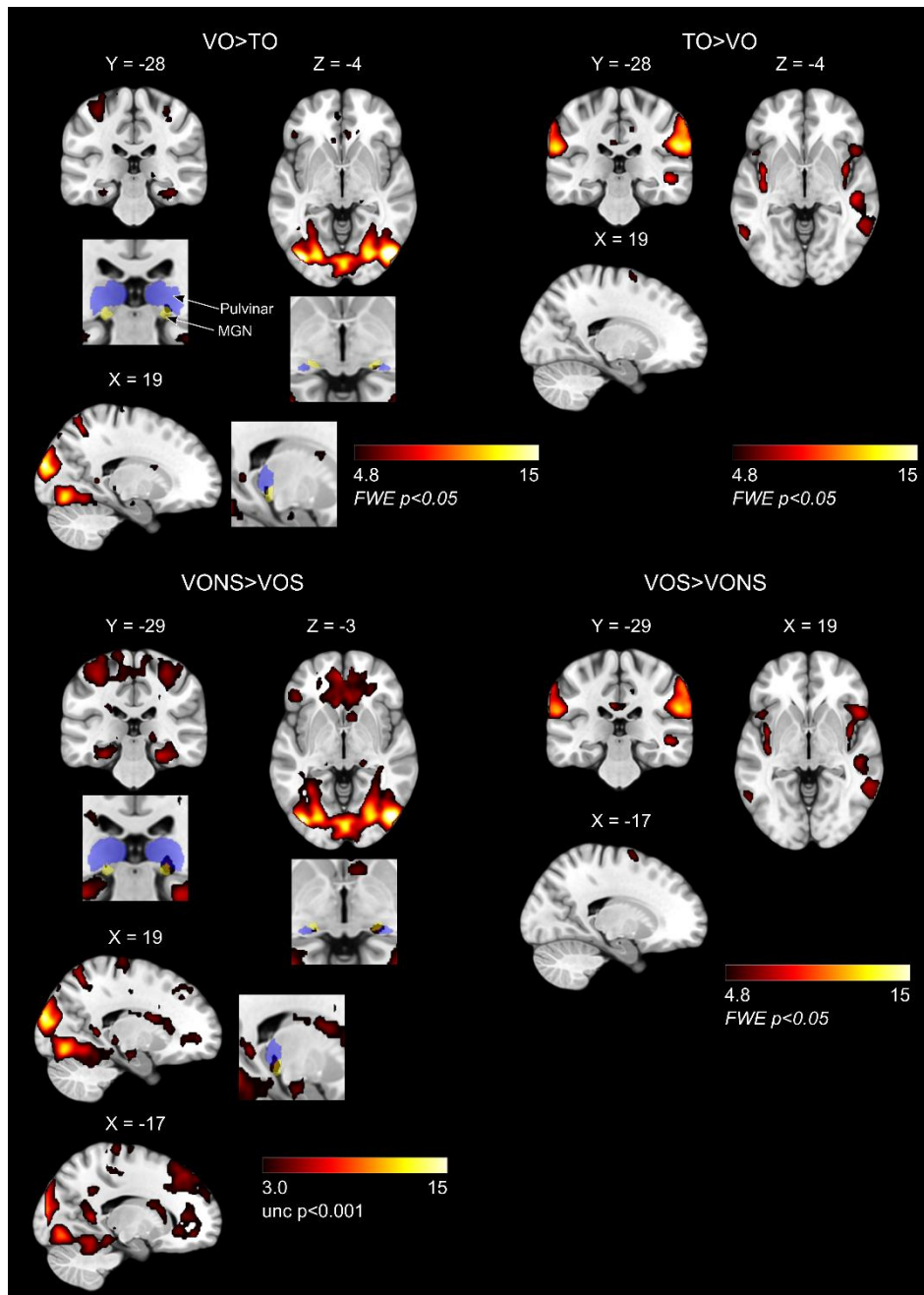


Figure 3: Unimodal visual and tactile and salience unimodal activations. Coronal, sagittal and axial view sections of group level activation maps thresholded at $p < 0.05$ FWE at voxel level and for VO>TO, TO>VO (Top), VOS>VONS (bottom) and at $p < 0.001$ uncorrected at voxel level for VONS>VOS (bottom). Pulvinar zoom for the three sections for VO>TO and VONS>VOS contrasts (left), blue: pulvinar, yellow: medial geniculate nucleus (MGN).

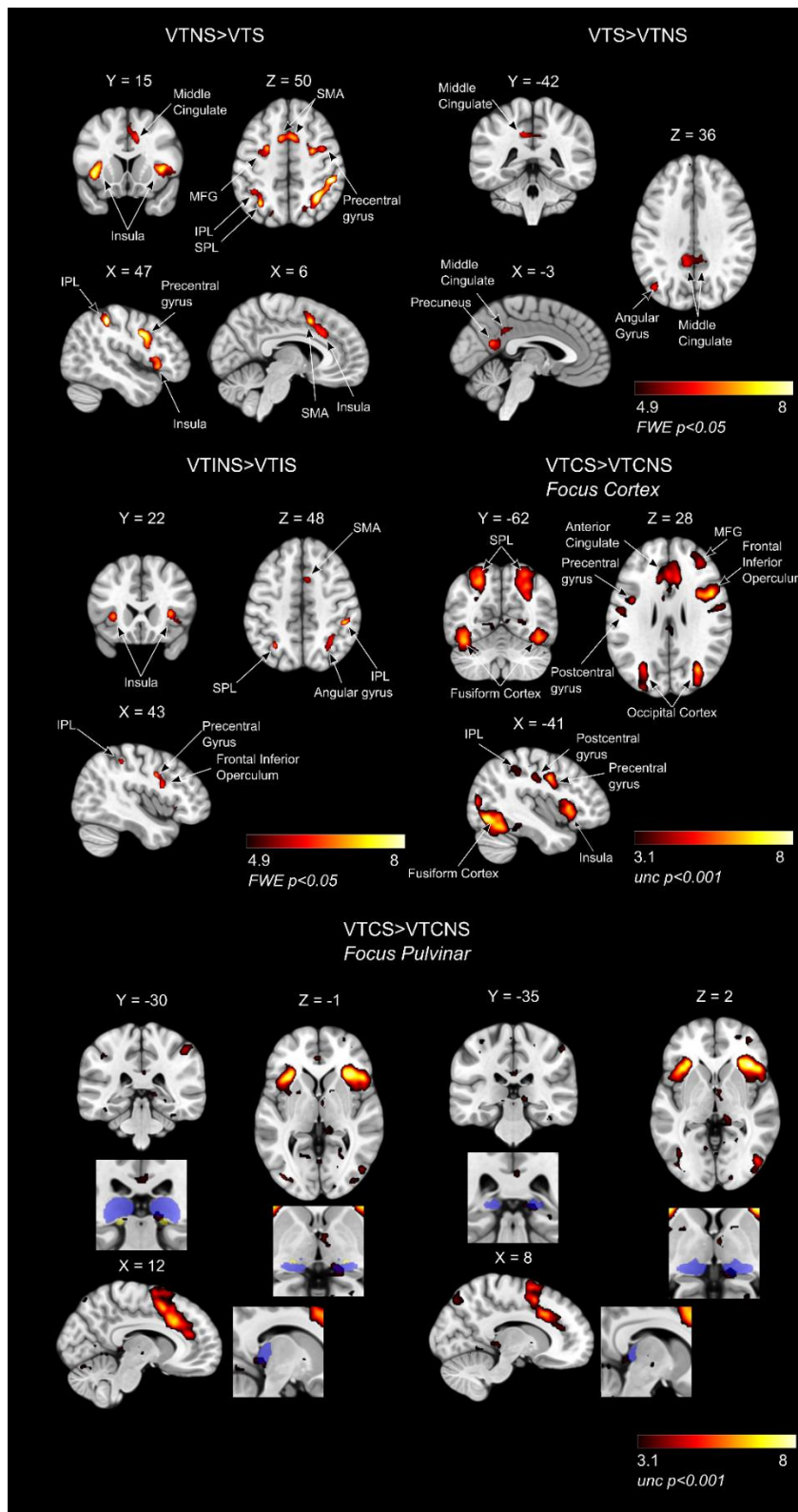


Figure 4: Multimodal visuo-tactile salience and congruency activations. Coronal, sagittal and axial view sections of group level activation maps thresholded at $p < 0.05$ FWE at voxel level and for VTNS>VTS, VTS>VTNS (Top), VTINS>VTIS (middle left), and at $p < 0.001$ uncorrected at voxel level for VTCS> VTCNS (middle right, bottom). VTCS>VTCNS contrast activations are shown with a focus on cortical domain and pulvinar area. A Pulvinar zoom for VTCS>VTNS contrast is shown for the three sections: blue: pulvinar, yellow: MGN.

Discussion

As this chapter is still in progress, the discussion will emphasize current results and propose the next steps for analysis.

Reaction times are reduced for salient and congruent

Shorter RTs were observed within salient trials compared to those non-salient emphasizing the importance of salience as a feature for discrimination and detection of visual stimuli. Also, we observed significant shorter RTs in the visuo-tactile congruent condition compared to incongruent condition. This observation provides strong evidence of potential multisensory integration facilitation, but it requires further analysis to be confirmed. . An efficient option would be to use a race model using cumulative distribution functions to compare RT between multisensory and unisensory trials (Stevenson et al., 2014). Investigation multisensory effect on accuracy by comparing the actual multisensory response to a modeled prediction of it (Stevenson et al., 2014) would be an important metric.

Insula activation in response to visual salience and tactile stimuli: The role of congruency We observed insula activation for tactile compared to visual unimodal stimulations (TO>VO contrast) but not for the inverse contrast (VO>TO). Insula has been highlighted to play a role in salient processing in both tactile and visual modalities (Chen et al., 2008; Menon and Uddin, 2010b). The lack of insula activation in VO>TO contrast could be explained by the fact that the salient and non-salient visual trials were pooled together. Salient versus non-salient visual trials contrast (VOS>VONS) showed insula activation. However, — the inverse contrast (VONS>VOS) did not. We observed the same pattern for visuo-tactile congruent salient versus non-salient trials (VTCS>VTCNS). These results suggest that tactile stimulation is salient as well as visual at unisensory processing scale. Interestingly, we did not observe the same pattern when the congruent condition is not analyzed in terms of congruent or incongruent trials. In visuo-tactile salient vs non salient (VTS>VTNS contrast), insula was not activated while it showed significant activation for non-salient vs salient in the inverse contrast (VTNS>VTS). This Finding suggest an interaction between saliency and congruency which has not been revealed behaviorally.

We did not observe interaction between saliency and congruency in behavioral results but did observed different saliency activation patterns for congruent and incongruent conditions. Suggesting an influence of multimodal visuo-tactile context on saliency processing and then a possible interaction effect.

Recruitment of attention network and pulvinar in difficult context to enhance sensory processing

For unimodal visual and visuo-tactile non-salient conditions, areas belonging to the attentional network (Behrmann et al., 2004; Macaluso and Doricchi, 2013; Mengotti et al., 2020) including superior parietal lobule (SPL), inferior parietal lobule (IPL), middle frontal gyrus (MFG) and frontal inferior (IFG) and

superior (SFG) gyri were activated as well as pulvinar and primary sensory areas. Visual discrimination of the target face is more difficult in non-salient condition than salient condition. Thus, we hypothesized that attentional network is recruited to enhance perception of visual stimuli (Desimone and Duncan, 1995; Pestilli et al., 2011) along with the pulvinar as a modulator nucleus (Benarroch, 2015a; Fiebelkorn et al., 2019a; Froesel et al., 2021a; Saalman et al., 2012a).

Interaction between saliency and congruency

Unimodal visual and multimodal visuo-tactile non-salient conditions versus salient conditions, as well as incongruent non-salient versus salient conditions, exhibited activation in IPL and SPL which make part of the dorsal attention network. But we did find the inverse pattern for congruent condition. IPL and SPL were activated for congruent salient versus congruent non-salient contrast and not for congruent non-salient versus salient. This result suggests an interaction between saliency and congruency effects which was not proved by the current RTs analysis. In fact, the dorsal attention network may have increased the speed and accuracy in task like the observed reduction of RTs for congruent trials. Attention has shown to improve performance by reducing RTs (Desimone and Duncan, 1995; Macaluso and Doricchi, 2013; Pestilli et al., 2011).

Activations in IFG, MFG and SPL are relevant with the task paradigm

Areas of the ventral fronto-parietal network, including IFG, MFG, intraparietal sulcus (IPS), angular gyrus and supramarginal gyrus have been found to be activated with right lateralization in attention studies for infrequent invalid trials (Macaluso and Doricchi, 2013). In our experiment, we observed activity in MFG and angular gyrus for incongruent conditions. It seems that incongruent trials could share same characteristics as invalid trials and logically activated partially the ventral fronto-parietal network.

SPL is an area of the attention network and has been demonstrated being involved in spatial attention shifting (Behrmann et al., 2004). In our experiment, its activation in unimodal visual or multimodal but not in unimodal tactile contexts is logical with the fact that the spatial position of visual target face is changing from one trial to the next.

Further analysis should investigate potential signal change PSC or use a model-based approach to characterize the exact nature of interaction.

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Cortical activations					Pulvinar					
					Left			Right		
	cortical area	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size	MNI (x, y, z)	Peak t-score	Cluster size
VO>TO FWE p<0.05	L Fusiform	-37/-56/-19	18.72	19577*	-	-	-	19/-28/-4	5.40	9
	R Fusiform	35/-54/-19	21.72	19577*						
	L Occipital	-29/-81/22	16.12	19577*						
	R Occipital	33/-76/24	15.67	19577*						
	L SPL	-24/-60/59	10.18	352						
	R SPL	24/-60/59	10.46	19577*						
	R Frontal Sup Med	6/57/22	7.02	723*						
	L Frontal Sup Med	-4/58/17	6.49	723*						
TO>VO FWE p<0.05	L Frontal Mid	-45/37/24	9.70	531	-	-	-	-	-	-
	L Insula	-31/18/9	8.75	258						
	R Insula	36/23/7	7.42	1363*						
	L Temporal Sup	-58/-32/24	13.57	6251						
	R Temporal Sup	63/-41/22	14.98	5938						
	L PostG	-52/5/19	9.41	528						
	R PostG	52/8/14	8.74	1363*						
VONS>VOS unc p<0.001	L Fusiform	-37/-56/-19	15.70	26551*	-5/-25/4	3.42	4	19/-28/-1	4.51	26551*
	R Fusiform	35/-54/-19	19.77	26551*						
	L Occipital Sup	81/194/-19	14.0	26551*						
	R Occipital Sup	24/-94/19	15.27	419						
	L SPL	-14/-60/59	8.15	26551*						
	R SPL	24/-60/59	8.24	26551*						
	L PreG	-29/-20/72	6.09	8058*						
	R PreG	28/-20/69	6.09	8058*						
	R Frontal Sup Med	1/57/14	7.09	9040*						
	R OrbitoFrontal Mid	6/29/-9	6.34	9040*						
VOS>VONS FWE p<0.05	AC	-8/27/-9	7.21	9040*	-	-	-	-	-	-
	R Temporal Sup	64/-40/24	14.73	5679*						
	L SupraMarginal	-55/-35/27	11.52	5663*						
	L Temporal Mid	-53/-60/9	10.91	5663*						
	R Temporal Mid	64/-51/7	9.65	5679*						
	L Insula	-31/18/9	9.35	429						
	R Insula	35/21/7	9.41	1824						
VTNS>VTS FWE p<0.05	Middle Cingulate	9/20/34	6.41	1004*	-	-	-	-	-	-
	L SMA	-8/10/49	6.98	1004*						
	R SMA	4/10/47	7.53	1004						
	L Insula	-34/20/2	9.40	662						
	R Insula	38/25/-1	8.38	1126						
	L MFG	-24/-1/52	7.17	612						
	R MFG	32/-3/57	6.35	523*						
	R PreG	27/-4/49	7.28	523*						
	L IPL	-40/-43/47	6.77	1149*						
	R IPL	32/-52/49	8.26	1342						
	L SPL	-31/-60/49	7.48	1149*						

VTS>VTNS FWE p<0.05	L Middle Cingulate	-8/-48/37	6.14	909*						
	Precuneus	-10/-57/19	6.31	909*						
	L Angular Gyrus	-40/-70/39	6.42	243	-	-	-	-	-	-
VTINS>VTIS FWE p<0.05	L Insula	-32/20/-2	7.09	234						
	R Insula	33/25/7	7.12	376						
	R SMA	4/10/47	5.95	85						
	R Frontal inf operc	44/8/22	6.31	268*						
	L PreG	-47/-1/37	6.19	93	-	-	-	-	-	-
	R PreG	41/0/34	6.85	268*						
	R IPL	47/-34/49	7.42	116						
	L SPL	-29/-59/47	6.69	152						
VTCS>VTCNS unc p<0.001	R Angular Gyrus	32/-56/49	6.30	243						
	L IPL	-40/-43/47	3.66	250						
	R IPL	49/-35/52	4.44	5537*						
	L SPL	-24/-65/54	6.52	4230*						
	R SPL	25/63/59	5.87	5537*						
	AC	1/26/32	5.70	10204*						
	L Insula	-34/21/2	7.56	1453						
	R Insula	40/25/-4	8.18	10204*						
	L PreG	-42/-4/37	6.63	883*	-	-	-	-30/12/-1	3.96	91*
	R PreG	41/4/32	7.14	10204*				-35/8/2	3.96	91*
	L PostG	-40/-20/37	4.16	883*						
	R PostG	49/-11/34	4.05	10204*						
	R MFG	35/39/27	4.70	499						
	R Frontal inf operc	48/8/29	7.27	10204*						
	L Fusiform cortex	-42/-60/-19	6.74	4230*						
	R Fusiform cortex	41/-65/-14	6.28	5537*						
VTCNS>VTCS FWE p<0.05	L Temporal Sup	-56/-32/24	8.01	622						
	R SupraMarginal	58/-28/27	9.13	564	-	-	-	-	-	-
	L Rolandic operc	-42/-16/19	6.59	68						
	R Rolandic operc	46/-14/19	7.07	131						
VTCS>VTIS FWE p<0.05	-	-	-	-	-	-	-	-	-	-
VTIS>VTCS FWE p<0.05	-	-	-	-	-	-	-	-	-	-
VTINS>VTIS FWE p<0.05	L Occipital Mid	-28/-67/34	5.77	152*						
	R Angular Gyrus	-29/-59/47	6.69	152*						
	R	32/-56/49	6.30	243						
	R IPL	47/-35/49	7.45	116						
	Middle Cingulate	-17/-1/37	6.19	93	-	-	-	-	-	-
	Pre G	41/0/34	6.85	268						
	R SMA	4/10/47	5.95	85						
	L Insula	-32/20/2	7.09	234						
VTIS>VTINS FWE p<0.05	R Insula	33/25/7	7.12	376						
	L Precuneus	-4/-59/22	5.53	44	-	-	-	-	-	-

Supplementary Table. fMRI activations. Cortical and pulvinar displaying cluster activations, for the VO>TO, TO>VO, VONS>VOS, VOS>VONS, VTNS>VTS, VTS>VTNS, VTINS>VTIS, VTCS>VTCNS. Cluster peak location, peak t-score and cluster size are indicated. Star (*) indicates presence of two local peaks within the same cluster. Star (*) indicates presence of two local peaks within the same cluster. AC: Anterior Cingulate Cortex, Angular Gyrus, Frontal Inf Oper: Inferior Frontal Gyrus, Opercular Part, Frontal Inf Operc: Inferior Frontal Gyrus, Opercular Part, Frontal Mid: Middle Frontal Gyrus, Frontal Sup Med: Superior Frontal Gyrus, Medial Part, IPL: Left Inferior Parietal

Lobule, IPL: Right Inferior Parietal Lobule, MFG: Left Middle Frontal Gyrus, Occipital Sup: Superior Occipital Cortex, OrbitoFrontal Mid: Middle Orbitofrontal Cortex, PostG: Postcentral Gyrus, PreG: Precentral Gyrus, SMA: Supplementary Motor Area, SPL: Superior Parietal Lobule, SupraMarginal: Left Supramarginal Gyrus, Temporal Mid: Middle Temporal Gyrus, Temporal Sup: Superior Temporal Gyrus.

Chapter II : Somatomotor transformations

Study 4 Spatial transformations underlying saccadic eye movements to tactile and visual target stimuli

Performing a motor action toward a tactile stimulus requires transforming the anatomical reference frame coding for stimulus position into an external reference frame. The neural basis underlying such process lack of clarity at large-scale brain activations as well as their dynamics. We investigated the whole brain activations in a task where participants had to perform saccades toward visual and tactile targets under cross and uncross arms body postures. Crossing the arms allows to create a conflict between anatomical and external representation of the tactile stimulus. When arms crossed, the left hand stimulated is in the right spatial position regarding of our body. We determined if both reference frame coding are carried by the same anatomical cortical areas and if these representations are dynamically evolving from the sensory perception of the stimulus to the saccade execution.

Spatial transformations underlying saccadic eye movements to tactile and visual target stimuli (work in progress)

Celia Foster^{1,2,*}, Maxime Gaudet-Trafit³, Valentin Marcon³, Franck Lamberton^{4,5}, Wei-An Sheng³,
Suliann Ben Hamed³, Tobias Heed^{1,6,7,*}

¹ Biopsychology & Cognitive Neuroscience, Bielefeld University, Bielefeld, Germany

² Center of Excellence in Cognitive Interaction Technology (CITEC), Bielefeld University, Bielefeld, Germany

³ Institut des Sciences Cognitives Marc Jeannerod, UMR5229, CNRS-University of Lyon 1, France

⁴ CERMEP-imagerie du vivant, 59 Boulevard Pinel, 69677 Bron Cedex, France

⁵ SFR Santé Lyon-Est, UMR3453, CNRS-University of Lyon 1, France

⁶ Cognitive Psychology, Department of Psychology, University of Salzburg, Salzburg, Austria

⁷ Centre for Cognitive Neuroscience, University of Salzburg, Salzburg, Austria

* Correspondence and requests for materials should be addressed to C.F. (email: celia.foster@uni-bielefeld.de) and T.H. (email: tobias.heed@plus.ac.at)

Abstract

Saccadic eye movements to tactile and visual target stimuli require different spatial transformations because tactile stimuli can be in different spatial locations depending on body posture, whereas the location of visual stimuli on the retina is independent of body posture. In this study, participants made eye movements to visual and tactile stimuli presented on their hands, while we recorded brain activity using fMRI. Participants' hands were positioned in uncrossed and crossed postures, allowing us to distinguish between coding of sensory stimuli in an anatomical reference frame (relative to the hands) and external-spatial reference frame (relative to the position in external-space). We further distinguished brain responses involved with sensory processing and saccade planning using a delayed anti-saccade paradigm, where participants only learned the final saccade target location (towards or away from the sensory stimulus) after a delay. We could decode the anatomical location of tactile stimuli (left vs right hand, regardless of side of space) from bilateral somatosensory, motor and insular cortex, and right inferior frontal cortex. External-spatial tactile location (left vs right side of space, regardless of stimulated hand) could be decoded from a medial region extending from posterior parietal to occipital cortex, and from bilateral lateral occipital cortical regions. External-spatial visual stimulus location could also be decoded from these regions, and classifiers trained to decode external-spatial tactile location could generalise to external-spatial visual location, and vice versa, suggesting a shared multisensory coding. Once the saccade movement goal was specified, a broad network of occipital, parietal and frontal brain regions encoded movement target location, in a common coding regardless of whether the movement target was tactually or visually defined. Altogether, our results demonstrate a multisensory, external-spatial coding of tactile and visual stimuli in posterior parietal and occipital cortex, and a dynamic transformation of these targets into a common movement goal.

Keywords

Touch, vision, multisensory, sensorimotor transformation, movement planning, tactile remapping, anti-task

Introduction

In our everyday life, we engage in a multitude of motor actions encompassing various categories such as reaching, pointing, grasping or mere visual exploration via eye movements. These actions are directed either towards elements in our environment or can be self-directed towards our own body. Self-directed actions include, for example, scratching ourselves or looking at a spot on our skin that has just been touched by an unidentified foreign body. This is the case following a mosquito bite which, once perceived on our skin, triggers a behavioural response that is expressed by looking at the bite location or hitting our arm at the expected location of the mosquito. In order to perform a self-directed motor action such as ocular saccades towards a tactile stimulus, a dynamic coordinates transformation of tactile stimulus from external space into 3D spatial coordinates anchored to the body is required. Indeed, the external coordinates of the stimulated body part might change if the body part is moved from its initial position (for example if I approach the hand on which I felt the mosquito bite to have a better look at my skin). This remapping has been studied and described over years in multiple behavioural experiments. A simple experimental design to test somato-motor remapping process is to manipulate body posture by crossing limbs relative to the body's frontal axis of symmetry. If we consider a tactile stimulation on one hand, stimulus location can be coded in two different reference frames: an anatomical frame of reference, which makes the distinction between left and right hands based on body representation and body's midline; an external-spatial reference frame, which considers the egocentric position of the stimulus in space, irrespective of bodily configuration of the arms. In the configuration where arms are crossed, a stimulation on the right hand will be represented in anatomical coordinates on the right part of the body whereas it will be represented on the left in external-spatial coding. Accordingly, such experiments highlight the conflict between skin-based and 3D external space framing, resulting in errors when arms are crossed (Badde et al., 2019; Heed and Azañón, 2014; Shore et al., 2002; Wada et al., 2004; Yamamoto et al., 2005; Yamamoto and Kitazawa, 2001). The coding of information in a specific reference frame depends on the sensory system involved in stimuli detection and effector system responsible in action preparation (Andersen and Buneo, 2002; Bencivenga et al., 2023; Cohen and Andersen, 2002). Thus, multiple spatial codes could co-exist and it may be mandatory to perform a coordinate transformation from one code to another in order to frame coordinates in an effector compatible system (Cohen and Andersen, 2002). Posterior parietal cortex (PPC) is an associative cortex located at the interface between somatosensory and visual primary cortex and functionally described to be an important region for sensorimotor planning of goal-directed movement and coordinates transformations (Andersen and Buneo, 2002; Rolls et al., 2023; Sereno and Huang, 2014). Klautke et al. (2022) mapped the areas during somato-motor transformation. In their study, participants had to point to the location of a tactile stimulus on one of their feet in either an uncrossed or a crossed leg configuration. They showed that the location of a tactile stimuli is represented in both an anatomical and an external frame of reference and that these representations are organized along an antero-posterior

axis within the PPC. Anterior PPC coded the somatosensory information in an anatomical reference frame in the primary sensory cortex S1 and the superior parietal lobule (SPL). Posterior PPC and more specifically medial intraparietal sulcus (mIPS) encoded the external location of the stimulus. They additionally identified a specific region in the SPL coding for anatomical stimulus location, task rule and movement goal. This region was thus considered as a good candidate for computing the conversion from tactile anatomical coordinates to motor spatial external location (aka somato-motor transformation). Finally, they identified a broad network for coding movement towards targets defined on one's own body. This network is similar to the network described in movement goal planning to visual targets, suggesting that this cortical network implements the same process for both somato-motor and visuomotor transformation. Previous studies independently focused on either visual or tactile modalities. Saccadic eye movements to tactile and visual target stimuli require different spatial transformations as tactile stimuli are initially coded in an anatomical, skin-based, reference frame, whereas visual targets are encoded in a retinotopic reference frame. Thus in order to better understand the network computations underlying movement planning towards tactile and visual stimuli, we proceeded as follows.

Participants made eye movements to visual and tactile target stimuli presented on their hands, while we recorded brain activity using fMRI. Participants' hands were positioned in uncrossed and crossed postures, allowing us to distinguish between coding of sensory stimuli in an anatomical reference frame (i.e. relative to the hands) and external-spatial reference frame (i.e. relative to the position in external-space). We further distinguished between sensory processing and saccade planning using a delayed anti-saccade paradigm. Participants only learned the final saccade target location (towards or away from the sensory stimulus) after a delay, allowing us to distinguish brain responses involved in sensory processing and movement planning (Gail and Andersen, 2006; Klautke et al., 2022). We report sensory tactile and visual brain activity as well as task rule related brain activity. Using multi-voxel pattern analysis (MVPA), we additionally decode anatomical location of a tactile stimulus bilaterally from somatosensory, motor and insular cortex, and right interior frontal cortex on the hands. Altogether, our results demonstrate a shared, multisensory, external-spatial coding of tactile and visual stimuli in posterior parietal and occipital cortex, and a dynamic transformation of these targets into a common movement goal once the movement goal is specified.

Methods

Participants

Twenty-two participants (11 female, 11 male), mean age 26.4 years old (range: 22–35 years old), completed both sessions of the experiment and were included in our data analyses. A further two participants each completed only one of the two sessions and were therefore excluded from our data

analyses. Seventeen participants were right handed, one participant was left handed and four participants were ambidextrous, as assessed by participant self-report using the Edinburgh Handedness Inventory questionnaire (Oldfield, 1971). All participants provided informed consent prior to the fMRI experiment. The experiment was granted ethical approval by the French Committee for the Protection of Persons (Comite de protection des personnes nord ouest III - IDRCB 2021-A03235-52).

Experiment setup

Participants lay supine with their head supported using cushions. Participants hands were positioned above and in front of their head, over the head coil, with the hands directly visible through the open spaces of the head coil (see Figure 1A). Participants arms rested on a plexiglass structure, and a second structure was placed behind the elbows to support the arms.

Each participant completed two experimental sessions on different days (range: 1-20 days apart). In each session, the participant was first positioned in an uncrossed or crossed arm posture (see Figure 1B) and performed 4 experimental runs (2 with tactile stimuli, 2 with visual stimuli) in this posture, before being taken out of the scanner and repositioned before performing 4 more experimental runs (2 with tactile stimuli, 2 with visual stimuli). Each participant completed one session with the uncrossed posture first and the other with the crossed posture first, and the starting order was counterbalanced across participants. Participants wore cotton arm sleeves to ensure there was no skin contact of the arms in the crossed posture to prevent conductive loops. The safety ball was taped between the participant's knees, and participants practiced pressing it prior to being positioned inside the scanner bore.

In each hand, participants held a custom-made 3D printed device, which was used to deliver tactile and visual stimuli (see Figure 1C). Tactile stimuli consisted of air puffs delivered to the thumb and ring finger of each hand. Visual stimuli consisted of optic fibres which lit up at positions closely adjacent to the thumb and ring finger of each hand. A small mirror was positioned between the participant's hands, slightly to one side of the central bar of the head coil, towards the participant's dominant eye (17 participants right eye, 5 participants left eye). This mirror showed a central optic fibre that was used to deliver task instructions to the participant via light flashes of differing lengths. The experiment was controlled using Psychtoolbox version 3.0.18 (Kleiner et al., 2007) and Matlab 2019 on a Windows 10 laptop.

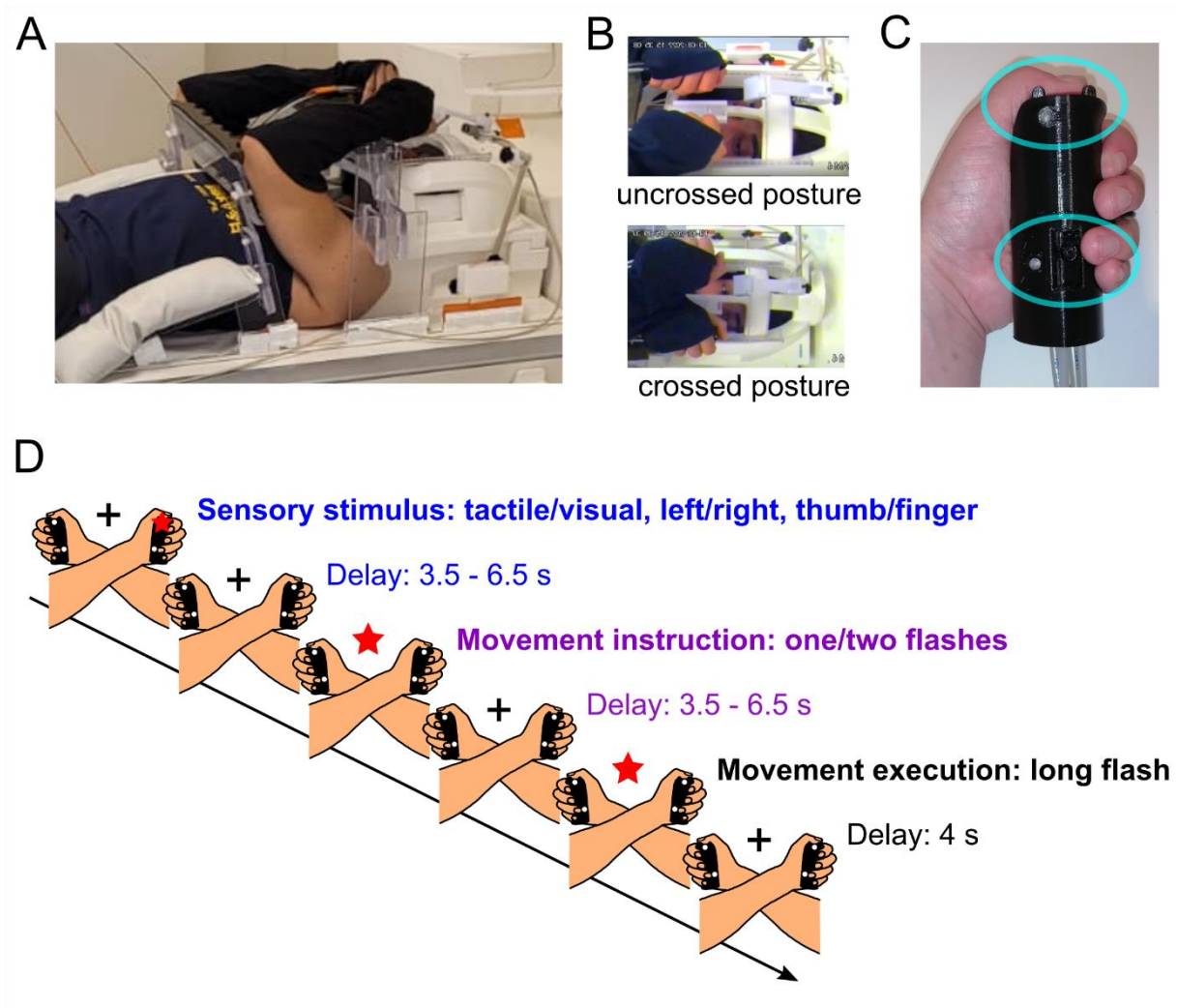


Figure 1. Experimental setup and trial procedure. **A** Participant setup in the uncrossed posture, viewed from the side. **B** Participant hand position in uncrossed and crossed postures, viewed from above. **C** One custom 3D-printed device held in the participant's hand. Cyan circles indicate the locations of air puffs and optic fibres used to deliver tactile and visual stimuli to the thumb and ring finger. **D** Trial procedure. Each trial began with a tactile or visual stimulus to the left or right thumb or finger. After a variable delay of 3.5-6.5 s, one or two short flashes of the central fixation point indicated to the participants whether they would make a pro- or anti-movement. After another variable delay of 3.5-6.5 s, a long flash of the central fixation point indicated to participants that they should immediately perform the saccadic eye movement, and then look back at the central fixation point to await the next trial. The next trial began after a 4 s delay.

Eye tracking

Eye tracking was conducted using an MRC systems 12M-I MR-compatible camera and MRC eye tracking software (MRC Systems GmbH, Heidelberg, Germany). We monitored monocular eye position at a frame rate of 60 Hz. The monitored eye differed across participants (16 sessions right eye, 28

sessions left eye) and was optimised to provide an optimal view of the eye and camera position relative to the position of the small mirror that was used to show a central optic fibre with task instructions, and the position of participants' hands.

At the start of each run, we performed a custom eye calibration procedure. Participants performed two repetitions of a procedure where they iteratively fixated the central optic fibre and then each of the four sensory stimulation points for 2 seconds each. For visual runs, visual stimulation was used as the sensory stimuli at the hands, for tactile runs, tactile stimulation was used at the hands, in each case to ensure a close matching of the calibration data to the sensory trial data. This calibration data was used to calibrate the pupil position data offline to determine whether a movement was made to the correct sensory target at the correct time on each trial.

Experiment procedure

Each participant completed a total of 16 experimental runs, conducted over two experimental sessions. Eight runs were conducted in an uncrossed posture, and eight in a crossed posture. Eight runs contained tactile stimuli and eight contained visual stimuli. Each run began with a short eye tracking calibration, before 33 experimental trials were presented. Trials were counterbalanced for left/right sensory stimulus and pro/anti movement task (Brooks, 2012). Thumb or finger stimulation was ordered randomly.

The trial procedure is illustrated in Figure 1D. Participants received a 200 ms tactile or visual stimulus to the thumb or finger of the left or right hand, followed by a variable delay of 3.5 -6.5 s (in steps of 1 s), together forming the sensory processing interval. Participants then saw one or two 50 ms flashes of the central fixation point, which indicated the task rule (pro- or anti-saccade), followed by a second variable delay of 3.5 - 6.5 s (in steps of 1 s), together forming the movement planning interval. After this delay, a long 500 ms flash of the central fixation point indicated to the participant that they should make the saccadic eye movement, and then move their eyes back to the fixation point to await the next trial. Participants were instructed to maintain fixation throughout the experiment, except for when they were instructed to make a saccade.

The number of flashes of the central fixation point at the start of the movement planning interval indicated to participants whether they would need to perform a pro-saccade to the sensory stimulus location, or an anti-saccade to the corresponding location on the other hand. For each participant, in one experimental session one flash indicated a pro-saccade and two flashes indicated an anti-saccade, and in the other experimental session one flash indicated an anti-saccade and two flashes indicated a pro-saccade. The starting order of the task rule cue was also counterbalanced across participants.

MRI parameters

We acquired MRI images using a 3T Siemens Prisma scanner with a 20-channel head coil. We acquired functional T2* echo-planar images using a sequence with the following parameters; multiband acceleration factor 2, TR 1.6 s, TE 30 ms, flip angle 80°, FOV 210.6 x 210.6 mm, 48 slices, isotropic voxel size of 2.7 x 2.7 x 2.7 mm. In each fMRI session, we recorded two gradient echo field maps, one during the uncrossed posture runs and the other during the crossed posture runs, which were used to correct the fMRI images for magnetic field inhomogeneities during preprocessing. We additionally acquired a high-resolution (1 x 1 x 1 mm) T1-weighted structural MRI scan for each participant.

fMRI preprocessing

fMRI data were preprocessed using SPM12 (<http://www.fil.ion.ucl.ac.uk/spm/>) and Matlab 2019a. Functional images were slice-time corrected, realigned and unwarped and coregistered to the anatomical image. For univariate fMRI analyses, images were additionally normalised to MNI (Montreal Neurological Institute) space and spatially smoothed with a 6 mm Gaussian kernel.

Univariate fMRI analysis

We modelled the fMRI data using general linear models (GLMs) with SPM12. For the univariate analyses, we modelled separate GLMs for tactile and visual experimental runs, and each run contained 4 predictors for the sensory processing interval (left/right, thumb/finger), 8 predictors for the movement planning interval (left/right, thumb/finger, pro/anti), 1 predictor modelling two 8s baseline sections at the beginning and end of each run, 2 predictors modelling the sensory processing and movement planning intervals of the first trial and one predictor for saccade movements, as well as 6 head motion regressors.

We performed whole brain group-level *t*-tests to test for anatomical sensory coding (sensory stimulus on left vs right hand) and external-spatial sensory coding (sensory stimulus on left vs right side of space) during activity from the sensory processing interval, and to test for movement planning coding (preparation of left vs right final movement) and task rule coding (preparation of pro- vs anti-saccade). We tested for significance using a voxelwise $p < .05$ False Discovery Rate (FDR) correction for multiple comparisons.

Multi-voxel pattern fMRI analysis

We modelled the fMRI data using general linear models (GLMs) with SPM12. For the multi-voxel pattern analyses (MVPA), we modelled separate GLMs for tactile and visual experimental runs, and each run contained a separate predictor for the sensory processing interval and movement planning interval of each trial, 1 predictor modelling two 8s baseline sections at the beginning and end of each

run, as well as regressors of no interest for the sensory processing and movement planning interval of the first trial, a predictor for the saccade movements, and 6 head motion regressors.

Beta estimates from the sensory processing interval and movement planning interval of each trial were then input into MVPA classification analyses. We conducted MVPA with The Decoding Toolbox version 3.999F (Hebart et al., 2015) using a searchlight approach with a radius of 6 voxels (Kriegeskorte et al., 2006). We used a linear support vector machine classifier (LIBSVM) and feature-scaled the classifier input data using z-score normalization, estimating the mean and standard deviation on the training data and applying these values to the test data, and setting any outlier values (greater than 2 standard deviations from the mean) to 2 or -2.

We used SPM12 to normalise participant classification accuracy maps to MNI space and spatially smoothed them with a 6 mm Gaussian kernel. We performed group-level analyses using SnPM13 (<http://warwick.ac.uk/snpm>) using nonparametric permutation tests (Nichols and Holmes, 2001) with 10,000 permutations and 6 mm FWHM variance smoothing. We tested for significance using a one-sided t-test with a threshold of $p < .05$ with FDR correction for multiple comparisons.

Results

Univariate fMRI Results

We first performed whole brain univariate fMRI analyses to determine if any brain regions showed higher activity to sensory stimuli based on their anatomical location (i.e. left vs right hand) or external-spatial location (i.e. left vs right side of space) during the sensory processing interval. Figure 2 shows results of whole brain analyses showing regions with higher activity to tactile stimuli on the left vs right hand (Figure 2A) and to tactile stimuli on the right vs left hand (Figure 2B). No regions showed differences in univariate activation between left vs right external-spatial tactile stimuli, and no regions showed differences in univariate activation for left vs right anatomical or external-spatial visual stimuli.

We further tested whether any regions showed differences in activation between saccade planning to the left vs right during the movement planning interval, or to pro- vs anti-saccade preparation during the same interval. No regions showed any significant univariate differences in activation between these conditions.

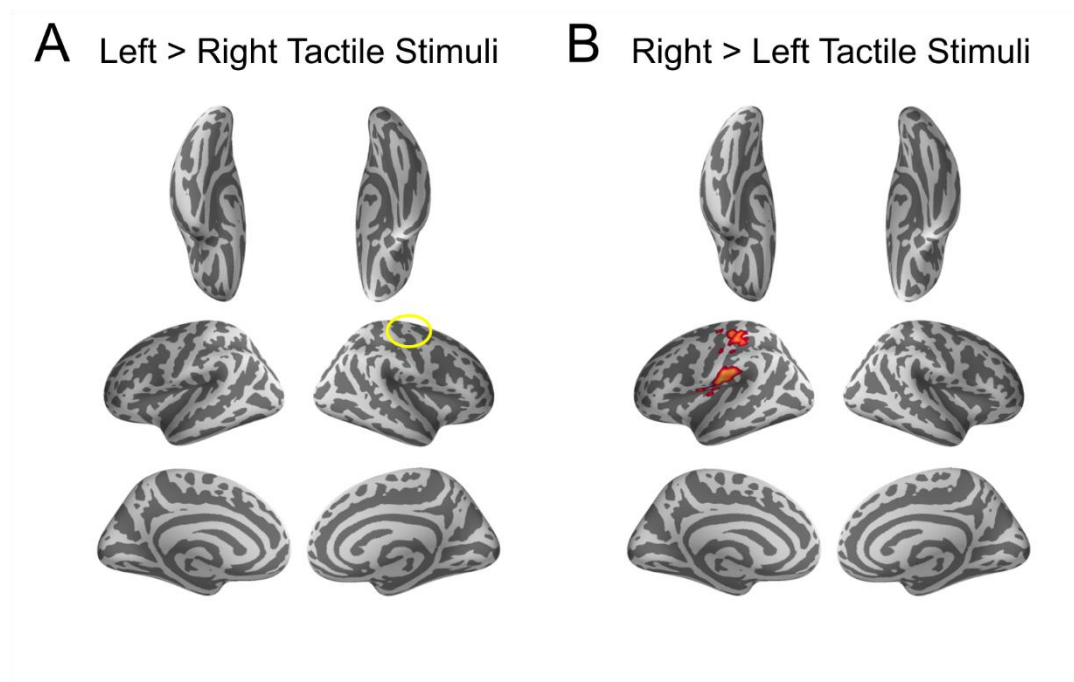


Figure 2. Univariate responses to anatomical tactile stimulus location (i.e. left vs right hand). **A** The region circled in yellow showed higher responses to tactile stimuli on the left hand compared to the right hand. **B** The activated regions showed higher responses to tactile stimuli on the right hand compared to the left hand. Whole-brain activation maps show significant regions defined using a $p < .05$ False Discovery Rate (FDR) correction.

MVPA fMRI Results

Anatomical and external-spatial coding of sensory information. We tested whether we could decode left vs right anatomical location of tactile or visual sensory coding during the sensory processing interval, using a whole brain searchlight approach. Figure 3A gives an overview of the conditions organised to decode anatomical location (i.e. left vs right hand). Areas showing successful anatomical decoding of tactile stimuli shown in Figure 3B. We could not decode anatomical location of visual stimuli from any regions. We further tested whether we could cross-classify anatomical location across posture, training the classifier on uncrossed posture trials and testing it on crossed posture trials, or vice-versa. We did not identify any regions that could cross-classify anatomical stimulus location across posture.

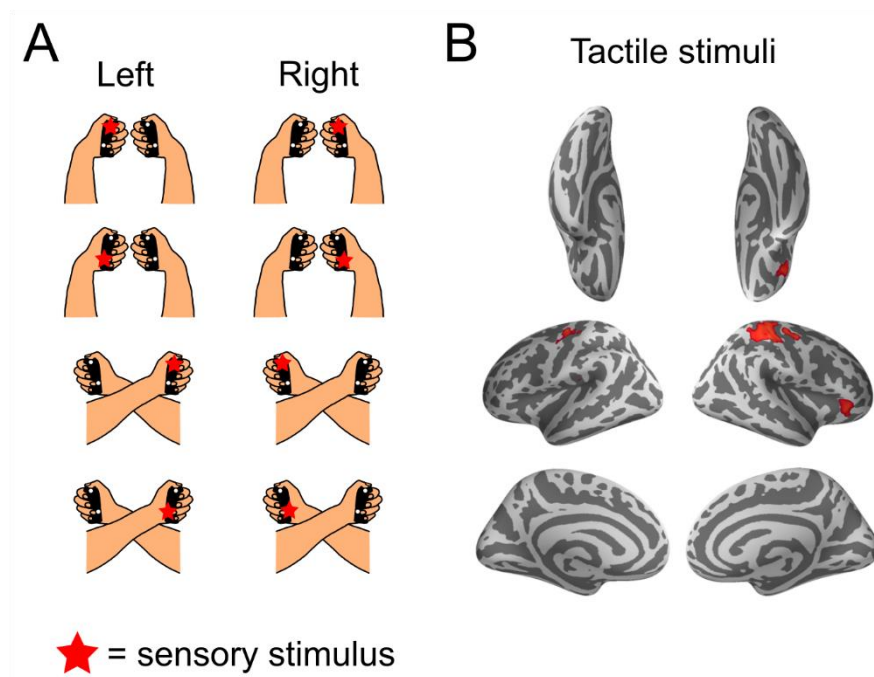


Figure 3. MVPA results for decoding anatomical touch location (i.e. left vs right hand). **A** Overview of conditions organized for anatomical decoding. **B** Whole brain searchlight decoding results for anatomical coding of tactile stimuli. Searchlight maps show significant regions defined using a $p < .05$ False Discovery Rate (FDR) correction.

We next tested whether we could decode external-spatial location of tactile or visual stimuli (i.e. left vs right side of space, regardless of stimulated hand). Figure 4A gives an overview of the conditions organised to decode external-spatial location. Areas showing successful decoding of external-spatial tactile location are shown in Figure 4B and areas showing successful decoding of external-spatial visual location are shown in Figure 4C. We further tested whether we could cross-classify external-spatial location across posture, training the classifier on uncrossed posture trials and testing it on crossed posture trials, or vice-versa. Results are shown in Figure 5.

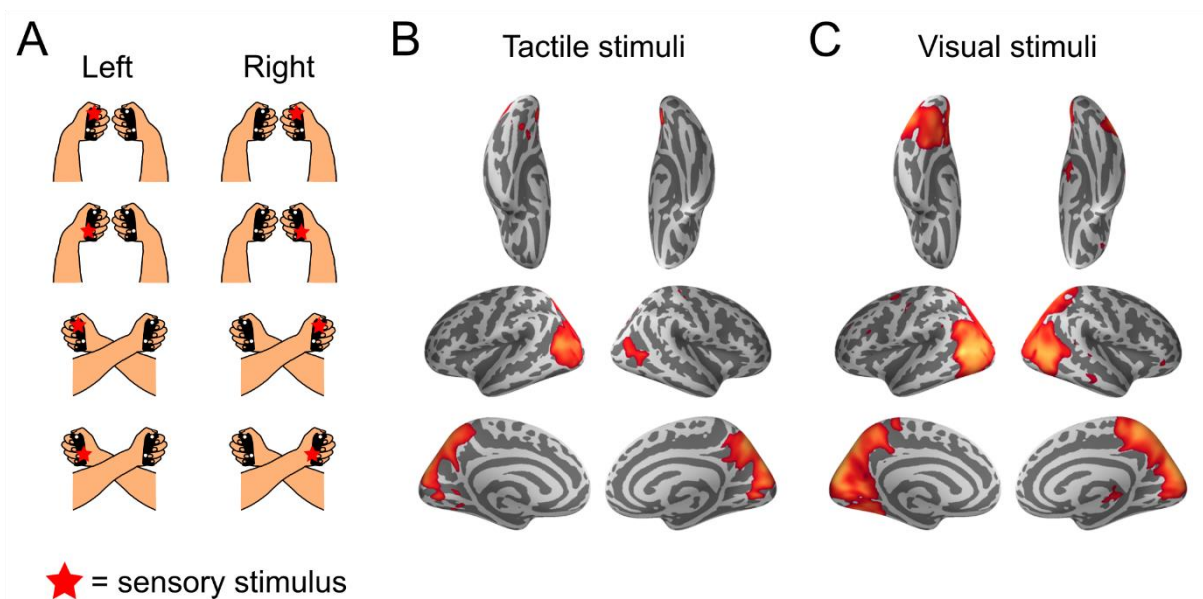


Figure 4. MVPA results for decoding external-spatial location of tactile and visual stimuli (i.e. left vs right side of space). **A** Overview of conditions organized for external-spatial decoding. **B** Whole brain searchlight decoding results for external-spatial coding of tactile stimuli. **C** Whole brain searchlight decoding results for external-spatial coding of visual stimuli. Searchlight maps show significant regions defined using a $p < .05$ FDR correction.

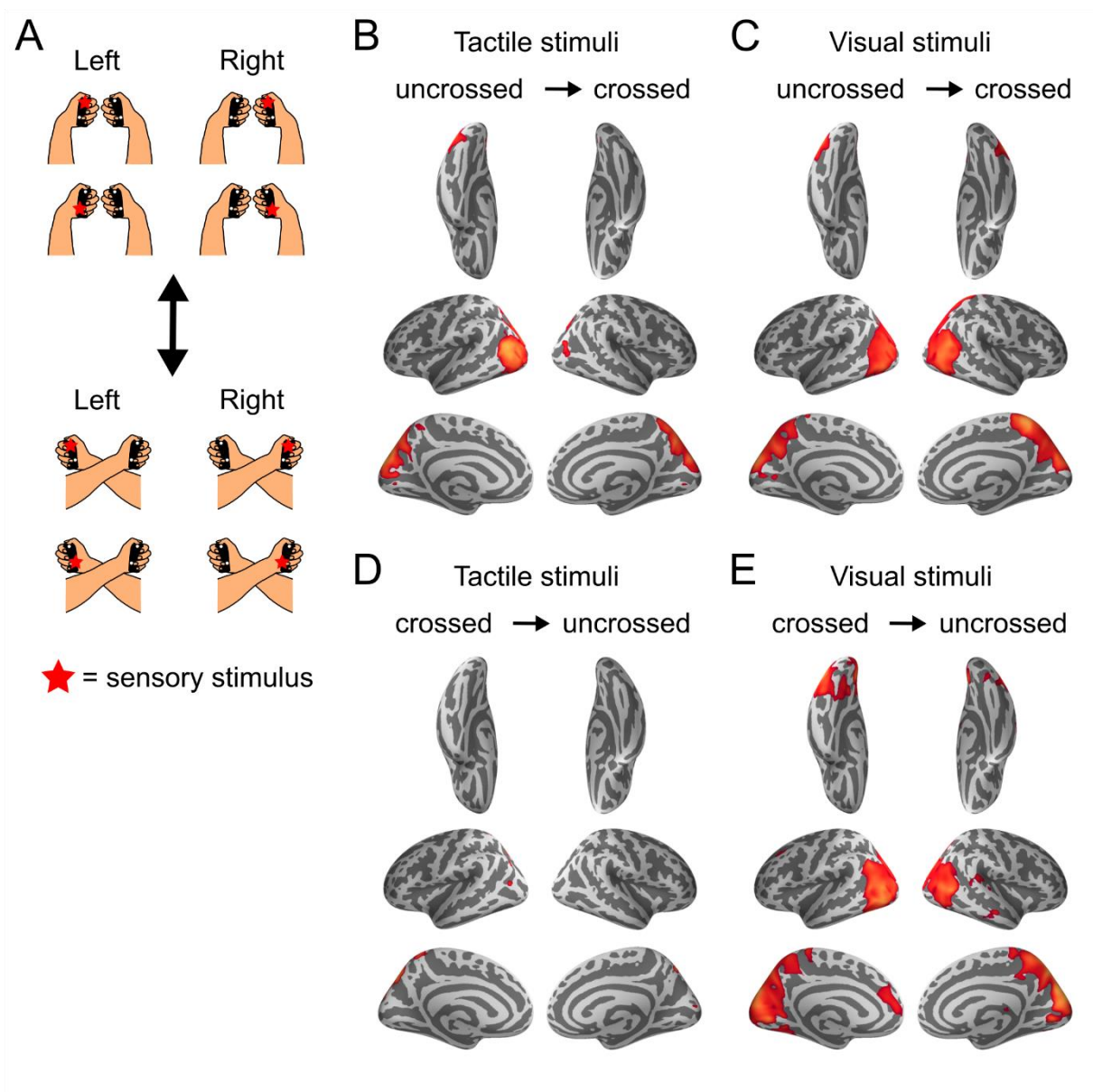


Figure 5. MVPA results for cross-classification of external-spatial stimulus location across posture. **A** Overview of conditions organized for external-spatial cross-classification, with classifiers trained on uncrossed posture trials and tested on crossed posture trials, or vice-versa. **B** Whole brain searchlight decoding results for cross-classification of tactile stimuli, training on uncrossed posture trials and testing on crossed posture trials. **C** Whole brain searchlight decoding results for cross-classification of visual stimuli, training on uncrossed posture trials and testing on crossed posture trials. **D** Whole brain searchlight decoding results for cross-classification of tactile stimuli, training on crossed posture trials and testing on uncrossed posture trials. **E** Whole brain searchlight decoding results for cross-classification of visual stimuli, training on crossed posture trials and testing on uncrossed posture trials. Searchlight maps show significant regions defined using a $p < .05$ FDR correction.

Lastly, we tested whether classifiers trained to distinguish tactile sensory location could decode visual sensory location, or vice-versa. We could not decode anatomical location across sensory stimuli from any regions. Figure 6 shows regions showing significant cross-classification of external-spatial sensory stimulus location across sensory stimulus modality.

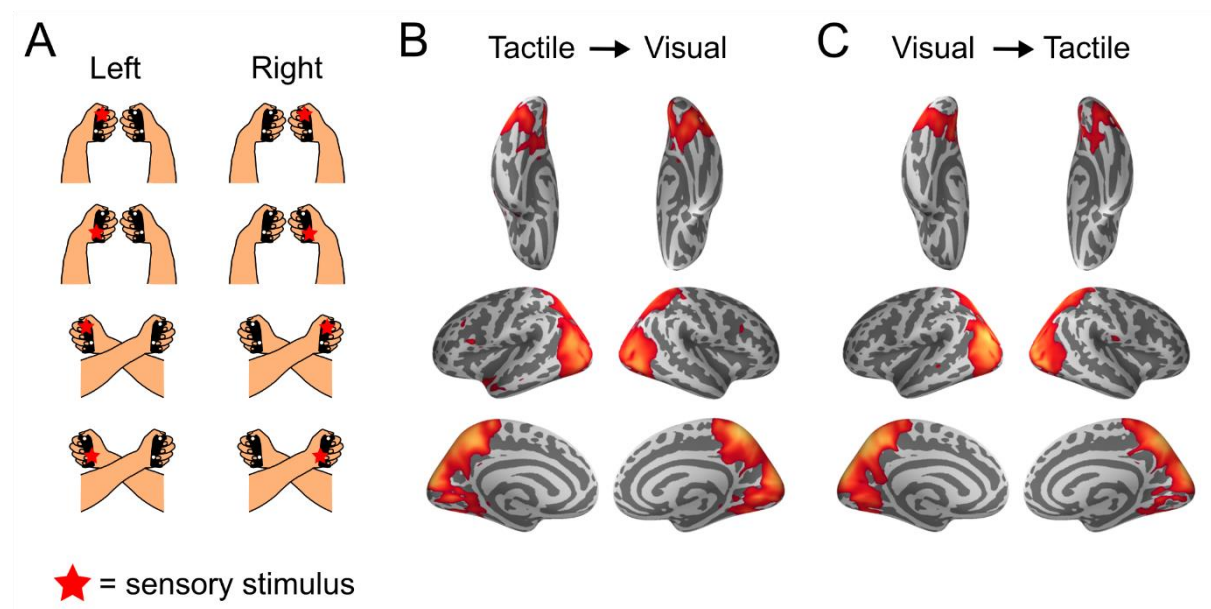


Figure 6. MVPA results for cross-classification of external-spatial stimulus location across stimulus sensory modality. **A** Overview of conditions organized for external-spatial decoding. **B** Whole brain searchlight decoding results for external-spatial cross-classification, training on tactile stimulus trials and testing on visual stimulus trials. **C** Whole brain searchlight decoding results for external-spatial cross-classification, training on visual stimulus trials and testing on tactile stimulus trials. Searchlight maps show significant regions defined using a $p < .05$ FDR correction.

Movement goal location. We tested whether we could decode the final movement goal location during the movement planning interval, after participants received the task rule (pro- or anti-movement). Figure 7A gives an overview of the conditions organised to decode the movement goal location. Areas showing successful decoding of movement goal defined by tactile stimuli are shown in Figure 7B and for the of movement goal defined by visual stimuli in Figure 7C.

We further tested whether classifiers trained to distinguish the movement goal defined by tactile stimuli could decode the movement goal defined by visual stimuli, or vice-versa. Figure 8 shows regions showing significant cross-classification of the final movement goal location across sensory stimulus modality.

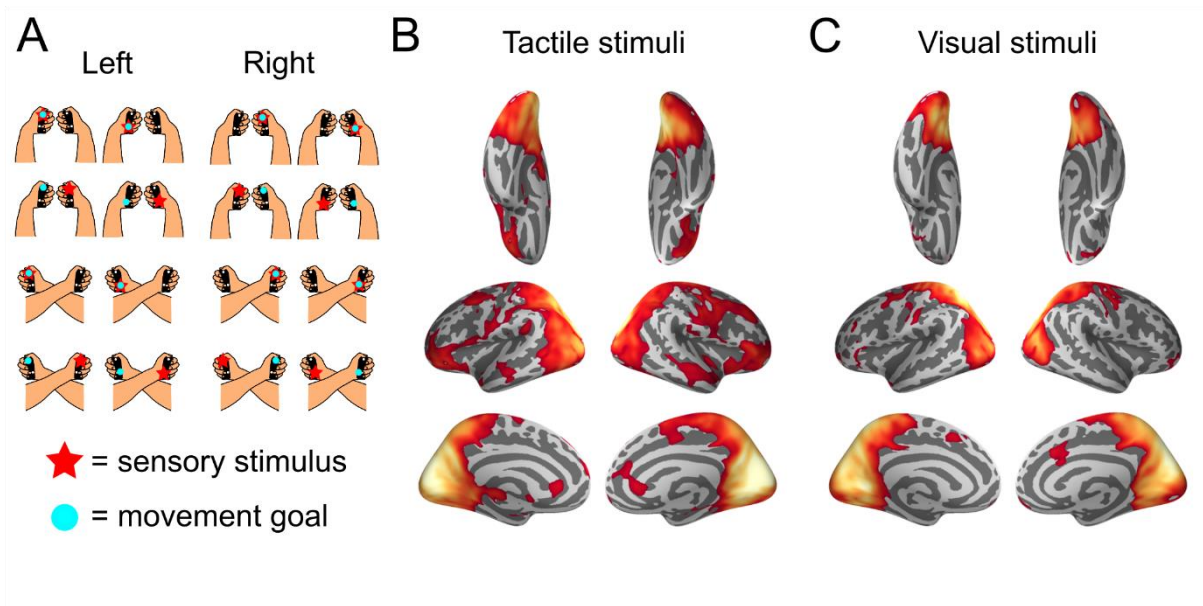


Figure 7. MVPA results for decoding final movement goal location, defined by tactile and visual stimuli (i.e. planning left vs right movement). **A** Overview of conditions organized for decoding movement goal location. **B** Whole brain searchlight results for decoding tactually defined movement goal location. **C** Whole brain searchlight results for decoding visually defined movement goal location. Searchlight maps show significant regions defined using a $p < .05$ FDR correction.

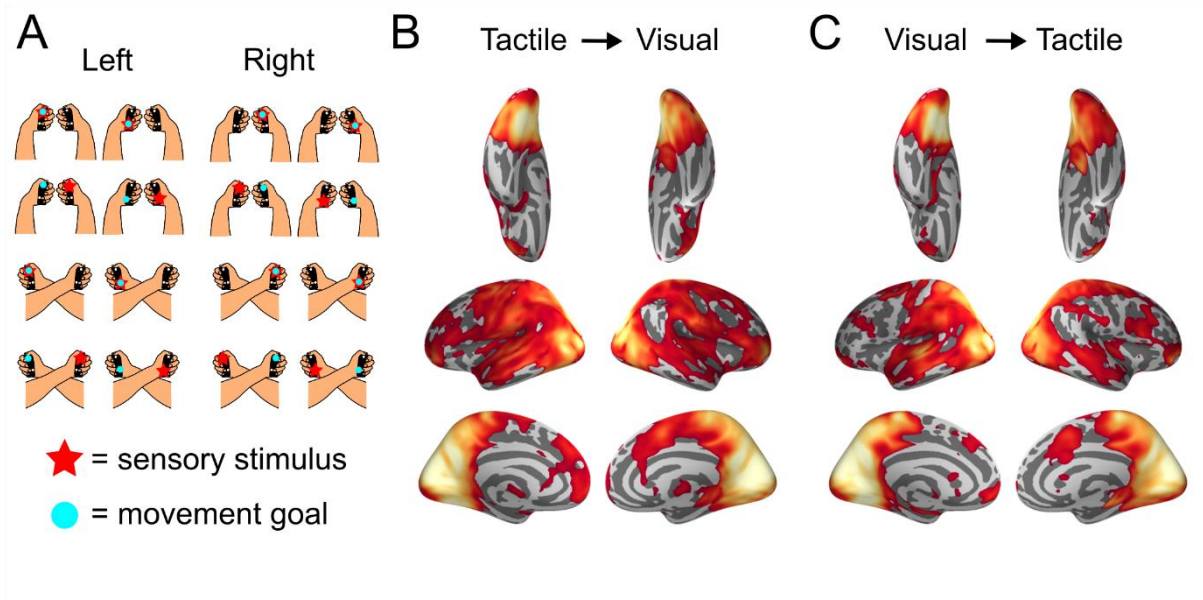


Figure 8. MVPA results for cross-classification of final movement goal location across stimulus sensory modality. **A** Overview of conditions organized for decoding movement goal location. **B** Whole brain searchlight decoding results for movement goal location, training on tactile stimulus trials and testing on visual stimulus trials. **C** Whole

brain searchlight decoding results for movement goal location, training on visual stimulus trials and testing on tactile stimulus trials. Searchlight maps show significant regions defined using a $p < .05$ FDR correction.

Task Rule. We tested whether we could decode the task rule (pro- vs anti-saccade) from any regions. Figure 9A gives an overview of the conditions organised to decode the task rule. Areas showing successful decoding of task rule for tactile stimulus trials are shown in Figure 9B and for visual stimulus trials in Figure 9C.

We further tested whether classifiers trained to distinguish task rule during tactile stimulus trials could decode the task rule during visual stimulus trials, or vice-versa. Figure 10 shows regions showing significant cross-classification of task rule across sensory stimulus modality.

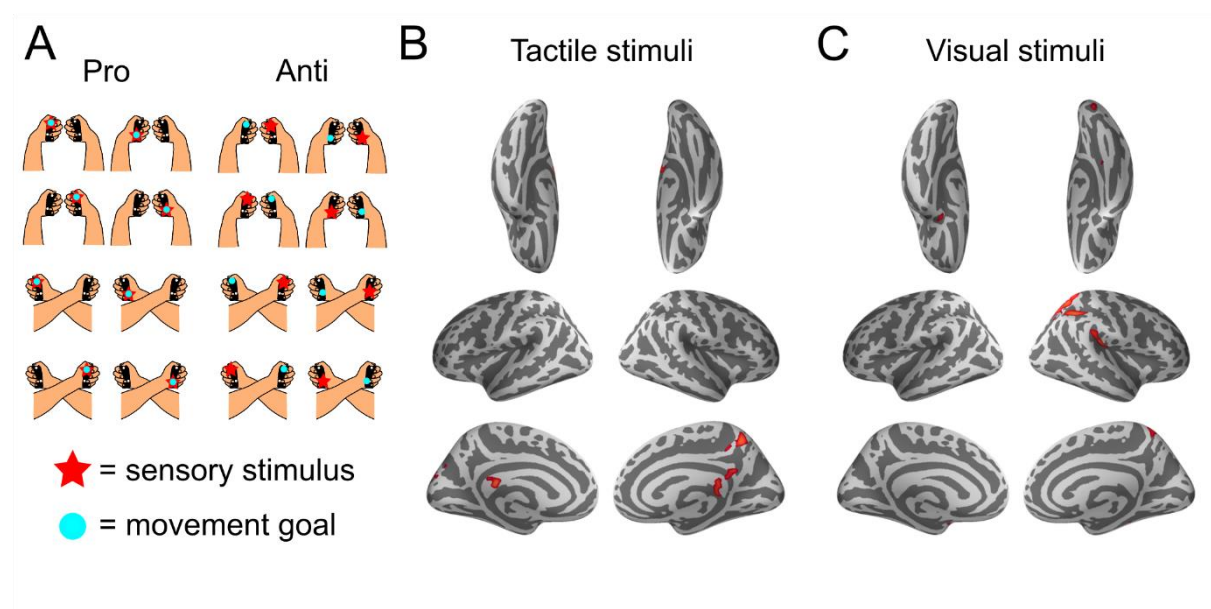


Figure 9. MVPA results for decoding task rule, defined by tactile and visual stimuli (i.e. pro vs anti movement). **A** Overview of conditions organized for decoding task rule. **B** Whole brain searchlight results for decoding task rule for tactile stimulus trials. **C** Whole brain searchlight results for decoding task rule for visual stimulus trials. Searchlight maps show significant regions defined using a $p < .05$ FDR correction.

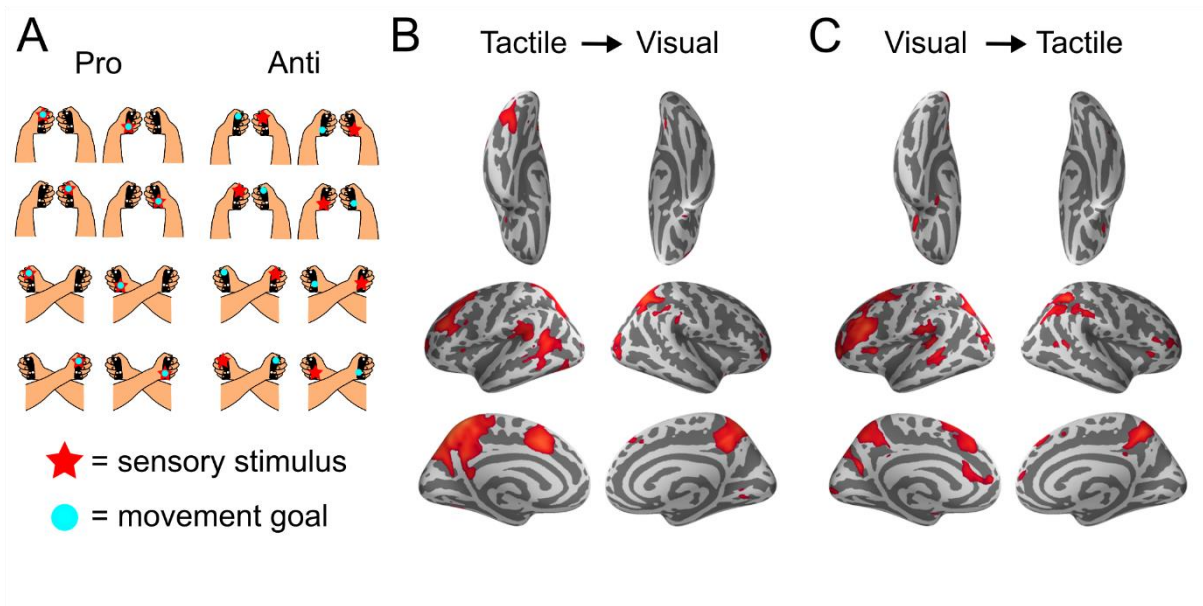


Figure 10. MVPA results for cross-classification of task rule across stimulus sensory modality. **A** Overview of conditions organized for decoding task rule. **B** Whole brain searchlight decoding results for task rule, training on tactile stimulus trials and testing on visual stimulus trials. **C** Whole brain searchlight decoding results for task rule, training on visual stimulus trials and testing on tactile stimulus trials. Searchlight maps show significant regions defined using a $p < .05$ FDR correction.

Discussion

In the present study, we studied how a tactile stimulus on the hands is spatially represented in the cortex and how this representation is dynamically remapped when subjects have to execute an ocular saccade towards or away from the stimulus' spatial location. We directly compared in the same task paradigm tactile and visual coding so as to contrast the cortical networks involved in visuo-motor and somatomotor transformation. Additionally, we manipulated hand posture, comparing crossed and uncrossed hand configurations in order to distinguish between coding of sensory stimuli in an anatomical reference frame (i.e. relative to the hands) and external-spatial reference frame (i.e. relative to the position in external-space). We could decode the anatomical location of tactile stimuli (i.e. left vs right hand, regardless of side of space) from bilateral somatosensory, motor and insula cortex, and right interior frontal cortex. External-spatial tactile stimulus location (i.e. left vs right side of space, regardless of stimulated hand) could be decoded from a medial region extending from posterior parietal to occipital cortex, as well as from bilateral regions in the lateral occipital cortex. Visual stimuli could also be decoded in an external-spatial reference frame from these regions, and classifiers trained to decode external-spatial tactile location could generalize to decode external-spatial visual location, and vice versa, suggesting a shared multisensory coding. Once the final saccade movement goal was specified, a broad network of occipital, parietal and frontal brain regions encoded movement target location, in a common coding regardless of whether the movement target was tactually or visually defined.

Specific anatomical coding identified during early sensory processing of lateralized tactile stimulations

We identified lateralized cortical univariate activations in response to tactile hand stimulation in secondary somatosensory central regions, matching with a coding of this tactile information in an anatomical frame of reference, in agreement with expectations based on previous experiments (Chen et al., 2008; Huang et al., 2012). No regions showed differences in univariate activation between left vs right external-spatial tactile stimuli, and no regions showed differences in univariate activation for left vs right anatomical or external-spatial visual stimuli.

The absence of activations to visual stimuli in external frame of reference could be due to the weakness of the stimulus which is a red LED light flashing for 200ms whereas classical studies working in visual field used wider visual stimuli (Roger B. H. Tootell et al., 1998). Likewise, we did not find any specific activation between saccade planning to the left vs right, or to pro- vs anti-saccade preparation during the movement planning interval. Using multivariate statistics, a broad network of occipital, parietal and frontal brain regions was shown to encode movement target location, in a common external coding regardless of whether the movement target was tactually or visually defined, including frontal eye field, intraparietal sulcus and dorso-lateral prefrontal cortex (DeSouza et al., 2003; Furlan et al., 2016).

Different cortical regions code for tactile stimuli in anatomical and external reference frame

We used MVPA to decode both anatomical and external locations of tactile stimuli on the hands. We identified bilateral somatosensory, motor and insular cortices, and right interior frontal cortex that encoded tactile information in an anatomical reference frame. We identified a large region extending from posterior parietal to occipital cortex, and from bilateral lateral occipital cortical regions that encoded tactile information in an external-spatial reference frame.

Shared cortical network for coding tactile and visual stimulation in an external reference frame

We found a cortical network that coded for both tactile and visual stimulus location in an external reference frame. Indeed, a decoder trained on tactile information can be employed to decode visual stimulation from these regions, and vice versa. This successful cross-classification strongly suggests a shared spatial representation of visual and tactile location in the same reference frame in parietal and occipital cortex. Our result is in agreement with the hypothesis of a somatomotor remapping, whereby the tactile location of stimulus is remapped into an external location closely associated with the corresponding visual stimulus. Decoding anatomical locations for visual stimulation was not possible. This is most probably due to the fact that the transformation from visual to tactile external to anatomical frames of references was not required in the task performed by the subjects.

PPC codes for anatomical and external reference frame in a gradient-like organization

In a previous study, Klautke et al., (2022), showed that anterior PPC (covering somatosensory cortex and SPL) coded for anatomical reference frame whereas posterior PPC (covering medial IPS) codes for external reference frame. Our results are congruent with these previous observations, and the view of a gradient-like organization of PPC holding anatomical representation of tactile information more anteriorly and external representation more posteriorly.

Automatic transformation or post-hoc reconstruction

Our results showed that external location of a tactile stimulus is expressed early on during sensory processing interval and before movement planning phase. A study from (Klautke et al., 2022) also described both anatomical and spatial coding during sensory interval in a hand pointing task. This is in contrast with a previous study from Maij et al., (2020) which hypothesized a post-hoc reconstruction on demand. The precise dynamics of remapping has been assessed by Maij et al. (2020) using a temporal order judgement protocol (TOJ). Their results suggest that, rather than being automatically remapped

from an anatomical to external frame of reference, the external location of the tactile stimulus is built on demand according to the limb chosen to perform the motor action. In this experiment protocol, participants were asked to point to the spatial location of the first tactile stimulus among two consecutive stimuli delivered on both hands under dynamic bimanual posture change (crossing, uncrossing). In other words, participants started trials in a given hand posture (cross or uncross) and moved their hand to end up in a different or in same posture. Once the trial over, they had to lift up the perceived first stimulated finger and point to the perceived location of stimulation. The authors observed that when the second stimulated hand was incorrectly chosen, the pointed location was the location of this hand at the moment of the presentation of the first stimuli (on the other hand). In other words, they pointed towards a location where no tactile stimulation occurred. This result thus questioned the classical view of automatic reconstruction of tactile stimuli in space and supported a time reconstruction hypothesis in which stimulus time and history of body posture are integrated to reconstruct spatial position afterwards.

A first hypothesis possibly accounting for such a difference is the influence of the context. As participants were instructed to perform a motor action, remapping could be computed directly after tactile perception to prepare the action. It is possible that post-hoc reconstruction phenomenon only exist in pure sensory contexts when motor action is not planned prior to sensory stimulus presentation. In the temporal order judgment task from (Maij et al., 2020), participant did not have to respond by pointing to an external location but just raised the stimulated finger. This is in line with the observation by Bencivenga et al., (2023) of different fronto-parietal networks in a visuomotor task depending on the effector to be used. It could be interesting to run an experiment which tests two different effectors intermixing ocular saccades and hand pointing towards tactile stimuli. Another possibility is that the number of possible targets affects the remapping process. In our task, the number of targets is limited to four locations whereas the targets are more spread along a continuous distribution in Maij et al., (2020) experiment. It is possible that participants quickly learn the spatial positions of targets for both hand postures and construct external representations in the beginning of each run which they keep and use for each trial without requiring to compute a new tactile to external remapping for each trial. This implies a learning effect that quickly disappears once participants master the task. This would thus indicate that the process involves a selection process between already established spatial locations rather than dynamic coordinate transformations. If true, it may be possible that number of targets is limited by memory and if the number of possible targets exceed the memory threshold, then the system shifts to relying on dynamic coordinates transformations to localize tactile stimulus position. Comparing early trials in our experimental design to later trials might also help disambiguate this possibility.

Conclusion

Altogether, our results demonstrate a multisensory, external-spatial coding of tactile and visual stimuli in posterior parietal and occipital cortex. This transformation is computed at early process stages of

sensory perception. This remapping is performed by a large network composed by areas in frontal, parietal and occipital cortical areas and along anterior-posterior axis along the PPC. It would be extremely interesting that future experiments concurrently manipulate sensory modality (visual vs. tactile), response effector (eye, hand, foot), body configuration (aligned, crossed, midline, or more complex configurations) as well as task instruction (perceptual judgment, direct reach to sensory stimulation, or more complex effector to body interaction).

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Competing interests

The authors declare that they have no competing interests.

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General Discussion

The perception of the world and the interaction with it relies on complementary multisensory processes. We first sense it through independent sensory modalities outside of the brain, in the peripheral nervous system. It is within the central nervous system that the projections of the senses are integrated and the percept is built thanks to cross-modal interactions. Thus, to have a precise understanding of brain functions, it is important to characterize the neural basis underlying both unisensory and multisensory processing. Cortical areas have been highlighted to carry a majority of the sensory and cognitive functions but less has been investigated concerning the role of subcortical areas, classically described as sensory relays. Among them, the pulvinar nucleus of the thalamus is a good candidate to play a role of cortical areas modulator (Arcaro et al., 2018a; Benarroch, 2015a; Cortes et al., 2023; Froesel et al., 2021a; Saalmann et al., 2012a). Indeed, the pulvinar is highly connected to the cortical and subcortical domains (Arcaro et al., 2015a; Cappe et al., 2009b; Froesel et al., 2021a; Guedj and Vuilleumier, 2020a) and is implicated in various cognitive functions such as attention, visual saliency, social processing, multisensory processing (I. Arend et al., 2008; Benarroch, 2015a; Froesel et al., 2024b, 2024a; Kaas and Lyon, 2007a; Vittek et al., 2023). Several recent studies showed its role in face processing in both macaques (Froesel et al., 2024b; Nguyen et al., 2016) and humans (Arcaro et al., 2015a; Wen et al., 2023) with different results. Froesel et al identified the ventral part of the lateral pulvinar, Arcaro et al identified ventral part of medial pulvinar and Wen et al identified ventral and dorsal parts. The pulvinar is thought to be a hub for modulation of cortical activity due to its large functional connectivity with the brain (Arcaro et al., 2018a; Barron et al., 2015a; Cortes et al., 2023; Eradath et al., 2021; Froesel et al., 2024a; Guedj and Vuilleumier, 2020a; Saalmann et al., 2012a). The pulvinar has been divided in subnuclei according to cytoarchitectonic parcellation (Anne Morel et al., 1997) or functional properties (Guedj and Vuilleumier, 2020a). However, how the pulvinar adjusts its connectivity according to the ongoing task is still less described. Additionally, its role in multisensory processing has been investigated mainly in audio-visual studies showing integration properties for complex stimuli like faces and vocalizations (Froesel et al., 2024b; Vittek et al., 2024, 2023). Few studies described its properties in visuo-tactile context. A clearer view of its functions, properties, intrinsic organization as well as connectivity profile with the rest of the brain is missing.

The perception of the environment is often linked with the production of a behavioral response, mainly motor action. In order to produce a motor action toward an external object, we have to align the spatial coordinates systems to the effector with the position of the object (Bencivenga et al., 2023; Cohen and Andersen, 2002; Klautke et al., 2022). For example, reaching a cup of coffee with our hand that we see in our environment requires a visuo-tactile transformation. Reaching or looking at an itch or insect bite on our arm require a soma-motor transformation. This transformation process is carried out by the PPC

(Azañón et al., 2010; Bolognini and Maravita, 2007; Ruzzoli and Soto-Faraco, 2014). For the somato-motor transformation with visual effector (saccade), the tactile position firstly coded in anatomical coordinates has to be coded in external spatial frame (eye's reference frame). Several questions remain, do anatomical and external locations of the same target shares the same neural substrate? Are both anatomical and external representations kept during the motor action or the external replace the anatomical one not needed to perform a saccade? In order to study such questions, we used a paradigm where subjects crossed their arms. Such body configuration allows to create a conflict between anatomical and external reference frames. A tactile stimulation on the left hand will be coded anatomically on the left but spatially on the right. A recent study in which subjects had to point toward tactile stimuli on the foot, the authors showed that tactile stimulation representations in anatomical and external reference frames are carried out by different areas within the PPC. SI and SPL for anatomical coding and mIPS for external coding. They also showed that anatomical reference frame is not retained after movement instruction phase, only the external code for movement goal was kept (Klautke et al., 2022). We investigated the same process with tactile stimulation on the hands instead of feet and saccade (different effector) toward it.

The aim of this thesis was to investigate the perception-action cycle through the implication of the pulvinar, firstly, in unimodal perception of high order stimuli, faces and voices and low order tactile stimuli. For tactile processing, the objective was to question the existence of a topographical organization, to do so we stimulated multiple body parts (face, hands, feet). Secondly, its implication in multisensory visuo-tactile face detection and discrimination. Thirdly, we investigated the pulvinar as a hub for cortical modulation by characterizing pulvino-cortical functional connectivity at rest and how it dynamically adjusts according to ongoing task. We investigated the neural basis underlying dynamic somato-motor transformation to produce saccades toward tactile stimulus. To assess these questions, we designed and recorded several fMRI tasks in humans (4 for pulvinar and 1 for somato-motor transformation) and one fMRI in NHP for somato-motor transformation (recorded, not shown in the manuscript).

In the first chapter, in study 1, we showed the pulvinar role in face and tactile but not for voice processing. We showed that it doesn't follow a somatotopic organization for tactile processing. We showed a dynamic pulvino-cortical connectivity of the medial part of the medial pulvinar with different cortical areas according to the stimulated body part. We notably showed that during face processing the pulvinar connects with the visual ventral pathway and disconnects with low-level visual areas. In study 2, we identify the functional connections between pulvinar subdivisions at rest and its dynamic changes modulated by face, voice and tactile processing. We specified different functional connectivity patterns for each subdivision, based on its connectivity at rest and during face, voice and tactile processing: Anterior part for somatosensory, motor, lateral part for complex visual, inferior part for visual, auditory

and default mode network and medial part for several modalities. For the latter, we provided a precise description of functional connectivity for 8 subdivisions. In study 3, we showed that pulvinar is probably recruited along attention network area to enhance sensory processing, we still are under investigation to determine if it show integration properties. In the second chapter, we showed that anatomical and external representation of a tactile stimuli are carried by different areas and that the somato-motor transformation of anatomical tactile to external reference frame to execute a saccade is probably performed during sensory processing of tactile stimulus.

In chapter I, we investigated the role of pulvinar in face, voice, tactile and visuo-tactile face detection and discrimination processing. We asked the following questions, which subdivision(s) of the pulvinar is(are) carrying face, voice and tactile processing in unimodal context? How is the pulvinar functionally connected to the rest of the brain depending on ongoing task? Is the pulvinar showing multisensory processing for visuo-tactile face detection and discrimination?

Which subdivision(s) of the pulvinar is(are) carrying face processing?

In chapter I, study 1, . along with previously identified pulvinar face clusters in both human and macaques, our results showed an activated cluster of voxels in the right ventro-medial part of the pulvinar. Activation clusters identified in MRI studies are schematically represented in figure 6. Anterior, inferior and medial part of pulvinar (Froesel et al., 2024c), medial and ventral part of pulvinar (Arcaro et al., 2018b), ventral and dorsal part of pulvinar (Wen et al., 2023). Our results showing ventro-medial activation of the pulvinar for face processing are in line with the literature. The reason for this discrepancy between the different studies is likely attributable to methodological differences, including variations in experimental paradigm, the models used (human and NHP) as well as fMRI sequence parameters. For example, the resolution used in our study was higher (i.e. smaller voxels) than standard resolution which can reduce the signal to noise ratio and lead to smaller activation cluster or no cluster at all. In the third study, we identified a similar pulvinar activation cluster for the unimodal visual condition showing faces only. Both face activations of the first and third study were located in the ventral part of the right PuM. This reproducibility across experiment confirms the role of ventral PuM in face processing. The fact that we did not find activation cluster located dorsally questioned the specificity of dorsal part of the pulvinar for face processing.

Given these diverse activation patterns, it may be possible that pulvinar in humans is part of a visual object categorization network, probably by connecting with the visual ventral stream (Kaas and Lyon, 2007a) and enhancing or synchronizing cortical activities of face areas like it does at rest (Erath et al., 2021) or during attention selection processes (Greenwood and Ward, 2024).

Which subdivision(s) of the pulvinar is(are) carrying voice processing?

Surprisingly, in study 1, we did not observe activations for voice processing contrary to literature which reported activations in the medial and lateral pulvinar for complex and socially meaningful sounds (Froesel et al., 2024b; Gattass et al., 2005; Jafari et al., 2023; Vittek et al., 2023). Audio-visual multisensory properties of the pulvinar have been highlighted in NHP in both electrophysiological recordings and fMRI in previous studies (Froesel et al., 2024b; Vittek et al., 2023, 2024). In these studies, the pulvinar responded to both unimodal auditory and multimodal stimulations. The first study we presented in this manuscript was investigating brain activations in a voice localizer in humans. We observed logical activations in the identified voice patch system (Belin, 2006; Belin et al., 2000; Bodin and Belin, 2020; Formisano et al., 2008; Pernet et al., 2015; Poremba et al., 2004; Zhang et al., 2021). We did not observe activation in the pulvinar whereas we were able to detect another subcortical and close nucleus, the MGN. Pulvinar shares anatomical connections with auditory cortex (Froesel et al., 2021), possibly supporting a direct activation. Given the gating properties of pulvinar, it is possible that pulvinar is recruited to enhance auditory detection and discrimination in difficult situations (Isabel Arend et al., 2008; Bourgeois et al., 2020a; Desimone and Duncan, 1995; Pestilli et al., 2011; Saalmann et al., 2012a). Our main hypothesis is that the passive nature of the voice localizer task did not require the recruitment of pulvinar. A general role attributed to pulvinar is to enhance sensory perception when it is meaningful (Froesel et al., 2021a). The activation of pulvinar in audio-visual context has been tested in a task highly similar to visuo-tactile of study 3. This task has not been presented here, the same visual stimuli were used, lateralized auditory “beeps” were used instead of tactile air-puffs. The data has been recorded and the analysis started. It will be important to test the behavioral facilitation effect on reaction times and accuracy as well as auditory response of the pulvinar in unisensory and multisensory contexts. We expect unisensory activations of pulvinar in the medial and inferior pulvinar for their responses to simple auditory stimuli but not lateral as it responds to complex auditory stimuli (Froesel et al., 2021a; Ricardo Gattass et al., 1978; Magariños-Ascone et al., 1988a; Yirmiya and Hocherman, 1987a). Ross et al demonstrated multisensory processing in the pulvinar for high order audio-visual processing (speech processing) (Ross et al., 2022b). But not specific enough of pulvinar. Our auditory stimuli were not complex as they don’t carry social information, it may be possible that pulvinar is not activated for such low-level processing. It may be activated to perform multisensory processing such as integration and enhance cortical responses to the detection and discrimination of faces in the task.

Which subdivision(s) of the pulvinar is(are) carrying tactile processing?

The tactile activation is located to the medial part of the medial pulvinar, which partially match with the anatomic connections between anterior pulvinar and anterior parietal cortex in NHP (Delhay et al., 2018). Functionally, activations also partially matched previous results in humans (Charyasz et al., 2023) and electrophysiological recordings (Ricardo Gattass et al., 1978). We did not observe any topographical organization as the stimulations of face, hands and feet activated the same locations in the

medial part of the medial pulvinar. The pulvinar has been considered as a higher order nuclei, the fact that it doesn't represent body maps is not surprising (Benarroch, 2015a; Froesel et al., 2021a; Saalman et al., 2012a). Alternatively, the spatial resolution of our fMRI mapping was not sufficient to highlight a putative body map.

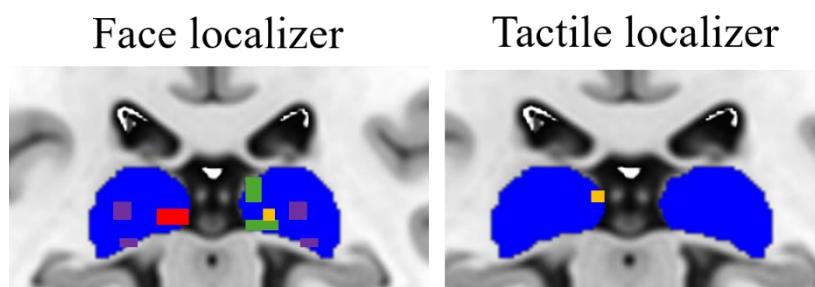


Figure 6: Schematic representation of fMRI pulvinar activity across studies for face and tactile processing, coronal section. Activation cluster for Face (left) and Tactile (right) localizers tasks. Blue: Left and Right Pulvinar (Morel atlas), purple: (Froesel et al., 2024c), red: (Arcaro et al., 2018b), green: (Wen et al., 2023), yellow: our activation cluster.

How is the pulvinar functionally connected to the rest of the brain depending on ongoing task?

In study 1, we saw that pulvinar was dynamically connecting with cortical areas specific to the ongoing task, visual face areas for face processing and sensorimotor areas for tactile processing. In study 2, we showed that pulvinar is functionally connected to different cognitive networks including, visual, sensorimotor, salience, fronto-parietal networks. We showed that subdivisions of the pulvinar (anterior, medial divided in 8, lateral and inferior) have specific functional connectivity with the cortex. Moreover, we showed that during face, voice and tactile perception, the pulvinar subdivisions increased their connectivity with cortical areas specific of the ongoing processing. The analysis revealed the possible organization according to multiple functional gradients for pulvinar where a dorso-ventral and antero-posterior and medio-lateral gradients have been identified in humans (Basile et al., 2024) and NHP (Froesel et al., 2024a). In the third study, our preliminary results suggested that areas belonging to the attention network (MFG, SPL, IPL, inferior and superior frontal giri) were activated with the pulvinar to enhance sensory processing. The results for the 3 studies showed that the pulvinar is implicated in face, voice, tactile, visuo-tactile and dynamically and specifically connected to the cortical regions important for these processes. This large implication reinforces the current view of the pulvinar as a hub for cortical processing.

Is the pulvinar showing multisensory processing for visuo-tactile face detection and discrimination?

In the third study, in order to investigate multisensory integration within the pulvinar, it was important to first observe behavioral enhancement for the multisensory conditions. In this paradigm, we expected that the spatial congruency conditions, the target face and tactile stimuli on the same side, would show a faster RTs than spatial incongruent. We did observe such a decrease in RTs specifically for the congruency factor. We observed the same decrease in RTs for the visual saliency factor, the salient (target face in color) condition showed lower RTs than non-salient (all faces in black and white). We confirmed the independence (with a random anova) of congruency and saliency factors for the behavior results. Thus, both congruent and salient conditions had an effect on RTs. Further analysis should model more accurately the multisensory interactions on RTs by using a race model of cumulative RTs distribution functions like proposed in (Stevenson et al., 2014). Additionally, accuracy of correct responses (not shown) should be investigated by using methodology described in (Stevenson et al., 2014). To do so, it would be interesting to compute the predicted multisensory response rate and see if the actual response rate is greater, which would account for a integration process, as in (Stevenson et al., 2014). Our accuracy performance was close to 100% as the task was totally feasible, so it would be interesting to change the level of difficulty of the task to be able to use maximum criterion to assess multisensory integration in accuracy. In order to assess multisensory integration in fMRI analysis, we firstly observed 2 peaks of activity in the pulvinar for the congruent salient condition. Two peaks were located in the ventral part of the PuM, one posterior and one more anterior. The activation in PuM during multisensory task is congruent with literature (Cappe et al., 2009b; Froesel et al., 2024a, 2021a; Ricardo Gattass et al., 1978; Vittek et al., 2023). According to the described antero-posterior gradients, the anterior peak could be associated with tactile and motor processing (Delhay et al., 2018; Froesel et al., 2021a; Ricardo Gattass et al., 1978) and posterior more to visual processing (Froesel et al., 2024a). This needs further investigation to be confirmed. The next step would be to analyze the percent signal change (PSC) and maximum criterion to find additive or subadditive effects would be a first step to test. Basically, we will compare the PSC in the two identified clusters for multimodal congruent and incongruent conditions as well as salient and non-salient in unimodal and multimodal conditions. Integration mechanisms could be inferred if the PSC is higher in multimodal condition than unimodal. Then it would be interesting to compare congruent and incongruent conditions, we expect higher PSC for congruent. Similarly for salient and non-salient, we expect higher PSC for salient. Nonetheless, additional methods could be explored too, fMR-adaptation, additive factors and BOLD dynamics (Stevenson et al., 2014).

To sum up, it is highly probable that pulvinar is contextually activated according to the ongoing task. For a low demanding task that requires low priority, the pulvinar may not be needed to play any modulation role. On the other hand, when the task has a high priority or saliency, the pulvinar may be recruited to enhance the perception and then general response (for example motor action). This has been assessed in visual face perception where a fast subcortical network composed of pulvinar and amygdala

has been proposed to work independently from the visual cortex to process fearful faces (McFadyen et al., 2017).

Interaction between pulvinar, salience and attention

In study 3 of chapter I, we assessed the interaction between pulvinar, salience and attention networks using a multisensory visuo-tactile task. Participants had to detect and localize a salient target face among 4 faces. Our purpose was to describe the pulvinar activations in response to multisensory and salient contexts. Figure 7 represents the activation profile of pulvinar we hypothesize regarding the multimodal and the salient condition. Figure 7 additionally presents our experimental observations, the top panel illustrates the predicted activations whereas bottom panel shows observed activations for pulvinar, salience and attention networks. We classified trials according to their difficulty. For the salient condition, salient target face is easier to perceive than non-salient. For the multimodal condition, the visuo-tactile congruent condition is easier to perceive than the incongruent which is easier than unimodal visual. This classification is supported by the RTs results showed in study 3. For the salience condition, we predicted to observe activation of cortical salience network as well as pulvinar activation given its important implication in salience processing (Benarroch, 2015a). We predicted a stronger and larger activation in salient congruent trials compared to incongruent and unimodal visual because of probable recruitment of pulvinar for both saliency and multisensory processing. In non-salient condition, we predicted activation of the cortical attention network as well as pulvinar for all multimodal and unimodal visual conditions. We predicted stronger and larger pulvinar activity for the most difficult type of trials (unimodal visual non-salient) where pulvinar would be recruited to support and potentialize attention cortical network. Previous studies showed the importance of the pulvinar in visual attention (Isabel Arend et al., 2008; Bourgeois et al., 2020b; Fiebelkorn et al., 2019a; Guedj and Vuilleumier, 2023b; Saalmann et al., 2012a).

The preliminary results of study 3 showed activations partially matching with the predictions (figure 7, bottom). Firstly, pulvinar activations for the multimodal congruent salient and the unimodal non-salient trials were confirmed. No pulvinar activity was identified for the other conditions. In the cortical domain, both attention and salience networks were activated for multimodal congruent salient trials. This result suggests an interaction between saliency and attention networks. In this study we also showed activation within the insula. Altogether, these results are in line with literature showing interaction between salience and attention networks via insula activation (Menon and Uddin, 2010b). The activation of pulvinar together with these two cortical networks enhances the view of pulvinar has a cortical modulator as part of a larger perception functional network. The pulvinar could play a role in shaping insula activity to enhance its response to salient stimuli. The activation of pulvinar for salient multimodal congruent condition and the absence of activation for incongruent and unimodal visual conditions could be due to an interaction between multisensory processing in pulvinar and salience (figure 7). The

pulvinar could process visuo-tactile information and perform multisensory processing, such as integration, rapidly before transmitting an excitatory output to the insula, thereby enhancing salience processing. The insula shares anatomical connection with oral and medial parts of the pulvinar which can be anatomical substrate for this transfer of information (Mufson and Mesulam, 1984). In addition, the medial pulvinar could play its role as it has been demonstrated to exhibit multisensory integration in monkeys (Froesel et al., 2024; Vittek et al., 2024, 2023). The pulvinar could share the same kind of interactions with areas of the attention network. It would be interesting to investigate functional connectivity of pulvinar subnuclei with the rest of the brain during the visuo-tactile task in order to describe the pulvino-cortical connection during ongoing task.

In this PhD work, we identified several activation sites within the pulvinar in different perception contexts. In study 1, both visual face processing and tactile processing showed pulvinar activation. In study 3, both unimodal visual non-salient and multimodal visuo-tactile congruent salient contexts showed pulvinar activation. Figure 8 illustrates the activation sites for each context. The pulvinar exhibited specific activation sites for different contexts. These results show the implication of pulvinar in multiple processes from low demanding sensory processing tasks to higher demanding tasks such as detection and discrimination of a visual target in multisensory context. Thus, it is crucial to identify a clearer segmentation and specialization of pulvinar to identify its fine-grained functional organization. This refinement is initiated in the second study of chapter I. In this study, we segmented the largest subnuclei of the pulvinar, the medial pulvinar, into 8 subdivisions and investigated their functional connectivity with multiple cortical networks and subcortical structures during rest and visual, auditory and tactile processing. We provided a large description of pulvinar subnuclei functional connectivity profile. This should provide support for future task-based experiments.

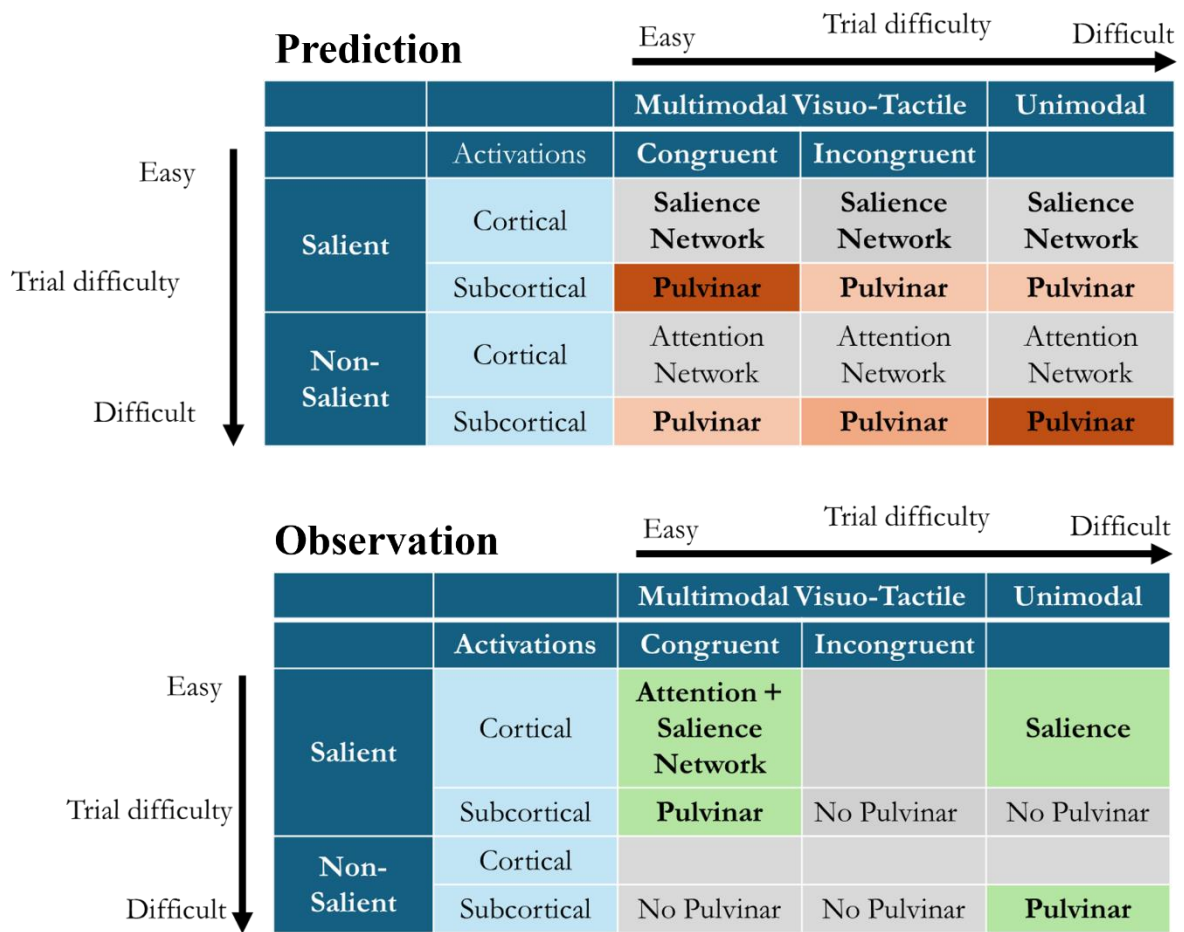


Figure 7: Pulvinar activation in visuo-tactile context (study 3, chapter I). Top Panel. Prevision of pulvinar activation. Orange: Gradient scale for prediction, darker means greater or stronger predicted activation, Gray: binary status of the activation (activation, no activation), Bottom panel.. Green: scale for observed activation, Gray: binary status of the activation (activation, no activation)

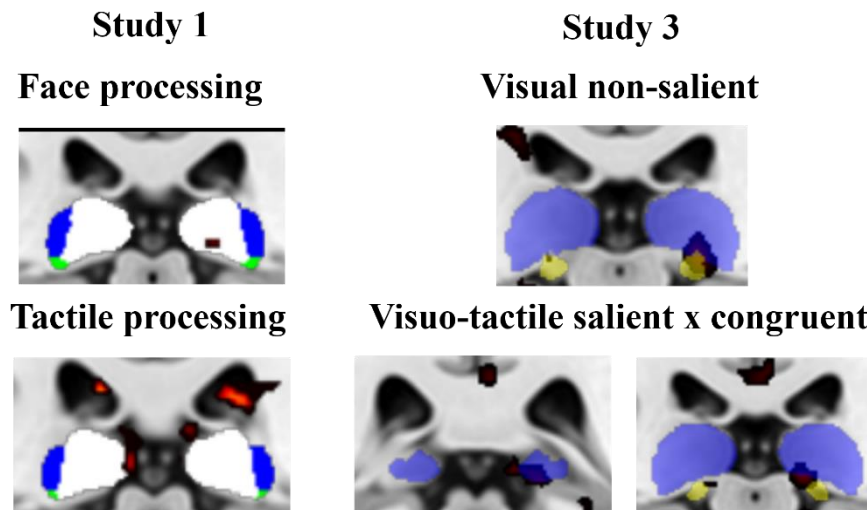


Figure 8: Summary of pulvinar activations sites across studies and contexts.

.Left Panel. White: medial pulvinar, blue: lateral pulvinar, green: inferior pulvinar, red: fMRI activation.

Right Panel. Blue: pulvinar, yellow: medial geniculate nucleus, red: fMRI activation

In chapter II, we investigated the neural substrate underlying somato-motor transformations, we asked two main questions: Anatomical and external reference frames share the same neural substrate? Are anatomical and external reference frames kept when motor action start?

Anatomical and external reference frames share the same neural substrate?

We showed that tactile stimulus was coded in an anatomical reference frame in bilateral somatosensory, motor, insular and right interior frontal cortices. We showed that the external location of tactile stimuli was coded in a medial region encompassing posterior parietal and occipital cortices. However, the identified regions lack in precision, mainly for external location.

Are anatomical and external reference frames kept when motor action start?

At this stage, we were not able to directly answer the question. Yet, our results suggested that the external location of the tactile stimulus is built automatically, contrary to another study suggesting a post-hoc reconstruction on demand (Maij et al., 2020). In fact, in our study, the external representation of the tactile stimuli was already present during sensory phase and before movement instruction phase, meaning that performing the coordinates transformation from anatomical to external reference frame does not require to plan the motor goal beforehand. However, this observation could be due to a contextual effect. As the subject are engaged in a task where they will perform saccades or anti-saccades, they may anticipate the preparation of the saccade already during the sensory phase. To answer this question, a third condition of no saccade would have allowed to determine if the subject anticipate the saccade preparation. Indeed, after sensory processing phase, the subjects would have to wait the

instruction phase to prepare a saccade, anti-saccade or no movement. Even with a contextual effect, we show that the brain can perform the sensory-motor transformation even early during sensory processing as conscious anticipation or not of the following motor action.

As a further analysis, it would be important to determine more precisely the regions coding for tactile and visual anatomical and external reference frames. Once done, we can assess the dynamics of remapping within these regions by comparing the sensory (visual or tactile) interval with the movement planning interval. In a previous study, (Klautke et al., 2022), showed that an overlap of activations within the same areas conducted these areas to code for both stimuli anatomical or spatial location and to task goal. They showed that these areas replace the sensory codes by the motor code. We can hypothesize that the same process occurs in our task which required saccade after tactile or visual stimulation when their task required hand pointing after tactile stimulation.

We know that effector selection differently shapes functional connectivity of fronto-parietal areas in visuo-motor task (Bencivenga et al., 2023). The IPS subregions effective connectivity with frontal areas changes given saccade or hand, foot pointing is used as effector. As a long-term perspective, it could be interesting to design an experiment in which subjects have to use several effectors with at least eye-saccade, pointing and reaching under crossed body configuration. Comparing different effectors coding for the same or different stimuli within the same task could allow us to determine which effectors share the same reference frames and if this is carried out by the same structures. It would be also interesting to test whether the same effector reference frame is always carried out by a structure or if this could be dynamically changed. For example, we saw that external location was coded in PPC and occipital cortices for a saccade. Would it be carried by MIP for a reaching movement?

Finally, the second study in NHP fMRI, which has not been presented in this manuscript, has been recorded and is work in progress. In this study, macaque monkeys performed a similar task as the humans, they performed saccades toward visual or tactile stimuli under cross or uncross arms configurations. This study will allow us to compare human and macaque species and to link similar or different areas to anatomical or external coding. This will allow us to compare if the same dynamics occur in both species. Finally, fMRI will allow us to identify important areas for somato-motor transformation at large-scale brain activation that could be used to determine regions of interest for further electrophysiological recording.

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Annexes 1 : Macaque detection and tracking using deep learning methods

Study 5 : Automatic Detection of Laboratory-housed Macaques

This project is a side project developed concomitantly to the two main projects presented in chapter I and II. The aim of the project was to develop an automatic tool to detect, track and categorize behaviors of housed-lab monkeys. This tool has the purpose of filling the gap of automatization of ethological monitoring being often made manually. The consequences are reduced reproducibility between scientists and lab as well as within the same scorer. Thus, uniformization and reduction of time for ethological analysis is an urgent need. More such a tool has been developed with the future purpose to be deployed and generally used for animal welfare monitoring or study neural basis of primate cognition, such as ethological psychiatric macaques models. We developed a pipeline using markerless deep learning methods and tested different existing algorithms.

Automatic Detection of Laboratory-housed Macaques

Genevieve Moat^{1,+,*}, Maxime Gaudet-Trafit^{2,+}, Julian Paul³, Jaume Bacardit^{1,*}, Suliann Ben Hamed², and Colline Poirier^{4,*}

¹School of Computing, Newcastle University, Newcastle upon Tyne, United Kingdom

²Institut des Sciences Cognitives Marc Jeannerod, UMR5229, CNRS-Universite Claude Bernard Lyon I, Bron, France

³Free lance

⁴Biosciences Institute Centre for Behaviour and Evolution, Faculty of Medical Sciences, Newcastle University, Newcastle upon Tyne, United Kingdom

*g.j.moat@newcastle.ac.uk; colline.poirier@newcastle.ac.uk; jaume.bacardit@newcastle.ac.uk

+These authors contributed equally to this work

ABSTRACT

Despite advancements in video-based behaviour analysis and detection models for various species in scientific and safety contexts, there remains a lack of efficient models specifically designed to detect macaques in complex laboratory environments. To address this gap, we present MacqD, a modified Mask R-CNN model that robustly detects macaques in their home-cage under challenging scenarios such as occlusions, glass reflections, and overexposure to light. To evaluate MacqD against pre-existing macaque detection models, we collected and analysed video frames from 20 caged rhesus macaques at Newcastle University, UK. Our results demonstrate MacqD's superiority, achieving a median F1-score of 99% for frames with a single macaque in the focal cage—21% higher than the next-best model, SegNet. Generalisation tests on frames from another set of macaques yielded an F1-score of 95%, surpassing the two next-best models, which both achieved the same F1-score, by 17%. Furthermore, MacqD achieved F1 scores of 90% for frames where macaques were paired in the focal cage and 81% for frames involving a different macaque set. To further improve detection performance, we applied a tracking algorithm to the detected results, which led to an enhanced F1-score of 0.90 when tested on videos from a different facility. This study highlights MacqD's effectiveness in accurately detecting macaques across diverse research settings and conditions.

1 Introduction

Non-human primates (NHPs) are crucial models for studying human cognition, neurobiology, and neuropathology due to their biological and cognitive similarities to humans¹⁻³. Rhesus macaques, the most commonly used NHPs in biomedical research^{4,5}, are often monitored to assess behaviour in fields such as neuroscience⁶ and animal welfare^{7,8}. Current methods for animal behaviour monitoring typically rely on either video recording or on-site observation, both of which are labor-intensive and prone to observer bias. These manual approaches limit the scalability of analysis, particularly for large datasets, and can introduce inconsistencies due to subjectivity across different observers' interpretations⁹.

Recent advances in deep learning offer scalable, efficient, and consistent tools for automating animal behaviour analysis from video footage^{10,11}. Accurate detection of animals in video frames is critical, as subsequent behaviour recognition depends on reliably identifying the animals.

Most research on automated animal detection focuses on small lab animals, such as mice and flies^{9,12}. However, detecting

NHPs like macaques presents unique challenges: (1) lab environments are often uncontrolled, with occlusion and lighting variations; (2) macaques' flexible joints create a wide range of body postures;

and (3) distinguishing individuals in overlapping scenes is difficult due to the lack of distinctive fur patterns.

Few studies have focused on automatically detecting macaques in laboratory settings. One study on tracking Japanese macaques used deep learning but was limited to wild environments. In contrast, research on macaques in controlled lab settings, such as visual hull reconstruction for behaviour observation¹³, achieved promising results but may not generalise to typical, more variable lab conditions. A recent development, SIPEC¹⁴, integrates segmentation, identification, pose estimation, and behaviour classification for primates, achieving a mean average precision (mAP) of 0.91 ± 0.03 for segmentation tested on 191 video frames.

In this study, we introduce MacqD, a Mask R-CNN-based model specifically designed to detect rhesus macaques in complex laboratory cages from video footage. A series of experiments were conducted to evaluate the robustness of the model. First, we assessed MacqD's detection performance in two conditions: a single macaque in the focal cage (Experiment 1) and pairs of macaques (Experiment 2). Its performance was compared to other existing models. In both experiments, after training the models (when appropriate) on images of specific individuals, we evaluated their performance on new data—both from the same animals (not used in training) and from different animals. In the third experiment, we tested whether incorporating a tracking algorithm would improve detection accuracy. Finally, in the fourth experiment, we assessed the model's ability to generalise by testing it on footage featuring paired macaques from a different institution. Together, these experiments offer a thorough evaluation of MacqD's performance, its generalisation to new individuals, challenging single and pair conditions, and its potential to adapt to different environments, such as footage from other institutions.

2 Materials and Methods

2.1 Data collection

A collection of video recordings, subsequently referred as *Macaque* data, were acquired at the macaque research facility of Newcastle University, UK, between 2014 to 2020. The facility complies with the NC3Rs Guidelines for 'Primate Accommodation, care and use'¹⁵, and comprises cages 2.1 m wide, 3 m deep, and 2.4 m high, exceeding the minimal requirement of the UK legislation, and where animals are housed in pairs. Besides the presence of a social partner, the cages are enriched by a multitude of structural elements and objects (e.g. shelves, swings, ropes) to promote the well-being of the animals. The recording of the data was approved by Newcastle University Animal Welfare and Ethical Review Body (project number: ID 928).

Video recordings from 20 focal macaques were collected using a remotely-controlled on-wall digital camera (Cube HD

Y-cam, 1080p and Axis M1065-L, 1080p), positioned in front of each cage. Data were stored in .mp4 or .mov formats, with a spatial resolution of 1280 x 720 pixels, and sampled at 15 frames per second. The number of macaques visible on the videos was variable. While by default the two cagemates were present, in some videos, one animal was temporarily absent (e.g. when it was in the experimental laboratory). Due to the positioning of some cages back-to-back, some videos also included non-focal animals in neighbouring cages. Examples of video frames are shown in Figure 1.

To further evaluate the model's generalisation capabilities, a video from the Institut des Sciences Cognitives Marc Jeannerod, referred to as the ISC dataset, was also used. The ISC dataset has a frame rate of 24 frames per second and a spatial resolution of 2880 x 2160 pixels, providing a different dataset from the *Macaque* data.

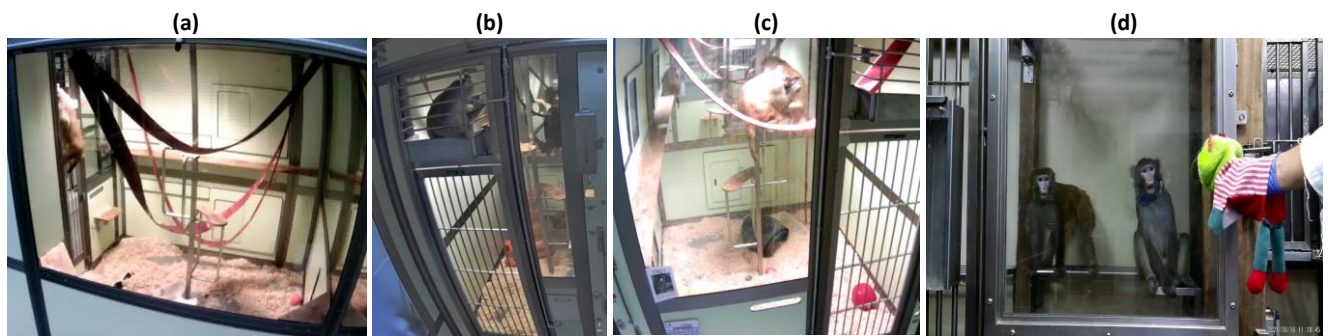


Figure 1. Example video frames used in this study. (a) A single macaque in the cage with side view (from the *Macaque* data); (b) A single macaque in the cage with cage railing occlusion and (from the *Macaque* data); (c) A pair of macaques in the cage, with cage enrichment occlusion and one macaque from a neighbouring cage appearing in the background (from the *Macaque* data); (d) A pair of macaques in the cage with a stimulus toy appearing (from the *ISC* dataset).

2.2 Data description

Macaque data were divided in several training and testing datasets (see table 1) to be used in Experiments 1 and 2. Experiments

1 and 2 only differ by the number of macaques in the focal cage, respectively one and two. For the training datasets, video recordings from 10 individuals (either alone or in pairs) were selected. The same individuals were used for both experiments. Video frames were pseudo-randomly selected from recordings spanning various dates and times of day, ensuring that the datasets encompassed a wide range of cage settings, macaque postures, and positions within the cage. This approach aimed to ensure the training dataset were representative of the macaques housed at the Newcastle facility.

For the testing datasets, five-minute videos were used for each individual in Experiment 1 and each pair in Experiment 2. In both experiments, two different testing datasets were employed: (1) the 'Same' dataset, comprising new video recordings of the same 10 individuals used in training, and (2) the 'Different' dataset, consisting of recordings from 10 new individuals. The 'Different' dataset was included to assess the generalisability of the models to macaques not encountered during training. The use of video data for testing is essential to mirror real-world scenarios, where behaviour recognition applications require dynamic information rather than isolated frames. However, because consecutive video frames are often highly similar in terms of macaque posture and position, this reduced variability between frames was addressed by increasing the number of frames in the testing datasets by a factor of 20 (see number of frames in Table 1).

Both the training and testing datasets included instances of occlusion, reflections, rapid motion, and overexposure. These diverse challenges were incorporated to enhance the comprehensiveness of the study and ensure that the analysis was conducted under realistic and varied conditions.

Additionally, a 17-second video from the Institut des Sciences Cognitives Marc Jeannerod (ISC dataset) containing 420 frames was used to further test the model. This video presented additional challenges, including occlusion, the presence of objects like toys used for stimuli, and human reflections on the glass.

Single or paired macaques	Training or test dataset	Same or different macaques as training dataset?	Number of frames
Single	Training	-	2,160
	Test	'Same'	45,000
		'Different'	45,000
Paired	Training	-	2,400
	Test	'Same'	22,500
		'Different'	22,500

Table 1. Description of our *Macaque* data sub-setting into different training and testing datasets. For paired animals (Experiment 2), the same videos were used to assess the detection of each pair member (see [Results](#) section for details).

2.3 Data annotation

Training and testing datasets were annotated using the VIA image annotator¹⁶. In the training dataset, each video frame was annotated with pixel-level masks, a technique known as segmentation, where each pixel is assigned to an individual macaque. For the testing datasets, including the ISC dataset, annotations were made with bounding boxes, with rectangles drawn around each macaque to include all body parts while minimising the box area. This approach was selected to facilitate the comparison of different algorithms, some of which output only bounding boxes (see [Performance Metrics](#)). Annotations were performed by ten different research assistants, with each annotation verified by at least one other assistant and subsequently validated by the first authors. In the training datasets, macaques visible in neighbouring cages were also annotated to maximise learning, whereas for the testing datasets, macaques in the background were excluded to focus on evaluating how well models detected animals in the focal cage.

2.4 Macaque detection algorithms

In this study, macaque detection was assessed using three different algorithms. The first two implemented a deep learning approach using Mask R-CNN as the framework, training a neural network through supervised learning (where the network learns from labeled examples). The third algorithm was based on background elimination, an approach that does not require any training.

2.4.1 Mask R-CNN

Mask R-CNN¹⁷ is a state-of-the-art framework widely used for instance segmentation, which extends Faster R-CNN¹⁸ by adding a mask branch to predict the instance mask. Mask R-CNN adopts a two-stage procedure to detect objects. In the first stage, Region Proposal Network propose candidate object region. In the second stage, features are extracted using RoIAlign from each Region of Interest (ROI) and the class label, bounding box offset and mask for each candidate object are produced

(Fig. 2). In Mask R-CNN, the classification of the ROI does not depend on the mask predictions. However, during training, Mask R-CNN utilises a multi-task loss on each sampled ROI, considering the classification loss, bounding-box loss and the mask branch loss. It is different to object detection methods such as Faster R-CNN since it distinguishes the instances in the image, and it gives richer information.

In this study, a modified Mask R-CNN model was created specifically for detecting macaques in video frames, incorporating SWIN¹⁹, a state-of-the-art transformer-based feature extractor. Throughout this paper, the term 'MacqD' refers specifically to this modified model. As a benchmark, MacqD was compared with SegNet, another variant of Mask R-CNN proposed by Marks and colleagues¹⁴, using ResNet101²⁰ as feature extractor.

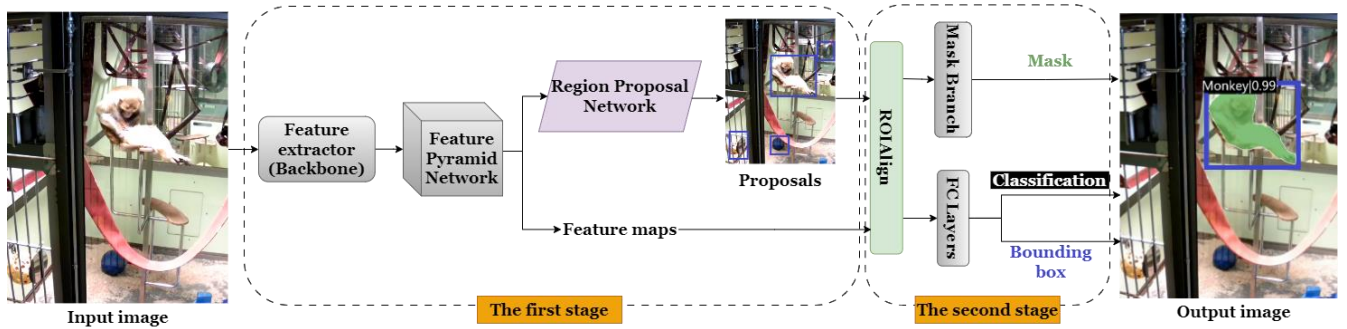


Figure 2. Overview of the Mask R-CNN framework.

2.4.2 Background Elimination (BE)

Unlike deep-learning techniques, which require extensive training and computational resources, the background elimination method is simpler to implement. In this research, an optimised Background Elimination (BE) pipeline was designed specifically for extracting macaques from video footage.

The pipeline (Fig. 3) takes a video as input and computes sub-background images at eight-second intervals. These sub-background images are amalgamated to create a unique background image for the entire video. Each frame of the video is then compared with this background image, using the MOG2²¹ algorithm in the OpenCV library²² to facilitate background elimination. Even when the background is successfully removed, noise around or holes within the foreground cluster may persist. To mitigate these imperfections, morphological operations are applied, including grouping adjacent pixels. A series of convex hulls are then calculated for each group of clusters, and overlapping convex hulls are merged. Ultimately, clusters smaller than 3000 pixels are filtered out, resulting in the isolation of the primary macaque cluster.

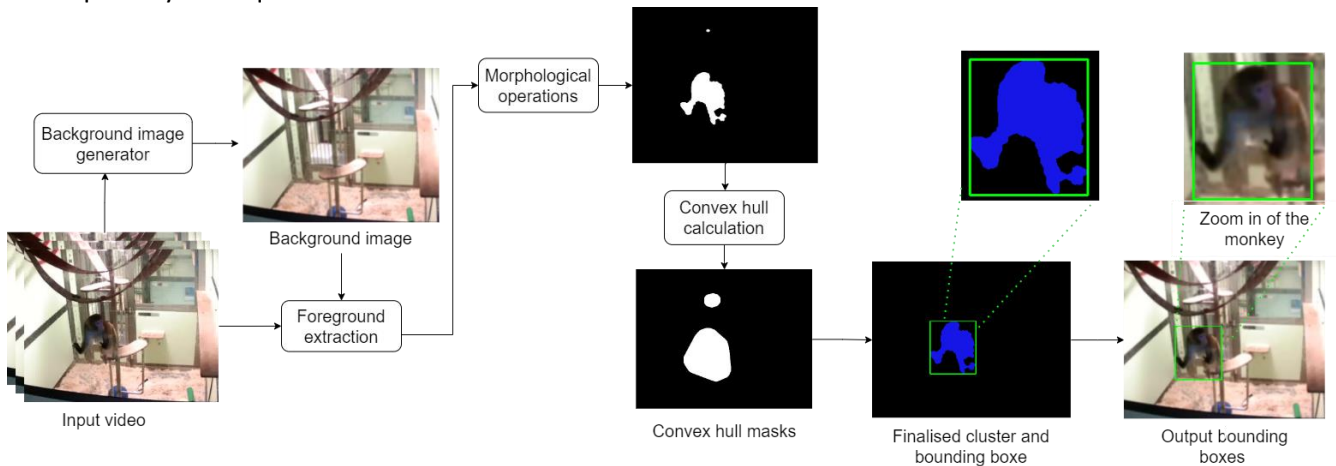


Figure 3. Overview of the background elimination pipeline.

2.5 Model comparison (Experiments 1 and 2)

Figure 4 illustrates the models used in two experiments: Experiment 1 tests models on single macaque recognition, and Experiment 2 on paired macaques. In Experiment 1, the MacqD model was trained on a dataset where a single animal was present in the focal cage (*Macaque Single* dataset) and compared with background elimination (BE) and three SegNet variants: SegNet - Primate, from the original paper¹⁴, trained on a primate dataset with macaque images from the authors' research facility; SegNet - Macaque Single, trained exclusively on our *Macaque Single* dataset; and SegNet - Primate + Macaque Single, which used SegNet - Primate as a starting point and was further trained with our *Macaque Single* dataset.

In Experiment 2, the MacqD model trained on the *Macaque Single* dataset was compared to two other MacqD derivatives: MacqD - Macaque Curriculum, which used MacqD - Macaque Single as a starting point and was further trained with the dataset featuring paired macaques in the focal cage (*Macaque Paired*); and MacqD - Macaque Combine, which was trained on a merged training dataset combining *Macaque Single* and *Macaque Paired* datasets. MacqD - Macaque Curriculum demonstrates curriculum learning²³, a strategy where models are trained by gradually increasing task complexity, mimicking human learning by starting with simpler concepts and progressing to more difficult ones.

Three SegNet variants trained on different datasets were tested in Experiment 1 (Fig. 4). However, due to poor results

(see [Experiment 1: detection of single macaques](#)), they were excluded from Experiment 2, where only MacqD models were compared to BE. Each model was tested on both the Macaque 'different' and, where applicable, the Macaque 'same' datasets (Fig. 4), with the final model selected based on the minimum validation loss.

Experiment 1 			
Model	Trained?	Tested with 'Same' dataset?	Tested with 'Different' dataset?
MacqD - Macaque Single	✓	✓	✓
SegNet - Macaque Single	✓	✓	✓
SegNet - Primate + Macaque Single	✓	✓	✓
SegNet - Primate	✓	✗	✓
BE	✗	✗	✓

Experiment 2 			
Model	Trained?	Tested with 'Same' dataset?	Tested with 'Different' dataset?
MacqD - Macaque Single	✓	✓	✓
MacqD - Macaque Curriculum	✓	✓	✓
MacqD - Macaque Combine	✓	✓	✓
BE	✗	✗	✓

Figure 4. Overview of experiments 1 and 2 illustrating how the models compared in this study differed in terms of training and testing datasets. 'Same' and 'Different' correspond to the datasets described in Table ??, with the labels referring to the fact that the model was tested with videos of individual macaques 'seen' during the training phase ('same') or not ('different') (see subsection [Data description](#) for more details).

2.6 Tracking algorithm

In computer vision, tracking algorithms monitor object movement across consecutive video frames by estimating the target object's states in subsequent frames, given its initialized state^{24,25}. In Experiment 3, results from detection models were compared before and after implementing a tracking algorithm to test performance improvement. The tracking algorithm aims to maintain detection continuity across frames by estimating the location and motion of macaques, thereby reducing instances where the macaque is not detected in subsequent frames.

After obtaining the bounding boxes and masks or clusters information in Experiments 1 and 2 for each tested video, the centroid of the detected segment or cluster is used as input to the Kalman filter²⁶ to update the object's position and predict its position in the next frame. If the macaque is not detected, the Kalman filter uses its previous location to estimate the current position, maintaining the same bounding box size as the previous frame with the corresponding centroid. The prediction process continues until a match is found between the prediction and detection in the current frame, or a maximum of 20 frames is reached. The Hungarian algorithm²⁷ associates the current detection with the prediction to determine the object to which this detection belongs (see Fig. 5).

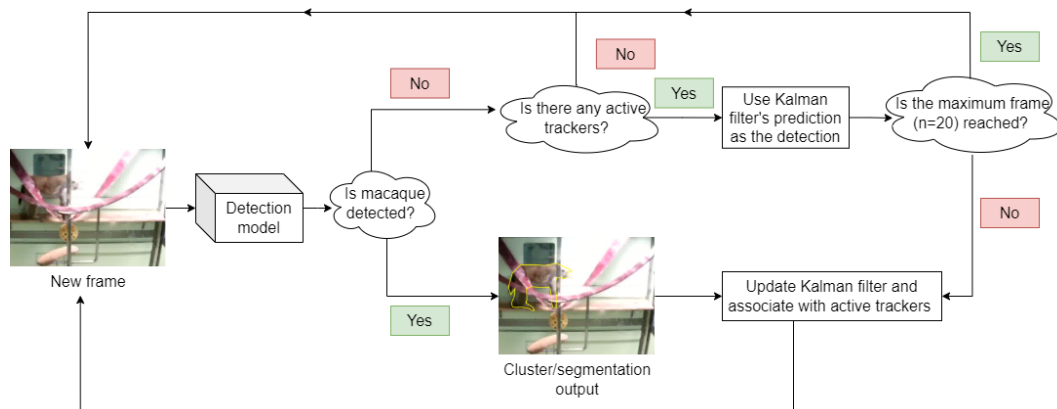


Figure 5. Tracking algorithm pipeline.

2.7 Performance Metrics

Bounding boxes were used for evaluation to standardise comparisons across all models. While both MacqD and SegNet (based on Mask R-CNN) provide pixel-based masks, bounding boxes were chosen for consistency, as BE only outputs bounding boxes. For the MacqD and SegNet models, each bounding

box was assigned a confidence score indicating the likelihood of correct identification. Typically, a confidence score threshold of 0.5 is used to filter out false positives, but in this study, the optimal threshold was determined to maximise performance metrics, specifically precision and recall. Confidence scores were assessed from 0.5 to 0.95 in 0.05 increments, and the median precision vs. recall was plotted to identify the optimal threshold (see supplementary material Fig. 10). For MacqD and SegNet, confidence scores were applied to bounding boxes to filter out false positives and optimise performance. Confidence scores were tested from 0.5 to 0.95 in 0.05 increments, and the optimal threshold was determined by plotting the precision versus recall curve (see supplementary material Fig. 10). The Intersection over Union (IoU) metric was used to measure the overlap between predicted and ground truth boxes, with an IoU of 0.5 or higher considered a true positive (TP) and below this threshold classified as a false positive (FP).

$$IoU = \frac{\text{Ground truth} \cap \text{Detected box}}{\text{Ground truth} \cup \text{Detected box}} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (1)$$

Performance was assessed using precision, recall, and the F1 score. Precision indicates how accurately the model identifies a macaque when it is present, recall measures the model's ability to detect all macaques present, and the F1 score is the harmonic mean of precision and recall. The metrics calculated in Experiments 1 and 2 are based on individual-level assessments. In both experiments, only the metrics for the focal macaque are considered, and the presence of macaques in the non-focal cage is not accounted for in the results. Furthermore, in Experiment 2 and 4, where paired macaques are present in the focal cage, the metrics are calculated for each macaque individually.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad (3)$$

$$F1score = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

2.8 Statistical test

The metrics precision, recall, and F1 score are non-independent, meaning changes in one metric can influence the others. Hence, statistical analyses were restricted to the F1 scores. Initial attempts to fit linear mixed-effects models indicated non-normally distributed residuals, violating parametric assumptions. Consequently, two non-parametric tests were utilized: the Friedman test compared the performance of multiple models across datasets, while the Wilcoxon (signed-rank) test assessed the performance of pairs of models. The Benjamini-Hochberg procedure²⁸ was employed to control the false discovery rate for multiple comparisons. Additionally, Wilcoxon tests were conducted to compare results before and after implementing the tracking algorithm.

2.9 Computing Environment

The MacqD models were implemented from the open-source framework MMDetection version 2.25²⁹, and SegNet from the open-source pipeline³⁰. We used Python (3.7.13), CUDA toolkit (10.1) and GPU-accelerated library (CUDNN 7.6.3) distributed on 2 GPUs, one is NVIDIA GeForce RTX 2080 Ti, the other one is NVIDIA GeForce GTX 1080 Ti. Python (3.6.13), CUDA (10.2.89) and CUDNN (7.6.5) were used to develop BE.

3 Results

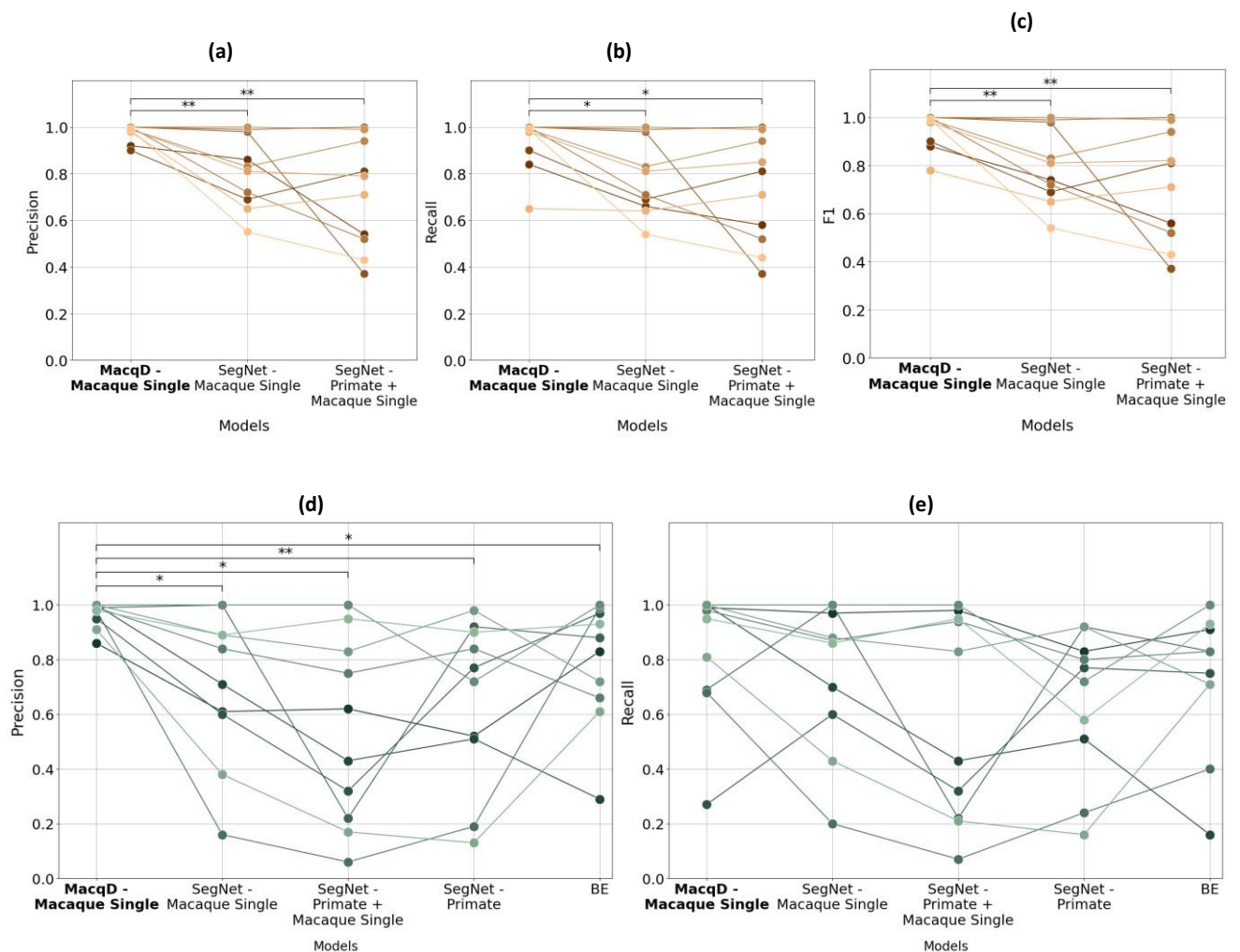
3.1 Experiment 1: detection of single macaques

Model comparison of videos featuring the same animals as in the training dataset ('Same' dataset) revealed a significant differences in precision, recall and F1 score [Friedman test: $\chi^2(2)=13.00$, p-value=0.002 for precision; $\chi^2(2)=10.16$, p-value=0.006 for recall; and $\chi^2(2)=12.79$, p-value=0.002 for F1 score; Fig. 6a, 6b and 6c]. Wilcoxon pair-wise tests revealed that MacqD - Macaque Single was significantly better than other models in precision ($n=10$, $T^+=0$, corrected p-value=0.008 for all comparisons), recall [$n=10$, $T^+=0$, corrected p-value=0.015 compared to SegNet - Macaque Single, and $n=10$, $T^+=3$, corrected p-value=0.021, compared to SegNet - Primate + Macaque Single] and F1 score ($n=10$, $T^+=0$, corrected p-value=0.008 for all comparisons). The performance of MacqD - Macaque Single was particularly high, with a median precision of 0.99, recall of 0.99, and F1 score of 0.99 (Table 2).

In evaluating videos featuring different animals than those used for the training dataset ('Different' dataset), significant differences between the models were found in precision and F1 score [Friedman test: $\chi^2(4)=14.56$, p-value=0.006 for precision; and $\chi^2(4)=10.57$, p-value=0.032 for F1 score; Fig. 6d and 6f], but no significant differences were found in recall (Fig. 6e). Wilcoxon tests revealed that MacqD - Macaque Single was significantly better than any other tested model in precision [$n=10$, $T^+=1$, corrected p-value=0.014 compared to SegNet - Macaque Single; $n=10$, $T^+=0$, corrected p-value=0.014 compared to SegNet - Primate + Macaque Single; $n=10$, $T^+=0$, corrected p-value=0.008 compared to SegNet - Primate; and $n=10$, $T^+=3$, corrected p-value=0.021 compared to BE] and significantly better

than SegNet - Primate + Macaque Single [$n=10$, $T^+=0$, corrected p-value= 0.031]. MacqD - Macaque Single also demonstrated notably high performance, achieving a median precision of 0.99, recall of 0.97 and F1 score of 0.95 (Table 2, see also the two first columns of Fig. 8).

In addition to performance metrics, training and inference times are crucial parameters for evaluating and comparing models. SegNet models required the longest training time (10 hours) and inference time (7.2 hours for a 5-minute video), followed by MacqD models (train: 4 hours; inference: 17 minutes for a 5-minute video). BE was the fastest overall, requiring only 20 minutes for inference for a 5-minute video. Considering both performance and time factors, only MacqD and BE were used for analysing the paired macaque datasets in the subsequent experiment.



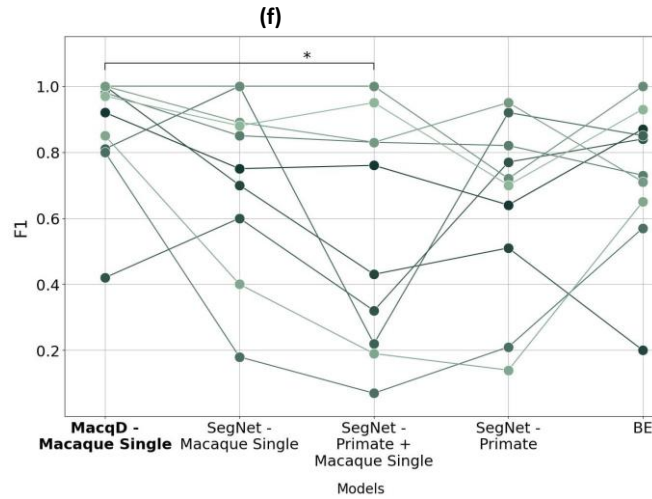


Figure 6. Model performance in Experiment 1, where the test datasets contain only a single macaque in the focal cage. (a), (b), and (c) represent precision, recall, and F1 scores, respectively, evaluated on datasets featuring the same macaques (but different videos) as the training dataset ('Same' dataset). (d), (e), and (f) represent precision, recall, and F1 scores, respectively, evaluated on datasets featuring macaques not featuring in the training dataset ('Different' dataset). Markers represent results for individual macaques, with lines connecting the markers across models to illustrate performance variations. Wilcoxon test: * $p < 0.05$, ** $p < 0.01$.

Model	Dataset	Precision	Recall	F1
MacqD - Macaque Single	'Same'	0.99	0.99	0.99
SegNet - Macaque Single		0.82	0.76	0.78
SegNet - Primate + Macaque Single		0.75	0.76	0.76
MacqD - Macaque Single	'Different'	0.99	0.97	0.95
SegNet - Macaque Single		0.78	0.87	0.78
SegNet - Primate + Macaque Single		0.53	0.63	0.60
SegNet - Primate		0.75	0.75	0.71
BE		0.86	0.79	0.78

Table 2. Median metrics of the different models compared in Experiment 1.

3.2 Experiment 2: detection of paired macaques

Model comparison of videos featuring the same animals as in the training dataset ('Same' dataset) revealed significant differences in precision, recall and F1 score [Friedman test: $\chi^2(2)=8.16$, p-value= 0.017 for precision; $\chi^2(2)= 6.53$, p-value= 0.038 for recall; and $\chi^2(2)= 7.59$, p-value= 0.023 for F1 score; Fig. 7a, Fig. 7b and 7c]. Wilcoxon pairwise tests indicated that the performance of MacqD - Macaque Curriculum and MacqD - Combine was significantly better than MacqD - Macaque Single in precision (n=10, $T^+= 6.5$, corrected p-value= 0.041; and n=10, $T^+= 0$, corrected p-value= 0.035, respectively), and in F1 score (n=10, $T^+= 2$, corrected p-value= 0.038; and n=10, $T^+= 3.5$, corrected p-value= 0.038, respectively). MacqD Curriculum and MacqD - Combine achieved similar results, with median F1 scores of 0.9 and 0.87, respectively. MacqD Curriculum had the highest recall of 0.87, while MacqD - Combine had the highest precision of 0.97 (see Table 3).

In the model comparison of videos featuring animals not featured in the training dataset ('Different' dataset), significant differences emerged between the models in precision, recall and F1 score [Friedman test: $\chi^2(3)=25.56$, p-value <0.001 for precision; $\chi^2(3)= 10.16$, p-value= 0.017 for recall; and $\chi^2(3)= 11.42$, p-value= 0.010 for F1 score; Fig. 7d, Fig. 7e and 7f]. Wilcoxon pairwise tests revealed that MacqD - Macaque Curriculum and MacqD - Combine performed significantly better than MacqD - Macaque Single in precision (n=10, $T^+= 4.5$, corrected p-value= 0.016; and n=10, $T^+= 0$, corrected p-value= 0.004, respectively). Additionally, these models significantly outperformed BE in precision (n=10, $T^+= 1$, corrected p-value= 0.006 compared to MacqD - Macaque Single; and n=10, $T^+= 0$, corrected p-value= 0.002 compared to MacqD - Macaque Curriculum and MacqD - Combine). Significant differences in F1 score were also noted, with MacqD - Macaque Single and MacqD - Macaque Curriculum outperforming BE (n=10, $T^+= 3$, corrected p-value= 0.041; and n=10, $T^+= 4$, corrected p-value= 0.041, respectively). MacqD - Curriculum and MacqD - Combine achieved very similar results, with median F1 scores of 0.81. MacqD - Curriculum had the highest recall of 0.72, while MacqD - Combine achieved the highest precision of 0.95 (see Table 3; see also the last two columns of Fig. 8).

Although the Friedman test revealed significant differences between the models in recall, the Wilcoxon pairwise tests did not find any significant differences in recall after p-value correction for both testing the 'Same' and 'Different' datasets. There were no significant difference between MacqD - Macaque Curriculum and MacqD - Macaque Combine in any of the metrics.

(a)

(b)

(c)

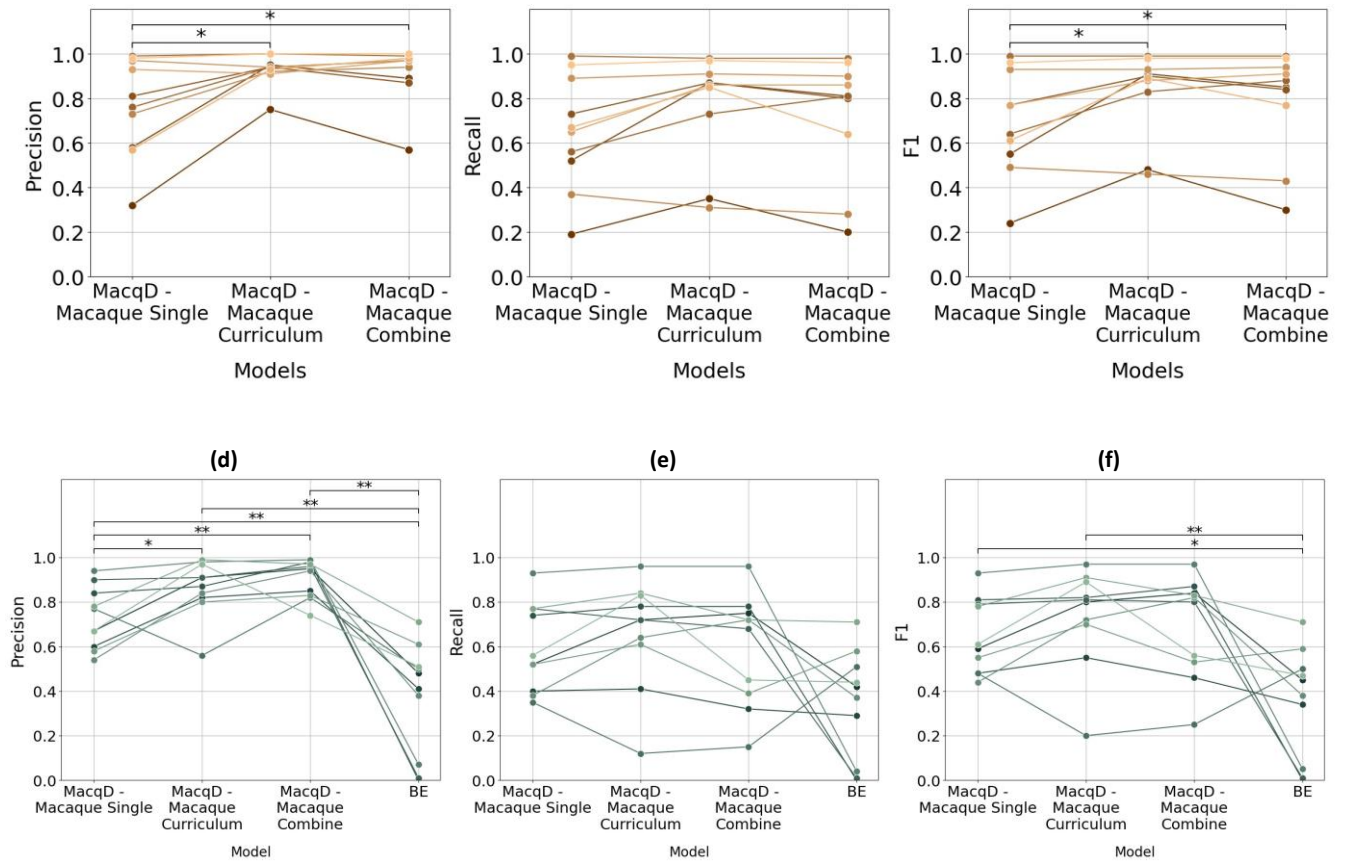


Figure 7. Model performance based on F1 scores in Experiment 2: Models tested on datasets with individual macaques paired in the focal cage. (a) F1 scores evaluated on datasets featuring the same macaques as the training dataset ('Same' dataset). (b) F1 scores evaluated on datasets featuring different macaques from the training dataset ('Different' dataset). Markers denote results for individual macaques, with lines connecting the same symbol across models to illustrate performance variations. Wilcoxon test: * $p<0.05$, ** $p<0.01$.

Model	Dataset	Precisio n	Recall	F1
MacqD - Macaque Single	'Same'	0.79	0.66	0.71
MacqD - Macaque Curriculum		0.94	0.87	0.90
MacqD - Macaque Combine		0.97	0.81	0.87

MacqD - Macaque Single		0.72	0.54	0.6
MacqD - Macaque		0.89	0.72	0.81
Curriculum MacqD - Macaque Combine	'Different',	0.95	0.7	0.81
BE		0.45	0.40	0.42

Table 3. Median metrics of the different models compared in Experiment 2

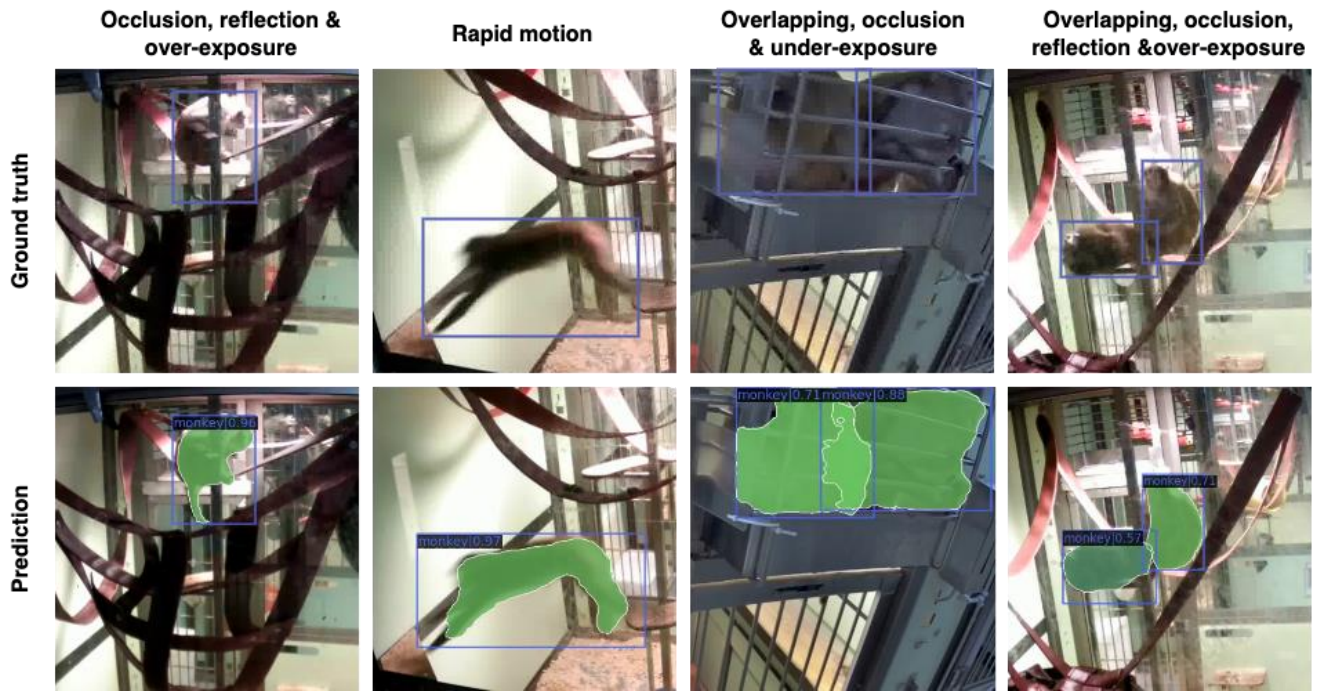


Figure 8. Comparison of ground truth versus predicted segmentations/bounding boxes for frames from unseen animals. The first two columns show MacqD - Macaque Single's predictions on frames featuring single animals not present in the training dataset (Experiment 1, 'Different' dataset). The third and forth columns show MacqD - Combine's predictions on frames featuring pairs of macaques not present in the training dataset (Experiment 2, 'Different' dataset).

3.3 Experiment 3: Impact of tracking algorithm on detection performance

To further evaluate potential improvements in model performance, a tracking algorithm was incorporated into the results from Experiments 1 and 2. In Experiment 1, the application of the tracking algorithm resulted in a statistically significant increase in recall for the background elimination (BE) model when tested on datasets with single macaques not included in the training dataset ('Different' dataset; Wilcoxon pair-wise test: $n=10$, $T^+=0$, $p\text{-value}=0.042$; Fig. 11b), where the recall improving from 0.79 to 0.82 (Table 5). In Experiment 2, the tracking algorithm led to a statistically significant decrease in precision for the MacqD - Macaque Single and MacqD - Macaque Combine models when tested on datasets with paired macaques included in the training dataset ('Same' dataset; Wilcoxon pair-wise test: $n=10$, $T^+=0$, $p\text{-value}=0.027$; and $n=10$, $T^+=0$, $p\text{-value}=0.042$, respectively; Fig. 13a). A similar decrease was observed for the F1 score with MacqD - Macaque Single in the same setting

(Wilcoxon pair-wise test: $n=10$, $T^+=0$, $p\text{-value}=0.016$; F1f. 13c). Although the metric decrease were not ideal. However, these decreases were minor, around 1-2%. (Table 6).

3.4 Experiment 4: performance on ISC dataset

To evaluate the generalization of the MacqD models, we tested MacqD - Curriculum and MacqD - Combine on a video from another facility (ISC dataset) featuring paired macaques. As shown in Table 4, both models achieved the same F1 score of 0.87. After applying the tracking algorithm, MacqD - Curriculum's F1 score improved to 0.90 (see Fig. 9 for example frame).

Model	Before tracking implementation			After tracking implementation		
	Precision	Recall	F1	Precision	Recall	F1
MacqD - Curriculum	0.89	0.86	0.87	0.91	0.89	0.90
MacqD - Combine	0.92	0.82	0.87	0.9	0.82	0.86

Table 4. Results of ISC dataset evaluation.

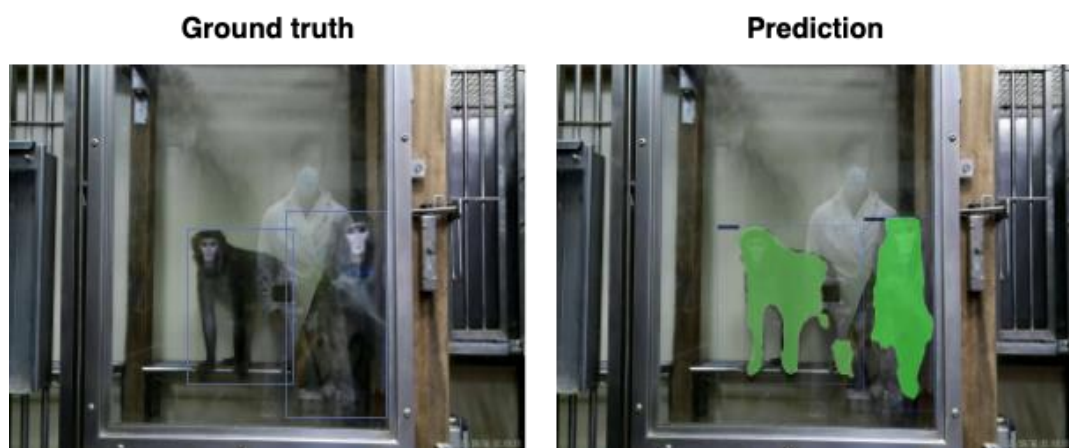


Figure 9. Comparison of ground truth versus predicted segmentations/bounding boxes for frame from another facility (ISC dataset) used MacqD - Curriculum.

4 Discussion

4.1 Summary of Key Findings

In this study, we introduced MacqD, a Mask R-CNN-based model for detecting rhesus macaques in complex laboratory environments from video footage. Our model effectively detected both single macaques and paired macaques in the focal cage (Experiment 1 and Experiment 2, respectively), and demonstrated generalisation to new individuals and cage settings within those datasets. Additionally, the model successfully generalised to footage from a different institution, indicating its adaptability to new environments and unseen video data.

4.2 Challenges

Numerous studies have explored animal detection and behaviour classification using deep learning^{10,31,32}. However, detecting non-human primates (NHPs) presents greater challenges than detecting smaller lab animals like rodents or *Drosophila*, due to their complex and varied body shapes. Research on automatic NHP detection in complex laboratory settings is limited. Many studies are conducted in settings with minimal occlusion, unlike real-world conditions where cameras outside the cages encounter visual obstructions such as bar occlusions, reflections, and light overexposure. Additionally, UK legislation requires large cages with enrichment (e.g., swings) and pair housing of NHPs, which introduces visual clutter and overlapping subjects, making detection more difficult. Our study addresses these complexities, providing insights into handling real-world challenges (see Fig. 8 for examples).

4.3 Interpretation of results

To evaluate our model and compare it with existing ones, we created novel datasets comprising video frames of rhesus macaques in laboratory settings, including frames with either single macaques (*Macaque single* dataset) or pairs (*Macaque Paired* dataset). We tested the models on datasets where the macaques were either the same as those in the training set ('Same' dataset) or different ('Different' dataset).

In Experiment 1, with the *Macaque Single* dataset, MacqD consistently outperformed other models in both precision and recall across 'Same' and 'Different' datasets, showing robust learning and generalisation. The performance drop in the 'Different' dataset was expected due to new macaques and cage settings. SegNet - Primate + Macaque Single had low precision (0.53) and recall (0.63) in the 'Different' dataset (Table 3), possibly due to the need for parameter tuning, though we used default parameters due to time constraints. For the Background Elimination (BE) models, high precision (86%) but lower recall (79%) was observed, due to difficulties in detecting static macaques, which were often misclassified as background, leading to missed detections.

In Experiment 2, with the *Macaque Paired* dataset, both MacqD - Macaque Curriculum and MacqD - Combine outperformed MacqD - Macaque Single, which struggled with overlapping macaques due to a lack of training on video frames where they appear overlapped. Curriculum learning, a deep learning approach where the model is trained on simpler datasets (*Macaque Single*) before progressing to more complex ones (*Macaque Paired*), was applied to improve performance. However, the results showed minimal differences between MacqD - Macaque Curriculum and MacqD - Macaque Combine, with one showing slightly higher precision and the other higher recall. This indicates that curriculum learning did not provide significant advantages in this study, likely due to the lack of a clear progression in dataset complexity. The BE models performed poorly in this scenario due to their inability to distinguish between closely positioned macaques, often merging adjacent clusters. SegNet was excluded due to its poor performance on the single macaque dataset and long training and testing times (see subsection [Model comparison \(Experiments 1 and 2\)](#)).

In Experiment 3, we applied a tracking algorithm to assess whether it would improve detection results. While it significantly increased recall for one model, it reduced precision for others, with changes of 1-3%. The tracking algorithm reduced false negatives by detecting missed macaques between consecutive frames, improving recall. However, it also propagated false positives, where an incorrect detection in one frame was carried forward, lowering precision. Notably, the tracking algorithm was applied to test detection performance improvement; its actual tracking ability was not measured. This highlights the trade-off between improving recall and risking reduced accuracy from

false detections. While tracking enhances detection stability, future work should focus on refining the approach to mitigate these effects.

In Experiment 4, the evaluation of MacqD models on the ISC dataset demonstrated their ability to detect paired macaques from a different facility. Both MacqD - Curriculum and MacqD - Combine initially achieved an F1 score of 0.87. After applying the tracking algorithm, MacqD - Curriculum's score improved to 0.90, while MacqD - Combine's score declined slightly. While the model successfully generalised to this new setting, if performance in other facilities or new environments is suboptimal, further training with local data may be required to improve detection accuracy.

4.4 Limitations and Future work

A key limitation of this study is the focus on evaluating models using bounding boxes, despite MacqD and SegNet offering both bounding box and segmentation outputs. Segmentation provides richer information, which could be leveraged for more detailed analysis. Future work could explore the benefits of incorporating segmentation data for a more comprehensive assessment.

Further refinement of the tracking algorithm is needed to balance recall and precision. Exploring more advanced tracking techniques or hybrid approaches may help mitigate the propagation of false positives while maintaining improved detection stability.

A potential avenue for future work is the integration of MacqD with behaviour classification models to enable automatic behavioural analysis in addition to detection, offering a more comprehensive tool for research on primate behaviour.

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10 Undergraduate students at Newcastle University carried out the data annotation and manual refinement for this study.

6 Author contributions statement

Must include all authors, identified by initials, for example: A.A. conceived the experiment(s), A.A. and B.A. conducted the experiment(s), C.A. and D.A. analysed the results. All authors reviewed the manuscript.

7 Supplementary materials

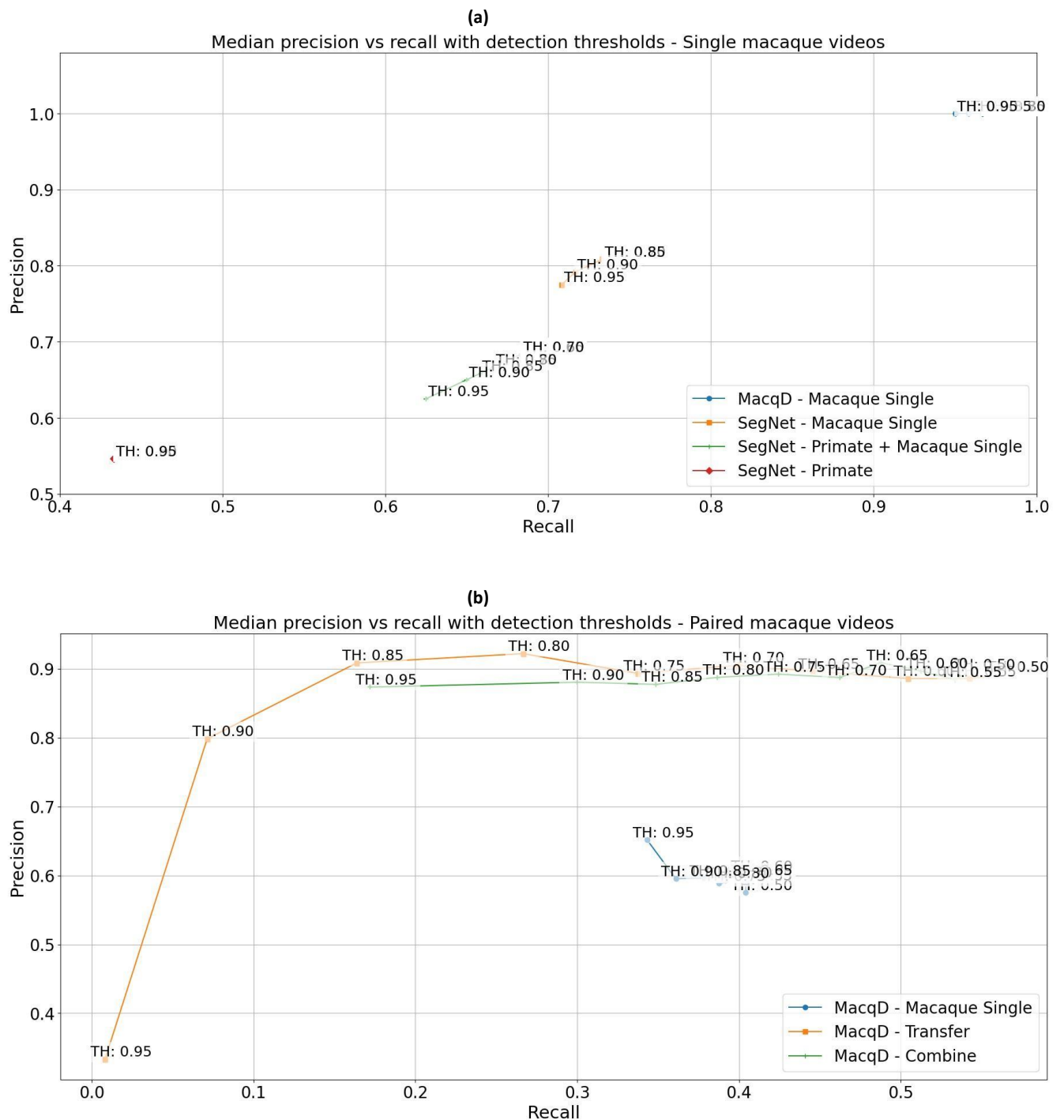


Figure 10. (a) Median precision vs. recall across various confidence score thresholds on the validation dataset with a single macaque in the focal cage. Optimal thresholds are: SegNet - Primate (0.95), SegNet - Primate + Macaque Single (0.7), SegNet Macaque Single (0.85), and MacqD - Macaque Single (0.8). (b) Median precision vs. recall across various confidence score thresholds on the

validation dataset with a pair of macaques in the focal cage. The optimal threshold for MacqD - Macaque Single, MacqD - Curriculum, and MacqD - Combine is 0.5.

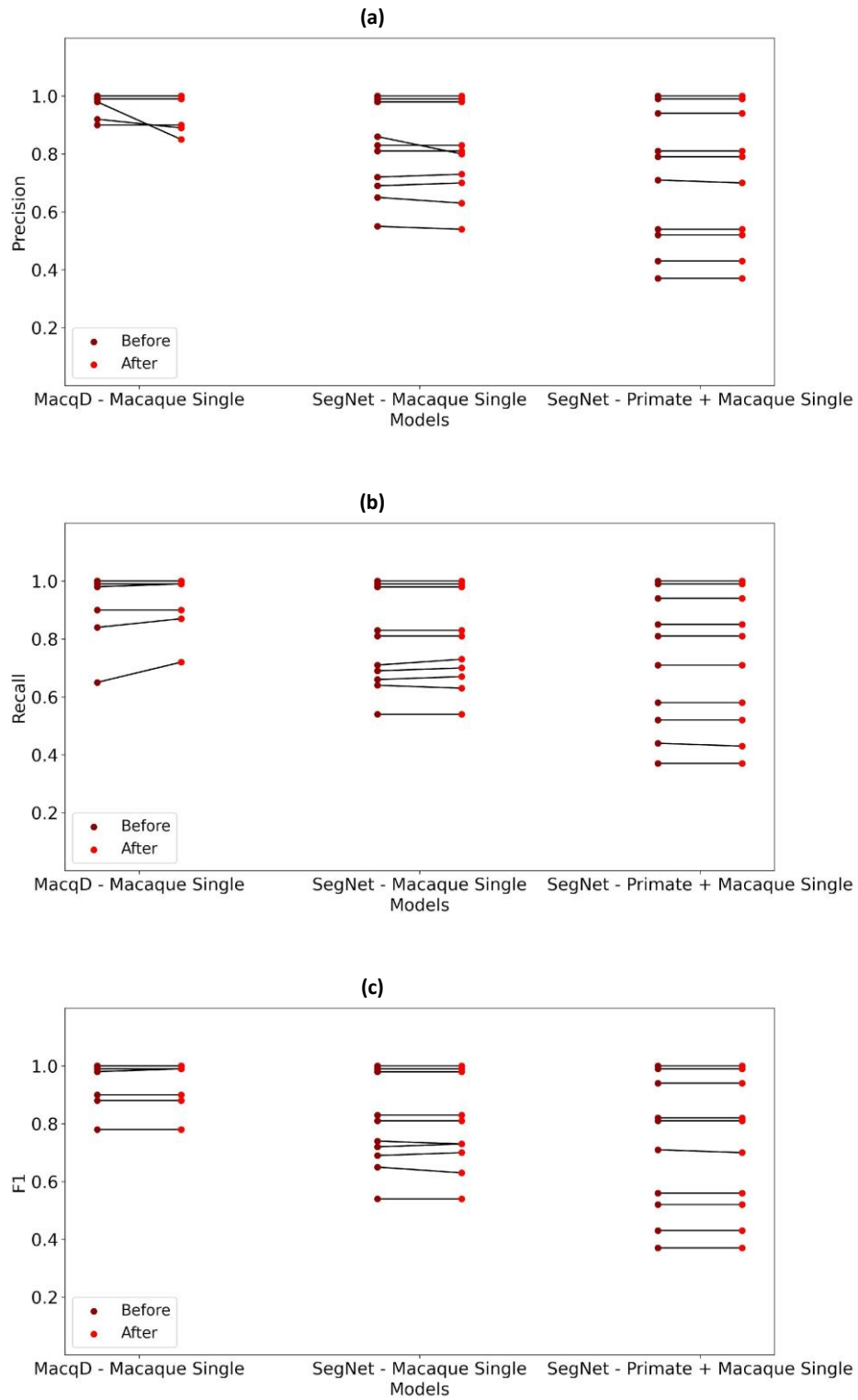
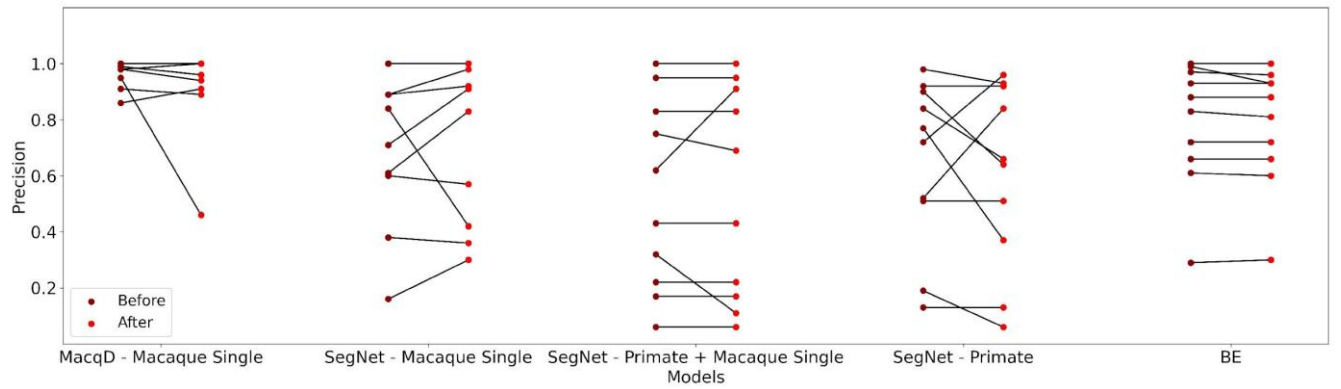
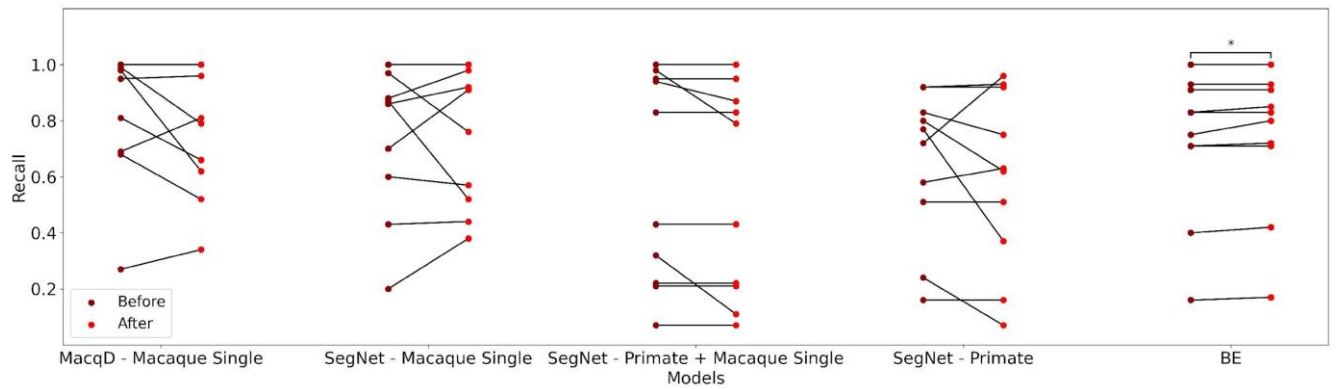


Figure 11. Visual representation of model predictions on frames featuring single macaques in the focal cage, which are the same individuals from the training dataset (Experiment 1, 'Same' dataset), before and after applying the tracking algorithm. (a) Precision; (b) Recall; (c) F1 Score.

(a)



(b)



(c)

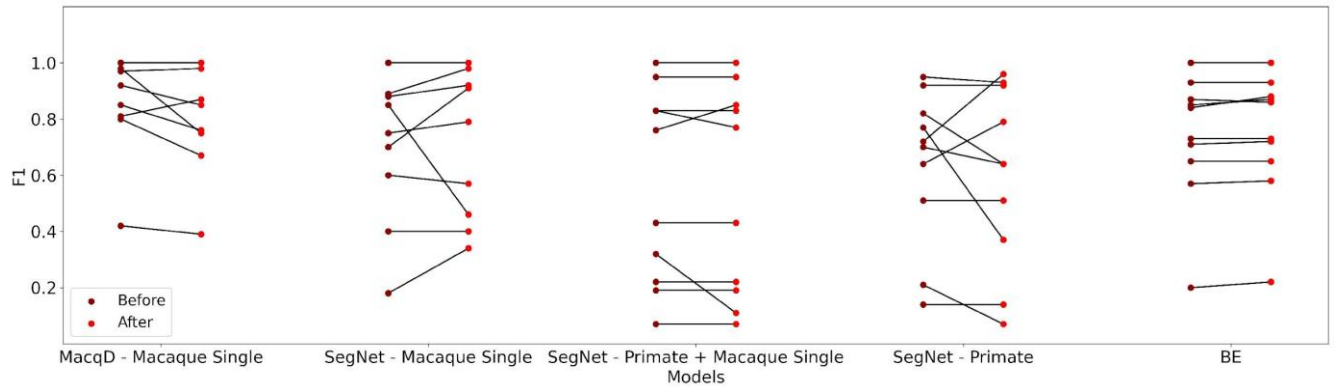


Figure 12. Visual representation of model predictions on frames featuring single macaques in the focal cage, which are the different individuals from the training dataset (Experiment 1, 'Different' dataset), before and after applying the tracking algorithm. (a) Precision; (b) Recall; (c) F1 Score. Bar indicates significant differences before and after applying the tracking algorithm (Wilcoxon test: $p < 0.05$).

Model	Dataset	Precisio n	Recall	F1
MacqD - Macaque Single	'Same'	0.99	0.99	0.99
SegNet - Macaque Single		0.81	0.77	0.77

SegNet - Primate + Macaque Single	0.75	0.76	0.76
MacqD - Macaque Single	0.96	0.8	0.86
SegNet - Macaque Single	0.87	0.84	0.85
SegNet - Primate + Macaque Single	'Different ' 0.56	0.61	0.6
SegNet - Primate	0.65	0.53	0.64
BE	0.85	0.82	0.80

Table 5. Summary of model predictions on frames featuring single macaques in the focal cage after applying the tracking algorithm.

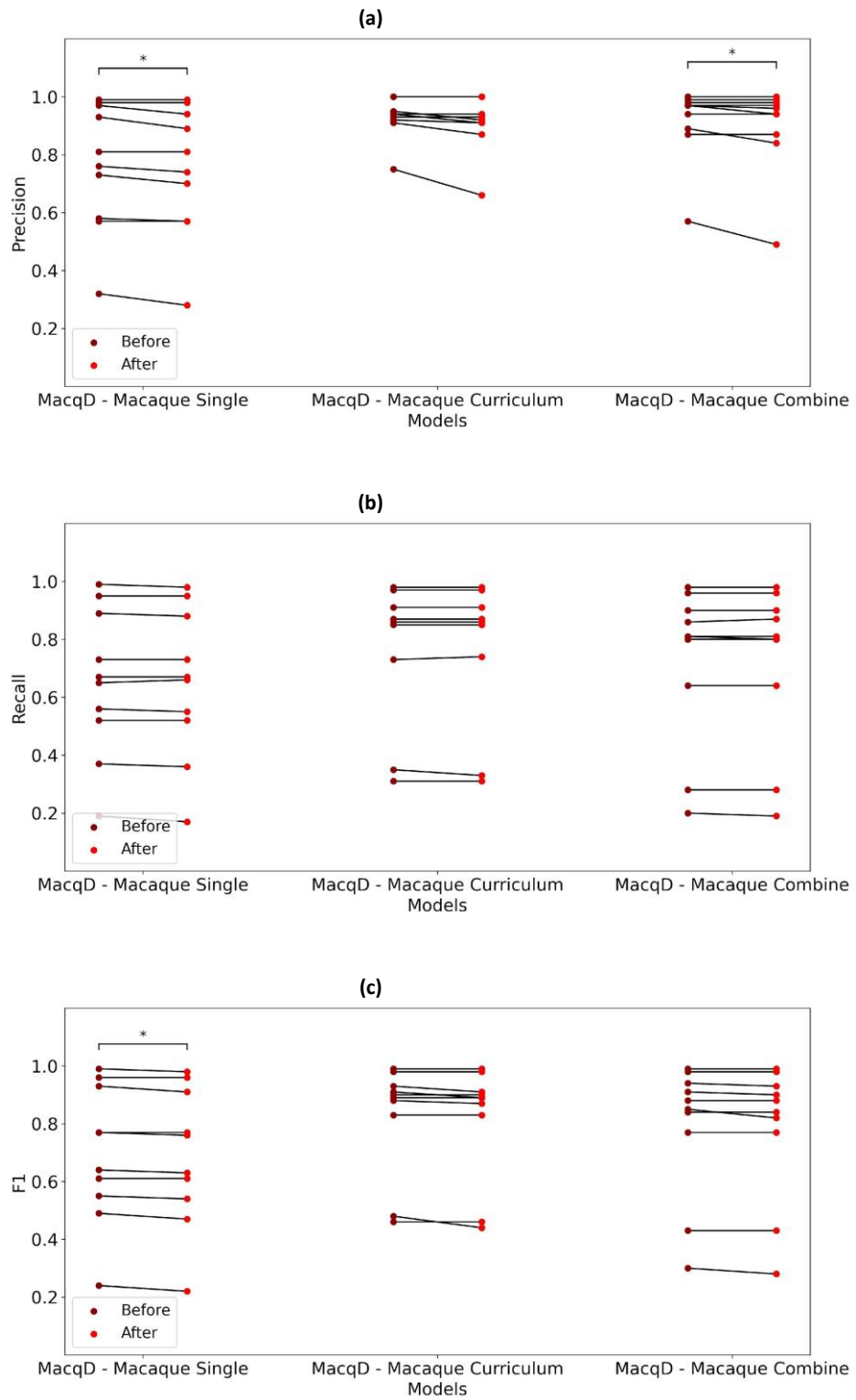


Figure 13. Visual representation of model predictions on frames featuring paired macaques in the focal cage, which are the same individuals from the training dataset (Experiment 2, 'Same' dataset), before and after applying the tracking algorithm. (a) Precision; (b) Recall; (c) F1 Score. Bars indicate significant differences before and after applying the tracking algorithm (Wilcoxon test: $p < 0.05$).

(a)

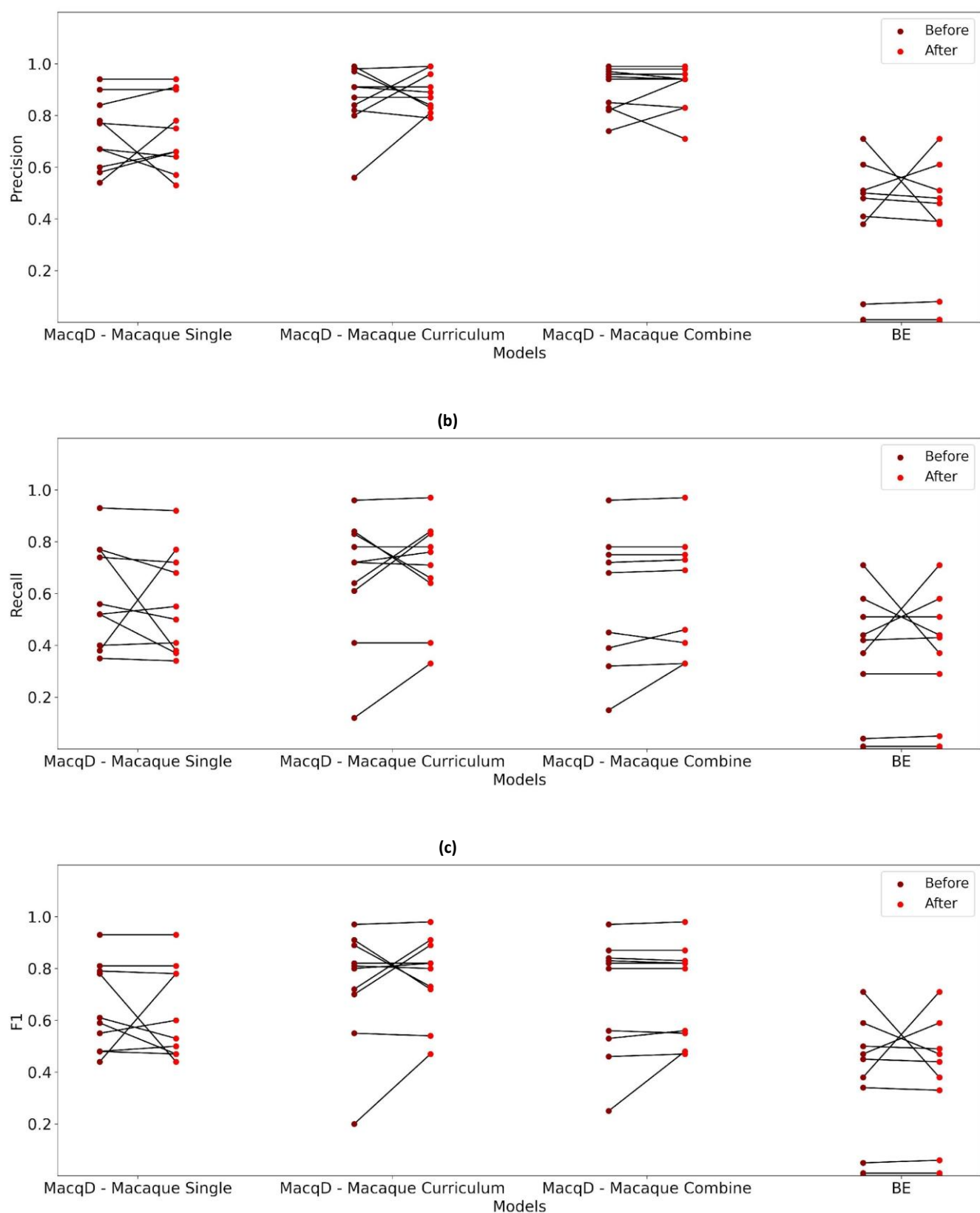


Figure 14. Visual representation of model predictions on frames featuring paired macaques in the focal cage, which are the different individuals from the training dataset (Experiment 2, 'Different' dataset), before and after applying the tracking algorithm. (a) Precision; (b) Recall; (c) F1 Score.

Model	Dataset	Precision	Recall	F1
MacqD - Macaque	'Same'	0.78	0.67	0.70
Single		0.93	0.87	0.89
MacqD - Macaque				
Transfer				
MacqD - Macaque		0.95	0.81	0.86
Combine				
MacqD - Macaque	'Different'	0.71	0.53	0.57
Single		0.88	0.74	0.81
MacqD - Macaque		0.94	0.71	0.81
Transfer				
MacqD - Macaque				
Combine				
BE		0.43	0.40	0.41

Table 6. Summary of model predictions on frames featuring paired macaques in the focal cage after applying the tracking algorithm.